WORLD FORUM FOR WOMEN IN SCIENCE EGYPT - 2026

Envisioning the Future: Empowering Science Innovation and Entrepreneurship

FORUM BOOK

National Research Centre

September 8-10, 2026

under the auspices of

Prof. Mamdouh Moawad
President of the National Research Centre (NRC)
Acting President of the Academy of Scientific Research and Technology (ASRT)

Judge Mrs. Amal Ammar
President of the National Council for Women (NCW)

under the supervision of

Prof. Ahmed Youssef
Dean of Chemical Industrial Research Institute, National Research Centre
Forum Founding Chair

Prof. Amal Amin
Founding Chair for World Forum for Women in Science, National Research Centre

WOMEN IN SCIENCE WITHOUT BORDERS INITIATIVE

WORLD FORUM FOR WOMEN IN SCIENCE, EGYPT 2026

Forum BOOK

Dates	8–10 September 2026
In-Person Venue (8–9 Sept)	National Research Centre, Egypt
Virtual Day (10 Sept)	Virtual Online Day

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Prof. Mamdouh Moawad

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Prof. Ahmed Youssef
Dean of Chemical Industrial Research Institute, National Research Centre, Egypt

Forum Founding Chair

Prof. Amal Amin

Professor for nanotechnology/polymer technology at National Research Center, Egypt, founding chair of World forum for women in science.

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8–9 September	National Research Centre, Egypt
10 September	Virtual Online Day

MASTER PROGRAM OVERVIEW

Date	Time	Program
Tuesday, 8 September 2026	**09:30–10:00**	Registration
Tuesday, 8 September 2026	**10:00–10:30**	Opening Remarks
Tuesday, 8 September 2026	**10:30–12:30**	Opening / Governing Session
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Tuesday, 8 September 2026	**14:15–15:00**	Lunch Break
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Wednesday, 9 September 2026	**09:30–11:00**	General Session: From Knowledge to Impact
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Wednesday, 9 September 2026	**14:30–15:00**	Lunch Break
Wednesday, 9 September 2026	**15:00–17:00**	Parallel Scientific Sessions
Thursday, 10 September 2026	**10:00–17:00**	ONLINE DAY

Foreword of Professor Mamdouh Moawad

President of the National Research Centre (NRC), Acting President of the Academy of Scientific Research and Technology (ASRT)

World Forum for Women in Science – Egypt 2026

Envisioning the Future: Empowering Science, Innovation and Entrepreneurship

Distinguished guests, dear colleagues, scientists, researchers, innovators, entrepreneurs, ladies and gentlemen,

It is a great pleasure to welcome you to the World Forum for Women in Science – Egypt 2026. On behalf of the National Research Centre, I am delighted to see this diverse community gathered here—bringing together different generations, disciplines, cultures and experiences, united by a shared belief in the power of science to improve lives and shape a better future

Our theme, "Envisioning the Future: Empowering Science, Innovation and Entrepreneurship," reminds us that the future is not something we simply wait for; it is something we create together.

Women have made remarkable contributions to science, innovation and society. Yet true progress requires an environment where everyone—regardless of gender, background, discipline or career stage—has the opportunity to learn, contribute, lead and transform ideas into meaningful solutions.

Creating such an environment requires inclusive institutions, equal opportunities, supportive networks, access to resources, mentorship and strong partnerships across academia, industry, government and society.

As the National Research Centre, we are committed to fostering scientific excellence, encouraging innovation and entrepreneurship, supporting young researchers, and strengthening collaboration across borders and disciplines. We believe that diversity is not only a value; it is a source of creativity, resilience and scientific progress.

I hope this Forum will be a space where voices are heard, ideas are exchanged, connections are made and inspiring conversations lead to meaningful action.

Together, let us build a future in which science is open to all, innovation serves humanity, and every individual has the opportunity to contribute to a more sustainable and inclusive world.

Welcome to Egypt, and welcome to this shared journey toward the future.

Thank you.

Biography of Profesoor Mamdouh Moawad

President of the National Research Centre (NRC)

Acting President of Academy of Scientific Research and Technology (ASRT)

Professor Dr. Mamdouh Moawad Ali Hassan is a distinguished scientist and academic leader, currently serving as President of the National Research Centre (NRC) and Acting President of Academy of Scientific Research and Technology (ASRT). He is a Professor of Biochemistry at the Department of Biochemistry, Biotechnology Research Institute, NRC. He earned his Bachelor of Science degree in 1989, followed by a Master's degree in Biochemistry in 1995, and a PhD in Biochemistry in 2000 from the Department of Biochemistry, Faculty of Science, Ain Shams University.

Professor Mamdouh began his career at the National Research Centre as an Assistant Researcher (17 July 1994 – 24 January 1996), then served as Assistant Researcher (25 January 1996 – 25 December 1999), Researcher (29 February 2000 – 29 April 2005), and Associate Professor (30 April 2005 – 2 May 2010). Since 3 May 2010, he has held the position of Professor of Biochemistry at the Department of Biochemistry, Biotechnology Research Institute, NRC.

Throughout his career, Professor Mamdouh has held several key administrative and leadership positions at the NRC. These include Deputy of the Department of Biochemistry (2002–2010), Deputy of the Biotechnology Research Institute (2010–2015), Head of the Biotechnology Research Institute (10 February 2018 – 7 November 2018), Vice President of the NRC for Scientific and Research Affairs (27 July 2020 – 26 July 2024), and currently President of the NRC (since 25 November 2024) and Acting President of ASRT (since 11 May 2026). These leadership roles have significantly enriched his administrative expertise, enabling him to oversee institutional development, research governance, and strategic planning, while gaining comprehensive insight into national and international research systems.

Professor Mamdouh has conducted numerous international scientific visits to strengthen cooperation with universities, research centres, and institutes worldwide, resulting in the signing of several bilateral and multilateral cooperation agreements. He has also chaired and served as a member of many scientific, administrative, and advisory committees at both national and international levels, and has participated in various leadership and capacity-building programs, enhancing his understanding of the regulatory frameworks governing scientific research.

In terms of scientific achievements, Professor Mamdouh has published more than 175 research articles in leading international peer-reviewed journals. His work has received over 3,990 citations with an H-index of 37 (Scopus), and more than 5,350 citations with an H-index of 42 (Google Scholar). He has authored 12 national and international books, supervised over 25 Master's and PhD theses, and participated in more than 25 funded research projects supported by national and international agencies.

Notably, he served as a Visiting Professor through a joint project funded by the German Research Foundation (DFG) in collaboration with Germany, Canada, France, and Switzerland at the Department of Oncology, National Cancer Institute, and Germany. He is also the holder of two patents.

Professor Mamdouh has received numerous awards and honors, including the National Research Centre Encouragement Award in Chemistry (2004), the Scientific Excellence Award in Basic Sciences (2012), and the Distinguished Research Output Award for achieving the highest H-index and impact factor between 2010 and 2015. In 2020, he was awarded the Best Personality Award by the Egyptian Arab Foundation for Investment, Innovation, and Industrial Development. He has also received several certificates of appreciation from academic institutions, professional associations, and civil society organizations.

In addition, Professor Mamdouh has made significant contributions as a peer reviewer for numerous national and international journals, research projects, theses, academic promotions, and scientific awards. He is a member of more than 100 national committees and approximately 25 international scientific associations, including the International Society of Infectious Diseases, the International Society of Nephrology, and the Middle East Cancer Society. He is listed in Who's Who in the World and serves on the editorial boards of more than 190 prestigious international scientific journals.

Professor Mamdouh has also actively contributed to capacity building, having attended over 45 national and international training courses and workshops, participated in more than 80 international and national conferences and seminars, and organized or taught more than 110 training programs for young researchers, technicians, and research staff both within and outside the NRC.

Foreword of Professor Ahmed Youssef

Dean of Chemical Industrial Research Institute

It is a great honor to participate in the World Women Forum, a distinguished international platform that brings together women scientists, researchers, innovators, and leaders from different countries to exchange knowledge, share experiences, and promote scientific collaboration.

Women have always played a fundamental role in advancing science, technology, and innovation, and their contributions are increasingly essential to addressing the complex challenges facing our world, including climate change, sustainable development, food security, environmental protection, and the transition toward a green and circular economy.

The World Women Forum provides an important opportunity to recognize these achievements while creating new pathways for cooperation, mentorship, and knowledge exchange among women scientists and institutions worldwide. I strongly believe that empowering women in science is not only a matter of equality, but also a strategic investment in the future of scientific progress and sustainable development.

As Dean of the Chemical Industrial Research Institute, I am pleased to support initiatives that encourage scientific excellence, innovation, and the meaningful participation of women in research and industrial development. I sincerely congratulate the organizers and participants of the World Women Forum and wish this important event continued success in building a stronger, more inclusive, and sustainable scientific future.

Biography of Professor Ahmed Youssef

Prof. Ahmed Youssef is a Dean of Chemical Industrial Research Institute and previous Head of Packaging Materials Department, NRC, Giza, Egypt. He also is working as Secretary of the Industry Technology Council, Academy of Scientific Research and Technology. He has completed his PhD from Ain Shams University and postdoctoral studies from Arkansas University, USA and Grenoble INP-France, He so far won the State Award for Excellence in Advanced Technological Sciences that Serve Basic Sciences (2023), Ford Foundation Scholarship (2004) for PhD, the AU-TWAS Young Scientist National Awards in field of Basic Sciences, Technology and Innovation (2013), and the NRC Award of Encouragement in Applied Chemistry (2015) and Excellence award in Applied Chemistry (2021) and NRC award for Pioneer Award in applied chemistry (2016).. Moreover, He awarded the State Award for Advanced Technological Sciences, which serves the basic sciences (2016). He received the Order of Merit, first class, for Sciences & Arts, from the President Abdel-Fatah Al-SiSi, President of the Arab Republic of Egypt)2017 (and the Egyptian Academy of Science and Technology awarded for Plastic and Rubber Technology (2017), and Dream award for the best applied article in industrial nutrition (2015, 2019, 2020, 2021, 2022, 2023). He published more than 233 papers with H-index 55 citation 10410 (Google scholar), and 55 H-Index and 10250 citations on Scopus site. He currently engaged in several research grants in the field of polymer nanotechnology, preparation of nanomaterial's, packaging technology and solar cell. He has coupled research experience with industrial work with National and International Counterparts.

Foreword of Professor Amal Amin

Professor at Chemical Industries research Institute-NRC

Founding Chair of Women in Science Without Borders Initiative (WISWB) & World Fora for Women in Science

It is my distinct honor, as the Founder of Women in Science Without Borders (WISWB) and the World Forum for Women in Science, to organize the World Forum for Women in Science – Egypt 2026 at the National research Centre (NRC-my workplace), where it convenes under the theme "Envisioning the Future: Empowering Science, Innovation and Entrepreneurship" under the auspices of Prof Dr Mamdouh Moawad, the president of National Research Centre and the acting president of the Egyptian Academy of scientific research and technology. The forum is organized by the chemical industries research institute and under supervision of Prof Dr. Ahmed Youssef, the dean of the institute. The forum is under patronage of women national council, sponsored by Sanofi and Misr Elkeir and supported by many organizations and high-level scientists and stakeholders worldwide. WISWB is a global initiative spreads over 92 countries now and recognized by UNESCO as global gender inclusive scientific platform with the theme of science for sustainable development. The Forum is grounded in the recognition that scientific advancement is fundamental to sustainable development, societal resilience, and the creation of knowledge-based economies. At the same time, achieving the full potential of science requires gender inclusive research ecosystems in which all have equitable opportunities to participate, lead, innovate, and translate knowledge into societal impact. The contributions presented in the current forum reflect diverse perspectives on the evolving relationship between science, innovation, entrepreneurship, leadership, and sustainable development. They highlight both the achievements of women and men in scientific and technological fields and the structural challenges that continue to influence participation, career progression, research leadership, and innovation pathways. A central objective of this Forum is therefore to move beyond discussion toward evidence-informed action. Strengthening the interface between research and application, fostering entrepreneurial capacities, supporting interdisciplinary and international collaboration, and creating enabling environments for scientific leadership are essential components of a future-oriented research ecosystem. The Forum also emphasizes the importance of collaboration across disciplines, institutions, generations, and geographical boundaries. Through WISWB, we seek to advance a global scientific community founded on knowledge exchange, partnership, inclusion, and mutual respect. I hope the forum will provide a valuable scholarly resource for researchers, policymakers, educators, innovators, and emerging scientific leaders especially by participation of about 25 nationalities. More importantly, I hope they will stimulate further research, dialogue, partnerships, and practical initiatives that contribute to a more inclusive and innovative global scientific landscape. Finally, I like to mention Louis Pasteur quote that:

"Science knows no country, because knowledge belongs to humanity."

Biography of Professor Amal Amin

Dr. Amal Amin is a professor for nanotechnology/polymer technology at national research Center-Egypt. She studied in, worked at and travelled to +30 countries including Spain, Germany (PhD-DAAD), USA, France having numerous scientific activities. She was cofounder and executive committee member of the global and Egyptian young academies (GYA, EYAS). She was cofounding president and coordinator of the Egyptian society and Arab network for nanotechnology & TWAS young affiliate, TWAS-TYAN member, TWAS-AAAS science diplomacy alumni, executive committee member of (science in Exile) global initiative to support refugee scientists and at-risk scholars, advisory board member of ISTIC-UNESCO and Ex-member of ORCID board of directors. She is one of the coordinator lead authors of UNESCO-science report 2026 and member of the UNESCO International Experts Group on Scientific Literacy and Science Popularization 2026. She is a fellow of World academy of sciences (TWAS), International science council (ISC), founding chair of women in science without borders' (WISWB) initiative & World forum for women in science series, science for humanity foundation-Egypt, cofounding chair for science for humanity-global society-Italy, world forum for science innovation and entrepreneurship and science diplomacy for the future initiative. She is cofounding chair of Northern African Research and Innovation Management Association (NARIMA) initiative. She has received outstanding women in tech award of Africa (Africa), national award as distinguished woman in science and was classified by the global gender scan representing Egypt among few women in S&T in Africa who (make the difference). She got 2024 peace science and technology leadership award for exceptional women for peace global award.

Dr. Amal achievements were featured in (NASAC books 2017, 2020 and 2026), scientific African (2019), Nature (2020), Royal society for chemistry (2020), and others.

Opening Remarks

Prof. Amal Amin

Professor for nanotechnology/polymer technology at National Research Center, Egypt, founding chair of World forum for women in science.

Prof . Ahmed Youssef

Dean of Chemical Industrial Research Institute, National Research Centre, Egypt

Judge Mrs Amal Ammar

The President of the National Council for Women (NCW)

HRH Princess Dr. Nisreen El-Hashemite

Executive Director of the Royal Academy of Science International Trust (RASIT),

Prof. Quarraisha Abdool Karim

President of The World Academy of Sciences (TWAS)

Mr. Adrien Delamare-Deboutteville

International Pharma MCO Head for Africa, Sanofi'

Prof. Mamdouh Moawad Ali Hassan

President of the National Research Centre (NRC), Acting President of the Academy of Scientific Research and Technology

Judge Mrs. Amal Ammar

President of the National Council for Women (NCW)

Judge Mrs Amal Ammar is one of the prominent female leaders in the Arab Republic of Egypt. She has served as President of the National Council for Women (NCW) since 2024, pursuant to a decision issued by H.E. President Abdel Fattah El-Sisi in November 2024. She was also appointed as a member of the Supreme Council of the Egyptian Red Crescent. This appointment marked a significant milestone, making the National Council for Women, for the first time, represented among the members of the Supreme Council of the Egyptian Red Crescent. From 2020 to 2024, Judge Ammar served as Assistant Minister of Justice for Human Rights, Women, and Children Affairs, where she contributed to the development of policies and legislation aimed at advancing women’s rights and protecting the most vulnerable groups. Judge Amal Ammar was among the first 30 female judges appointed to the Egyptian judiciary in 2007. This historic appointment represented an important milestone in promoting women’s participation in the Egyptian judicial system and strengthening their role within the justice sector. At the regional level, she was elected Vice President of the Advisory Council of the African Union Advisory Board Against Corruption in 2019. She had previously served as a member of the Council from 2018 to 2022, contributing to efforts to promote integrity, good governance, and the fight against corruption across the African continent. Judge Ammar has also been a member of the National Council for Women since 2012 and served as Vice Chair of the Council’s Legislative Committee. In this capacity, she contributed to the development and review of a number of laws and policies related to women’s rights and empowerment. She has represented Egypt at a number of high-level international and regional forums, including meetings of the United Nations Committee on the Elimination of Discrimination against Women (CEDAW). She has also participated in regional workshops addressing the fight against corruption and human trafficking, as well as the promotion of judicial cooperation. Judge Amal Ammar holds an LL.B. and a Master’s degree in International Business Law from Cairo University, in cooperation with the University of Paris 1 Panthéon-Sorbonne.

HRH Princess Dr. Nisreen El-Hashemite

Executive Director of the Royal Academy of Science International Trust (RASIT)

Champion behind the International Day of Women and Girls in Science, observed annually on 11 February. HRH Princess Dr. Nisreen El-Hashemite is an internationally recognized medical scientist, humanitarian leader, and global advocate for science, technology, innovation, gender equality, cultural diplomacy, and sustainable development. A direct descendant of Prophet Muhammad and granddaughter of King Faisal I of Iraq and Syria, she is distinguished as the first Royal Princess to qualify professionally in biomedical sciences, medicine, and human genetics. Her scientific career includes research at University College London, where she contributed to advances in pre-implantation genetic diagnosis, and at Harvard Medical School, where her pioneering work on tuberous sclerosis and sex-based differences in cancer contributed to developments in immunotherapy and early cancer detection. A longstanding champion of women in science, Princess Nisreen founded the Women in Science International League in 1997 and established an international award recognizing women in science in 1998. As Executive Director of the Royal Academy of Science International Trust (RASIT), she has promoted science, culture, education, and business diplomacy as essential instruments for sustainable development, peace, and international cooperation. Her global leadership has been closely associated with the United Nations and its agencies, particularly in advancing reproductive health, education, gender equality, infectious-disease prevention, and the Sustainable Development Goals. She is widely recognized for her role in advancing the establishment of the International Day of Women and Girls in Science, observed annually on 11 February, and for promoting women's leadership in science diplomacy. She has also contributed to initiatives supporting women in diplomacy, cultural diplomacy, science-policy engagement, and international peacebuilding.

Princess Nisreen founded and leads numerous initiatives, including Science Café, Women in Arts, Culture for Peace, and programs connecting science, culture, and social development. She has authored scientific, socio-economic, literary, and poetic works and has received numerous international honors, honorary doctorates, and three Lifetime Achievement Awards.

Beyond her professional achievements, she is deeply committed to mentoring young people, particularly girls, encouraging them to pursue scientific careers. Her lifelong vision emphasizes dialogue across cultures, disciplines, and regions, positioning science, education, culture, and innovation as powerful foundations for peace, inclusion, and sustainable human development.

Professor. Dr. Quarraisha Abdool Karim

President of The World Academy of Sciences (TWAS)

Quarraisha Abdool Karim, is a world renowned South African infectious diseases epidemiologist and co-founder of the Centre for the AIDS Programme of Research in South Africa (CAPRISA), where she is the John C. Martin Chair in Global Health and Associate Scientific Director. She is Professor of Clinical Epidemiology at Columbia University, New York and Pro Vice-Chancellor Health Sciences at the University of KwaZulu-Natal, Durban South Africa and Distinguished Visiting Professor, VinUniversity, Hanoi.

Abdool Karim's work has profoundly influenced the global HIV prevention landscape over the past three decades, particularly in enhancing understanding of the evolving HIV epidemic and developing prevention technologies for women. Her groundbreaking research has transformed HIV prevention for women in Africa. As co-leader of the CAPRISA 004 trial, she demonstrated the efficacy of tenofovir gel in reducing HIV acquisition, establishing proof of concept for antiretroviral-based prevention. Her research also highlighted the role of genital inflammation in enhancing HIV acquisition. She also provided key insights into HIV-tuberculosis (TB) co-infection and the impact of COVID-19 on HIV.

She is President of The World Academy of Sciences (TWAS) (the first woman in its 40year history) and UNAIDS Special Ambassador for Adolescents and HIV. Abdool Karim is a member of the WHO Science Council and serves on the Boards of the Global Health Innovative Technology (GHIT) Fund, Global Virus Network, and the U.S. Friends of the Global Fight Against AIDS, Tuberculosis and Malaria. She is a member of the Steering Committees of the World Science Forum, Science and Technology for Society Forum, and the Forum on Big Data and Sustainable Development, as well as the WHO Executive Groups for the COVID-19 Solidarity Therapeutics and Vaccines Trials.

Professor Abdool Karim has authored more than 300 peer-reviewed publications in leading journals, including *Science*, *Nature*, *The New England Journal of Medicine*, and *The Lancet*, and co-edited key reference works such as the *Oxford Textbook of Global Public Health* and *HIV in South Africa*. Through the Columbia University–Southern African Fogarty AIDS International Training and Research Programme, she has played a pivotal role in strengthening southern Africa's science base by training over 600 researchers in HIV and TB.

She is an elected member of the U.S. National Academy of Medicine, the American Academy of Arts and Sciences, and the Indian Academy of Sciences; Fellow of: TWAS, the Royal Society, International Science Council, the Chinese Academy of Sciences Centre for Big Data and AI (CBAS), the African Academy of Sciences, the Royal Society of South Africa, and the Academy of Science of South Africa. She has received multiple honorary doctorates and has been inducted to the Order of Mapungubwe, South Africa's highest national honour. Her contributions have been recognised with more than 40 major awards, including the inaugural African Academy of Science Olusegun Obasanjo

Prize for Scientific Breakthrough and/or Technological Innovation, the TWAS-Lenovo Science Award; L’Oréal-UNESCO Women in Science Award (MENA region), the Canada Gairdner Global Health Award, the Noguchi Africa Prize, the Virchow Award and the Lasker-Bloomberg Public Service Award.

Mr. Adrien Delamare-Deboutteville

The International Pharma MCO Head for Africa :Adrien Delamare-Deboutteville is the International Pharma MCO Head for Africa within Sanofi's General Medicines International Pharma department, based in Cairo, Egypt.

The International Pharma MCO Head for Africa :Adrien Delamare-Deboutteville is the International Pharma MCO Head for Africa within Sanofi's General Medicines International Pharma department, based in Cairo, Egypt. He leads strategic pharmaceutical operations and business initiatives across the Africa Market Cluster Organization's international pharma portfolio.

General information

National Research Centre (NRC), Egypt

The National Research Centre (NRC) is Egypt's largest multidisciplinary research and development institution dedicated to advancing both basic and applied scientific research. Established as an independent public organization in 1956, the NRC was founded to promote scientific research addressing national priorities, particularly in industry, agriculture, public health, and other sectors of the national economy.

Located in Dokki, Giza, the NRC has evolved into a major national hub for scientific research, innovation, technology development, and capacity building. According to its official information, the Centre currently comprises 14 research institutes and 109 departments, covering diverse fields including pharmaceutical and chemical industries, biotechnology, agriculture, food and nutrition, medicine, environmental and climate research, engineering, energy, physics, and other fundamental sciences.

A major strength of the NRC is its multidisciplinary character, which enables scientists from different fields to collaborate on complex national and global challenges. Its research activities aim not only to generate scientific knowledge but also to translate research outcomes into products, technologies, services, and practical solutions that support economic and societal development. The Centre also provides scientific consultancy, training, and opportunities for cooperation with industry and other national and international institutions.

Over its seven decades of development, the NRC has played an important role in strengthening Egypt's scientific and technological capacity. Today, it continues to promote innovation, technology transfer, international collaboration, and knowledge-based sustainable development, positioning scientific research as a key driver of Egypt's future.

National Council for Women (NCW), Egypt

The National Council for Women (NCW) is Egypt's principal national institution dedicated to promoting and advancing the rights, empowerment, and participation of women in society. It was established in 2000 under Presidential Decree No. 90 as a national mechanism to support women's advancement and integrate their contributions into Egypt's comprehensive development efforts. The Council was subsequently regulated by Law No. 30 of 2018, which affirmed its independence and its financial, administrative, and technical autonomy.

The NCW works across the political, economic, social, cultural, educational, and legislative dimensions of women's empowerment. Its responsibilities include proposing policies and national plans for women, monitoring their implementation, contributing to the development of legislation, promoting equal opportunities and non-discrimination, and representing Egyptian women in regional and international forums.

A major focus of the Council is strengthening women's participation in decision-making, leadership, entrepreneurship, employment, education, and scientific research, while supporting initiatives that address violence and discrimination against women. The Council also collaborates with governmental institutions, civil society organizations, universities, international organizations, and other stakeholders to advance gender equality and sustainable development.

The NCW has played an important role in developing and supporting Egypt's National Strategy for the Empowerment of Egyptian Women 2030, which provides a framework for advancing women's political, economic, social, and legal empowerment. Through its policies, awareness programmes, research, training, advocacy, and partnerships, the Council contributes to creating an enabling environment in which Egyptian women can participate fully in national development and help shape Egypt's future.

Sanofi Egypt

Sanofi Egypt is an affiliate of Sanofi, a global healthcare company R&D driven dedicated to improving people's lives through science, innovation, and greater access to healthcare while accelerating its goal of reducing the environmental impact of its products and operations worldwide while building a diverse and inclusive workplace.

With more than six decades of presence in Egypt, Sanofi has developed a strong role within the Egyptian healthcare ecosystem, providing innovative medicines and vaccines across major therapeutic areas, including diabetes, immunology, oncology, rare diseases, cardiovascular conditions, and other significant health needs.

Beyond pharmaceutical products, Sanofi Egypt contributes to strengthening healthcare through partnerships with governmental institutions, universities, healthcare organizations, and local stakeholders. Its initiatives support disease awareness, patient care, healthcare professional education, and access to essential treatments.

Sanofi Egypt has a long term sustainable corporate social responsibility programs emphasis on access to healthcare through education

Through scientific collaboration, innovation, strategic partnerships, and responsible business practices, Sanofi Egypt continues to contribute to advancing healthcare, strengthening local capabilities, and building a healthier and more sustainable future for Egypt.

Misr El Kheir Foundation

Misr El Kheir Foundation is a leading Egyptian non-governmental, non-profit development organization established in 2007, dedicated to developing human potential through sustainable and integrated solutions that create lasting impact. Guided by the belief that investing in people is the foundation of development, the Foundation works across key areas including health, education, scientific research and innovation, social solidarity, and life necessities, reaching millions of beneficiaries across Egypt through strong partnerships with government institutions, the private sector, academia, and civil society. The Foundation's Scientific Research and Innovation Sector plays a pivotal role in bridging the gap between scientific research and sustainable development by supporting applied research, technological innovation, and entrepreneurship, and by transforming promising research and innovative ideas into practical solutions that address pressing societal and environmental challenges. Through funding research, supporting prototype development and validation, facilitating implementation and scale-up, and fostering incubation and startup creation, the Sector connects researchers and innovators with communities, industry, and development partners to ensure that knowledge and innovation translate into tangible and sustainable impact. Through this integrated approach, Misr El Kheir Foundation contributes to building a stronger innovation ecosystem in Egypt and advancing solutions that improve lives and support sustainable development for present and future generations.

Forum Coordinators

Professor Dr. Hossam El Nazer

Board Director of the International Association for Hydrogen Energy (IAHE), of Renewable Energy and Water Technology at the National Research Centre (NRC), Egypt

Prof. Hossam El Nazer is an internationally active scientist and academic specializing in water technologies, renewable energy, hydrogen, and sustainable development. He serves as a Board Director of the International Association for Hydrogen Energy (IAHE) and is involved in international networks and initiatives including RES4CITY, Global Reach Hub, UMJAE, and the Water Embassy (Ambassade de l'Eau), representing Egypt. Prof. El Nazer is a Professor of Renewable Energy and Water Technology at the National Research Centre (NRC), Egypt, with more than 20 years of research and professional experience. His expertise includes water treatment and desalination, renewable-energy applications, hydrogen technologies, photocatalysis, and climate-resilient water solutions. He has coordinated and contributed to international research and innovation projects with partners across the UK, Europe, and other regions, and is actively engaged in strengthening collaboration between academia, research institutions, industry, and policymakers. He also serves as member of the Entrepreneurship Committee of the Chemical Industries Research Institute, NRC supporting the translation of scientific research into innovation, entrepreneurship, and practical applications.

Professor Mahmoud Essam Abd El-Aziz Shaaba

Polymers and Pigments Dep., Chemical Industries Research Institute, National Research Centre (NRC), Cairo, Egypt

Mahmoud Essam Abd El-Aziz Shaaba is a Professor in the Polymers and Pigments Dep., Chemical Industries Research Institute, National Research Centre (NRC), Cairo, Egypt. He obtained his B.Sc. in Chemistry from Cairo University in 2001, an M.Sc. in Organic Chemistry from Cairo University in 2009, and a Ph.D. in Organic Chemistry from Ain Shams University in 2015.

His research interests encompass polymer and biopolymer nanocomposites, nanotechnology, sustainable materials, agricultural waste valorization, nanofertilizers, water and wastewater treatment, eco-friendly packaging materials, and functional nanomaterials. He has served as PI or Co-PI and as a member of several national and international research projects addressing sustainable materials, nanotechnology, water treatment, agricultural waste utilization, and eco-friendly packaging. He has published over 81 scientific papers, with an h-index of 30 and more than 2,600 citations.

He has also contributed to numerous national and international scientific conferences and has participated in the organization of several scientific conferences and workshops. He serves as a reviewer for a range of international journals in polymer science, nanotechnology, materials science, agriculture, and related fields, and has contributed as a guest editor and editorial board member for scientific journals. He received several awards, including the Scientific Excellence Award in Basic Sciences and Their Technological Applications from the Members' Club of Research Institution (2025), the NRC Scientific Encouragement Award in Basic Sciences (2019) and the Prof. Yehya Abd El-Latif Award in Applied Nanotechnology (2018).

Members of Organizing committee

(Alphabetically arranged)

Dr. Ahmed Hegazy

National Research Centre, Egypt

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Photochemistry Department, Chemical Industries Research Institute, National Research Centre (NRC), Egypt

Prof. Eman E. El Shanawany

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Physics Research institute, National Research Centre, Egypt

WORLD FORUM
FOR WOMEN IN SCIENCE

Envisioning the Future: Empowering Science Innovation and Entrepreneurship

EGYPT 2026

Under auspices of

PROFESSOR MAMDOUH MOAWAD
President of the National Research Centre (NRC), Acting President of the Academy of Scientific Research and Technology (ASRT)

JUDGE MRS. AMAL AMMAR
President of the National Council for Women (NCW)

Under supervision of

PROFESSOR AHMED YOUSSEF
Dean of Chemical Industrial Research Institute, National Research Centre

Forum Founding Chair

PROFESSOR AMAL AMIN
Founding Chair for World Forum for Women in Science, National Research Centre

FORUM COORDINATORS

PROF. DR. HOSSAM EL NAZER

Professor of Renewable Energy
and Water Technology, National Research Centre, Egypt
National Diploma International Association for
Hydrogen Energy (IAHE)

PROF. DR.MAHMOUD ESSAM ABD EL-AIZIZ SHAABA

Professor, Polymers Department
Chemical Industries Research Institute
National Research Centre, Egypt

MEMBERS OF THE ORGANIZING COMMITTEE
(ALPHABETICALLY ARRANGED)

DR.
AHMED HEGAZY

National Research Centre, Egypt

DR.
ELZAHRAAH AHMED RAMADAN ELGOHARY

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PROF.
EMAN E. EL SHNANWAY

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HEBA ABDELAZIZ MOUSSA ABDALLA

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ASS. PROF.
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ASS. PROF.
DR. SAFAA SAEID MOHAMED ALI

Physics Research Institute, National Research Centre, Egypt

NATIONAL RESEARCH CENTRE

SEPTEMBER 8–10 , 2026

Plenary opening session

Plenary Opening session:

Title: Envisioning the Future: Empowering Science, Technology, Innovation, Entrepreneurship and the Governance Interface for Sustainable Development

The session Theme

This session explores how science, innovation, entrepreneurship, and good governance can work together to accelerate sustainable development. It is organized into two complementary panels. The first focuses on transforming scientific knowledge into practical solutions, while the second examines how effective governance and evidence-informed policies can help those solutions create lasting impact.

Moderators

Mr Ashraf Amin

Head of the Science Desk, Al-Ahram Newspaper

Mr Ashraf is a Senior Editor, media trainer, Science and Health journalist of 25 years' experience. I am responsible for a weekly Science & Health page every Wednesday in AlAhram Daily newspaper in Arabic. I also publish my stories in Ahram online English service. He is a board member of local and regional associations in the fields of science policies and journalism. His experience varies in the fields of media training, managing media conferences & workshops, creating, assessment and production of media content for newspapers and TV. He has worked as a freelancer for several media outlets like: Scientific American Arabic Edition, SciDev.net, Environment& Development media outlet, the Saudi Press Agency and Dubai TV. He has attended and covered several climate change and environmental conferences and workshops around the world like COP27 in Sharm El-Sheikh, Egypt, UNEP conference in Bali, UNE Assemblies in Nairobi, Kenya, and UNFCCC workshop on deforestation in Inner Mongolia, China. For the past years, he has worked as project manager for regional and international conferences on science and environmental journalism. He has also designed and moderated workshops and training programs for science journalists, public officers on science communication and production of scientific and environmental content. These projects were supported and sponsored by companies and universities and public institutions like: the Goethe Institute, Ford Foundation, Candid Academy, the GIZ, German Academic Exchange Service DAAD, Sanofi, SicommX, Sandoz, Cairo Climate Talks, the American University in Cairo AUC, Al-Ahram Regional Institute for Journalism, Al-Ahram Canadian University, Cairo University, Discovery Education and RiseUp Summit. Mr Ashraf is a Co-Author of the "Science Journalism Manual". a project of the Goethe Institute and supported by the German Federal Foreign Office. (Published in 3 languages English, French and Arabic- September 2021). He is the Editor of the book: "Letters from Egypt" HIV/AIDS: Testimonials of Stigma & Discrimination" a project supported by the Ford Foundation (published in May 2011). He held a Master degree with distinction in Sciences & Technology for Sustainability (STS), from Science & Technology Policy Research Unit (SPRU), Sussex University, Brighton, UK. (2006-2008). His dissertation topic was on: "Technology Transfer of bioethanol production to Egypt: Stakeholders' analysis of values and policy options using multi-Criteria Mapping". He got French Diploma in Media from Cairo University & Paris 2 University (2000-2001). He had B.Sc. in Chemistry & Physics, Faculty of Sciences, Cairo University.

Mr. Mohamed El-Hawary

Executive Director and Editor-in-Chief of Al-Fanar Media

Mr.Mohamed El-Hawary is the Executive Director and Editor-in-Chief of Al-Fanar Media, an independent platform dedicated to higher education, scientific research, and media and information literacy in the Arab region. He serves as the Arab States Representative on the Global Board of the UNESCO Media and Information Literacy Alliance and as a Board Member of the Arab Science and Technology Foundation, where he oversees the media sector. He has extensive experience in journalism, newsroom management, media development, capacity-building, and moderating regional and international conferences. His work focuses on strengthening the connections between scientific knowledge, public policy, responsible communication, and sustainable development.

Panel Discussion 1

Title: Turning Science and Innovation into Sustainable Solutions

Focus

Scientific discoveries improve lives only when they are translated into practical solutions. This panel will discuss how research, technology, entrepreneurship, and collaboration among governments, academia, industry, and civil society can drive innovation for sustainable development.

Guiding Questions

- How can science and innovation better address today's global challenges?
- What creates a strong innovation ecosystem?
- How can entrepreneurship help transform research into real-world impact?
- How can artificial intelligence be used responsibly for sustainable development?
- How can partnerships support future innovators and entrepreneurs?

Speakers

Prof Dr Adla Ragab

Member of Parliament, Tourism and Aviation Committee

Prof Adla is a Professor of Economics, Director of the Center for Economic and Financial Research and Studies, Coordinator of Migration Research Unit MRU and Director of Economic Observatory Using Artificial Intelligence EOUAI at Faculty of Economics and Political Science FEPS, Cairo University. She is a member of Parliament since January 2026, Tourism and Aviation Committee. She is a member of House of Experts for Social Issues and Development of Egyptian Society, Cairo University, since January 2026. She was nominated as first Chairperson of Misr Travel in July 2019. She was Economic Advisor to Minister of Tourism from 2006-2017, then she became the first Vice Minister of Tourism from 2017-2018. She was the Vice Dean for Community Services and Environment Development at FEPS from 2010-2012. She has experience for more than 30 years in Research, Training, and Awareness in addition to providing consultancy in the field of travel and tourism with several publications. She was General Coordinator of Tourism Satellite Account TSA Unit in Egypt from 2008- 2018. She participated in Tourism Strategic Sustainable plan in Egypt 2020 and the revision of Egypt Vision 2030. She is a member of Economics and Political Science Committee at the Supreme Council of Culture, Ministry of Culture, the Supreme Council of Universities and the ministry of Higher Education. She is a board Member of Tourism & Antiquities Fund, Egyptian Saudi Business Council; and Arab Investments for Urbanization Co. She was graduated from a French school and obtained her BSC in Economics, Master and PhD degree from FEPS Cairo University.

Mrs. Ghada Michel

Head of Communications and CSR at Sanofi

A communications and CSR leader with over 25 years of experience in multinational pharmaceutical companies across the Middle East and Africa. Holding a BSc from Ain Shams University, a Postgraduate Diploma in Public Relations from The American University in Cairo, an MBA from Ain Shams University, and a certificate in Measuring Sustainable Development from the SDG Academy, she brings a uniquely multidisciplinary perspective to her work. Based in Egypt, she currently serves as Head of Communications and CSR at Sanofi, where she oversees corporate and internal communications and designs strategic partnerships that drive public health awareness and sustainable impact. Recognized among the top 50 women in Egypt, honored by Ain Shams University for pioneering research on telemedicine's role in reducing CO_2 emissions by 74%, and awarded by the International Diabetes Federation for a program reaching over 56,000 beneficiaries, Ghada is a three-time recognized leader at the intersection of health, sustainability, and social impact. She is fluent in Arabic, English, French, and Sign Language.

Prof. Dr. Maged Ghoneima

Founder and General Partner of M Empire Angels

Chairman of Startup Egypt

Prof Maged is an angel investor, professor, and ecosystem builder. Through M Empire Angels, he has invested in more than 200 startups across over 30 countries. He is the Founder and General Partner of M Empire Angels, and Chairman of Startup Egypt, a community he founded to help entrepreneurs grow through access to networks, mentorship, programs, partnerships, and real ecosystem support. A scholar with a PhD from Northwestern University and industry experience at Intel and NVIDIA, Maged is also a Professor of Mechatronics at Ain Shams University.

There, he founded the ASU Innovation Hub, creating a university-based platform to support students, researchers, and entrepreneurs in turning ideas, research, and technologies into real ventures and innovation-driven impact.

Mrs. Nagwa A. ElSayed

Head of the Scientific Research and Innovation Sector at Misr- elkheir Foundation

Nagwa A. ElSayed has been working in the field of Scientific Research Project funding since 2008. Nagwa is currently working as the Head of the Scientific Research and Innovation Sector at Misrelkheir Foundation. She is responsible for many programs to fund world-class scientific research projects with high societal impact and promote the culture of scientific research and innovation and support innovators and technology startups solving social

challenges. Previously, she worked in the Science and Technology Development Fund (STDF) in the Planning and Monitoring Department from 2008 – 2011, where she was responsible for reporting about STDF's status to the management and monitoring STDF achievements in alignment with the plan. From 2002 to 2008 she worked in Science and Technology Center, in the Academy of Scientific Research and Technology in the Information Technology field. From 1990 till 2002 she worked in several USAID projects in the field of Information Technology with the Academy of Scientific Research and Technology, Egyptian Electricity Authority, and the Ministry of Industry and Foreign Trade. Nagwa holds a Doctor of Business Administration (DBA) in Strategic Management from the American International Theism University, Master's in Business Administration (MBA) from the International Business Administration of Switzerland (IBAS) and a Diploma in Management of Technology from Nile University. She holds Microsoft Certified System Engineer and has B.Sc. in Computer Science from the American University in Cairo. She took several courses in Project Management Professional (PMP), Millennium Manager, and Strategic Planning for NGOs.

Panel Discussion 2

Title: Building Stronger Governance for Science, Innovation and Sustainable Development

Focus

Science has its greatest value when it informs policy and decision-making. This panel will explore how governments, parliamentarians, researchers, and international partners can strengthen science-policy interfaces, promote responsible innovation, and enhance international cooperation through science diplomacy.

Guiding Questions

- How can governments make better use of scientific evidence?
- What role should parliamentarians play in supporting science and innovation?
- How can science diplomacy strengthen international cooperation?
- What governance systems encourage innovation while maintaining ethics and public trust?
- How can women leaders contribute to more inclusive governance?

Mrs. Rania Sadeky

Member of the Egyptian Senate

Member of the Defense and National Security Committee

Mrs Rania is a vice president of the Shaf Center for Future Studies and Crisis Management, Middle East and Africa. She is a member of Foreign Relations Committee at the National Council for Women. She is a member of the Egyptian-African Solidarity Council and member of the Egyptian Council for Foreign Affairs in addition to several councils affiliated with the African Union, some European countries and the United Kingdom working towards peacekeeping, women's and youth empowerment, and entrepreneurship. She is a member of several initiatives in environmental conservation and climate change. She is a member of the Advisory Committee for the National Youth Project for Arts, Culture, Media, and Advertising.

Prof. Heba M. Sharobeem

Member of the Egyptian Senate

Vice chair of the Committee of Human Rights and Social

Solidarity

Prof Heba is a senator, vice chair of the Committee of Human Rights and Social Solidarity in the Egyptian Senate, and member of the Committee of Education at the National Council of Women. She is also a professor of modern British and American Literature and the former chair of the English Department at the Faculty of Education/ Alexandria University, Egypt. Her research spans feminist, post-colonial and cross -cultural studies, as well as comparative literature. She has given presentations in these fields at conferences in Egypt, the United States, Russia, England, Scotland, Morocco, and India. Prof, Sharobeem is also a civic education advocate and teacher trainer. She has published extensively in academic journals and magazines both in Egypt and abroad. She contributed chapters and translations to the books, Women Writing Africa and Joke and Performance in Africa. She also serves on the editorial boards of three International Journals.

Ms. Mona keshta

Egyptian House of Representatives, Member of the Foreign Affairs Committee

Ms Mona is a member of the Foreign Affairs Committee at the Egyptian House of Representatives. She is a researcher Specializing in Regional Security Affairs. Ms Mona is a Ph.D. Candidate in Political Science – Faculty of Economics and Political Science, Cairo University. She holds master Degree in Political Science* – Faculty of Economics and Political Science, Cairo University (2025). She has bachelor's Degree in Political Science – Faculty of Economics and Political Science, Cairo University (2019). She is a senior Researcher – Security and Defense Program at the Egyptian Center for Strategic Studies (ECSS). She published numerous research papers and articles focusing on extremism and terrorism issues. She is the author of several books as: "Gender and Security Challenges: Complex Threats." & "Asia: Internal and External Transformations." She is a contributing Researcher to peer-reviewed academic and strategic journals, including: Journal of Strategy and National Security (Published by the Nasser Higher Military Academy for Strategic Studies). Al-Siyassa Al-Dawliya (International Politics) and Al-Dimuocratiyah (Democracy) journals (Published by Al-Ahram Establishment).

Prof. Dr. Nevine Makram Labib

Chair of the Computer and Information Systems Department, Sadat Academy for Management Sciences (SAMS)

Prof. Dr. Nevine Makram Labib is a globally distinguished researcher, academic leader, and strategist operating at the vanguard of Artificial Intelligence and Data Science. With a career dedicated to breaking institutional silos and driving ethical digital transformation, Dr. Labib is widely celebrated for her ability to translate complex computational frameworks into lifechanging, real-world applications across healthcare, economics, and national policy. In recognition of her exceptional contributions to global technology and societal development, she was honored with the prestigious international "Woman of the Year in AI Leadership" Award by the International Royal Diplomatic Club of the United Nations. She has also received high-level national accolades, including the Women in Science in Developing Countries Recognition Award from the National Research Center. She contributed as a keynote speaker in the conference entitled "Women in Science Empowering: Artificial Intelligence" (WISE-AI) in 2025. Prof. Labib served as the Egyptian consultant of the international project titled: "Towards Advancing and Securing Higher Education Legislation for the Representation of Females in Executive Leadership in Egypt", in partnership between Coventry University, the British council and the Arab Academy for Science, Technology and Maritime Transport. Currently, Dr. Labib serves as the Rapporteur of Scientific Culture, Innovative Thinking, and Artificial Intelligence at the Supreme Council of Culture, and is the Chair of the Computer and Information Systems Department at the Sadat Academy for Management Sciences (SAMS). On the global stage, she drives international healthcare technology policy as the Chair of the "AI Governance and TeleHealth" working group for the International Society for Telemedicine and eHealth (ISfTeH) in Switzerland, and as a Board Member of the International Institute of Telemedicine (IITM) in Italy. An influential policy advisor, Dr. Labib has lent her expertise as a Data Science Consultant to the Egyptian Cabinet Information and Decision Support Center (IDSC), the Egyptian Board of Health Informatics (Ministry of Health and Population), and the AI-based Economic Observatory at Cairo University. Her prolific research portfolio includes over 30 published papers in AI, data mining, and deep learning, with a specific focus on predictive medical analytics, including pioneering work in breast cancer metastasis prediction and intelligent systems for oncology. A passionate advocate for gender equity and youth empowerment, Dr. Labib has spent over a decade as the Academic Advisor for Enactus SAMS, mentoring the next generation of innovators to harness technology for sustainable development. She is a frequent television commentator and a highly sought-after international

keynote speaker, having presented her insights at premier forums across the USA, Europe, Africa,

8and the Middle East. She holds a Ph.D. in Information Technology from Alexandria University and a Higher Education Diploma from the Université de Paris-Sorbonne, France

Concluding Discussion

From Ideas to Action

This session will conclude with an interactive discussion bringing together all speakers. Together, they will identify practical ways to strengthen partnerships among governments, academia, industry, civil society, and international organizations. The discussion will focus on translating ideas into collaborative actions that advance the Sustainable Development Goals.

Biology & Medical Sciences

SCIENTIFIC PROGRAM & SPEAKER PROFILES

Section: Biology & Medical Sciences

--- SESSION 1 - BIOLOGY & MEDICAL SCIENCES ---

Prof. Mohamed Salama

Professor at the American University in Cairo (AUC) and Senior Fellow for the Global Brain Health Institute (GBHI)

PRESENTATION: Risk and Resilience in Dementia: Towards a More Proactive Approach

Biography

Prof. Mohamed Salama is a professor at the American University in Cairo (AUC) and senior fellow for the Global Brain Health Institute (GBHI). He established the Egyptian Network for Neurodegenerative Diseases (ENND) in 2013 and the Egyptian Dementia Registry in 2023 as platforms to study disease determinants. Currently, he is leading the NIH-funded Egyptian Survey of Healthy Aging (AL-SEHA) to study Egyptian seniors to identify determinants of healthy aging. Additionally, funded by the Davos Alzheimer's Collaborative (DAC), Dr. Salama is leading the North African Dementia Registry (NADR) aiming at studying the various aspects of dementia in the region. He is also leading the Health and Wellbeing across Lifespan (HWLS) strategic research program within the African Population Cohorts Consortium (APCC).

Abstract

The interplay of protective and risk factors influences healthy aging, particularly in high-inequality regions. This led to introducing the biological age gaps (BAGs) based on clocks, a novel framework for measuring the combined effects of protective/risk factors on delaying/accelerating aging, as well as their modulation by disparities and exposomes. Furthermore, scientific advances have unveiled several promising blood-based markers of health and aging. To have a useful model for studying dementia and brain health, more inclusive research is needed. Recently, Davos Alzheimer's Collaborative (DAC) has supported the initiation of an aging cohort in Egypt (modelled after HRS studies). The existing cohort is composed of 2,000 subjects above the age of 50 who provided detailed history of exposure to risk factors, medical history, and socio-economic status. Additionally, the cohort has been evaluated using recent digital biomarkers (speech and olfactory) aligned to clinical performance. DAC cohort study subjects provided blood samples, which will help generate blood-based molecular (proteomics/epigenetics/metabolomics) data that explores the association of blood markers with Biological Clock calculated from cognitive, environmental, and other available social determinants of health (SDOH)/clinical phenotyping data. This Egyptian cohort uniquely represents a high-inequality population with distinct lifestyle patterns, genetic background, and environmental exposures, critical for developing culturally and metabolically relevant aging algorithms. The existing infrastructure provides an exceptional opportunity to integrate multi-omics with longitudinal behavioral and cognitive outcomes. We are developing a multimodal Biological Clock by integrating peripheral blood biomarkers, functional/social/digital phenotyping, environmental data, and comprehensive clinical assessments to identify novel aging targets and validate clock algorithms in the Egyptian population to quantify, monitor, and slow/reverse biological aging through identifying resilience determinants/model.

Prof. Dr. Nadia M. Hamdy

Prof. of Clinical Pharmaceutical Biochemistry and Molecular Biology, Faculty of Pharmacy, Ain Shams University | Dept. Head (Feb. 2013–July 2015 & Sept. 2019–Sept. 2023) | GBD Senior Collaborator

PRESENTATION: (Epi)genetics Integration into Medicine for Advancing Precision via Chem/Bio-informatics Piloting/Validation

Biography

Prof. Hamdy was a member in projects funded by NIH/AHA, European PRIMA, Erasmus+, ASRT-Egypt, STDF-Egypt, STDF-Germany, Higher Ministry of Education Egypt ('HME & Scientific Research'), ASU since 2004 till now. Dr. Nadia has 140+ publications, 2 edited books, and 14+ book chapters, with Scopus h-index 52 and 40+ Masters' and Ph.D. supervised students. Prof. Hamdy is one of the world's top 2% influencing scientists; she earned the Obada International Prize for Lifetime Achievement Scientists and 2 International BSI 2025 conference awards by Harvard University, honorary awards, Ain Shams University Appreciation award in Technological Sciences (2024), the ASRT State Award Dr. Shawky El-Hadad award in cancer (2023), International 'Wendy Harvan' Top Honorary Award & Certificate (February 2023), 'Paola Sassone-Corsi' Top Award (February 2021), and ASU first 'Best Scientific Project Research Topic' (2021) & Honored by Women of Egypt (WOE) 2020 to 2023 yearly. Dr. Nadia is an Ambassador of the EACR and Bentham Publisher. She is a member of the Biochemical Society & FEBS, IFCC, ISPE MENA Chapter, ESMO, AAAS, and more. Dr. Hamdy is a fellow of the GBD, MHE Egypt, ASU, ASRT-Egypt. Prof. Nadia is a member of the editorial board of 10 Scopus (Q1 & Q2) journals and reviewer for hundreds of international and regional grants, awards, and scientific journals. Profile: https://www.asu.edu.eg/staffPortal/en/staffProfile/1010

Abstract

Based on 'disease-(epi)genetic profile', facilitating the unraveling of disease mechanism(s) and disease stratification via individuals' (epi)genetic picture. State-of-the-Art Study-for-Precision ncRNAs-downstream protein or gene axis identified and predicted cellular molecular processes via in silico and chem- or bio-informatics search and then verified experimentally and/or confirmed clinically, prompting 'Good Precision-Diagnosis or -Prognosis Practice' (GPDP or GPPP). Finally, 'Good Precision Medicine Practice' (GPMP) via targeted therapies development or repurposing promising potential-to-repurpose drugs as the 'Old-is-Gold' approach examining natural or chemically-modified small molecules drugs.

Prof. Dalia Osama Saleh

Pharmacology Department, Medical Research and Clinical Studies Institute, National Research Centre (NRC), Egypt

PRESENTATION: **Natural Products as Multi-Target Therapeutics for Polycystic Ovary Syndrome**

Biography

Dr. Dalia Saleh obtained her BSc, MSc, and PhD degrees from the Faculty of Pharmacy, Cairo University, in 2002, 2009, and 2012, respectively. She is currently a Professor of Pharmacology at the Institute of Medical Research and Clinical Studies, National Research Centre (NRC), Giza, Egypt (33 El-Bohouth Street, Dokki). She has also served as a course instructor at MSA University and Sadat City University. Dr. Saleh's research interests encompass molecular pharmacology, biochemistry, cardiovascular diseases, in vitro pharmacology, and hepatic disorders, with a strong focus on mechanistic and translational research. She received the NRC Scientific Pioneers Award in 2019 and the NRC Scientific Encouragement Award in 2015. She has published over 88 original research articles in high-impact international journals and holds a Scopus H-index of 21. Dr. Saleh has supervised more than nine Master's and PhD theses. She served as Co-PI for an STC/ASRT-funded project (2019-2021) and contributed to several national and international projects including EG-KTA (2017-2019), US-Egypt Fund/STDF (2010-2014), and Egypt-USA Nano-Gold Materials project (2009-2014). She is a Member of the Scientific Committee for Herbal Medicines at the Egyptian Drug Authority (EDA) and the National Committee for Women in Sciences.

Abstract

Polycystic ovarian syndrome (PCOS) is one of the most prevalent endocrine and metabolic disorders affecting women of reproductive age, characterized by hyperandrogenism, chronic anovulation, insulin resistance, obesity, and menstrual irregularities. The pathophysiology of PCOS is multifactorial and involves oxidative stress, low-grade chronic inflammation, mitochondrial dysfunction, and dysregulation of key metabolic and inflammatory signaling pathways. Increasing evidence implicates the aberrant activation of NF-κB, NLRP3 inflammasome, and PI3K/AKT pathways, alongside impaired AMPK signaling and downregulation of the Nrf2-mediated antioxidant defense system, as central contributors to ovarian dysfunction and follicular arrest in PCOS. Recent preclinical and clinical studies highlight the therapeutic potential of natural products, including flavonoids, polyphenols, terpenoids, and alkaloids, in mitigating PCOS-associated reproductive and metabolic disturbances. These bioactive compounds exert pleiotropic actions by restoring hormonal balance, improving insulin sensitivity, regulating lipid metabolism, and preserving ovarian morphology. Mechanistically, natural products activate AMPK and Nrf2 signaling pathways, enhance endogenous antioxidant capacity, and suppress inflammatory cascades through inhibition of NF-κB and NLRP3 inflammasome activation, ultimately reducing pro-inflammatory cytokine production. Collectively, accumulating evidence supports the role of natural products as promising adjunct or alternative therapeutic strategies for PCOS management. Targeting the crosstalk between oxidative stress, metabolic dysfunction, and inflammation through natural bioactive compounds may offer a safer, multi-targeted approach for improving ovarian function and metabolic health in women with PCOS.

Dr. Mawell Sameh, MD, CMD

Medical Affairs Head for Vaccines Egypt and Sudan, Sanofi

PRESENTATION: **Enhancing Women's Healthcare by Vaccinations**

Biography

Dr. Mawell Sameh, a physician with two decades of clinical and pharmaceutical leadership experience, serves as Sanofi's Medical Affairs Head for Vaccines, where she has expanded access to life-saving medicines across Egypt and Sudan and earned over 10 Global & Regional Best Practice Awards, guided by her unwavering belief that no patient should be left behind. Some careers follow a path. Dr. Mawell Sameh built one from the wards of public hospitals to the forefront of vaccines leadership at one of the world's most innovative pharmaceutical companies.

Abstract

Vaccination is a cornerstone of preventive healthcare and a key strategy for protecting women throughout their lives, particularly during pregnancy. Maternal immunization helps safeguard both mothers and their infants against serious vaccine-preventable diseases, improving health outcomes for families and communities. As innovation continues to expand the impact of preventive medicine, the commitment of healthcare partners such as Sanofi to developing vaccines that address the needs of pregnant women reflects the vital role of science in advancing maternal and newborn health. Together, research, innovation, and equitable access to vaccination can help build healthier futures for women and the generations they nurture.

Dr. Sherehan Khalil, MD

Medical Operations Lead for Egypt within Sanofi's Africa Medical Operation department, Cairo

PRESENTATION: Diversity and Inclusion in Clinical Trials

Biography

Sherehan Khalil, MD is an internal medicine physician who transitioned to the pharmaceutical industry, bringing 8 years of clinical expertise and patient-centered perspectives to Sanofi. Based in Egypt, she has developed deep expertise in multi-market pharmaceutical environments and now leads with a focus on operational excellence and strategic impact in life sciences.

Abstract

Women remain significantly under-represented in clinical trials, yet they experience diseases differently and respond to treatments differently than men — a gap that perpetuates inequities in medical knowledge and care. Advancing diversity and inclusion in clinical research requires intentional action: recruiting more women participants, supporting women investigators as leaders in science, and designing trials that reflect the real-world needs of women across all populations.

Dr. Eman Abd El-Razek Metawea Abbas

Microbial Biotechnology Department, Biotechnology Research Institute, National Research Centre (NRC), Egypt

PRESENTATION: Host Genetic Biomarkers Associated with Disease Progression in HCV-Related Hepatocellular Carcinoma

Biography

Dr. Eman Abd El-Razek Metawea Abbas is a Researcher at the Microbial Biotechnology Department, Biotechnology Research Institute, National Research Centre (NRC), Egypt. She specializes in medical biotechnology, molecular genetics, and translational cancer research, with a primary focus on hepatitis C virus, hepatocellular carcinoma (HCC), precision medicine, and molecular diagnostics. Her research aims to bridge the gap between molecular discoveries and clinical practice through the identification and validation of innovative biomarkers for hepatitis C virus-associated liver diseases and hepatocellular carcinoma. Her work encompasses host genetic polymorphisms, circulating cell-free DNA (cfDNA), liquid biopsy, genomic alterations, and non-invasive molecular biomarkers for disease susceptibility, early diagnosis, prognosis, survival prediction, and therapeutic response. Dr. Eman Abd El-Razek has authored numerous peer-reviewed publications in high-impact international journals covering host genetic biomarkers, liquid biopsy, and precision oncology. Her current research focuses on integrating genomic biomarkers with minimally invasive diagnostic approaches to support personalized medicine and improve clinical decision-making for liver cancer patients. In addition to her research activities, she serves as a reviewer for several international scientific journals and actively participates in multidisciplinary national and international research collaborations. She is committed to developing innovative molecular diagnostic strategies that enhance early cancer detection, prognostic assessment, and individualized treatment of liver cancer.

Abstract

Authors: Samar Samir Youssef[1], Eman Abd El Razek Abbas[1], Asmaa M. Elfiky[2], Sameh Seif[3], Mohamed Mahmoud Nabeel[4], Hend Ibrahim Shousha[4], Ashraf Omar Abdelaziz[4]

Affiliations: [1]Microbial Biotechnology Dept., NRC; [2]Environmental and Occupational Medicine Dept., NRC; [3]National Hepatology and Tropical Medicine Research Institute; [4]Endemic Medicine and Hepato-gastroenterology Dept., Faculty of Medicine, Cairo University.

Background: Hepatocellular carcinoma (HCC) is a leading cause of cancer-related mortality worldwide. The marked biological heterogeneity of HCC highlights the need for reliable molecular biomarkers to improve disease characterization and personalized patient management. Aim: This study aimed to investigate the clinical significance of PNPLA3 rs738409 and TM6SF2 rs58542926 polymorphisms and their association with disease progression and clinicopathological characteristics in patients with HCV-related HCC. Methods: Genomic DNA was extracted from peripheral blood samples of Egyptian patients with HCV-related HCC. Genotyping was performed using TaqMan allelic discrimination real-time PCR. Associations between genetic variants and clinicopathological parameters, tumor stage, radiological treatment response, and overall survival were analyzed. Results: TM6SF2 rs58542926 showed significant associations with advanced tumor stage and radiological treatment response, suggesting a role in disease progression. In contrast, PNPLA3 rs738409 demonstrated more limited clinical associations but contributed to the molecular characterization of HCC. Neither polymorphism was significantly associated with overall survival. Conclusion: Host genetic biomarkers may improve molecular risk stratification in HCV-related HCC. TM6SF2 rs58542926 demonstrated greater clinical relevance for disease progression, while combined evaluation of PNPLA3 and TM6SF2 may support future precision medicine strategies.

--- SESSION 2 - MEDICAL SCIENCES AND BIOLOGY ---

Prof. Reham Dawood

Professor of Microbial Biotechnology, National Research Centre (NRC), Egypt

PRESENTATION: Evaluation of lncRNA PVT1 rs13255292 Variant and Serum E-Cadherin Levels in Breast Cancer

Biography

Professor Reham Dawood is a Professor of Microbial Biotechnology at the National Research Center (NRC), Cairo, with more than 25 years of experience in molecular biology, virology, and translational biomedical research. She earned her Ph.D. in Molecular and Cellular Biology from the University of Jean Monnet, Saint-Étienne, France (2011), focusing on broadly neutralizing monoclonal antibodies against HIV-1. Prof. Dawood has authored over 69 scientific publications with an H-index of 15 and >600 citations. Her research spans HCV, COVID-19, cancer biomarkers, genome editing, and vaccine development. She is Technical Supervisor of the

DNA Fingerprinting Service at NRC and recipient of the Prof. Mona Mohamed Rashoud Award in Biotechnology (2024) and the State Encouragement Award in Health and Pharmaceutical Sciences (2020).

Abstract

Breast cancer (BC) is the most frequent malignancy and a prime cause of cancer lethality worldwide. Genetic research, particularly on lncRNA, shows prospects for BC management. PVT1 can activate many tumorigenic pathways, contributing to angiogenesis and pathological progression. This study examined the association between PVT1 rs13255292 variants and serum E-cadherin levels with BC risk, hormone receptor status, and tumor grade. Genotyping was performed on 120 blood samples (20 controls, 100 BC patients stratified by histological grade: G1=3, G2=74, G3=23) using TaqMan assays. Patients were classified as luminal A (n=79) or non-luminal A (luminal B, HER2-enriched, TNBC; n=21). Serum E-cadherin was evaluated by ELISA. Findings revealed a statistically significant association between the PVT1 rs13255292 non-risk CC genotype and luminal A subtype patients, suggesting a potential protective effect. E-cadherin levels significantly declined in BC patients compared to controls, with notable reduction in advanced G3 compared to G2. Combining genetic screening of PVT1 variants with E-cadherin surveillance could enhance prognostic stratification in BC management.

Prof. Dr. Menha Mahmoud Swellam

Head of High Throughput Molecular & Genetic Technology Lab, Center of Excellence for Advanced Sciences, Biochemistry Dept., Biotechnology Research Institute, NRC

PRESENTATION: Clinical Impact of MiRNAs in Women Malignancies: What Do We Know So Far?

Biography

Menha Swellam is a Professor of Biochemistry at Biochemistry department- Biotechnology Research Institute, National Research Centre, Egypt. She is a member of editorial board and a peer reviewer of international journals. Supervised many MsCs and PhDs, a principal investigator for national projects, member of the Egyptian permanent promotion council. Has published over 50 peer reviewed manuscripts in PubMed and Scopus cited journals (Scopus ID: https://www.scopus.com/authid/detail.uri?authorId=56065210800). She has been recognized in 2008 by Encouraging National Research Centre prize from National research Centre, and in 2009 by Encouraging State prize from the by the Academy of Scientific Research and Technology, then in 2017 a prize of National Centre for Research in Advanced Sciences. She is the Founder and the Head for High Throughput Molecular and Genetic Technology lab in the Centre of Excellences and Advanced Sciences, National Research Centre which is facilitated with next generation sequencers and microarray systems.

Abstract

MicroRNAs (miRNAs) are endogenous, small non-coding RNAs which have an important function in regulating RNA stability and gene expression. Considerable proofs have demonstrated that miRNA expression is dysregulated in human cancer through several mechanisms, involving abnormal transcriptional control of miRNAs, amplification or deletion of miRNA genes, dysregulated epigenetic variations and limitations in the miRNA biogenesis machinery. MiRNAs may perform as either oncogenes or tumor suppressors through certain circumstances. The dysregulated miRNAs have been shown to influence the hallmarks of cancer, including escaping growth suppressors, supporting proliferative signaling, opposing cell death, triggering invasion and metastasis, and provoking angiogenesis. Cumulative numbers of studies have recognized miRNAs as potential molecular markers for human cancer diagnosis, prognosis and therapeutic targets, which desires more investigation and validation.

Assoc. Prof. Ghada Salum

Department of Microbial Biotechnology, Biotechnology Research Institute, National Research Centre, Egypt

PRESENTATION: Genetic Dysregulation in Breast Cancer: Impact on Immune Surveillance and Infection Control

Biography

Dr. Ghada Maher Salum is an Associate Professor of Medical Biotechnology at the NRC, Egypt, specializing in virology and microbial biotechnology. She earned her Ph.D. in Microbiology (Virology) from Ain Shams University (2016) focusing on IFN-alpha related genes in HCV liver diseases, an M.Sc. from Cairo University (2011), and B.Sc. from Zagazig University (2004). With 27 publications and an H-index of 9, her research outputs cover hepatic fibrosis, COVID-19 liver injury, and breast cancer microRNAs. She holds an STDF registered patent for measuring neutralizing antibodies in convalescent plasma for COVID-19 patients.

Abstract

Authors: Ghada M. Salum[1], Mai Abd El Meguid[1], Basma E. Fotouh[1], & Reham M. Dawood[1] ([1]Microbial Biotechnology Dept., NRC, Cairo, Egypt).

Breast cancer (BC) is one of the main causes of cancer deaths in women worldwide, despite improvements in treatment. The understanding of genetic dysregulation in BC—specifically the alterations in tumor suppressors and oncogenes—provides crucial insights into how immune surveillance is compromised in these patients. This compromised immunity increases their susceptibility to infections, including those acquired in healthcare settings. Herein, we evaluated the expression levels of two circulatory miRNAs (miR-124-3p, and miR-155-5p) were assessed by using qRT-PCR in 75 BC patients (Luminal A: n = 24, Luminal B: n=12, HER2: n=28 and basal like: n = 5) and 20 healthy women. A computational analysis was carried out to identify the possible network between the two miRNAs and their related mRNAs involved in BC, which could help improve therapeutic options. Comparing to healthy controls, miR 124 3p levels were significantly lower in the serum of BC patients, however, miR 155 5p showed a significant elevation in BC patients specially in basal group. In-silico analysis suggested that CCND1 is an oncogene that is inhibited by miR-155-5p, was significantly overexpressed in luminal A & luminal B patients groups compared to controls. Moreover, CCND1 overexpression in basal like BC and HER2-positive subtypes do not achieve statistical significance. PRKD1 as a tumor suppressor was shown to be inhibited by miR-124-3p. PRKD1 showed a significant downregulation in the four BC subtype groups compared to controls. ROC curve analysis demonstrated that miR-124-3p possesses high diagnostic accuracy for distinguishing BC patients from controls. By integrating current findings into clinical risk assessments, infection control strategies can be personalized to identify high-risk BC patients, optimize preventive measures, and improve monitoring and targeted interventions. Thus, unveiling genetic dysregulation data can significantly enhance infection control effectiveness and patient outcomes in BC care.

Assoc. Prof. Ahmed Hassan Hassan Afifi

Department of Pharmacognosy, Pharmaceutical and Drug Industries Research Institute, National Research Centre, Egypt

PRESENTATION: **Marine Environment as a Potential Source for Small Molecule Targeted Anti-Cancer Agents**

Biography

Dr. Ahmed Hassan Afifi is an associate professor in the Department of Pharmacognosy at the Pharmaceutical and Drug Industries Research Institute, National Research Centre, Egypt. He holds a Ph.D. in Pharmaceutical Sciences (Pharmacognosy) from Kumamoto University, Japan, and Cairo University, Egypt, under the Channel System Program, along with an M.Sc. and B.Sc. in Pharmaceutical Sciences from Cairo University and Ain-Shams University, respectively. Dr. Afifi has extensive research experience spanning natural products chemistry, marine pharmacognosy, and drug discovery. He has held research and academic positions in Egypt, Saudi Arabia, and Japan, including a visiting researcher role at Kyushu University. His work has been published in leading journals such as the Journal of Natural Products, Tetrahedron Letters, RSC Advances, Scientific Reports, and Biomedical Chromatography. His research focuses on the isolation and characterization of bioactive natural compounds from plants and marine organisms, computer-assisted drug design and molecular modeling, and the discovery of novel anticancer, antiviral, and neuroprotective agents. He possesses strong expertise in chromatographic separation, spectral analysis, and metabolic profiling using LC-MS and GC-MS. Dr. Afifi has participated in numerous national and international research projects and has presented his work at conferences in Egypt and Japan, including meetings of the Japanese Society of Pharmacognosy and the Asian Natural Product Conference. He continues to contribute to advancing drug discovery through interdisciplinary, technology-driven approaches to natural product research.

Abstract

Cancer remains one of the leading causes of death globally, responsible for approximately 10 million deaths in 2022, and is projected to claim 18 million lives by 2050. While conventional approaches such as chemotherapy and radiation therapy have long been employed for cancer treatment, their non-specific nature often results in severe side effects and the development of resistance in cancer cells. Targeted cancer therapy has emerged as a new generation of cancer treatment, providing a more precise approach by interfering with certain molecular targets unique to cancer cells while sparing healthy tissues. Marine environments, known for their exceptional biodiversity, have yielded a rich array of bioactive compounds with unique structures and potent anticancer properties. This presentation highlights the role of marine-derived small molecules in targeted cancer therapy, particularly those modulating key cancer specific molecular targets such as protein kinases, ubiquitin-proteasome system, epigenetics, and other molecular drivers of cancer. With several marine-derived agents already in clinical use, the continued exploration of these natural products holds great promise for developing novel, more effective, and less toxic treatments in the fight against cancer.

Assoc. Prof. Mona Mohammad Agwa

Department of Chemistry of Natural and Microbial Products, Pharmaceutical and Drug Industries Research Institute, NRC

PRESENTATION: **Active Tumor Targeting Exploiting Overexpressed Receptors: Emphasize Towards Harnessing Cellular and Subcellular Targeting Ligands**

Biography

Mona M. Agwa, Associate Professor at NRC Egypt, obtained her Ph.D. (2017) and M.Sc. (2011) from Alexandria University, and B.Sc. from Al-Azhar University (2002). She has contributed to 46 scientific papers (Scopus/WoS, H-index 20), authored 4 book chapters for CRC Press and Elsevier, and reviewed >20 manuscripts for Springer, Elsevier, Bentham, and Dove Press. She has co-supervised graduate theses, participated in >5 dosage form/drug delivery projects, and judged graduation projects at MSA University. Her teaching covers targeted delivery, overcoming chemotherapy hazards, and nanomedicine-based phototherapy.

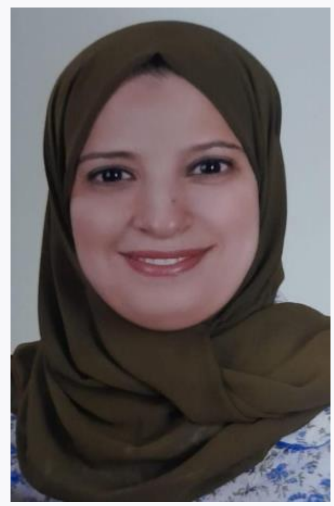
Abstract

Nano-delivery systems for actively targeting drugs against diseased tissues like tumors is a prime goal of nanomedicine, which ameliorates the therapeutic delivery to the aimed tissue while evading the off-target toxicity issue and the necessity for dosing several doses. An Active targeting approach is achieved via decorating the nanoparticle surface with various targeting ligands, harnessing chemical conjugation. At the cellular level, these targeting ligands enable selective tumor homing for the nanocarriers via recognizing the overexpressed cellular receptors or transporters, thereby guaranteeing excellent drug efficacy in the aimed tissue with negligible toxicity. In addition to cell-specific targeting, these ligands can also assist active subcellular therapeutic targeting to particular intracellular organelles, involving lysosomes nuclei, endoplasmic reticulum, and mitochondria, where multiple pathological processes originate. Integrating cellular and subcellular targeting creates a hierarchical delivery system that maximizes drug availability at the site of action, minimizes systemic toxicity, and holds strong promise for treating complex diseases.

Keywords: Active targeting, cellular and subcellular active target ligand, chemotherapy, overexpressed cellular and subcellular receptors.

Dr. Basma Elsayed Fotouh

Researcher at National Research Centre, Egypt (Molecular Biology and Medical Biochemistry)



PRESENTATION: Impact of KRAS Gene Expression on Ovarian Cancer Development and Progression

Biography

Dr. Basma Elsayed Fotouh is a Researcher at the National Research Centre, Egypt, with extensive experience in molecular biology, gene expression analysis, and medical biochemistry.

Abstract

Ovarian cancer (OC) is a leading cause of cancer deaths in women. Comprehensive molecular studies are required to understand OC pathogenesis. KRAS and NOXA genes are involved in tumorigenesis and disease progression. KRAS promotes tumor growth, while NOXA triggers the apoptotic signaling pathway. Fifty-six ovarian cancer patients and twenty healthy controls were enrolled in the study. Gene expression profiling was performed utilizing the qRT-PCR assay. Then, the proteomic levels of KRAS and NOXA genes were assessed by the ELISA technique.The results showed that KRAS gene expression was significantly increased in OC patients compared to controls ($p < 0.0001$). Additionally, KRAS overexpression was correlated significantly with more aggressive characteristics, including advanced stage, positive lymph invasion, and metastatic status ($p = 0.04$, 0.05, and 0.02, respectively). In contrast, NOXA expression was downregulated in OC patients compared to controls ($p = 0.001$), and this reduction was more pronounced in patients with more aggressive characteristics ($p = 0.001$, 0.01, 0.0008, 0.02, respectively). At the transcriptomic level, KRAS concentration was higher among the patient group ($p = 0.03$) and correlated with aggressive tumor features ($p = < 0.0001$, 0.001, and 0.03, respectively). Interestingly, no significant changes were detected in NOXA protein levels concerning ovarian cancer development and progression. These findings were further confirmed by receiver operating characteristic (ROC) curve analysis, validating the genetic and transcriptomic results. The differential expression of KRAS and NOXA genes holds promise as potential biomarkers for ovarian cancer development and progression. Specifically, KRAS transcriptomic levels serve as a reliable discriminator of ovarian cancer progression. In contrast, NOXA protein expression warrants further investigation to elucidate its role in the progression and pathophysiology of ovarian cancer.

Dr. Tânia dos Reis

Instituto de Higiene e Medicina Tropical (IHMT) and Universidade de Cabo Verde, Cape Verde

PRESENTATION: Nasal microbiota and its relationship with asthma and other atopic conditions in children from Cabo Verde

Biography

I am a Biotechnology graduate, Bioinformatics specialist, and MSc in Genetics and Molecular Biology, currently pursuing a PhD in Tropical Diseases and Global Health. My academic trajectory is rooted in interdisciplinary research, global health equity, and scientific innovation, with a strong commitment to leadership, capacity building, and social impact across Africa and the Global South. I have extensive experience in scientific research, having worked in multiple laboratories and participated in interdisciplinary projects addressing health challenges relevant to low and middle income countries. I have contributed to the scientific community through conference presentations and the organization of scientific symposia and workshops that promote knowledge exchange beyond geographical borders. I also bring over five years of teaching experience in biomedical sciences, alongside academic leadership experience as a course coordinator and founder of a health sciences research group. My doctoral research focuses on the respiratory microbiome in children, investigating differences among those with asthma, allergic rhinitis, and healthy individuals. This study employs innovative laboratory techniques, including advanced molecular biology approaches and high throughput sequencing, integrated with state of the art bioinformatics and computational analyses. By combining microbial diversity profiling, environmental exposure data, and clinical information, my research aims to advance understanding of microbiome–environment interactions and their role in allergic disease development, contributing to precision and preventive health strategies adaptable to resource limited settings. Beyond academia, I am committed to science communication, mentorship, and gender equity advocacy. Through the WFWS and WISWB Science Without Borders platform, I seek to strengthen international collaborations and amplify the voices of women scientists from Africa and the Global South.

Abstract

The role of the human microbiota in the pathogenesis of allergic diseases is well established, however, the specific contribution of the nasal microbiota remains less understood. Emerging evidence suggests that alterations in the composition and diversity of the respiratory microbiota may contribute to the development of asthma and other atopic conditions in susceptible individuals. Asthma is the most prevalent non-communicable disease in children, with a major burden in low- and middle-income countries. In Cabo Verde, a population-based study conducted involving 1,045 children reported a current asthma prevalence of 10.5%, exceeding the global average of 9.1%. A case-cohort study is currently in progress and aims to investigate the relationship between alterations in the nasal microbiota, atopic conditions, and the risk of upper respiratory tract colonization by pathogenic microorganisms among children living on Santiago Island, Cabo Verde. Here we report the preliminary characterization of the nasal microbiota of children from that study, based on analyses of two batches of sequenced samples in a total of 142 samples, including 67 cases (current or previous respiratory atopy) and 75 control samples (no atopic condition). Nasal swab samples collected from children were subjected to DNA extraction followed by amplification of the V3–V4 hypervariable regions of the 16S rDNA gene in the Illumina NovaSeq PE250 platform. After quality control analysis, the reads were clustered into OTUs by vsearch. Taxonomic assignments were based on the SILVA reference database. In the first batch of samples, 70k paired-end reads per sample were obtained. Preliminary taxonomy analyses revealed more than 200 families of bacteria and the presence of some bacterial species commonly associated with respiratory diseases in children (Staphylococcus aureus, Haemophilus influenzae). Alpha-diversity indices and beta-diversity will be evaluated to assess differences between case and control groups. Differentially abundant taxa will be identified through linear discriminant analysis.The study is expected to provide deeper understanding of the relationship between airway microbiota composition and asthma and

other atopic phenotypes. The findings may support the development of personalized therapeutic strategies for pediatric allergic diseases.

Dr. Walaa Abdallah Gad

Veterinary Research Institute, National Research Centre (NRC), Egypt

PRESENTATION: Epigenetics and Gene Therapy Technologies for Precision Veterinary Medicine

Biography

Dr. Walaa Abdallah Gad, a researcher at the Veterinary Research Institute, National Research Centre (NRC), Egypt, specializing in infectious diseases, molecular diagnostics, biotechnology, immunology, and bioinformatics. My research focuses on developing innovative diagnostic and therapeutic strategies for veterinary infectious diseases, antimicrobial resistance, molecular epidemiology, and One Health applications. I have extensive experience in molecular biology techniques, including PCR, real-time PCR, sequencing, gene expression analysis, immunological assays, and bioinformatics. During the past year, I have significantly expanded my scientific contributions through research, publishing, and academic editing. I currently serve as an Editor of an international scientific book and have authored eight book chapters, three review articles, and five original research articles, reflecting my commitment to advancing knowledge in veterinary medicine and molecular biosciences. My research interests include rapid molecular diagnostics, CRISPR-based technologies, isothermal amplification techniques, epigenetics, precision veterinary medicine, and artificial intelligence applications in biomedical research. Beyond research, I actively organize and deliver scientific workshops and professional training programs on molecular diagnostics, gene expression analysis, bioinformatics, and advanced laboratory techniques for researchers and veterinary professionals. I also serve as a peer reviewer for international scientific journals, contributing to the advancement of high-quality scientific publishing. My long-term goal is to bridge laboratory innovation with practical veterinary applications by developing affordable, rapid, and sustainable diagnostic solutions that improve animal health, reduce antimicrobial resistance, and support global One Health initiatives through interdisciplinary collaboration and translational research.

Abstract

Mastitis remains one of the most prevalent and economically significant diseases affecting dairy livestock worldwide, compromising animal welfare, milk quality, and farm profitability. Although antimicrobial therapy has long been the cornerstone of mastitis management, the emergence of antimicrobial resistance, recurrent infections, and treatment failures has emphasized the need for innovative molecular-based interventions. This review explores the rapidly evolving roles of epigenetics and gene therapy as promising strategies for the prevention, diagnosis, and treatment of mastitis in veterinary medicine. We discuss the molecular mechanisms underlying host pathogen interactions, mammary gland immunity, and inflammatory responses, highlighting how DNA methylation, histone modifications, and non-coding RNAs regulate susceptibility and resistance to infection. The review further examines the influence of the mammary microbiome and its interaction with epigenetic pathways in maintaining udder health or promoting disease. Recent advances in epigenetic modulators, including DNA methyltransferase, histone deacetylase, and bromodomain inhibitors, are evaluated for their therapeutic potential and limitations in food-producing animals. In addition, current gene therapy technologies, including CRISPR/Cas systems, viral and non-viral vectors, antimicrobial peptide delivery, and regulatory RNA-based approaches, are critically assessed with respect to efficacy, safety, and tissue-specific delivery challenges. The integration of epigenetic editing, multi-omics technologies, artificial intelligence, and bioinformatics is presented as a transformative framework for precision veterinary medicine, enabling biomarker discovery and individualized disease management. Finally, the review addresses regulatory, ethical, and practical considerations associated with implementing

these advanced technologies in livestock production. Collectively, this review highlights how combining epigenetic insights with gene therapy may provide sustainable, antibiotic-sparing solutions for mastitis control while advancing the future of precision livestock health management.

Dr. Amal Abdel-Bassir Draz

Phytochemistry and Plant Systematics Dept., Pharmaceutical and Drug Industries Research Institute, NRC, Egypt

PRESENTATION: From Authentication to Bioactivity: Metabolite Profiling, DNA Barcoding, and Antimicrobial Evaluation of Zygophyllum scabrum Christenh. & Byng.

Biography

Dr. Amal A. Draz is a researcher at the National Research Centre (NRC), Egypt, specializing in plant chemotaxonomy. She earned her Ph.D. in Plant Taxonomy and Flora from the Faculty of Science, Cairo University, in 2025, following a Masterâ€™s degree in the same field (2021). She also holds a Master degree in Microbiology (2014) from Benha University and graduated with a B.Sc. in Botany and Chemistry with Excellent with Honors (86.77%) in 2009. Dr. Amal Draz joined the National Research Centre as a Research Assistant (2017 “2021), was promoted to Assistant Researcher (2022“2025), and has served as a Researcher since 2026. She also held the position of ISO Quality Coordinator in the Phytochemistry and Plant Systematic Department (2024“2026). Earlier in her career, she worked as a Chemist in the field of Safety and Occupational Health (2012“2017). Her research interests focus on plant taxonomy, flora, systematics, and related biological sciences. She has published 11 research papers in international peer-reviewed journals and received the National Research Centre Best Master Thesis Award in 2021. Dr. Amal Draz has completed specialized training in artificial intelligence applications in research, and young researcher capacity building. She is also a certified TOEFL-ITP holder (score: 573) and has qualifications in digital transformation, reflecting her commitment to continuous professional development.

Abstract

The growing threat of antimicrobial resistance highlights the urgent need for novel plant-derived therapeutics supported by reliable authentication methods. This study integrates DNA barcoding, metabolite profiling, and antimicrobial evaluation to characterize Zygophyllum scabrum and assess its pharmaceutical potential. Species authentication was achieved using the rbcL DNA barcode, resulting in a novel registered NCBI GenBank accession (PQ403596), representing the first reported rbcL sequence for this species. Comprehensive phytochemical profiling of 70% methanolic extract of the plant by LC-ESI-MS/MS tentatively identified 97 metabolites, including flavonoids, phenolic acids, saponins, coumarins, and fatty acids. On the other side, in a first evaluation of the antimicrobial activity of this plant, the methanolic extract demonstrated notable antimicrobial activity comparable with that of the standard antibiotic disc, as determined by the agar well diffusion assay, against Helicobacter pylori (14.0 Â± 0.05 mm), Bacillus cereus (13.0 Â± 0.08 mm), and Staphylococcus aureus (11.0 Â± 0.09 mm). The minimum inhibitory concentrations (MICs), determined using the broth microdilution method, ranged from 1875 to 3750 Î¼g/ 300 Î¼L. In contrast, no antimicrobial activity was observed against Escherichia coli or Candida albicans. These findings demonstrate that Z. scabrum represents a promising medicinal plant with significant phytochemical diversity and antibacterial potential. The integration of molecular authentication with metabolomic profiling provides valuable data for the pharmaceutical quality control of medicinal plants and their potential pharmaceutical applications.

Assoc. Prof. Mai Abd El Meguid Ali Dawoud

Biotechnology Research Institute, National Research Centre (NRC), Egypt

PRESENTATION: **Evaluation of PNPLA3 genetic variant in Egyptian HBV-infected patients concomitant with metabolic-associated steatotic liver disease (MASLD)**

Biography

Dr. Mai Abd El Meguid, PhD, is an Associate Professor of Microbial Biotechnology at the Biotechnology Research Institute, National Research Centre (NRC), Dokki, Giza, Egypt—one of the region's leading multidisciplinary research institutions. She earned both her MSc and PhD in biological sciences from the Faculty of Science, Ain Shams University, Cairo, Egypt. Dr. El Meguid is a molecular biologist whose research focuses on viral hepatitis and liver diseases, particularly infections caused by Hepatitis C Virus (HCV), Hepatitis B Virus (HBV), and Cytomegalovirus (CMV), which represent major public health challenges in Egypt and worldwide. Her work integrates molecular biology techniques with clinical relevance to advance understanding of viral pathogenesis, host immune responses, and the genetic and epigenetic factors involved in disease progression and hepatocellular carcinoma (HCC). Her long-term research goals include the development of early diagnostic biomarkers and preventive strategies for liver diseases. Throughout her career, Dr. El Meguid has authored more than 25 peer-reviewed research articles and reviews, primarily as first or corresponding author, published in respected international journals. Her work has received over 250 citations, with an h-index of 9 according to Google Scholar. In recognition of her academic excellence, she received the National Research Centre's Best MSc Award in 2012. Dr. El Meguid is an active member of several scientific societies, including the Egyptian Society for the Study of Gastroenterology and Liver Disease and the Organization for Women in Science for the Developing World (OWSD), and is a postdoctoral member of the American Society for Microbiology. She also serves as a peer reviewer and editorial board member for multiple international journals. Committed to mentorship and capacity building, she actively supports young researchers while fostering national and international scientific collaborations.

Abstract

Hepatitis B virus (HBV) infection and metabolic-associated steatotic liver disease (MASLD) increasingly coexist, creating complex diagnostic and therapeutic challenges. This study examined the association between the PNPLA3 rs738409 genetic variants and the progression of hepatic steatosis and fibrosis in HBV patients who have and do not have MASLD. A total of 170 Egyptian HBV patients were enrolled and categorized into two groups: non-MASLD (n = 99) and MASLD (n = 71). The assessment of liver stiffness and hepatic steatosis was conducted using transient elastography (FibroScan®) and controlled attenuation parameter (CAP). Genotyping for PNPLA3 was performed via real-time PCR. The PNPLA3 GG genotype and G allele were significantly linked to the progression of hepatic steatosis in HBV-MASLD patients. Moreover, the frequency of CG+GG genotypes was higher in HBV-MASLD patients during both early (53.1% vs. 34.2%, p=0.035) and late fibrosis stages (73.7% vs. 33.3%, p=0.014) compared to non-MASLD patients. No association was found between PNPLA3 genotypes and HBV DNA levels. HBV viral load was notably associated with different fibrosis stages but was not linked to steatosis. HBV patients carrying the PNPLA3 GG genotype or G allele are more susceptible to developing MASLD and severe steatosis. Additionally, PNPLA3 variants may be implicated in the progression of fibrosis within the context of metabolic dysfunction; this association is not apparent in non-MASLD cases. Furthermore, viral dynamics appear to influence the progression of hepatic fibrosis but not steatosis. The concomitant MASLD in HBV patients was not associated with the severity of hepatic fibrosis.

Assoc. Prof. Ola Mohamed Mousa

Faculty of Nursing, Minia University, Egypt (Maternal and Women's Health Nursing)

PRESENTATION: From Awareness to Action: Exploring Risk Perception and Behavioral Patterns of Antibiotic Misuse Among Minia University Students

Biography

Dr. Ola Mousa is currently an assistant professor at the faculty of nursing, Minia University. She worked for more than 10 years (from January 2013 to September 2023) as an associate professor at King Faisal University, College of Applied Medical Sciences- Nursing Department, Saudi Arabia. She started her career in Minia University, Egypt and taught there for 13 years from 2000 to 2013. She earned her master's from Ain Shams University, Egypt, and PhD degree from Assiut University, Egypt with a specialization in Maternal and Women's Health Nursing. Her expertise in healthcare professions extends over 23 years, both nationally and internationally. Her area of interest in research is Maternity Nursing, Ethics, Women and Adolescent Health, and mothers of children with special needs. She participated in organizing many scientific days and conferences, as a resource speaker. In addition to publishing more than 55 papers in significant journals, including those indexed in Web of Science and Scopus, she has presented research articles at numerous national and international conferences. She served as an editorial board member and reviewer in many journals. In addition to being a thesis advisor, she is a research guide for MSN students. Over the past seven years, she has been responsible for community engagement and participation in activities with special needs pupils and caregivers. A distinguished faculty award was recently received (2020) from King Faisal University in Saudi Arabia. Presently, she is teaching at Minia University, Faculty of Nursing, the chairperson of the Quality Assurance Department Committee, and Faculty Development committee.

Abstract

Self-medication, defined as the use of pharmaceutical products without professional prescription to manage self-diagnosed or recurrent symptoms, has emerged as a major contributor to antibiotic misuse. This practice is particularly prevalent among adolescents and university students, raising concerns due to its irrational nature and long-term public health implications. Factors such as limited awareness, permissive attitudes, and unregulated access to antibiotics at pharmacies exacerbate the problem. The consequences include not only the proliferation of resistant bacterial strains but also adverse drug reactions and increased economic burden on national health systems. In the United States alone, antibiotic-resistant infections are estimated to cost $35 billion annually and account for over 8 million additional hospital days. To assess the level of awareness, attitudes, and behavioral practices regarding antibiotic misuse among medical and non-medical students at Minia University, and to identify the key factors influencing their knowledge and decision-making. This study employed a descriptive cross-sectional design to assess the knowledge, attitudes, and practices (KAP) related to antibiotic misuse among medical and non-medical undergraduate students at Minia University. A sample size of [400] students was calculated based on previous prevalence estimates of antibiotic misuse among university students, with a confidence level of 95% and a margin of error of 5%. A stratified random sampling method was used to ensure proportional representation from both medical and non-medical faculties.

Data Collection Tool

- Demographics: age, gender, faculty, year of study.
- Knowledge: 10 items assessing understanding of antibiotic use and resistance.
- Attitudes: 6 items exploring perceptions and concerns about misuse.

Practices: 8 items evaluating self-reported behaviors related to antibiotic use.

Prof. Dr. Nesrine Hegazi

Phytochemistry and Plant Systematics Department, Pharmaceutical and Drug Industries Research Institute, NRC, Egypt

PRESENTATION: From Sea to Table: The Power of Metabolomics in Marine Food Science and Facing Challenges

Biography

I am a graduate of the Faculty of Pharmacy, Cairo University, Egypt (2003). MSc (2008) and PhD (2014) degrees from Ain Shams University, Egypt, in Pharmaceutical Sciences. Professor at the National Research Centre in the Department of Phytochemistry and Plant Systematics since 21/4/2026. My research interests are natural products chemistry, with a special emphasis on recent advances in metabolomics tools and their application.

Abstract

Marine food science has gained increasing attention owing to its potential to offer valuable bioactive compounds beneficial for human health aside from nutritive value. Metabolomics, a cutting-edge analytical technology aims to provide a comprehensive insight into the chemical composition of marine food systems, ensuring quality and safety. This review explores the integration of metabolomics in marine food science, emphasizing its ability to profile metabolites for quality control, authentication, and safety determination in seafood products. Metabolomics can aid in identifying geographical origin, detecting contaminants, and monitoring freshness levels in seafood. Additionally, it plays a critical role in advancing aquaculture by monitoring the health and nutritional quality of farmed fish species. The use of advanced spectroscopy exemplified by nuclear magnetic resonance (NMR) and mass spectrometry (MS) allows for the discovery of novel bioactive compounds with therapeutic potential, enhancing the nutritional value of seafood. Despite the challenges in data interpretation due to the complexity of marine metabolites and their origin, the combination of metabolomics with other "omics" technologies held promise to revolutionize our understanding of marine ecosystems. Further research is needed to overcome current analytical challenges and translate these findings into practical applications for the food industry.

Assoc. Prof. Mona Abd El-Mouty Moustafa Raslan

Department of Pharmacognosy, Pharmaceutical and Drug Industries Research Institute, National Research Centre, Egypt

PRESENTATION: Beyond Random Screening: MS/MS Molecular Networking and the Evolution of Natural Product Drug Discovery

Biography

Dr Mona Raslan is an Associate Research Professor in the Department of Pharmacognosy at the Pharmaceutical and Drug Industries Research Institute, National Research Centre. Dr Raslan holds a Ph.D. in Pharmaceutical Sciences from Cairo University. She has extensive research experience in natural products chemistry and pharmacognosy. Dr Raslan has published numerous research articles in prestigious journals such as Phytochemistry Letters, Egyptian Journal of Chemistry, New Journal of Chemistry, and Chemistry and Biodiversity. Dr Raslanan is research focuses on various aspects of natural products research, including the isolation and characterization of natural compounds, evaluation of their biological activities, and development of novel therapeutic agents. Dr Raslan possesses extensive expertise in advanced spectroscopic techniques for the structural elucidation of natural compounds. She has actively participated in numerous research projects and scientific conferences.

Abstract

Landmark advances in bioinformatics tools have recently broken through long-standing bottlenecks in untargeted mass spectrometry, transforming the landscape of natural products research. Traditionally, discovery workflows have been severely hindered by the frequent rediscovery of known compounds. While modern liquid chromatography-tandem mass spectrometry (LC-MS/MS) captures thousands of metabolic profiles simultaneously, translating these massive datasets into actionable structural insights represents a major technical challenge. This presentation highlights MS/MS molecular networking as a transformative computational framework causing a true revolution in the field of natural product drug discovery. By employing vector-based spectral matching, molecular networking organizes and visualizes complex mass spectrometry data into distinct clusters, revealing underlying structural relationships based on fragmentation similarities. This powerful framework enables the rapid identification of known compounds within complex mixtures, significantly streamlining the dereplication process. Crucially, a cluster anchored by a known compound identified via spectral library matching simultaneously highlights associated nodes representing previously uncharacterized structural analogues. This inherent capability to visually map the intricate chemical landscape of complex samples allows researchers to expedite the prioritization of their isolation workflows toward entirely new molecular families. Ultimately, this talk will discuss the foundational concepts of molecular networking, its underlying construction metrics, and its diverse pharmaceutical applications. Furthermore, we will explore the Global Natural Products Social Molecular Networking (GNPS) platform, illustrating how this widely utilized ecosystem is practically deployed to turn raw spectral data into targetable chemical insights.

Assoc. Prof. Rehab Fikry Taher Fareed

Department of Pharmacognosy, Pharmaceutical and Drug Industries Research Institute, National Research Centre, Egypt

PRESENTATION: The Molecular Mechanisms Underlying the Antineoplastic Profile of Vitexin 2″-O-α-L Rhamnopyranoside From *Cordyline australis* (G.Frost.): Isolation and Constituent Profiling

Biography

Dr. Rehab Fikry is an associate professor of Pharmacognosy at the National Research Centre. She holds a Ph.D. in Pharmaceutical Sciences from Cairo University, specializing in natural products chemistry and pharmacognosy. Her research focuses on the isolation, identification, and biological evaluation of bioactive compounds from medicinal plants, with particular expertise in advanced spectroscopic techniques and chromatographic analysis. Dr. Fikry has extensive academic and research experience, having held positions at the National Research Centre and several universities, including Heliopolis University, Al-Ahram Canadian University and Egyptian Chinese University. She has authored numerous peer-reviewed publications in the fields of phytochemistry, pharmacology, and natural product-based therapeutics. Her work contributes to the discovery of plant-derived compounds with potential applications in treating diseases such as cancer, liver disorders, and metabolic conditions.

Abstract

Cordyline australis (G.Forst.) Endl. Red Star leaves have been used in traditional medicine for several disorders. This study investigated the anticancer potential of C. australis leaves extract and characterized its bioactive constituents. Three compounds, a steroidal saponin, fruticoside K 23, and two flavonoids, vitexin 2″-O-α-L-rhamnopyranoside 35 and helichrysoside 68,wereisolated and identified from its aqueous methanolic extract (CAME) using column chromatography. Seventy-nine additional compounds were tentatively identified using high-performance liquid chromatography coupled with electrospray ionization mass spectrometry (HPLC-ESI-MS/MS) analysis and The Global Natural Product Social Molecular Networking (GNPS-MN), with 57 compounds reported for the first time in the Cordyline genus. CAME and its isolated compounds were evaluated for cytotoxicity by MTT assay against seven different cancer cell lines. Vitexin 2″-O-α-L-rhamnopyranoside 35 exhibited significant cytotoxicity against HCT-116 colon cancer cells. Additionally, CAME, fruticoside K 23, andvitexin2″-O-α-L-rhamnopyranoside 35 demonstrated significant cytotoxicity against osteosarcoma (HOS) cells. Afterwards, the safety profile of CAME and all the isolated compounds were examined upon human normal cells BJ-1. At 100µg/mL, CAME and all isolated compounds showed a safe response on human normal BJ-1 cells (0.6%–8.5% cytotoxicity). Vitexin 2″-O-α-L-rhamnopyranoside 35 possessed the most significant selective anticancer response on osteosarcoma cells (HOS), with the least IC50 value of 43.7µg/mL. It induced apoptosis in HOS cells by modulating Bax, Bcl-2, and caspase-3 expression and caused G1 phase cell-cycle arrest. These results highlight C. australis as a source of potential anticancer agents, particularly vitexin 2″-O-α-L-rhamnopyranoside 35, which warrants further investigation for its therapeutic potential.

Assoc. Prof. Zeinab Abd Elkhalek Ali Elshahid

Chemistry of Natural and Microbial Products Dept., Pharmaceutical and Drug Industries Research Institute, NRC, Egypt

PRESENTATION: Investigation of Endophytic Fungi Secondary Metabolites with Potential Biological Activity

Biography

Associate professor of Therapeutic Biochemistry at Chemistry of natural and microbial products department, Natural Research center, PhD in Biochemistry, faculty of science, Ain Shams University.I'm Interested in the search for new drug candidates from natural resources with variable biological activities as antitumor, antimicrobial, antioxidant, the treatment of diseases like: Al-zaheimer, Diabetes and Skin Hyperpigmentation. I participated in several national and international research projects funded by NRC and ASRT. I'm the principal investigator of a research project, entitled (An approach towards the isolation of novel bioactive secondary metabolites from some endophytic and symbiotic fungi as a new therapeutic agent for the treatment of type II diabetes and its complications),funded by national research center. Member of several reputable organizations, Arab Medical Association Organization, the National Society of Human Genetics (NSHG), Arab Society for Medical Research, Alzheimer Association International Society to Advance Alzheimer's Research and Treatment (ISTAART), Women In Science Without Borders (WISB), OWSD, American Society for Microbiology, and the Egyptian Chemical Society. ORCID ID: https://orcid.org/0000-0001-8237-9568 . Publications: 38 original research articles in highly respected peer-reviewed international scientific journals. Scopus h-index = 10 and total citation of 250. Google Scholar h-index=11, total citation 323.

Abstract

Overview: The quest for novel compounds of natural origin has been a persistent ambition for pharmaceutical research. Natural compounds are the main source of active ingredients in medicines, and in spite of modern pharmaceutical synthesis techniques, natural products are still the basis of almost half of all approved drugs. This success is in part due to the fact that natural products usually display high bioavailability and high affinity to target, as well as a minor loss of entropy when binding to proteins. Many endophytes have the potential to synthesize various bioactive metabolites that may directly or indirectly be used as therapeutic agents against numerous diseases. Moreover, marine biological systems still represent an endless reserve of vast natural bioactivities and massive chemical entities that yield a characteristic source of chemical species exhibiting potential biological activities. Numerous studies have discovered bioactive marine-associated fungi and their metabolites Such bioactive products were evaluated for their antioxidant, radical scavenging, antimicrobial, and anticancer activities, among other biological activities. In this respect, the present work aimed to screen for bioactive secondary metabolites from fungal isolates isolated from some plants. These secondary metabolites will be assessed using various in vitro and in vivo systems, including antimicrobial, anti-inflammatory, and anti-cancer activities. The most active isolates that showed the highest bioactivity were selected for further studies. The potent isolates were subjected to large-scale fermentation on different media. Bioactivity measurements, chemical characterization, and structure elucidation will take place.

Assoc. Prof. Dr. Gehan Fawzy Abdel Raoof Kandeel

Pharmacognosy Department, Pharmaceutical and Drug Industries Research Institute, National Research Centre, Egypt

PRESENTATION: From Waste to Wellness: Bioactive Profiling and Multi-Target Therapeutic Potential of Vicia faba Peel By-Products

Biography

Dr. Gehan Fawzy Abdel Raoof, she completed her PhD in 2015 from Faculty of Pharmacy, Cairo University. She is an Associate Professor at Pharmacognosy Department, Pharmaceutical and Drug Industries Research Institute, National Research Centre Giza, Egypt. Her research focuses primarily on isolation, purification and identification of natural compounds from medicinal plants, and marine using advanced techniques for identification (UV, IR, 1H-NMR,1H-1H COSY, 13C-NMR, DEPT 13C-NMR, HSQC, HMQC), and bioactive assays in vivo and in vitro in natural products for use in treating different diseases. She has published many papers in international journals and has been serving as a reviewer and editor board member in many journals. She is a member of Organization for Women in Science for The Developing World (OWSD), a member of International Scientific Committee of World Academy of Science, Engineering and Technology (WASET).

Abstract

Plant by-products are increasingly recognized as sustainable sources of valuable bioactive compounds. The current study aims to investigate the bioactivities and metabolomic profile of the 70% methanol extract of Vicia faba peels. Comprehensive phytochemical profiling using ultra-high-performance liquid chromatography (UHPLC) hyphenated with quadrupole-time-of-flight tandem mass spectrometry (QTOF-MS) led to the tentative identification of 91 metabolites, most of which are reported in the tested plant for the first time. Furthermore, rhamnetin, quercitrin, rutin, and catechin were isolated from the tested extract and identified by different spectroscopic methods. Biological evaluations revealed an excellent safety profile, as both the crude extract and the isolated compounds exhibited no in vitro cytotoxicity against

normal skin fibroblast cells (BJ-1). The extract exhibited significant anticholinesterase activity, while rutin showed the highest antioxidant potential (90.54Â±0.73%) in the DPPH assay, alongside potent inhibition of carbohydrate-metabolizing enzymes. These findings suggest that V. faba peels represent a promising, safe agricultural by-product with multi-target therapeutic potential, particularly for managing diabetes, oxidative stress, and neurological disorders.

Assoc. Prof. Marwa N. Ahmed

School of Biotechnology, Nile University, Egypt

PRESENTATION: Evolutionary Trajectories of Antibiotic Resistance in *Pseudomonas aeruginosa*:Mechanisms, Dynamics, and Strategies for Constraint

Biography

Dr. Marwa Nabil Ahmed is currently an assistant professor of Microbiology at the Biotechnology School, Nile University. Her broad expertise includes microbial biotechnology, antimicrobial resistance, bacterial biofilms and industrial microbiology. She earned her master's degree in biotechnology from Georgia State University, USA, where she also worked as a teaching and research Assistant, contributing to undergraduate education and research training. She earned her PhD degree in 2019 in Immunology and Infectious Diseases from the Faculty of International Health and Immunology, University of Copenhagen, Denmark, where she studied the molecular mechanisms of biofilm resistance in Pseudomonas aeruginosa at the phenotypic, genomic, transcriptomic levels. Her research on the intersection of environmental and genetic factors on bacterial adaptation and pathogenesis under antibiotic stress has led to published works in high-impact journals. Her translational achievements include a U.S. patent, "Bactericidal Compositions and Methods for Treating Pathogenic Biofilms". In industrial biotechnology, Dr. Marwa has optimized microbial production of value added bioproducts from agro-industrial wastes. Additionally, she has implemented her bioinformatics expertise to elucidate the evolutionary trajectories of antibiotic resistance in pathogenic bacteria and to identify novel genetic pathways associated with the biosynthesis of valuable microbial bioproducts. As Co-Principal Investigator of the STDF One Health Research and Innovation Grant "Bacteriophages as a Potential Alternative Approach to Combating Extended Spectrum Beta-Lactamases Producing Bacteria," she continues advancing research on epidemiological surveillance, phage therapy and bioinformatic analysis. She has been supported by prestigious funding agencies, including the Georgia Research Alliance Venture Projects.

Abstract

Chronic bacterial infections, exacerbated by antibiotic resistance, pose a significant health challenge, particularly in the context of *Pseudomonas aeruginosa* (Pa) infections in Cystic Fibrosis (CF). The formation of biofilms by Pa complicates treatment, shielding bacteria and contributing to antibiotic resistance. Ciprofloxacin is a commonly used antibiotic for Pa in CF patients. Experimental evolution studies of PAO1 wild-type (WT) and ΔkatA in sub-inhibitory ciprofloxacin conditions revealed that biofilm growth promotes the development of low-level resistant mutants. Oxidative stress was found to increase mutagenesis and resistance development. Consequently, we explored the genome and transcriptome of the evolved PAO1 and ΔkatA colony biofilm and planktonic culture to elucidate the phenotypic, genomic and transcriptomic changes leading to antibiotic resistance evolution. Additionally, we investigated different strategies and formulations to constrain the evolution of antibiotic resistance including cellular hysteresis, antioxidants and postbiotics. The transcriptomic analysis of PAO1 and ΔkatA evolved populations indicates higher gene expression in CIP-evolved populations than CTRL populations. Similarly, biofilms (both CIP and CTRL) exhibit more expressed genes than planktonic populations, with minimal overlap between biofilm and planktonic conditions. Furthermore, cellular hysteresis strategy, application of postbiotics and antioxidants showed a significant reduction in the evolution rate of antibiotic resistance.

Overall, this comprehensive analysis enhances our understanding of the intricate genomic and transcriptomic changes in Pa populations under ciprofloxacin exposure, shedding light on both shared and population-specific adaptive responses.

Assoc. Prof. Rasha Mohamed Hassan

Medicinal and Pharmaceutical Chemistry Dept., Pharmaceutical and Drug Industries Research Institute, NRC, Cairo, Egypt

PRESENTATION: Synthetic Anti-Biofilm Agents as a Strategy in Combating Bacterial Resistance

Biography

Dr. Rasha is currently an Associate Professor of Pharmaceutical Chemistry at Medicinal and Pharmaceutical Chemistry Department, Pharmaceutical and Drug Industries Research Institute, National Research Centre, Cairo, Egypt. She obtained M.Sc. from Faculty of Pharmacy, Ain Shams University (2011) and Ph.D. from Faculty of Pharmacy, Cairo University (2015). Her publications cover a variety of drug-related areas in central nervous system, cancer, bacterial resistance, inflammatory and metabolism disorders in international peer-reviewed journals. She is in-charge of several valuable pharmaceutical research projects funded by National Research Centre and Science, Technology and Innovation Funding Authority (STDF), Egypt.

Abstract

The great increase in the bacterial resistance to the known antibiotics is considered a major obstacle in the control of many infections. Bacterial biofilm formation is regarded as a key element in bacterial resistance and has a substantial impact on the bacteria's ability to survive in host cells. Antibiotics affect bacterial cells in biofilms very differently than they do free-floating planktonic cells of the same strain. The anti-biofilm agents that could inhibit the biofilm production without affecting the bacterial growth, apply less selective pressure over the bacterial strains than the traditional antibiotics; thus the development of bacterial resistance would be less frequent. In this study, new quinazoline derivatives were synthesized and screened for their anti-bacterial effects. Some of the synthesized compounds inhibited biofilm formation in Pseudomonas aeruginosa, which is regulated by quorum sensing system, at sub-minimum inhibitory concentrations. These compounds decreased the exopolysaccharide production which is the major component of the matrix binding biofilm components together. Also, at sub-MICs Pseudomonas cells twitching motility was impeded by these compounds, a trait which augments the cells pathogenicity and invasion potential. Molecular docking study was performed to further evaluate the binding mode of the compounds as inhibitors of P. aeruginosa quorum sensing transcriptional regulator PqsR. The achieved results demonstrate that achieved compounds bear promising potential for discovering new anti biofilm and quorum quenching agents against Pseudomonas aeruginosa without triggering resistance mechanisms as the normal bacterial life cycle is not disturbed.

Assoc. Prof. Doaa Diab Khalaf

Microbiology and Immunology Dept., Veterinary Research Institute, National Research Centre (NRC), Egypt

PRESENTATION: Antimicrobial Resistance in the Animal Sector: The 'One Health' Approach as a Sustainable Framework

Biography

Dr. Doaa Diab Khalaf is an Associate Professor of Microbiology and Immunology at the National Research Centre (NRC) in Egypt. With a Ph.D. in Veterinary Medicine from Sadat City University, her professional focus lies in applied microbiology, food safety, and the development of diagnostic tools for veterinary and food industry applications. Throughout her career, Dr. Khalaf has contributed to various national research initiatives, participating in projects that explore the characterization of bacteriocins for food biopreservation, the development of diagnostic kits for meat adulteration, and the application of nanotechnology in managing cattle mastitis. Her research interests also extend to the study of antimicrobial resistance and the identification of pathogenic microbes in animal and food products, with a focus on translating laboratory findings into practical solutions for agricultural sustainability. In addition to her research, Dr. Khalaf is engaged in the institutional framework of the National Research Centre, participating in committees related to technology incubation and entrepreneurship. She is committed to the application of modern technical tools, such as AI, within the animal sector and actively participates in knowledge-sharing through scientific workshops and training sessions for graduate students. As a researcher, she is dedicated to the ongoing advancement of scientific methodology and the role of women in the scientific community. Her work aims to contribute to the development of animal wealth and public health through rigorous scientific inquiry and the exploration of new avenues in microbiology and innovation.

Abstract

Antimicrobial resistance (AMR) represents a growing global threat to food security and public health, necessitating the adoption of a "One Health" framework to integrate efforts across animal, human, and environmental health. The presentation addresses the scientific dimensions of microbial resistance in the animal sector, highlighting the importance of transitioning from total reliance on conventional antibiotics toward developing sustainable alternatives. The presentation examines innovative microbiological tools, such as the use of natural extracts and nanotechnology applications, as therapeutic alternatives that contribute to reducing the selection pressure leading to resistant strains. Furthermore, the abstract discusses the strategic role of rapid molecular diagnostics in enhancing herd management efficiency, thereby minimizing indiscriminate drug use and promoting sustainable animal production. Additionally, the presentation emphasizes that addressing this phenomenon requires an integrated scientific ecosystem that combines innovation in biological sciences with technological entrepreneurship. By fostering interdisciplinary collaboration and utilizing modern technologies, this talk aims to provide a practical roadmap to support the Sustainable Development Goals and ensure the preservation of biological resources for future generations.

Assoc. Prof. Islam Hany Ali Haridy

National Research Centre (NRC), Cairo, Egypt

PRESENTATION: **New Microtubule Modulators: Design, Synthesis and Anticancer Evaluation**

Biography

Dr. Islam Hany Ali Haridy obtained his Ph.D. (2018), M.Sc. (2012), and B.Sc. (2004) from the Faculty of Pharmacy, Cairo University. He specializes in medicinal chemistry, synthesis of heterocyclic anticancer agents, and tubulin targeting. ORCID: 0000-0002-5945-3472.

Abstract

New chalcone, and 4,5-dihydropyrazole compounds were designed and synthesized. A molecular docking study was done for the synthesized compounds at tubulin binding site. The newly synthesized compounds were tested for their in vitro anticancer screening to determine their inhibitory activities against a panel of cancer cell lines at 10 µM concentration at National Cancer Institute (NCI) - US. Tubulin polymerization assay was investigated for the synthesized compounds at 50 µM and compared with CA-4 and paclitaxel as standards. All the compounds showed microtubules targeting activity. Then, a tubulin immunohistochemical study was performed to the most active compounds to confirm their effects on tubulin polymerization via immunofluorescent detection. Cell cycle analysis and apoptosis were done via Flow cytometry for the tested compounds. The acute toxicity was performed to the most active compounds, in mice and then subjected to anticancer evaluation testing in rats. The anticancer parameters were determined in addition to histopathological examination. In conclusion, 4,5-dihydropyrazoles presented in vitro and in vivo anticancer activities without histopathological alterations.

Keywords: Chalcones, dihydropyrazoles, microtubules, modulators, tubulin polymerization.

Founder & CEO Yosra M. Morsy

Licensed Clinical Psychologist & Psychotherapist (Egyptian MoH), Ph.D. Candidate; Founder & CEO, Oasis Psychology Creations

PRESENTATION: **Mental Healthcare in 2035: From Clinics to Intelligent Care**

Biography

Yosra M. Morsy is a licensed Clinical Psychologist and Psychotherapist with the Egyptian Ministry of Health and a Ph.D. candidate in Clinical Psychology. She holds a Master's degree in Clinical Psychology and has extensive experience providing psychological assessment and psychotherapy for children, adolescents, adults, and older adults using evidence-based approaches. Yosra is the Founder and CEO of Oasis Psychology Creations, an initiative dedicated to advancing mental health through research, innovation, education, and interdisciplinary collaboration. Her professional interests focus on digital mental health, artificial intelligence in psychological practice, behavioral science, ethics in emerging technologies, and the future of mental healthcare. She is actively involved in international scientific collaborations and has presented her work at conferences focused on psychology, digital health, and innovation. Her current research explores how technology can enhance psychological assessment and improve access to evidence-based mental healthcare while maintaining ethical standards and human-centered practice. In addition to her clinical and research activities, Yosra is an international trainer and speaker who delivers workshops on psychology, mental health, and innovation. She is passionate about translating scientific knowledge into practical solutions that improve well-being and contribute to sustainable healthcare systems. Through her work, Yosra aims to bridge psychology with emerging technologies, fostering collaborations between clinicians, researchers, engineers, and entrepreneurs to shape the future of mental healthcare. Her long-term vision is to promote accessible, ethical, and innovative mental health services that create meaningful impact for individuals and communities worldwide.

Abstract

Mental health is becoming one of the world's most important public health priorities. However, the growing demand for mental health services is increasing faster than the availability of mental health professionals. This gap has encouraged researchers and innovators to explore new ways of delivering psychological care through technology while maintaining high standards of quality, ethics, and human connection. This presentation explores how mental healthcare may evolve by 2035, based on current scientific and technological trends. It discusses the growing role of artificial intelligence, digital mental health, virtual reality, wearable devices, digital therapeutics, and telepsychology in supporting prevention, early detection, assessment, treatment, and long-term follow-up. These innovations have the potential to make mental health services more accessible, personalized, and efficient for diverse populations. At the same time, the presentation emphasizes that technology should supportâ€"not replaceâ€"mental health professionals. Human empathy, ethical practice, clinical judgment, and the therapeutic relationship will remain the foundation of effective psychological care. The future of mental healthcare depends on successfully integrating technological innovation with evidence-based clinical practice. By bringing together psychology, healthcare, and innovation, this presentation provides a forward-looking perspective on how intelligent technologies can improve mental healthcare while preserving its human-centered values. The discussion contributes to Sustainable Development Goal 3 (Good Health and Well-Being) by highlighting innovative approaches to improving mental health services and expanding access to quality care.

Prof. Eman E. El Shanawany

Parasitology and Animal Diseases Dept., Veterinary Research Institute, National Research Centre (NRC), Egypt

PRESENTATION: Platyhelminth Glycans: From Host-Parasite Interactions to Diagnosis, Immunomodulation, and Vaccinology

Biography

Prof. Eman E. El Shanawany is Professor of Immunology and Parasitology at the Veterinary Research Institute, National Research Centre (NRC), Egypt, where she also serves as Technical Manager of the ISO/IEC 17025 Accredited Immunology and Parasitology Laboratory. She was a Visiting Research Fellow at the National Institute of Parasitic Diseases, Chinese Center for Disease Control and Prevention (China CDC), Shanghai, China, and completed an Erasmus International Credit Mobility teaching and research visit at Dublin City University, Ireland, where she taught undergraduate courses in cell culture and immunology. Her research focuses on host–parasite interactions, immunology, vaccine development, immunodiagnostics, cancer immunotherapy, tropical diseases, and One Health. She has served as Principal Investigator and Co-Investigator in several nationally and internationally funded projects, including vaccine development against toxoplasmosis, and international collaborative programs with China CDC. She also contributed to the development of a commercial glycoprotein-based diagnostic kit for fasciolosis. Prof. El Shanawany is the recipient of the National Research Centre Prize in Immunology Research (2023) and the National Research Centre Best PhD Thesis Award in Veterinary Sciences (2013). She has published extensively in international peer-reviewed journals, serves as a reviewer for several SCI-indexed journals, and is actively involved in scientific conference organization, postgraduate supervision, and laboratory quality management. She is a member of several professional scientific societies and is committed to advancing translational research in parasitology and immunology.

Abstract

Platyhelminths (Phylum Platyhelminthes, Subphylum Neodermata) comprise a diverse group of parasites of considerable veterinary, medical, and economic importance. Their glycans are expressed as components of O- and N-linked glycoproteins, glycolipids, and excretory/secretory products, as well as on the parasite surface, where they play essential roles in host–parasite interactions. These glycoconjugates interact with antigen-presenting cells through specific lectin receptors, particularly on dendritic cells, leading to modulation of the host immune response toward a T helper 2 (Th2)-dominant profile that promotes parasite survival. The adaptive immune response against platyhelminth glycans is characterized by the production of specific antibodies, including IgG and IgE, together with Th2-associated cytokines such as IL-4, IL-5, and IL-13. Anti-glycan antibodies represent valuable biomarkers and are widely exploited for the serodiagnosis of platyhelminth infections. Moreover, parasite glycoproteins and glycolipids have emerged as promising immunodiagnostic targets for the detection of infections in both humans and livestock, with their diagnostic performance largely depending on the structural characterization of glycan epitopes. Beyond diagnosis, the immunomodulatory properties of platyhelminth glycans have highlighted their potential in vaccine development and as novel therapeutic candidates for autoimmune and inflammatory diseases. This presentation discusses the biological significance of platyhelminth glycans and reviews their current and potential applications in diagnosis, immunomodulation, and vaccinology.

Dr. Aya Abdallah Seleem

Lecturer of Zoonoses, Faculty of Veterinary Medicine, Cairo University, Egypt

PRESENTATION: The Rise of Pan-Drug Resistance: Implications for Public Health and Zoonotic Diseases

Biography

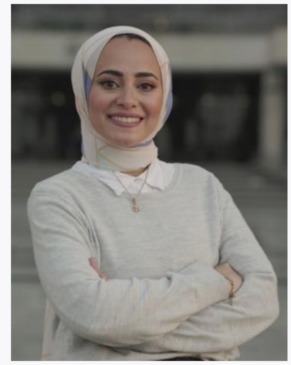
Dr. Aya Abdallah Seleem is a Lecturer of Zoonoses at the Faculty of Veterinary Medicine, Cairo University, Egypt. my expertise lies in molecular biology and antimicrobial resistance, with a research focus on zoonotic pathogens at the human–animal interface. My work explores the molecular epidemiology, detection, and characterization of infectious agents of veterinary and public health importance. My research interests include antimicrobial resistance profiling, molecular diagnostics of zoonotic bacteria, and surveillance of emerging and re-emerging zoonotic diseases within a One Health framework. She is particularly interested in understanding transmission dynamics between animals and humans and advancing evidence-based strategies for infection control and antimicrobial stewardship in veterinary settings. In addition to research, i teach zoonotic diseases and veterinary public health and is actively involved in training veterinary students in molecular and epidemiological approaches to infectious disease investigation. She is committed to promoting interdisciplinary collaboration and strengthening research capacity in zoonoses and antimicrobial resistance in the region. I am also engages in science communication and professional coaching initiatives that support awareness, resilience, and personal development among women in scientific and academic environments.

Abstract

Antimicrobial resistance (AMR) poses a global health threat. Pan-drug resistant (PDR) microbes, being resistant to all the available antimicrobial agents, are the most objectionable. Overuse and misuse of antibiotics in clinical, agricultural, and animal health environments have encouraged the emergence and spread of multidrug-resistant (MDR), extensively drug-resistant (XDR), and pandrug-resistant (PDR) strains of bacteria. Genetic mutations, plasmid-mediated gene transfer, enzymatic degradation of antibiotics, efflux pump overexpression, reduced membrane permeability, and modifications in target pathways all play a role in such mechanisms of resistance. Zoonotic bacteria only make it worse when strains of resistance spread further within humans as well as animals. The alternative options of classical therapy are declining, while new antibiotics are emerging at a slow pace. Hence, other methods are gaining prominence ever more. Examples include the utilization of artificial intelligence (AI) for the identification of drugs, use of antimicrobial peptides (AMPs) as the natural defense agents, and isolation of phytochemicals from medicinal plants possessing antibacterial and resistance-modulating activity. This work discusses the molecular mechanisms of resistance, global burden of antimicrobial resistance (AMR), and new interdisciplinary approaches to tackle pandrug-resistant (PDR) bacteria. To address this multifaceted issue, a need exists to enhance antimicrobial stewardship, augment surveillance efforts, and promote research on bioactive natural products. Global action is called for to maintain the efficacy of antimicrobial therapy, encompassing scientific discovery, regulatory changes, and public engagement in health initiatives.

Dr. Rana Ahmed Mohamed Eissa

Badr University in Cairo (BUC), Egypt



PRESENTATION: Antibacterial Film-Coated Stainless Steel Trays for Controlling *Klebsiella pneumoniae* and *Pseudomonas aeruginosa* in Intensive Care Units

Biography

Rana Ahmed Eissa is a Research Assistant at the Research Center, Badr University in Cairo (BUC). She earned her Bachelor's degree in Pharmacy from Ain Shams University in 2018. In 2020, she got her M.Sc. degree in Biotechnology from the School of Science and Technology, Nottingham Trent University, Nottingham, UK. She is currently a Ph.D. candidate in the Department of Microbiology and Immunology, Faculty of Pharmacy, Ain Shams University. In 2023, she was awarded the L'Oréal–UNESCO For Women in Science Egypt Young Talents Fellowship in the doctoral category. Her research interests focus on the formulation and evaluation of controlled-release drug delivery systems and microbiology, molecular biology, and biotechnology techniques. She contributed as co-author of several publications in peer-reviewed journals. She has also participated in research projects funded by the Egyptian Academy of Scientific Research and Technology and the Science, Technology and Innovation Funding Authority. She has been actively involved in delivering annual summer training programs for undergraduate students. She has also supervised two undergraduate graduation projects at the School of Biotechnology for the Classes of 2024 and 2025. In October–November 2024, she completed a one-month research visit to the Microbiology Laboratories at the School of Science and Technology, Nottingham Trent University. During this period, she gained hands-on experience in molecular biology techniques, including RNA extraction, cDNA synthesis, and real-time PCR to analyze. This experience further enhanced her skills in eexperimental optimization and gene expression data analysis.

Abstract

Infections caused by multidrug-resistant (MDR) bacteria in intensive care units (ICUs) are a leading cause of global morbidity and mortality, placing a significant burden on healthcare systems managing critically ill patients. Gram-negative pathogens, including *Klebsiella pneumoniae* and *Pseudomonas aeruginosa*, are highly prevalent on stainless steel trays in ICUs, and their ability to adhere to medical devices poses a serious threat to human health. Moreover, these pathogens often exhibit resistance to conventional antimicrobial therapies. Nanotechnology, combined with antibiotic therapy, has emerged as a promising alternative of conventional methods to manage MDR infections.

In this study, a novel system was developed to control the growth of *K. pneumoniae* and *P. aeruginosa* on stainless steel trays in ICUs. An antibacterial chitosan-silver nanoparticle (AgNP) film was synthesized and further loaded with ciprofloxacin and vancomycin. The nanocomposite film exhibited a mean number-averaged hydrodynamic diameter of 202 ± 3.46 nm, which was confirmed by scanning electron microscopy, showing well-defined nanovesicles evenly distributed throughout the film. In vitro characterization of the antibacterial film was then performed, including evaluation of swelling ratio, moisture content, and Fourier transform-infrared spectroscopy. The antimicrobial activity of the nanocomposite film loaded with vancomycin and ciprofloxacin demonstrated significant inhibition of *K. pneumoniae* and *P. aeruginosa* compared to the individual antibiotics alone. The successful application of this antibacterial chitosan-AgNP film could add new infection control strategies, improve patient outcomes and reduce the burden of healthcare-associated infections.

Assoc. Prof. Reham Amgad Mohamed-Ezzat

Chemistry of Natural & Microbial Products Dept., Pharmaceutical & Drug Industries Research Institute, NRC, Cairo, Egypt

PRESENTATION: Synthetic Strategies Towards Novel Sulfonamides as Potentially Active Agents

Biography

Reham Amgad Mohamed-Ezzat, PhD in Organic Chemistry, Associate Professor of Therapeutic Chemistry at Chemistry of Natural & Microbial Products Department, Pharmaceutical & Drug Industries Research Institute, National Research Centre, Cairo, Egypt. She got her BSc (2010) in Chemistry & Biochemistry-Faculty of Science. She was awarded her MSc (2015) & PhD degree of science in organic chemistry in 2019 from Helwan University. She has published books and articles in international journals in the field of bio-organic and medicinal chemistry. She also published many conferences presentations. She has experience in the field of design and synthesis of new molecules of medicinal application, as well as the implementation of novel strategies in the syntheses of bioactive compounds. She is a member in the Egyptian Chemical Society, and in the Egyptian Society for Basic Sciences. E-mail: reham_amgad_2010@yahoo.com; ra.mohamed-ezzat@nrc.sci.eg. Phone: 01220613360 Scientific Achievements: H-INDEX 14 No. of publications: 38 https://www.scopus.com/authid/detail.uri?authorId=55502779900 Last published book in Elsevier. New Strategies Targeting Cancer Metabolism Anticancer Drugs, Synthetic Analogues and Antitumor Agents 1st Edition - June 7, 2022 Latest edition Imprint: Elsevier Authors: Galal H. Elgemeie, Reham A. Mohamed-Ezzat Language: English Paperback ISBN: 9780128217832 - eBook ISBN: 9780128217849

Abstract

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New synthetic approaches were developed for the synthesis of novel sulfonamides as potent therapeutic agents, utilizing various sulfaguanidine derivatives. Additionally, a new class of pyrimidine sulfonamide thioglycosides was successfully designed and synthesized. The newly obtained substituted pyrimidine sulfonamides, reported here for the first time, were produced in high yields and demonstrated promising broad-spectrum biological activities, including antitumor, antimicrobial, and antiviral effects. The in vitro antiproliferative activity of most synthesized compounds was evaluated using the NCI-60 human cancer cell line panel. Furthermore, the antimicrobial potential of these compounds was also investigated. The anti- viral activity of the compounds against SARS CoV-2 virus was investigated utilizing MTT cytotoxicity assay to estimate the half-maximal cytotoxic concentration (CC 50) and inhibitory concentration 50 (IC 50). The most active compound revealed potent anti-viral potency against SARS-CoV-2 comparable to that of remdesivir. Our results demonstrate that these novel pyrimidine sulfonamides could potentially serve as novel anti-viral drugs.

Keywords:

Synthesis, pyrimidines, Sulfonamides, SARS-CoV-2.

Assoc. Prof. Iman Ahmed Youssef Ghannam

Chemistry of Natural and Microbial Products Dept., Pharmaceutical and Drug Industries Research Institute, NRC, Egypt

PRESENTATION: Development of Benzenesulfonamide-based PPARγ Agonists for the Treatment of Type II *Diabetes Mellitus*

Biography

Dr. Iman Ahmed Youssef Ghannam is an Associate Professor of Pharmaceutical Chemistry - Chemistry of Natural and Microbial Products Department - Pharmaceutical and Drug Industries Research Institute - National Research Centre (NRC) - Cairo - Egypt since January 2023. Dr. Ghannam has her area of expertise in the development and rational design of novel motifs targeting cancer and metabolic and neurodegenerative disorders including hyperlipidemia, Type II diabetes mellitus, and Alzheimer's disease (AD). Dr. Ghannam has joined as a principal investigator (PI), Co-PI, and member in several research projects funded from National Research Centre (NRC) and Science and Technology Development Fund (STDF) - Egypt since 2015. Dr. Ghannam has also participatedas a PI for a joint collaboration with Consiglio Nazionale delle Ricerche (CNR) - Italy through a joint NRC-CNR bilateral co-funded research project with Dr. Francesca Sciandra in 2019. Dr. Ghannam has co-supervised M.Sc. and PhD students from Faculty of Pharmacy - Cairo University since 2015 till present. Dr. Ghannam co-authored several publications including research papers and review articles indexed in (SCOPUS) in highly impacted journals in medicinal chemistry since 2012, and also peer-reviewed for several manuscripts in several Journals; Journal of Medicinal Chemistry, European Journal of Medicinal Chemistry, Bioorganic Chemistry, European Journal of Medicinal Chemistry Reports, Archiv der Pharmazie, Scientific Reports, and ACS Chemical Neuroscience, and Molecular Diversity.

Abstract

Iman A. Y. Ghannam [1], Islam H. Ali [1], Rasha M. Hassan [2], Heba M. I. Abdallah [3], Mahmoud T. Abo-elfadl [4,5], Ahmed M. El Kerdawy [6,7]

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Peroxisome proliferator-activated receptors (PPAR) isoforms (α, γ, and β/δ) play a crucial regulatory role in glucose, lipid metabolism. Thiazolidinone-benzenesulfonamide and chalcone-benzenesufonamide hybrids were designed, synthesized and tested *in vitro* for their PPARɣ agonistic activity comapred to pioglitazone via determination of their EC50 values. Immunohistochemical studies were performed in HepG-2 cells to confirm the PPARγ protein expression for the most active compounds. Morovere, the most active compounds were tested for their selectivity on PPARα/δ isoforms. In vivo studies revealed that the most compounds possessed antidiabetic activity better than that of pioglitazone. Furthermore,

molecular docking studies and molecular dynamic simulations were carried out for the newly synthesized compounds to study their predicted binding modes and energies in the PPARγ binding site.

Dr. Afnan Hamdy Elgowily

Chemistry Department, Biochemistry Section , Faculty of science , Tanta University

PRESENTATION: **Pathophysiological Significance of Mallory Denk Bodies: Addressing a Century-Old Mystery**

Biography

Afnan Elgowily, PhD, is a postdoctoral researcher at the Autophagy Research Center, Department of Physiology, Juntendo University Graduate School of Medicine, Tokyo, and a Lecturer of Biochemistry at Tanta University, Egypt. She earned her PhD in Biochemistry through a joint program involving Niigata University, Juntendo University, and Tanta University, following earlier MSc and BSc training in biochemistry and chemistry. Her research focuses on autophagy, selective cargo recognition, protein quality control, cancer biology, and disease-associated cellular stress. She has extensive experience in molecular and cell biology, including CRISPR Cas9 gene editing, molecular cloning, protein purification, stable and transient cell engineering, mammalian cell culture, real-time PCR, immunoblotting, flow cytometry, ELISA, and animal experimentation. Her current work investigates how NBR1 and p62/SQSTM1 cooperate in selective autophagy and how their interaction regulates the organization, composition, and maturation of p62 bodies. Afnan has contributed to research published in Nature Communications, the Journal of Biological Chemistry, Cell Biochemistry and Function, Biomedicines, Nanomaterials, and other international journals. Her publications include studies on UFM1-dependent ER-phagy, phase-separated p62 condensates, cancer-associated autophagy, apoptosis, drug delivery, and therapeutic repurposing. She received a Chugai Foundation postdoctoral fellowship and an international Egyptian mission scholarship for research training in Japan. Alongside research, she has taught practical biochemistry and chemistry courses to undergraduate students since 2010. Her long-term goal is to advance mechanistic discoveries in autophagy and proteostasis toward improved understanding and treatment of human disease through international collaboration. She also values scientific mentoring, interdisciplinary teamwork, and building research partnerships between Japan, Egypt, and global institutions.

Abstract

Afnan Elgowily and Masaaki Komatsu

Department of Physiology, Juntendo University Graduate School of Medicine, Tokyo, Japan

Mallory Denk bodies (MDBs) are cytoplasmic inclusions frequently found in hepatocytes in chronic liver diseases, including alcoholic hepatitis, metabolic dysfunction-associated steatotic liver disease, cholestatic liver disease, and hepatocellular carcinoma. They contain ubiquitin, p62/SQSTM1, and cytokeratin filaments and are thought to represent solidified p62 bodies membraneless condensates formed by p62 and polyubiquitinated proteins through liquid liquid phase separation that evade autophagic clearance. Beyond sequestering aberrant proteins for degradation, p62 bodies also regulate stress-responsive signaling, including NRF2 activation through KEAP1 sequestration. We previously showed that TBK1 phosphorylates p62 at Ser403, promoting condensate gelation and autophagic degradation. Here, we identify casein kinase 2 (CK2) as a previously unrecognized p62 kinase. AlphaFold3 modeling predicted a stable CK2 p62 interaction interface, and in vitro pull-down assays confirmed direct binding. Reconstitution assays demonstrated efficient recruitment of CK2 into p62 ubiquitin condensates. This recruitment was abolished by p62 interface mutations (L307A/M311A/I314A), despite preserved condensate formation. In vitro kinase assays showed that CK2A1 and CK2A2 phosphorylate p62 specifically at Ser349, but not Ser403; notably, only CK2A1 phosphorylated Ser349 within condensates.

Because Ser349 phosphorylation enhances KEAP1 binding, CK2 recruitment is expected to promote KEAP1 retention and NRF2 stabilization. Immunofluorescence further confirmed CK2 localization to p62 bodies in cells. These findings define a CK2 p62 NRF2 signaling axis distinct from TBK1Ser403-mediated regulation of condensate material properties. CK2-dependent Ser349 phosphorylation may contribute to hepatocyte adaptation, MDB persistence, and chronic liver pathology, providing a framework for in vivo studies and therapeutic evaluation of CK2 p62 signaling in MDB-associated liver disease and clarifying the century-old pathophysiological significance of MDBs in humans.

Assoc. Prof. Amira Samy Talaat Abou Taleb

Computers and Systems Department, Electronics Research Institute (ERI), Cairo, Egypt

PRESENTATION: Beyond the Black Box: How Explainable AI Can Support Trustworthy Healthcare Decisions

Biography

Amira Samy Talaat is an Assistant Research Professor at the Computers and Systems Department, Electronics Research Institute, Cairo, Egypt. Her research interests include artificial intelligence, machine learning, explainable artificial intelligence, computer vision, intelligent systems, and their applications in healthcare and engineering. Her work focuses on developing transparent, reliable, and human-centered artificial intelligence methods that support informed decision-making and improve understanding of complex machine-learning models. She is particularly interested in explainable artificial intelligence techniques that help researchers and professionals understand why an AI system produces a specific prediction. Her research also addresses model reliability, data quality, fairness, and responsible artificial intelligence. She has participated in multidisciplinary research projects and scientific publications involving medical data analysis, predictive modeling, and interpretable artificial intelligence. Through her academic and research activities, she aims to promote the responsible use of artificial intelligence and strengthen collaboration between computer scientists, engineers, healthcare professionals, and researchers from different disciplines. Her participation in the World Forum for Women in Science reflects her interest in international scientific exchange, knowledge sharing, and the contribution of women researchers to sustainable development and technological innovation.

Abstract

Artificial intelligence is increasingly used in healthcare to support disease diagnosis, risk prediction, medical imaging, treatment planning, and patient monitoring. However, many advanced artificial intelligence models operate as black boxes,•providing predictions without clearly explaining how those predictions were produced. This lack of transparency may reduce the confidence of healthcare professionals and patients, particularly when artificial intelligence is used in high-stakes medical decisions. This presentation introduces Explainable Artificial Intelligence (XAI) as an approach for making artificial intelligence predictions more understandable, transparent, and useful to human decision-makers. It explains the difference between global explanations, which describe the overall behavior of a model, and local explanations, which clarify an individual prediction for a specific patient. The presentation also introduces commonly used explanation techniques, including SHAP and LIME, and demonstrates how they can identify the clinical variables that influence an artificial intelligence prediction. Practical applications of XAI in radiology, pathology, genomics, clinical decision-support systems, intensive care monitoring, mental health, and remote patient monitoring will be discussed. The presentation also addresses important limitations, including misleading explanations, bias, the trade-off between predictive performance and interpretability, and the need for explanations that are meaningful to healthcare professionals rather than merely technically accurate. Explainable AI should not be considered a replacement for medical expertise. Instead, it can serve as a communication and decision-support tool that enables clinicians to examine, question,

and appropriately use artificial intelligence predictions. By promoting transparency, accountability, and informed human oversight, XAI can contribute to more trustworthy and responsible healthcare innovation.

Dr. Dina Magdy Ghanem

Department of Pharmacognosy, Pharmaceutical and Drug Industries Research Institute, National Research Centre, Egypt

PRESENTATION: Integrated Phytochemical Profiling and Biological Evaluation of Egyptian *Tribulus terrestris* L.: A Promising Source for Therapeutic Product Development

Biography

Dr. Dina M. Ghanem is a pharmacognosy researcher at the Pharmaceutical and Drug Industries Research Institute, National Research Centre (NRC), Egypt. She earned her Ph.D. in Pharmaceutical Sciences (Pharmacognosy) from Cairo University in 2025, following her M.Sc. degree in 2016 and B.Sc. degree with honors in 2005. Her doctoral research focused on the pharmacognostical study of *Carissa carandas* and *Carissa macrocarpa* cultivated in Egypt. With nearly two decades of research experience, Dr. Ghanem specializes in medicinal plants, natural products chemistry, ethnopharmacology, and pharmacognostic evaluation. Her expertise encompasses phytochemical screening, metabolite profiling, extraction techniques, chromatographic analysis, and the isolation and structural elucidation of bioactive compounds using advanced spectroscopic and analytical methods. She is also actively involved in biological activity assessment, metabolomics, molecular docking, and computer-aided drug discovery. Dr. Ghanem has contributed to several peer-reviewed scientific publications addressing the phytochemistry and pharmacological potential of medicinal plants. Her research has focused on identifying natural bioactive compounds with promising therapeutic applications and exploring sustainable plant-based resources for drug discovery and development. She is a member of the Special Unit of Pharmacognosy and Chemistry of Medicinal Plants, the National Laboratory for Algae, and the Incubator for Nutritional Supplements and Pharmaceutical Preparations from Medicinal Plants at the National Research Centre. In addition, she actively participates in scientific conferences, workshops, and educational programs, contributing to the training of university and school students. Her current research interests include medicinal plants, metabolomics, natural product chemistry, molecular networking, pharmacological evaluation of plant-derived compounds, and computational approaches for modern drug discovery.

Abstract

The genus Tribulus of the Zygophyllaceae family includes numerous species that grow as shrubs or herbs in subtropical regions around the world. Aerial parts of Tribulus terrestris L. growing in Egypt were successively extracted with solvents of increasing polarity. These successive extracts were studied phytochemically and biologically to discover the importance of this plant as well as the relation between its secondary metabolites and the proven biological activities. Isolation and identification of the pure compounds from the biologically active fractions were carried out using different chromatographic and spectroscopic techniques. The biological activities of polar, non-polar, and saponin fractions were evaluated. GC/MS analysis of the volatile constituents revealed that methyl linolenate (18.56%) is the major compound. GLC analysis of the lipoidal matter revealed that heptadecanoic acid (33.56%) is the main component. HPLC analysis of carbohydrate and amino acid contents revealed the presence of inulin (5.61%) and glutamic acid (2.85%) as the major components, respectively. Phytochemical investigation of the biologically active aqueous methanolic extract revealed the presence of phenolic acids (11.16%), flavonoids (6.076%), as well as steroidal saponins (7.38%). Purification of the aqueous methanolic extract led to isolation and identification of rutin, quercetin and diosgenin. The results of the biological investigation of both polar and non-polar fractions revealed their aphrodisiac, antimicrobial, cytotoxic and antioxidant activities with remarkable effects. The herb in this study showed potent and noteworthy biological activities due to the presence of various active secondary metabolites.

--- SESSION 5 - MEDICAL SCIENCES AND BIOLOGY ---

Prof. Rasha Mohamed Abdelraouf

Professor of Biomaterials & Postgraduate Course Director, Faculty of Dentistry, Cairo University, Egypt

PRESENTATION: From Static to Smart: 4D Printing in Dentistry

Biography

Rasha Mohamed Abdelraouf is Professor of Biomaterials and Postgraduate Course Director at the Faculty of Dentistry, Cairo University. Her research focuses on dental biomaterials and innovative dental technologies. She is the recipient of the State Encouragement Award for Women in Health and Pharmaceutical Sciences (2024; awarded in 2025). She holds an Egyptian patent and an international Patent Cooperation Treaty (PCT) application and has published extensively in international peer-reviewed journals.

Abstract

The 4D printing, being an additive manufacturing technique, reduces waste and allows the fabrication of complex structures compared to subtractive manufacturing. The 4D printing is an advancement of already-existing 3D printing by including time. The 4D printing involves the use of 3D printing techniques for printing smart materials. The 4D structures can respond with time to stimuli to which they are exposed. They can respond to heat, water, light, electric energy, magnetic energy, stress, strain, pressure, etc. Several materials could be printed with this technology. The 4D printing has several applications, introducing smart functional biomaterial.

Assoc. Prof. Dr. Tamer Mohamed Hamdy

Restorative and Dental Materials Dept., Oral and Dental Research Institute, National Research Centre (NRC), Egypt

PRESENTATION: Intelligent Dentistry: How Smart and 4D Materials Are Transforming Oral Care

Biography

Dr. Tamer Mohamed Hamdy Mahmoud Ibrahim is an Associate Researcher Professor in the Department of Restorative and Dental Materials Department, Oral and Dental Research Institute, National Research Centre (NRC). He obtained his PhD in 2016 in in Dental Materials Science from the Faculty of Dentistry, Cairo University. In 2019, he obtained certification as an international trainer from Ain Shams University, Cairo, Egypt. Serves as an international peer reviewer for several international dental journals. Dr. Tamer M. Hamdy received the National Research Centre Award for Best International Scientific Publication for Assistant Researchers in 2015, followed by the NRC Award for Best PhD Thesis in 2017. In 2021, he was also honored with the Oral and Dental Research Institute Award. In 2026, he obtined The Excellence Award in Research in the Field of Medical Sciences and Its Technological Applications from Members' Club of Research. In 2025, he contributed, in collaboration with others, to a nationally registered patent. He was also recognized among the top 2% of scientists worldwide according to the Stanford University ranking for the years 2024 and 2025.

Abstract

Smart dental materials represent a transformative shift in modern dentistry, characterized by their inherent ability to sense environmental changes and respond accordingly. Also known as “responsive materials,” these systems can alter their properties in a controlled manner when exposed to stimuli such as stress, temperature, moisture, pH, or electric and magnetic fields. Recent advances have introduced a wide range of smart materials into the dental market, including smart resin composites, glass ionomer cements, pit and fissure sealants, endodontic instruments, orthodontic wires, silicone elastomers, bite registration materials, and thermochromic toothbrushes. Notably, chromatic dental composites exhibit adaptive color-matching capabilities, enhancing aesthetic outcomes. Furthermore, the emergence of 4D printing adds a dynamic dimension, enabling materials to change shape or function over time in response to environmental triggers. Collectively, these innovations promise to revolutionize clinical practice by improving treatment precision, durability, and patient-specific outcomes, paving the way toward a new era of intelligent and adaptive dental care.

Assoc. Prof. Eman Samy Shalaby

Pharmaceutical Technology Dept., Pharmaceutical and Drug Industries Research Institute, NRC, Egypt

PRESENTATION: Bilosomes as a Promising Approach for Topical Application of Allantoin Intended Against Skin Aging

Bilosomes as a promising approach for topical application of Allantoin intended against skin aging

Biography

Dr. Eman Samy Shalaby is an Associate Professor of Pharmaceutics at NRC with 19 Scopus-indexed publications and active leadership in national/international formulation projects.

Abstract

Skin aging is one of the most common issues in the world, affecting nearly all women. Our study's objective was to evaluate allantoin bilosomes ability to prevent a rat model of aged skin. Allantoin bilosomes (ALL-BIL) were prepared and optimum formulations were selected that showed the highest encapsulation efficiencies, optimum particle size, and zeta potential. To improve application and industrial sustainability, the chosen formulation was included into a cream. The ALL-BIL loaded cream (ALL-BIL-C) showed pseudoplastic flow and shear thinning behaviour. The effectiveness of ALL loaded chosen formulation for aging healing was evaluated which is reflected in the histopathological study. These results support the ALL-BIL-C's further development and assessment as a viable dermatological or cosmeceutical intervention and offer insights into the potential mechanisms by which it can exert its anti-skin aging benefits.

Salma Elmaaz

Master's Student at Founder, Solvira Pharma for Pharmaceutical Industries, Egypt

PRESENTATION: Synergistic Topical Cream Formulation for Burn Scar Reduction Using Achatinoidea Shell Powder

Biography

Salma Elmaaz is STEM student graduated senior 2021, Bachelor degree (B.Sc.) holder from Faculty of Science, Chemistry Department, Menoufia University with Excellent with honored. Master degree (M.Sc) organic chemistry student. She was selected as the top 20 scientists in the UNESCO Nanotechnology Program in Africa and studied a Nanotech diploma at the British University in Egypt. National Patent Holder from the Ministry of Scientific Research for inventing an ointment to treat burns naturally. Founder of Solvira startup creating the solutions to burn patients' challenges. Salma was a national finalist in Intel ISEF and won 1st place in the SCIENCE FAIR competition. UNFCCC YOUNGO Ocean's Voice team and was listed among the top 100 Arab youth and 1000 junior academy youths in New York. She won 1st place at ICSSD for best applicable product and ranked 1st in scientific research competitions at Menoufia University, GIZ Greener Hackthon and Nile University (UGRF) at Egypt level. Salma represented Egypt at COP27 and the RCOY MENA in Jordan. She won 3rd place in the Arab world at the Creative Student Forum in Jordan and was named Ideal Student of Egypt 2023. Salma graduated from the EcoSciGen Climate Fellowship and was selected as a top 150 Max Thabiso Edkins Climate Ambassador.

Abstract

Burns are still one of the most important causes of death and disability in the world, where 800,000 people get burned per year. According to the World Health Organization, burns are the third-leading cause of death in the world. There are many ways to treat burns. One of these ways is by using a material called "betastestrol" that treats the burns without their effects. There are many ways to treat the effects caused by burns, such as lasers and surgical procedures, but these have negative effects like drying the skin. In this research, the project works to treat burns and eliminate their effects by using an ointment provided with nanotechnology to disinfect burns, treat infections, relieve pain, rejuvenate damaged cells, and repair body tissues; moisturize the skin and whiten it, give it freshness, remove burn-caused black spots, repair damaged tissues, regenerate damaged skin, and eliminate the effects of burns. The ointment treat burns and eliminate their deformities without using lasers or surgical procedures, and the treatment period is also very short; in the first degree, it takes 7 days to treat the burns and their effects. In the second degree, it takes 14 days. third degree, it takes about 22 days to treat the burns and their effects.

Assoc. Prof. Yousra Aly Ibrahim Sabry

Restorative and Dental Materials Dept., Oral and Dental Research Institute, NRC, Cairo, Egypt

PRESENTATION: Innovative Trends in Dentistry: A Shift Toward Conservative

Concept

Biography

My name is Yousra Aly Ibrahim. I am an Associate Professor of Endodontics, Restorative and Dental Materials Department, Oral and Dental Research Institute, National Research Centre, Cairo, Egypt. I am an Associate Professor in the field of endodontics, passionate about combining clinical practice with research innovation; with particular interests in regenerative endodontics and novel therapeutic strategies for hard tissue repair. I was a member in two Dental Projects Team at the National Research Centre in Egypt from (2016-2019) and from (2019-2022).

Abstract

Modern healthcare is moving toward treatments that preserve natural tissues whenever possible, and dentistry is following the same direction. One of the most promising innovations is adult pulpotomy, a minimally invasive procedure that aims to preserve the healthy part of the dental pulp instead of removing it completely through conventional root canal treatment. Recent advances in bioactive materials have significantly improved the success of this approach. These materials not only seal the treated cavity; but also stimulate natural healing and tissue repair, allowing the tooth to maintain its vitality and function. As a result, adult pulpotomy is becoming a reliable treatment option for carefully selected cases that were previously managed with more invasive procedures. This presentation highlights the scientific principles behind adult pulpotomy, the role of innovative biomaterials in its success, and its potential impact on the future of patient-centered dental care. It also discusses how this shift reflects a broader trend in healthcare toward regenerative, conservative, and biologically based treatment strategies.

Assoc. Prof. Salma Ahmed El Marasy

Department of Pharmacology, Medical Research and Clinical Studies Institute, NRC, Egypt

PRESENTATION: Anti-Osteoporotic Effect of Sitagliptin in an Osteoporosis Model of Ovariectomized Rats: Role of RUNX2 and RANKL/OPG Ratio

Biography

Associate Professor, Pharmacology Department, Medical Research Institute and Clinical Studies, National Research Centre (NRC), Egypt. PhD in 2015, in pharmaceutical sciences (Pharmacology and Toxicology), Faculty of Pharmacy, Cairo university. Published more than 25 articles related to experimental pharmacology in national and international journals. Participated as oral and Short talk (3 minutes) presenter in national and international conferences. Trainer in NRC Summer training courses for undergraduates 7 years from today's date. Coordinator and speaker in the workshops entitled "Ethics in Research" and "Tips and tricks in oral presentation skills" held at NRC, 2023. Coordinator and speaker of workshop entitled "Tips for Research Networking, writing an Impactful Research Paper and Grant Proposal" held at, 2022. Completed Nature Research Academics workshop entitled "Grant Writing", "Research Methodology" and "Writing a Research Paper" funded by STDF, 2022. Member in NRC Capital incubator funded by ASRT from March 2023 till now. Principle investigator of the project entitled: "Diabetes mellitus associated complications: Role of Advanced Nano-technology in the Delivery of Natural Flavonoids", NRC, 2020-2022. Principle investigator and workshop co-ordinator of ''Researcher Connect project" under Newton Mosharafa fund, 2017 and under STDF fund, 2020 delivered by the British council at the NRC. Participated as facilitator in "Responsible conduct of research (RCR)" workshop funded by National Academy of Sciences-USA (NAS) and hosted by NRC on 2015 and 2016. Co-ordinator of ''Researcher's Required Skills'' workshop held at the NRC, 2016 and 2017. Co-ordinator and speaker in various symposiums and workshops held at the premises of the NRC in 2016. Finalist in Famelab Egypt 2015, attended Masterclass presented by science communicator/ media trainer, Mr. Karl Byrne in science communication to improve presentation and communication skills, won the audience vote. Attendant and participated in various modules held within the DAAD Kairo Academie. Participated as organizer in the 1st Scientific forum for applied medical research marketing, National Research Centre ‘’ The Future is Now’’ held at Nile Ritz Carlton Cairo-Egypt on December 17, 2016. Obtained certificate of appreciation as organizer in the 12th Annual congress of Medical Sciences, under the theme ‘’Medical breakthrough technologies towards a healthy community’’ within the National Research Centre in 2014.

Abstract

Salma A. El-Marasya,*, Rehab F. Abdel-Rahmana, Reham M Abd-Elsalamb, Hanan A. Ogalyc, Rasha M. Allama

a Department of Pharmacology, Medical Research and Clinical Studies Institute, National Research Centre, Giza, Egypt

b Department of Pathology, Faculty of Veterinary Medicine, Cairo University, Giza, Egypt

c Department of Biochemistry and Chemistry of Nutrition, Faculty of Veterinary Medicine, Cairo University, Giza, Egypt

This study examines the potential anti-osteoporotic effect of sitagliptin in osteoporosis instigated by ovariectomy (OVX) in rats. Rats were assigned into 4 groups: Sham-operated, OVX group, and OVX rats orally treated with sitagliptin (10, 20 mg/kg), respectively, after 8 weeks of OVX for 4 weeks. Biochemical, real-time polymerase chain reaction, histopathological, and immunohistochemical analyses of bone resorption and formation were conducted. Sitagliptin ameliorated bone mineral density (BMD), restored calcium and phosphorus levels in OVX rats, elevated catalase and decreased malondialdehyde, reduced receptor activator of NF-κB ligand (RANKL), elevated osteoprotegerin (OPG), and reduced tartrate-resistant acid phosphatase (TRAP) femur contents. Sitagliptin mitigated variations in mRNA expressions of RUNX2 and protein kinase B (AKT) in femur tissue. Moreover, sitagliptin reduced caspase-3 protein expression and improved bone histomorphology and mechanical properties. Sitagliptin’s anti-oxidant activity mediated its anti-osteoporotic effect in OVX rats via modulation of RUNX2, downregulation of RANKL/OPG, AKT pathways, apoptosis, and histomorphometry alterations revealing attenuation of osteoclastogenesis and promotion of osteoblast formation.

Dr. Toka A. Ahmed

Faculty of biotechnology, Badr University, Cairo, Egypt

PRESENTATION: Human Adipose-Derived Pericytes in the Pathophysiology of Type 2 *Diabetes Mellitus*

Biography

Dr. Toka A. Ahmed is an accomplished researcher with 11 years of experience in stem cell biology, tissue engineering, and regenerative medicine. She holds a Ph.D. in Biochemistry from Cairo University (2024), along with an M.Sc. (2019) and B.Sc. (2014) from Helwan University. Her expertise spans stem cell reprogramming, differentiation, 3D spheroid models, and biomimetic vascularized scaffold systems (e.g., amniotic membrane, Matrigel). Notably, her M.Sc. pioneered generating human iPSCs from adipose-derived pericytes, while her Ph.D. unravelled pericyte-diabetes crosstalk driving vascular complications. Dr. Ahmed led an STDF-funded project developing 3D lung spheroid models using custom microwell plates, yielding high-impact publications. Her work differentiating stem cells into neural-like cells directly supports brain organoid research. A proven grant winner and recipient of national and international awards, Dr. Ahmed frequently publishes in leading journals and serves as an ISSCR member and Q1 journal reviewer. Complementing her research is a 9-year teaching record in stem cell biology and chemistry, extensive mentorship of undergraduate and graduate students, and an IBCT international trainer certification.

Abstract

Pericytes (PCs) are multipotent perivascular cells that wrap around endothelial cells (ECs) to regulate blood flow and maintain vascular integrity. Type 2 diabetes mellitus (T2DM) induces PC detachment and dysfunction, contributing significantly to diabetic vascular complications. This study investigated the deleterious effects of the diabetic microenvironment on the regenerative capacities of human adipose tissue-derived PCs. PCs were cultured in diabetic serum (DS-PCs) or normal serum (NS-PCs) for 6, 14, and 30 days. Morphological analysis revealed that DS-PCs exhibited increased aggregation, distinct topographical invaginations, fragmented mitochondria, and thickened nuclear membranes. Functionally, DS impaired angiogenic differentiation, marked by compromised tube formation, downregulated VEGF-A, IGF-1, and COL21A1, and upregulated anti-angiogenic factors (TSP1, PF4, and the actin-related protein 2/3 complex). DS-PCs also demonstrated heightened apoptosis, evidenced by elevated expression of Clic4, Bcl-2-like protein, serine/threonine protein phosphatase, and caspase-7. Genomic and inflammatory profiling revealed dysregulated DNA repair genes (CDKN1A, SIRT1, XRCC5, TERF2), upregulated pro-inflammatory markers (ICAM1, IL-6, TNF-Î±), and altered expression of focal adhesion proteins (TSP1, TGF-Î², fibronectin, PCDH7). Interestingly, despite these micro environmental insults, DS-PCs preserved baseline intracellular reactive oxygen species (ROS) levels and upregulated key extracellular matrix (ECM) organizing proteins (vinculin, IQGAP1, and tubulin beta chain) as an adaptive resistance mechanism. Overall, while a diabetic microenvironment impairs the regenerative potential of PCs, their intrinsic cytoprotective mechanisms underscore their potential as a viable cell therapy.

Dr. Ragaa Fikry Fathy

Veterinary Research Institute, National Research Centre (NRC), Egypt

PRESENTATION: Ecotoxicological Effects of Emerging Pharmaceuticals on African Catfish: Biomarkers for Neurotoxicity, Immunotoxicity, and Histopathology

Biography

Dr. Ragaa Fikry Fathy, a researcher at the Veterinary Research Institute, National Research Centre (NRC) in Dokki, Egypt, holds a PhD in biochemistry and biotechnology from Cairo University and is a Fulbright Postdoctoral Program nominee. Her academic research focuses on the impact of environmental pollutants on aquatic teleosts, aiming to develop sustainable solutions to mitigate such effects. Beyond her core research, she has contributed to numerous national and international conferences as both a speaker and an attendee, with her work published in prestigious Q1 and Q2 scientific journals.

Abstract

Pharmaceutical residues have emerged as critical emerging contaminants, posing substantial threats to aquatic biodiversity and sustainable aquaculture. However, their sublethal effects on nontarget aquatic species remain underexplored. This study evaluates the comparative toxicity of three widely detected pharmaceuticals: gabapentin (GPT, anticonvulsant), valsartan (VAL, antihyper tensive), and codeine (COD, opioid analgesic)â€"on African catfish (Clarias gariepinus), a species of ecological and aquacultural importance. Fish were exposed for 15 days to environmentally relevant concentrations of GPT (79.86 Âµg/L), VAL (28.22 Âµg/L), and COD (5.27 Âµg/L). Neurotoxicity was assessed via acetylcholinesterase (AChE), monoamine oxidase (MAO), and cortisol levels; immunotoxicity was evaluated through lysozyme (LZM) activity, immunoglobulin M (IgM), and phagocytic activity (PA), while histopathological and histochemical changes were examined in the kidney, intestine, and spleen. All pharmaceutical exposures induced significant neuroendocrine and immune disruptions. AChE activity was reduced, while MAO and cortisol levels increased, indicating neurotoxic stress. Immunosuppression was evidenced by decreased LZM, IgM, and PA. Histological examination revealed tissue degeneration, mucosal damage, and lymphoid depletion, with PAS staining showing glycogen depletion across all treated groups. Among the tested compounds, VAL induced the most severe multisystemic effects, followed by GPT and COD. These findings highlight the overlooked ecological risks of pharmaceutical pollutants, underlining their potential to compromise f ish health, disrupt aquaculture productivity, and threaten aquatic biodiversity even at low environmental concentrations. Such environmental challenges encourage the exploration of sustainable and environmentally safe methods for the disposal of these substances.

Ass. Prof. Radwa Hassan El-Akad

National Research Centre (NRC), Egypt

PRESENTATION: Marine Metabolomics for Bioactive Compound Discovery in Sea Cucumbers

Biography

She has authored over 20 peer-reviewed publications in reputable indexed journals, with an h-index of 7 and her contributions extend to participating in international collaborations. She presented several workshops and hands-on training sessions focusing on practical approaches in metabolomics for researchers and post-graduate students. She continues to play a pivotal role in advancing metabolomics applications in natural product research.

Abstract

Sea cucumbers are marine invertebrates with substantial bioactive compounds and considered food resource with high nutritional value. Metabolomic approaches enabled chemical profiling of two species of sea cucumbers collected in 2 different seasons and assigning biomarkers for each sample. In vitro cytotoxic activity assays against prostate and colon cancer cell lines showed its potent activity (IC50 6-12 microgram/millilitre) that was attributed to their triterpene content as revealed by multivariate data analyses Researcher at the Pharmacognosy Department, National Research Centre, Egypt. She earned her M.Sc. and Ph.D. in Pharmacognosy from Faculty of Pharmacy, Cairo University. With 10-years of experience, Dr. El-Akad has specialized in the metabolomic profiling of natural resources applying advanced analytical tools such as UPLC/MS/MS and GC/MS which contributes to understanding the chemical diversity and therapeutic potential of natural resources.

Ms. Naglaa Ramadan Atta Mohamed

Researcher, Faculty of Science, Cairo University, Egypt

PRESENTATION: Exploring the Therapeutic Potential of Postbiotics: Antimicrobial and Anticancer Properties from Sheep and Goat Sources

Biography

Naglaa Ramadan Atta is a Master's graduate in Microbiology from the Faculty of Science, Cairo University, with over five years of experience in medical and food microbiology laboratories. has strong research experience, including experimental design, microbiological analysis, and scientific writing, and has contributed to peer-reviewed publications derived from master's research. My research interests include microbiology, environmental sustainability, and the application of scientific innovation to address public health and environmental challenges.

Abstract

Postbiotics, the metabolic byproducts of probiotic microorganisms, have gained attention due to th, and consistent bioactivity, making them suitable for pharmaceutical and nutraceutical development. This study investigates the postbiotic potential of mixed Lactobacillus strains isolated from the microbiota of goats and sheep, focusing on their antibacterial and anticancer properties. Lactobacillus strains were isolated through subculturing on MRS agar under anaerobic conditions. Probiotic potential was screened using Gram staining, catalase tests, and multiplex PCR for species identification. Postbiotics were extracted via sonication. Antimicrobial activity was assessed using the well diffusion

method against key pathogens, and efficacy was measured through inhibition zones, Minimum Inhibitory Concentrations (MICs), and Minimum Bactericidal Concentrations (MBCs). Cytotoxicity was tested using the Sulforhodamine B (SRB eir promising therapeutic potential. Unlike live probiotics, postbiotics offer enhanced safety, stability) assay, and IC_{50} values were determined for potent extracts. Postbiotics derived from goat and sheep microbiota exhibited significant antibacterial and anticancer activity. All formulations showed inhibitory effects against Staphylococcus aureus, Streptococcus epidermidis, *Escherichia coli*, and *Klebsiella pneumoniae*, with MIC values as low as 8 µg/mL. The formulations from goat buccal swabs (Postbiotic 2) and sheep nasal swabs (Postbiotic 3) were most effective. All postbiotics induced dose-dependent cytotoxicity against MCF-7 and HT-29 cancer cell lines, with minimal toxicity toward normal fibroblasts. Postbiotics from mucosal sources demonstrated superior bioactivity compared to fecal sources. These findings highlight the potential of postbiotics as safe and stable agents with dual antimicrobial and anticancer properties, supporting their development for pharmaceutical and nutraceutical applications.

Ms. Gehad Kamel Abdelal

Faculty of Nursing, Minia University, Egypt

PRESENTATION: Exploring Future Healers' Views: Knowledge and Attitudes Toward Stem Cells Among Minia University Health Sector Students

Biography

My name is Jihad Kamel Abdelal, and I am a fourth-year undergraduate student at the Faculty of Nursing, Minia University, Egypt. I began my studies in nursing at the age of fifteen at a nursing school, where I completed the diploma program. After finishing the diploma, I pursued further studies at the nursing institute before joining the Faculty of Nursing, maintaining excellence in all semesters and achieving a distinction in every academic term. Throughout my academic journey, I have developed a strong interest in scientific research, particularly in regenerative medicine and stem cell therapy, due to their potential to significantly improve patient care and healthcare outcomes. I have participated in previous research projects organized by my faculty, which provided me with valuable experience in study design, data collection, analysis, and interacting with research participants. My current research focuses on exploring the knowledge and attitudes of health sector students toward stem cells, aiming to understand their awareness, engagement, and readiness to adopt emerging therapies in clinical practice.

Abstract

This study aimed to investigate the depth of understanding and the attitudinal landscape surrounding stem cell science and its clinical relevance among medical and dental students at Minia University. It further sought to explore the interplay between knowledge levels and attitudinal dispositions, offering insights into future healthcare professionals' readiness to engage with regenerative medicine. Stem cells are unique cells capable of self-renewal and differentiation into specific cell types. They are classified as embryonic stem cells, which are pluripotent and can develop into any cell type in the human body, and adult stem cells, which have a limited differentiation potential restricted to their tissue of origin. Stem cells play a vital role in tissue maintenance, repair, and regeneration, and have therapeutic potential in treating various diseases, including blood cancers and neurological disorders. Stem cell research also raises ethical and legal considerations in some countries. This cross-sectional study aimed to assess the knowledge and attitudes of undergraduate students in medical and allied health faculties at Minia University regarding stem cells, their clinical applications, and ethical implications. Data were collected over four months from students aged 18 years and above across Medicine, Nursing, Pharmacy, Dentistry, and Allied Health faculties. A validated structured questionnaire was distributed offline and online via social media platforms, and incomplete responses were excluded. The questionnaire

covered demographics, medical information, knowledge predictors, and attitudes. Data were analyzed using descriptive statistics.

Mr. Amr Abdelsattar Gamal

Faculty of Nursing, Minia University, Egypt

PRESENTATION: Evaluating the Accuracy of Blood Pressure Measurement Knowledge and Skills Among Nursing Students

Biography

Amr Abdelsattar Gamal is a fourth year nursing student at the Faculty of Nursing, Minia University. He has a strong academic interest in clinical nursing skills, patient safety, and evidence based nursing practice. Throughout his undergraduate studies, he has developed a solid foundation in nursing sciences, research principles, and clinical assessment skills essential for professional nursing practice. During his academic training, Amr participated in clinical placements across different hospital departments, including medical, surgical, and community health settings. These experiences enhanced his competencies in patient assessment, vital signs monitoring, infection prevention, therapeutic communication, and teamwork within multidisciplinary healthcare environments. His clinical exposure increased his awareness of the importance of accuracy, professionalism, and adherence to standardized clinical guidelines. Amr has a particular interest in improving the quality of nursing education and evaluating clinical performance among nursing students. He is involved in academic research focusing on knowledge and skills acquisition in fundamental nursing procedures, especially accurate blood pressure measurement. His research interests aim to bridge the gap between theoretical knowledge and safe clinical practice. In addition to his academic and clinical activities, Amr is interested in health promotion, continuous professional development, and the application of research findings to improve patient outcomes. He aspires to pursue postgraduate studies in nursing and contribute to nursing education, clinical research, and healthcare quality improvement. His long term goal is to support evidence based practice and enhance patient care standards within healthcare institutions through commitment, ethical practice, lifelong learning, and professional responsibility.

Abstract

Accurate blood pressure measurement is a fundamental clinical skill in nursing practice and plays a critical role in diagnosis, treatment decisions, and patient safety. Inaccurate measurement may lead to misdiagnosis and inappropriate management of various health conditions. Therefore, ensuring that nursing students possess adequate knowledge and practical skills in blood pressure measurement before entering clinical practice is essential. This study aimed to evaluate the knowledge and skills related to blood pressure measurement among nursing students at the Faculty of Nursing, Minia University, and to assess the accuracy of their performance during clinical practice. A descriptive cross-sectional observational design was utilized. The study was conducted at the Faculty of Nursing, Minia University, and included first-, second-, third-, and fourth-year nursing students who agreed to participate. Data were collected using two tools: a practical observational checklist and a theoretical assessment tool. The practical assessment evaluated students' performance during blood pressure measurement, including patient preparation, positioning, cuff selection and placement, palpatory and auscultatory techniques, timing, and accurate reporting of results. The theoretical assessment consisted of a 20-item test and picture-based questions developed according to the American Heart Association recommendations. To reduce potential bias, the practical assessment was conducted prior to the theoretical evaluation. A pilot study was conducted on 10% of the sample to evaluate the clarity and applicability of the developed tools. Content validity was assessed by a panel of five nursing experts, and reliability was determined using Cronbach's alpha coefficient, which demonstrated acceptable internal consistency. The findings of this study are expected to provide valuable insights

into nursing students' competencies and to support the development of targeted educational strategies aimed at improving blood pressure measurement accuracy and enhancing clinical practice outcomes.

Ms. Aya Dahy

Faculty of Nursing, Minia University, Egypt

PRESENTATION: Bridging the Knowledge Gap: PCOS Awareness Among Female Students at Minia University

Biography

Aya Dany is an undergraduate student at the Faculty of Nursing, Minia University. She graduated from the Technical Institute of Nursing with an "Excellent" grade, reflecting her academic dedication. Aya is deeply passionate about scientific research, with a particular focus on women's health and reproductive issues. Through her work, she aims to contribute to the field of nursing research by addressing critical health challenges, such as Polycystic Ovary Syndrome (PCOS). Her goal is to bridge the gap between clinical excellence and evidence-based practice to improve health outcomes for women in her community.

Abstract

Background: Polycystic Ovary Syndrome (PCOS) is a prevalent endocrine disorder starting in adolescence, characterized by hormonal imbalances and ovarian dysfunction. Despite its impact, significant challenges persist in diagnosis due to gaps in awareness and health literacy among young women.

Objectives: This study aims to assess the awareness of Minia University students regarding PCOS symptoms, causes, and complications, while identifying primary information sources and misconceptions.

Methods: A cross-sectional study was conducted among 395 female undergraduate students at Minia University. The participants' age distribution was 67.8% (20-25 years), 31.4% (18-20 years), and 0.8% (above 25 years). The sample included students from medical faculties (55.3%) and non-medical faculties (44.7%), using a structured questionnaire administered electronically and in hardcopy.

Results: Preliminary findings revealed that 53% of the participants were unaware of PCOS, while only 47% had previous knowledge. Statistical analysis is currently underway to further explore the correlation between academic background and specific knowledge levels.

Conclusion: The results highlight a significant gap in PCOS awareness. This underscores the urgent need for targeted educational programs and health campaigns, especially in non-medical faculties, to ensure early detection and better health outcomes.

Ms. Basmalaa Ahmed Thabet Hassan

Kasim Amein Language School, AWS AI Scholar, IBM AI Engineering Certified, Egypt

PRESENTATION: An In-Silico Computational Simulation Framework for Glioblastoma Progression Modeling: Integrating 3D Transformer Segmentation,

Computational Radiogenomic Feature Fusion, Simulation-Based Treatment-Response Modeling and Phantom-Based In-Silico Geometric Optimization

Biography

Basmalaa Ahmed Thabet Hassan is an Egyptian Biomedical Engineering researcher specializing in artificial intelligence for healthcare, computational medicine, and digital twin technologies. Her work focuses on developing AI-driven solutions for complex medical challenges, particularly in precision oncology and personalized healthcare. She is the founder of the Egyptian Biomedical Engineering Research Initiative (EBERI), a student-led platform promoting biomedical innovation and research training. Her research interests include AI-based disease modeling, computational neuroscience, and emerging healthcare technologies. Basmalaa has developed the GBM-Twin framework, an in-silico digital twin approach for glioblastoma research, integrating artificial intelligence, biomedical data analysis, and computational modeling to support personalized cancer research and clinical decision-making. Her work has been presented through international scientific forums and is progressing toward intellectual property protection. She has received recognition through scientific programs, innovation initiatives, and research opportunities, and actively contributes to STEM education and youth research empowerment. Through her interdisciplinary work, she aims to bridge artificial intelligence and medicine to advance the future of personalized healthcare.

Abstract

Glioblastoma multiforme (GBM, WHO CNS5 grade 4) is the most lethal primary brain tumour and provides a demandingtestbed for computational modeling. Existing computational workflows for GBM are typically (i) fragmented — segmentation, growthmodelling, treatment simulation and risk stratification are developed and evaluated in isolation, allowing errors to propagate withoutstatistical accounting; (ii) non-identifiable — the diffusion (D) and proliferation (ρ) parameters of the Fisher–Kolmogorov (FK) reaction–diffusion model are known to be structurally non-identifiable from standard MRI, yet are routinely reported as point estimates; (iii) static —most published computational tumor-progression pipelines do not implement bidirectional belief updating from new imaging within aretrospective sequential-Bayesian sense; and (iv) computationally unconstrained in the geometric-optimization layer, in which residualtumour volume is minimized without regard to brain organisation.

Methods.

We propose an In-Silico Computational Tumor ProgressionModel: a fully retrospective, computationally validated, non-clinical research framework integrating seven coupled computationalcomponents: (1) a hybrid nnU-Net (v2) + Swin-UNETR multi-task segmentation backbone with deep supervision and test-timeaugmentation; (2) computational radiogenomic feature fusion that explicitly combines a 256-d imaging deep-feature vector, 107 IBSI-compliant hand-crafted radiomics, and a 6-d molecular vector (IDH, MGMT, 1p/19q, TERT, ATRX, EGFR) via a gated cross-modal attentionblock; (3) an Identifiable Fisher–Kolmogorov growth model in which the original (D, ρ) parameterisation is replaced by the front-velocityreparameterisation $v = 2\sqrt{D\rho}$, and identifiability is certified by four convergent tools (profile likelihoods, FIM eigenvalue rank, Sobol'indices, and Simulation-Based Calibration); (4) a retrospective sequential Hamiltonian Monte Carlo (HMC/NUTS) update loop that ingestsfollow-up MRI to refresh the posterior over v in a Bayesian sense; (5) a simulation-based treatment-response module combining linear-quadratic (LQ) radiobiology and Hill-equation pharmacodynamics with a dedicated pseudo-progression (PsP) disambiguation sub-modulederived from RANO 2.0 image-derived features; (6) a Fine–Gray competing-risks risk-stratification module with isotonic-regressioncalibration; and (7) a phantom-based in-silico geometric-optimization module on 20 synthetic phantoms using the HCP-MMP1 cortical atlasand HCP-1065 tract templates to compute eloquence-aware computational penalties.

Results.

On 3,378 retrospective public cases (BraTS2023/2024, UPenn-GBM, UCSF-PDGM, TCGA-GBM, IvyGAP), the hybrid backbone reaches Dice 0.921 / 0.889 / 0.847 (WT/TC/ET) withvoxel-wise ECE = 0.018. The re-parameterised FK model

passes a four-tool identifiability certificate; sequential HMC reduces 90-dayvolumetric MAE from 12.4% to 7.1% ($p < 0.001$). The PsP module lowers false-progression labelling from 17.2% to 6.4% versus a single-timepoint RANO baseline. Fine–Gray + isotonic calibration achieves Uno C-index 0.761 (vs DeepSurv 0.743; $\Delta = +0.018$, $p < 0.001$) withtime-dependent Brier ≤ 0.18. Atlas-aware in-silico geometric optimization reduces simulated functional-region encroachment by 78% versusa volume-only objective on synthetic phantoms.

Conclusions.

To our knowledge this is the first in-silico GBM simulation framework that (a)resolves FK identifiability with a formal certificate, (b) integrates gated radiogenomic fusion and a PsP disambiguation sub-module, and (c)replaces volume-only objectives with atlas-aware computational penalties — all within a strictly retrospective, hypothesis-generating, non-clinical scope.

Mr. Mohammad Ahmed Ismael Ali Teleba

Faculty of Nursing, Minia University, Egypt

PRESENTATION: Awareness of Mental Health Disorders Among Students at Minia University, Egypt: A Cross-Sectional Study

Biography

Mohamed Ahmed Ismael is a final-year undergraduate nursing student at the Faculty of Nursing, Minia University, Egypt. He is currently pursuing his Bachelor's degree in Nursing, with a strong academic interest in community health nursing, mental health promotion, epidemiology, and evidence-based healthcare. Throughout his undergraduate education, Mohamed has participated in academic research and health education activities, with a particular focus on mental health awareness among university students. His current research explores mental health literacy and the factors influencing students' knowledge, attitudes, and help-seeking behaviors. He has also contributed to developing research proposals, survey instruments, and community-based health education initiatives. Beyond his academic work, Mohamed is actively involved in creating and sharing educational medical content for nursing students and the public through digital platforms. His interests include health promotion, preventive healthcare, and scientific communication, with the goal of improving public awareness and supporting better health outcomes. Mohamed is committed to continuous professional development through scientific research, conferences, and interdisciplinary collaboration. He aspires to pursue postgraduate studies and build a career in nursing research, community health, and healthcare education, contributing to the advancement of nursing practice and public health in Egypt and beyond.

Abstract

Mental health disorders represent a growing public health concern among university students, affecting academic performance, social relationships, and overall well-being. Despite the increasing burden of mental illness, mental health literacy remains limited in many developing countries, including Egypt. This study aims to assess the level of mental health literacy among undergraduate students at Minia University and identify the demographic and academic factors associated with it. A descriptive cross-sectional design will be conducted among approximately 400 students selected using stratified random sampling from medical and non-medical faculties. Data will be collected using the Mental Health Literacy Scale (MHLS), a validated instrument assessing knowledge, recognition of mental disorders, help-seeking confidence, attitudes, and social willingness toward individuals with mental illness. Descriptive and inferential statistical analyses will be performed using SPSS version 26, with statistical significance set at $p < 0.05$. The findings are expected to identify knowledge gaps, misconceptions, and stigma related to mental health, providing evidence for developing targeted awareness and educational programs within the university. Enhancing mental health literacy may promote early

recognition of mental disorders, improve attitudes toward affected individuals, and encourage timely help-seeking among university students.

Ms. Nada Hamada Elsayed

Biotechnology student, Ain shams university

PRESENTATION: Integrating 16srRNA with multi-omics for high resolution and reproductible microbiome profiling

Biography

I am an undergraduate Biotechnology student at the Faculty of Agriculture, Ain Shams University, with a strong academic standing and practical expertise in molecular biology, bioinformatics analysis, and genetic research methodologies. As a recipient of the Egypt Scientists Research Scholarship, I have undergone intensive scientific training focused on research methodology, scientific thinking, and skill development. Additionally, I have completed specialized practical training programs covering PCR, qPCR, SDS-PAGE, and sequence alignment tools (BLAST/NCBI) at prominent institutions such as the National Research Center and Ain Shams University. Passionate about bridging molecular science and real-world applications, I have engaged in collaborative research projects, demonstrating strong skills in teamwork, scientific visualization, and research communication.

Abstract

The 16S ribosomal RNA (rRNA) gene sequencing has emerged as a cornerstone molecular marker for profiling bacterial communities within complex microbial ecosystems. However, significant methodological challenges such as primer biases, a lack of standardized protocols, and reproducibility issues persist and constrain clinical and environmental translation. Here we show a comprehensive synthesis of current advancements, critically evaluating both technical bottlenecks and emerging computational solutions in microbiome profiling. Our literature review reveals that integrating long-read sequencing technologies, multi-omics frameworks, and artificial intelligence-driven platforms significantly enhances taxonomic resolution and data accuracy. Furthermore, standardized data pipelines are essential to overcome inter-study variability and ensure reliable comparative analyses. These findings advance the field by establishing a clear roadmap toward standardized, high-resolution microbiome profiling systems, ultimately bridging the gap between genomic innovations and clinical applications.

Ms. Sayeda Abdelrazek Abdelhamid

National Research Centre, Egypt

PRESENTATION: Ecofriendly silver nanoparticles biosynthesis from Penicillium commune NRC 2016-3 as antimicrobial agents for textile materials

Biography

Researcher in Microbial Biotechnology with more than 12 years of experience in industrial microbiology, microbial biotechnology, green nanotechnology, and sustainable bioprocess engineering. My research focuses on developing microbial technologies for the sustainable conversion of agricultural and industrial residues into biofuels, industrial enzymes, and functional nanomaterials. Through multidisciplinary research, I integrate microbial biotechnology, process optimization, and green nanotechnology to develop environmentally sustainable solutions for renewable energy, agriculture, environmental remediation, and biomedical applications and supporting the Circular Bioeconomy and Egypt Vision 2030.

Abstract

The biological synthesis of silver nanoparticles (AgNPs) using fungi has received considerable attention, given that they are environmentally friendly, non-toxic, and inexpensive. Therefore, the current work investigates the biological synthesis of silver nanoparticles (AgNPs) using the extracellular filtrate of Penicillium commune NRC 20163. Physicochemical analyses confirmed the formation of AgNPs with a surface plasmon resonance peak at 420 nm. Transmission electron microscopy (TEM) and Scanning electron microscopy (SEM) revealed spherical particles ranging in size from 3.69 to 9.61 nm, with a zeta potential of -30.5 mV, indicating good stability. The biosynthesized AgNPs exhibited strong antibacterial and antifungal activity against various test microorganisms, with inhibition zones ranging from 30 to 33 mm for Gram-positive and 26 to 30 mm for Gram-negative bacteria. The minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC) values were found to be 25 Î¼g/mL and 2030 Î¼g/mL, respectively. Furthermore, the AgNPs-treated fabrics revealed good antimicrobial properties and demonstrated their durability even after multiple washing cycles. This finding highlights the potential of green-synthesized AgNPs for creating durable antimicrobial textiles.

Ikram Benmakhloufi

Ph.D. Student in Computer Science, Algeria

PRESENTATION: Toward Accessible AI Healthcare: A Roadmap for Offline-First, On-Device Deep Learning in Medical Image Screening

Biography

Ikram Benmakhloufi is a PhD student in Computer Science from Algeria. Her re- search focuses on artificial intelligence, deep learning, medical image analysis, edge AI, and on-device machine learning. She is particularly interested in developing lightweight AI models for smartphone- and edge-based medical image screening to support acces- sible healthcare in resource-limited settings. Her work aims to bridge the gap between advanced AI technologies and real-world clinical applications through efficient and de- ployable solutions.

Abstract

Artificial intelligence (AI) has demonstrated remarkable potential in medical image anal- ysis; however, its adoption remains limited in many low-resource healthcare settings due to unreliable internet connectivity, limited computing infrastructure, concerns over data privacy, and restricted access to specialist expertise. Recent advances in lightweight deep learning have enabled AI models to execute directly on smartphones and other edge devices, offering a practical alternative to cloud-dependent solutions. This presentation proposes a roadmap for the design and deployment of offline-first, on-device deep learning systems for medical image screening. The roadmap is structured around five key compo- nents: selecting lightweight neural network architectures suitable for resource-constrained devices; applying model optimization techniques, including pruning and INT8 quanti- zation, while preserving diagnostic performance; designing offline-first workflows with optional synchronization capabilities; evaluating systems under realistic deployment con- ditions using device-level performance metrics alongside predictive accuracy; and incorpo- rating usability, privacy, and clinical safety considerations into application development. In addition, the presentation discusses current challenges, including dataset bias, model generalizability, performance variability across mobile hardware, and the need for rigor- ous external and clinical validation before real-world adoption. Rather than introducing a new diagnostic model, this work synthesizes recent advances and practical implemen- tation strategies to support researchers developing

accessible, privacy-preserving, and sustainable AI-assisted medical screening tools. By bridging advances in edge AI with healthcare accessibility, the proposed roadmap contributes to the development of equi- table digital health solutions aligned with Sustainable Development Goal 3 (Good Health and Well-being).

Materials-Engineering

SCIENTIFIC PROGRAM & SPEAKER PROFILES

Section: Material Engeneering

---Session 1: Materials-Engineering---

Prof. Dr. Ahmed A. Abdel Moneim (Moderator)

Founding Dean of the Faculty of Basic and Applied Sciences at the Egypt-Japan University of Science and Technology (E-JUST), Egypt

PRESENTATION: Turning CO_2 into Opportunity: The New Carbon Economy

Biography

Prof. Ahmed A. Abdel Moneim is an internationally recognized research leader, materials scientist, and higher education executive with more than 25 years of experience leading institutional transformation, building globally competitive research ecosystems, and advancing innovation-driven research and technology commercialization. His research interests encompass advanced materials, graphene and other two-dimensional materials, heterogeneous catalysis, printed electronics for IoT applications, flexible sensors, clean energy technologies, and sustainable manufacturing. Throughout his career, he has established academic programs, multidisciplinary research centers, advanced laboratories, and strategic international partnerships that have strengthened institutional research capacity, fostered interdisciplinary collaboration, and translated scientific discoveries into technologies and solutions with tangible industrial and societal impact.

As Founding Dean of the Faculty of Basic and Applied Sciences at the Egypt-Japan University of Science and Technology (E-JUST), Prof. Abdel Moneim also founded the Department of Materials Science and Engineering, the Graphene Center of Excellence for Energy and Electronic Applications, and the Industrial Catalysis Laboratory. Under his leadership, E-JUST expanded its research capabilities through the establishment of advanced research infrastructure, the development of high-impact research programs, and strategic collaborations with leading universities and research institutions, including Virginia Tech (USA); Tohoku University, Tokyo Institute of Technology, and the National Institute for Materials Science (NIMS) (Japan); TU Dresden and the University of Kassel (Germany); as well as leading research institutions in China and South Korea. He also established long-term partnerships with major industrial organizations, including EZZ Steel, Pharco Pharmaceuticals, Abu Qir Fertilizers and Chemical Industries, ETHYDCO (The Egyptian Ethylene and Derivatives Company), and MIDO Coatings, accelerating the translation of research into advanced materials, catalysis, printed electronics for IoT applications, clean energy technologies, sustainable manufacturing, and commercially relevant innovations.

Prof. Abdel Moneim has authored more than 160 peer-reviewed publications, holds four granted international patents, has filed more than ten patent applications, secured over EGP 65 million in competitively awarded national and international research funding, and supervised more than 67 M.Sc. and Ph.D. graduates. His contributions to research and innovation have been recognized through the State Excellence Award in Advanced Technological Sciences (2024), inclusion among Stanford University's World's Top 2% Scientists (2020–2025), and the UNDP Invention for Humanity Award (2025).

Prof. Abdel Moneim believes that research excellence is measured not only by publications and patents, but by its ability to generate innovation, develop the next generation of scientific leaders, strengthen industrial competitiveness, inform public policy, and create lasting value for society.

Abstract

The transition from carbon-intensive processes to a circular carbon economy requires technologies that can transform CO_2 from an emissions burden into a valuable carbon resource. Catalytic hydrogenation of CO_2 offers a promising pathway for converting captured CO_2 into hydrocarbons and chemical feedstocks while integrating renewable hydrogen into carbon recycling processes. In this study, alkaline-promoted $ZnFe_2O_4$ catalysts were developed to investigate how surface basicity, reducibility, and process configuration can be engineered to improve CO_2 conversion and hydrocarbon production. A 1.5%Na–$ZnFe_2O_4$ catalyst exhibited strong basic character, with a total surface basicity of 0.261 mmol g^{-1}, and demonstrated effective CO_2 activation under high-pressure reaction conditions. At 330–340 °C, 25 bar, a gas hourly space velocity of 2245 mL g^{-1} h^{-1}, and an H_2/CO_2 ratio of 3:1, the catalyst achieved approximately 41% CO_2 conversion, with hydrocarbons dominated by olefins (~63%), highlighting its potential for producing higher-value carbon products rather than predominantly methane.

Dual alkaline promotion through the incorporation of potassium further increased CO_2 conversion to approximately 49%; however, the enhanced hydrogenation activity shifted product distribution toward lighter hydrocarbons and increased methane and C_1–C_4 selectivity. Beyond catalyst design, process intensification through tail-gas recycling provided a substantial improvement in carbon utilization. Recycling the unreacted reactor effluent through a secondary catalytic stage increased overall CO_2 conversion from approximately 41% to 75% and enhanced the hydrocarbon yield from 1.04×10^{-2} to 1.91×10^{-2} mol g^{-1} h^{-1}, while reducing CO and methane formation and maintaining high olefin production.

These findings demonstrate that the emerging carbon economy can be advanced through the integration of catalyst engineering and process-level carbon management. Alkaline-promoted $ZnFe_2O_4$ enables the selective transformation of CO_2 into value-added hydrocarbons, while tail-gas recycling substantially improves carbon conversion and resource efficiency. The combined strategy provides a promising foundation for moving beyond CO_2 mitigation toward CO_2 valorization, carbon circularity, and the production of marketable carbon-based products, positioning captured CO_2 as a feedstock for a new generation of sustainable chemical and energy technologies.

Prof. Dr. Nour F. Attia (Moderator)

Gas analysis and fire safety lab, chemistry division, National Institute of standards (NIS)

PRESENTATION: Sustainable and Scalable Nanomaterials for Energy and Environmental Applications

Biography

Prof. Dr. Nour F. Attia is Full professor at gas analysis and fire safety lab, chemistry division, National Institute of standards (NIS), Prof. Attia's research contributions in the field of nanochemistry, energy gasses (H2 & CH4) storage, gas separation and greenhouse gases capture, smart textile, flame retardants and water treatments. This is in addition to sensors fabrication and thermal energy storage media. Prof. Attia published more than 116 international papers in high impact factor journals and presented in more than 60 in international conferences. Additionally, Dr. Nour published 5 book chapters, 6 international patents (2 USA and 4 Korean) and six Egyptian patents were recently filled. His H-index is 42. Also, he has been awarded several national and international prizes such as State Prize, National Institute of Standards Prize, Medal of Excellence from the first class from President of the Republic 2020 , Obada-Prize and Climate Change Alleviation Award 2022 from Mansoura University. He has been elected as a member of the Egyptian Young Academy of Sciences for 2021-2024. Additionally, he has selected among the highest distinguished top 2% worldwide scientists in the following connective years 2021,2022,2023,2024 and 2025. He has elected as member of National committee of new and advanced materials 2022-2025 and elected for another term 2025-2028. He is PI of several research projects funded from ASRT and STDF. Interestingly, he organized and chaired various international conferences and workshops in the filed of hydrogen energy, greenhouse gases and nanotechnology. Additionally, Prof. Attia academically supervised more than 20 postgraduates' students and teaching in different Egyptian universities. Recently, he has been selected as member of Electricity and Energy Research Council 2025-2028, ASRT, Egypt. Also, he got the Prof. Dr. President of The Research Staff Club Award in Basic Science and Technological Applications, NRC, Egypt (Oct. 2025).Also, he is editor-in-chief of Journal of Climate Change Action and editorial board members for many international Journals.

Abstract

Energy and environmental pollution crises were anticipated as result of rapid consumption of limited resources fossil fuels over the world in various purposes. Hence, the demand for clean and renewable alternative source of energy is necessity. Hydrogen (H2) is considered as alternative green, emission less, renewable and high-density energy source for vehicles. Nevertheless, the promising merits of hydrogen as future fuel for vehicles, its commercialization is still tied due to its gases state which technically, restricted its safe and efficient storage. Hence, the need for efficient and safe solid-state hydrogen storage media is the main challenge for commercialization of hydrogen energy. Thus, the scalable production of efficient nanomaterials is crucial for overcoming this energy crisis and other environmental challenges such as climate change crisis and other water crisis. Therefore, this lecture will cover the different routes for scalable synthesis of nanomaterials and their utilization for hydrogen storage applications, CO2 capture and utilization and other environmental applications.

Prof. Dr. Marwa S. M. I. Shalaby

Professor of chemical engineering, National Research Centre, Egypt

PRESENTATION: **Next-Generation Composite Membranes for Water and Wastewater Treatment**

Biography

Marwa Shalaby is a Professor of Chemical Engineering at the National Research Centre in Cairo, Egypt, with a Ph.D. from Cairo University. She has extensive teaching experience and has conducted research in areas like water desalination, nanotechnology, and wastewater treatment. Marwa has published over 90 research articles and holds several patents related to membrane technology. Her professional experience includes roles as a researcher and project leader in various international collaborations. She has led significant research projects funded by various national and international bodies, emphasizing sustainable development in water resources. She has received multiple awards for scientific excellence and actively supervises Ph.D. and M.Sc. students. Prof. Shalaby is also an invited/keynote speaker at numerous national and international conferences.

Abstract

Rapid industrialization, urban expansion, and the intensification of agricultural and pharmaceutical activities have accelerated the release of emerging pollutants into ecosystems, posing significant threats to environmental and public health. Among these contaminants, nicotine—a toxic and highly soluble alkaloid originating from tobacco production, cigarette waste, and pharmaceutical residues—has increasingly been detected in surface water, raising urgent concerns regarding aquatic pollution and long-term ecological impacts. This study investigates the enhancements in the removal of emerging pollutants through the utilization of advanced membrane technologies. The incorporation of chitosan into polyethersulfone (PES) membranes demonstrates significant benefits, including improved dye removal efficiency, filtration performance, and fouling resistance. Chitosan-modified membranes maintain a stable dye removal efficiency over time, while increasing permeability to support higher water flow rates and reduced pressure requirements, crucial for large-scale applications like desalination. In addition, novel mixed-matrix nanofiltration membranes have been developed to effectively separate and recover nicotine from aqueous media. These membranes, integrating hydrophilic functional composites into a polyvinyl chloride (PVC) matrix, leverage operational parameters such as nicotine concentration, temperature, and pH to optimize removal efficiency. Membranes modified with tannic acid (TA) and polyvinylpyrrolidone (PVP) achieved up to 75% nicotine removal at an initial feed concentration of 80 mg/L, with a remarkable flux recovery rate of 97.65% following humic acid fouling. The study also analyses three membrane types: M1, M2, and M3. M1 shows strong initial antifouling performance but declines quickly, making it suitable for short-term applications. In contrast, M2 offers a balanced decrease in antifouling effectiveness, ideal for moderate to long-term use. M3 exhibits high initial resistance but lacks durability, limiting its prolonged effectiveness. Structural and physicochemical characterization confirms the enhancement in functional performance, with mechanical strength increasing from 40.7 to 58.4 MPa and contact angle decreasing to 54.1°. Overall, the advancements in both chitosan-modified PES membranes and functional composite membranes highlight their potential as sustainable and efficient materials for addressing emerging pollutants like nicotine and dyes in surface waters. Future research should focus on optimizing these materials and assessing their long-term stability under various environmental conditions.

Prof. Dr. Ahlam M. Fathi

Physical chemistry dept., NRC, Egypt

PRESENTATION: Progress in using metal oxide-based electrode materials as safe and sustainable electrode for electrochemical supercapacitors

Biography

Prof. Dr. Ahlam M. Fathi is currently working as a professor of applied physical chemistry at physical chemistry department, National Research Centre (NRC), Egypt. She received his PhD (2004) from Chemistry department, faculty of science, Cairo University. She has over 25 years of research experience, teaching, training and student supervision in the field of electrochemistry, corrosion protection, electrocatalysis, photochemistry, supercapacitors and Energy production. In 2009, she worked as associate prof. in chemistry dept., faculty of science, King Faisal university, KSA. She is author/ co-author of over 54 scientific articles in peer-reviewed journals. She is a potential reviewer of more than 40 scientific articles and student projects. She is a supervisor for 4 MSc and pHD students.

Abstract

In the modern world, energy drives all the daily services starts from production to transportation, though generation of renewable energy sources grows. Although renewable sources can generate substantial energy, large-scale storage remains a major challenge. Economic pressures have accelerated the development of effective electrochemical energy storage devices (EESDs), such as rechargeable batteries, fuel cells, and supercapacitors (SCs). Among these, researchers have recently highlighted SCs as a highly viable alternative due to their unique performance characteristics, which include high power density (PD), scalable energy density (ED), rapid charge-discharge capabilities, and excellent cyclic stability. Modifying metal oxides (MOs) with using nanotechnology is reviewed to obtained high performance electrode materials for SCs. Therefore, several feasible synthesis processes for preparing metal oxide (MO) nanoparticles and their composites, and discussing how MO morphology affects electrochemical performance are highlighted. The novelty of these MO electrode materials stems from their rich redox activity, excellent conductivity, and strong chemical stability, as well as the synergistic effects present in their hybrid forms, making them outstanding candidates for SC applications.

Ass. Prof. Tamer Sami Mansour

Associate Professor, National Research Centre, Egypt

PRESENTATION: Diatomite-based electromagnetic wave absorption and shielding materials: Current research and future possibilities

Biography

Dr. Tamer Sami Mansour is an Associate Research Professor in the Department of Refractories, Ceramics, and Building Materials at Egypt's National Research Centre (NRC). He earned his Ph.D. in Chemistry from Ain Shams University, focusing his pioneering work on utilizing local Egyptian diatomite for heat-insulating materials. With more than two decades of academic and industrial expertise, Dr. Mansour is a leading researcher in advanced composites, porous ceramics, and sustainable waste recycling. His extensive contributions to the scientific community include a robust portfolio of international publications dedicated to ecological preservation and advanced materials innovation.

Abstract

Diatomite is a soft, siliceous sedimentary rock formed naturally from the fossilized remnants of diatoms, which are unicellular algae with a high natural amorphous silica content (80 to 90% silica). Diatomite pores are naturally occurring, fine, and plentiful apertures that contribute to the siliceous rock's high porosity (60-80%) and large specific surface area (30-160 m2/g). These pores, which range in size from nanometers to microns (1.7 to 800 nm), are what give diatomite its outstanding absorption, filtration, and insulating capabilities .

In recent decades, diatomite has found new applications in sophisticated biomaterials, environmental science, and biomedicine, in addition to its conventional functions. Its distinct characteristics make it a candidate as a base biomaterial for electromagnetic wave-absorbing composite materials. This critical review article provides an overview of progress in recognizing the potential of diatomite composite materials, including 2D and 3D materials, and the different techniques for fabricating diatomite-electromagnetic wave-absorbing material composites (electromagnetic interference shielding materials), their applications, and prospects.

Ass. Prof. Heba Elsayed Ali Hassan

Associate Professor, National Research Centre, Egypt

PRESENTATION: Advanced Ternary Nanocomposite Heterojunctions for Solar-Driven Hydrogen Evolution and Environmental Remediation

Biography

Dr. Heba Elsayed Ali Hassan is an Associate Professor of Physical Chemistry at the National Research Center (NRC) in Egypt, where she has established a robust career in advanced materials and environmental sustainability. She earned her PhD in Chemistry from Ain Shams University in 2016, specializing in the synthesis of modified TiO2 arrays for solar energy conversionâ€”a research endeavor that was honored with the Best PhD Thesis Award in 2017. Her research systematically explores the kinetics and isotherms of pollutant adsorption, the photocatalytic degradation of organic dyes under solar illumination, and the fabrication of functional polymer nanocomposite membranes. Dr. Hassanâ€™s research expertise lies at the intersection of nanotechnology, photocatalysis, and renewable energy. Her work focuses on the strategic design of 0D, 1D, and 2D nanocomposites, such as CdO/Yâ‚‚Oâ‚ƒ/graphene oxide and g-Câ‚ƒNâ‚„/CdS/ZrOâ‚‚, to facilitate solar-driven hydrogen production and the degradation of hazardous industrial pollutants. A prolific contributor to the scientific community, Dr. Hassan has authored numerous peer-reviewed publications in prestigious international journals and serves as a regular reviewer for prominent publishers, including Elsevier, Springer, and Wiley. Her professional leadership extends to organizing international workshops and mentoring undergraduate students. She is also an active member of several esteemed organizations, including the Society for Women in Science in Developing Countries (OWSD) and the Egyptian Society of Advanced Materials and Nanotechnology.

Abstract

Addressing the intersecting challenges of global energy scarcity and widespread water pollution requires the engineering of highly efficient, visible-light-driven photocatalysts capable of facilitating sustainable fuel production and environmental remediation. While traditional single-semiconductor photocatalysts are vital, they are fundamentally constrained by restricted light absorption and the rapid recombination of photogenerated charge carriers. This lecture explores the design of high-performance and reusable ternary photocatalysts; including 0D/1D/2D CdO/Y2O3/graphene oxide, 2D/0D g-C3N4/CdS/ZrO2, and Ag3PO4/NiO/TiO2 mesoporous p-n heterojunctions. These advanced systems utilized transformative potential for green fuel production and wastewater treatment. Experimental results demonstrate exceptional synergy, achieving near-complete degradation of hazardous industrial dyes (Crystal Violet, Methylene Blue, and Malachite Green) and impressive solar hydrogen evolution rates. Specifically, CdO/Y5/GO40 exhibited a dye degradation rate constant of ~2.4, 22, 1.5, and 4.9 times higher than CdO, Y2O3, CdO/Y5, and P25, respectively. g-C3N4/CdS/ZrOâ‚‚ system exhibits a rate constant 13.5 times greater than pure ZrOâ‚‚ and 1.8 times greater than its binary counterpart. Agâ‚ƒPOâ‚„/NiO/TiOâ‚‚ triheterojunction achieved a degradation rate constant 70 times higher than pristine TiOâ‚‚ and 20 times higher than the binary NiO/TiOâ‚‚ mixed oxide. Quantitative analysis reveals that the CdO/Y2O3/GO system achieves a remarkably narrow bandgap of 1.45 eV, a high surface area of 160.57 mÂ²/g and significantly quenched photoluminescence intensity, confirming

inhibited recombination, consequently facilitating a hydrogen evolution rate of 2410 Î¼mol/g/h. g-C3N4/CdS/ZrO2 heterojunction, utilizing a dual S-scheme charge transfer pathway, yields an Hâ‚‚ production rate of 3928.32 Î¼mol/g/h. Notably, the Ag3PO4/NiO/TiO2 triheterojunction demonstrated the highest Hâ‚‚ evolution rate of 4642.56 Î¼mol/g/h.

Prof Giovanna Avellis

Director of International Consortium of Research Staff Associations (ICoRSA), Italy

PRESENTATION: Entrepreneurship and Science innovation by tackling the Women Digital Divide Gap

Biography

Giovanna Avellis is leading researcher in ICT at InnovaPuglia SpA, Italy, and has been involved as project manager in a number of European projects in Software Engineering and Mobile Telecommunication. She served the EU Commission as independent expert evaluator. Her current research interest is in Gender Equality and Artificial Intelligence. Currently scientific reviewer of several International Journal on mobile technologies.

She is Delegate of Women20 Italy since 2019, an engagement group of G20 on Gender Equality policies. Director of iCORSA (International Cornsortium of Research Staff Associations) www.icorsa.org , she founded and is chairing the Italian branch of ICoRSA ItalianRSA (Italian Research Staff Association). Past President of ITWIIN (Italian Women Innovators and Inventors Network) www.itwiin.org and currently vice- President of ITWIIN APS (Associazione di Promozione Sociale). Founder of two Working Groups on Gender Equality in Marie Curie Fellows Association and Marie Curie Alumni Association Advisory Board member of the Italian CNR Observatory GETA (Gender and Talent).Member of Italian women associations Donne&Scienza and CREIS (European Center for Sustainable Innovation).

Abstract

Entrepreneurship and Financial Inclusion of women are key issues to be tackled: improving gender balance in entrepreneurship could have a 6% return on investment up to $5 trillion. This will increase local job creation, tax revenues, boost innovation, drive macroeconomic gains and contribute to community well-being by reinvesting in education, health, and family welfare. Another important issue is to accelerate entrepreneurship ecosystems with inclusive legal, policy, and regulatory frameworks to provide enhanced opportunities for women entrepreneurs from start-up to scaling through growth. To address this issue, we need to secure equal rights to asset ownership and finance and develop tailored programs for women in business training centres to include education and mentoring in finance, entrepreneurship, and emerging sectors (artificial intelligence, blue and green economies, care economy, and space). Science innovation can be facilitated investing in Education, STEM and the Digital Divide: implementation and financing gaps persist. These gaps are underpinned by harmful social stereotypes and norms that impact women and girls at all levels of education, especially in the STEM fields, as well as in an increasingly AI-driven digital economy. Finally, it is advising to mandate targeted national scholarships and funding schemes for women and girls -especially those from marginalised communities - pursuing high-growth STEM fields such as AI, electronic engineering and informatics, and digital innovation spanning the course of education to employment.

"Degree in Computer Science, about 100 papers in peer reviewed conferences and journals, 30 years in industrial research and 4 years in academic research, and 18 months at Imperial College, London, UK, Distributed Software Department, funded by a Marie Curie Individual Fellowship .

Prof. Dr.Shimaa El-Hadad

Central Metallurgical Research and Development Institute (CMRDI)), Egypt

PRESENTATION: **Titanium Alloys: Recycling Technologies and Challenges**

Biography

Prof.Dr. Shimaa El-Hadad is a Professor and Vice Dean of the Manufacturing Technology Institute at Central Metallurgical Research and Development Institute (CMRDI), Egypt. Dr. Shimaa has 26 years of experience in Manufacturing generally and metal casting specifically. She has obtained her Ph.D. in materials engineering from Nagoya Institute of Technology, Japan, and her M.Sc. from the University of Quebec, Canada. She published more than 85 articles and managed several projects that targeted the localization of the spare parts industry. She used to teach training courses to engineers from the industry on casting and heat treatment of metallic alloys.

Dr. Shimaa El-Hadad is a Professor and Vice Dean of the Manufacturing Technology Institute at Central Metallurgical Research and Development Institute (CMRDI), Egypt. Dr. Shimaa has 26 years of experience in Manufacturing generally and metal casting specifically. She has obtained her Ph.D. in materials engineering from Nagoya Institute of Technology, Japan, and her M.Sc. from the University of Quebec, Canada. She published more than 85 articles and managed several projects that targeted the localization of the spare parts industry. She used to teach training courses to engineers from the industry on casting and heat treatment of metallic alloys.

Abstract

With the rising global demand for clean and efficient energy solutions, there is a growing need for materials capable of performing under extreme conditions while improving both performance and durability. Titanium has become a key material in the energy sector due to its excellent corrosion resistance, high strength-to-weight ratio, and ability to withstand severe temperatures and pressures. Its use spans nuclear reactors, offshore oil and gas platforms, and renewable energy technologies, including solar, wind, and hydrogen systems, highlighting its critical role in advancing energy infrastructure and shaping the future of power generation. This review presents a market analysis of global titanium production, its various applications, and the state of titanium scrap worldwide. It also provides a brief overview of the technologies and challenges involved in recycling titanium, particularly in the context of renewable energy. Lastly, promising future opportunities in titanium recycling are introduced.

Prof. Dr. May M. Eid (Moderator)

Physics Research institute, National Research Centre, Egypt

PRESENTATION: **Green-Synthesized Magnetic-Plasmonic**

Biography

Prof. Dr. May M. Eid is a Professor of Biophysics in the Spectroscopy Department of the Physics Division at Egypt's National Research Centre (NRC) in Cairo, where her research sits at the intersection of vibrational spectroscopy, green nanotechnology, and cancer nanomedicine.

She trained as a biophysicist at Cairo University before earning a master's degree in physics at Ain Shams University. She went on to complete a Ph.D. in Radiation Oncology and Biology through a joint program between the University of Oxford's Gray Institute and Ain Shams University, supported by a Ministry of Higher Education fellowship. Her postdoctoral work took her to the Cancer Centre at the University of Medicine and Dentistry of New Jersey on an STDF fellowship, and later to the SISSI beamline at the Elettra Synchrotron in Trieste, Italy, where an ICTP TRIL fellowship allowed her to train in synchrotron-based SR-FTIR microscopy under Lisa Vaccari.

Dr. Eid joined the NRC in 2005 and has advanced through its research ranks to Professor, a position she has held since 2023. Her work centers on using FTIR and SR-FTIR spectroscopy as a diagnostic lens on biological and nanomaterial systems — tracking how green-synthesized, chitosan-functionalized nanoparticles (silver, zinc oxide, and superparamagnetic iron oxide core-shell structures) interact with cells and tissue. This spectroscopic approach underpins her broader research program in nano-enabled cancer therapeutics, including sorafenib- and microRNA-loaded nanocarriers developed and evaluated for hepatocellular and colorectal cancer, as well as immunogenicity studies of nanocomposite vaccine-adjuvant platforms. A parallel line of her research applies the same green-synthesis and characterization methods to agricultural and environmental problems, including bio-fungicides and nanomaterial risk assessment.

Dr. Eid has authored more than 30 peer-reviewed publications and book chapters, holds two Egyptian patents for nanocomposite synthesis methods, and has served as principal investigator on STDF- and NRC-funded grants, as well as a joint CNR Italy–NRC Egypt project. She has supervised multiple doctoral and master's students at the NRC and co-organized several international spectroscopy conferences hosted by the Centre. She is fluent in Arabic, English, French, and Italian.

Abstract

The use of Fourier Transform Infrared (FTIR) microscopy as a label-free, non-destructive method for identifying molecular and chemical alterations in biological samples treated with superparamagnetic iron oxide nanoparticles (SPION) nanocomposites will be discussed in this talk. We examined cellular and tissue samples after treatment to find notable biomolecular changes, such as changes in lipid, protein, and nucleic acid content, by utilizing the sophisticated spatial resolution and spectral sensitivity of FTIR microscopy. Our approach combines multivariate data analysis with hyperspectral imaging to provide a comprehensive chemometric evaluation of nanoparticle-cell interactions and their biological effects. The findings support

SPION nanocomposites' potential use in determining their effectiveness in certain therapeutic and diagnostic applications by showing that they cause quantifiable biochemical changes. These results highlight the usefulness of FTIR microscopy in preclinical research for assessing the biocompatibility of new nanomaterials and in nanotoxicological evaluations.

Ass. Prof. Nihal El-Mehalawy

Institute of Advanced materials technology and mineral resources researches, National Research Centre, Egypt

PRESENTATION: Dielectric Evaluation of the fabricated Alumina Composites doped with Different Oxides for Electrical Insulation Applications

Biography

Dr. Nihal Ahmed El-Mehalawy is an associate professor at Refractories, Ceramics and Building Materials department, Advanced Materials Technology and Mineral Resources Research Institute, National Research Centre, Egypt. She gained her B. Sc., M. Sc. and Ph. D degrees from Faculty of Sciences, Ain Shams University. Her fields of interests include Ceramic Science and Technology, Nanotechnology, Biomaterials, Natural Raw Materials, Dielectric Ceramics, Advanced Ceramic Characterization and Applications of nano ceramics in Electric and Radiation fields. She is a member in many local and international research projects. She is a reviewer for many international journals. She has an awarded patents, published book entitled in “Phosphate-containing wastes for fine ceramics production”, and many research papers in international journals. She is a member in the Social Committee of the National Research Center.

Abstract

Dielectric ceramics are the basic building unit of every electronic device such as ceramic capacitors and microwave dielectric components. This study focused on the fabrication of ceramic composites with good dielectric characteristics and the improvement of the physico- mechanical properties of alumina insulating bodies by doping with different oxides and rare earth elements. Doping the ceramic materials such as alumina with different percentages of ZnO, MgO, spodumene or small content of rare earth elements such as Gd2O3 and Nb2O3 was successful in the enhancement of densification behavior, mechanical properties and the improvement of dielectric characteristics of the fabricated alumina bodies. Such fabricated ceramic materials are promising for utilization in the high voltage transmission line insulator, and for high power circuits that operate at high speed. The doping of alumina with MgO improved the breakdown voltage and resistivity. Further improvement was obtained by co-doping the alumina with Gd2O3 and MgO. The lower the Gd2O3 mass % addition, the better the dielectric properties. Doping the alumina bodies with 7 % ZnO improved both the bodies’ densification behavior and mechanical properties. The results positively confirmed the impact of the addition of ZnO on the resistivity of the samples. Doping the alumina bodies with spodumene and Nb2O3 was successful which promoted the early sintering of the composites at lower firing temperature in addition to

the improvement in the dielectric characteristics. The best electrical properties were obtained which are considered satisfying to meet the requirements of high voltage insulating materials.

Ass.Prof. Rim Gasmi

University Center of Illizi, Algeria

PRESENTATION: Artificial Intelligence as a Catalyst for Science-Based Innovation and Entrepreneurship in Desert Regions

Biography

Dr. Rim Gasmi is an academic researcher and innovation leader specializing in Artificial Intelligence, intelligent systems, and science-based entrepreneurship. She currently serves as Director of the Artificial Intelligence House and the Institute of Sciences at the University Center of Illizi, Algeria, where she plays a key role in developing research, training, and innovation strategies.

Her work focuses on applying AI and IoT technologies to solve real-world challenges, particularly in desert and resource-constrained environments. She has led and contributed to several interdisciplinary research projects addressing road safety, animal vehicle collision prevention, smart monitoring systems, and sustainable development. One of her flagship initiatives involves the design of an AI-based smart collar system aimed at reducing traffic accidents involving animals in desert regions.

Dr. Gasmi is actively involved in empowering young researchers and students by integrating research, entrepreneurship, and responsible innovation into academic programs. She has supervised student-led research projects that resulted in scientific publications and international conference participation.

She was selected as an Emerging Leader in STEM by the International Institute of Education (USA) through the TechWomen program, recognizing her contributions to innovation, leadership, and technology-driven social impact. Her academic interests include ethical AI, inclusive innovation, and the translation of scientific research into sustainable entrepreneurial solutions. Through her work, she seeks to strengthen the role of science and technology in achieving sustainable development goals in Africa and beyond.

Abstract

Desert regions face complex and persistent challenges, including road safety risks, limited infrastructure, environmental constraints, and restricted access to technological solutions. Traditional approaches often fail to address these issues effectively due to harsh climatic conditions and vast geographical areas. This contribution explores how Artificial Intelligence (AI) can serve as a powerful catalyst for science-based innovation and entrepreneurship in desert environments.

The proposed approach emphasizes transforming academic research into practical, scalable, and impact-driven solutions by integrating AI, Internet of Things (IoT), and data analytics. AI-driven systems enable predictive analysis, real-time monitoring, and intelligent decision-making, allowing the development of innovative

applications in areas such as road safety, smart agriculture, environmental monitoring, and resource management. These technologies not only address critical societal problems but also create new opportunities for entrepreneurship and job creation.

Special attention is given to the role of young researchers and emerging innovators in bridging the gap between scientific research and market-oriented solutions. By fostering collaboration between universities, startups, policymakers, and industry stakeholders, desert regions can become living laboratories for sustainable and inclusive innovation.

This work highlights that desert areas should not be viewed solely as vulnerable or disadvantaged zones, but rather as strategic spaces for scientific experimentation and technological entrepreneurship. Empowering science-based innovation in these regions contributes directly to sustainable development, economic resilience, and societal well-being, while reinforcing the importance of ethical and responsible AI for long-term impact.

Assis. Researcher Mona S. H. Gaber

British University in Egypt (BUE), Egypt

PRESENTATION: Advanced Functional Materials for Biosensing and Biomedical Applications

Biography

Mona Samir Hassan is an Assistant Researcher at the Nanotechnology Research Center (NTRC), The British University in Egypt, with a research background in Organic Chemistry and Nanotechnology. She holds a master's degree in Organic Chemistry and has multidisciplinary research experience in the design, synthesis, functionalization, and characterization of advanced functional materials for Solar Cell, biosensing and biomedical applications.

Her research interests focus on the development of functional materials with tailored chemical and physicochemical properties for advanced applications. Her research experience includes working with graphene-based materials, Metal Oxides, gold nanoparticles, carbon-based nanomaterials, and covalent organic frameworks (COFs), with a particular interest in their integration into functional sensing platforms.

Mona has practical experience in the synthesis and characterization of nanostructured and functional materials, as well as in the investigation of their structure–property relationships. Her technical expertise includes Fourier-transform infrared spectroscopy (FTIR), thermal analysis, and various lab techniques for material preparation, purification, and characterization.

Her work reflects a strong interest in interdisciplinary research, combining organic chemistry, materials science, and nanotechnology to develop functional materials with potential applications in advanced biosensing and biomedical technologies.

Abstract

Advanced functional materials have emerged as powerful tools for the development of innovative biosensing technologies and biomedical applications. Their unique physicochemical properties, tunable surface chemistry, and ability to interact with biological molecules make them highly attractive for the design of sensitive, selective, and efficient sensing platforms.

This presentation will highlight the development and application of functional materials for biosensing, with a focus on the relationship between material structure, surface properties, and sensing performance. Different classes of functional materials, including graphene-based materials, gold nanoparticles, and covalent organic frameworks (COFs), will be discussed as versatile platforms for the development of advanced sensing systems. The role of material design, synthesis, functionalization, and characterization in tailoring the properties of these materials for specific applications will also be addressed.

Particular emphasis will be placed on how the integration of functional materials with appropriate recognition and sensing strategies can enhance analytical performance and enable the detection of biologically relevant targets. The potential of these materials in biomedical applications, including the development of innovative diagnostic and monitoring platforms, will also be explored.

Overall, the presentation aims to provide an overview of the opportunities and challenges associated with the development of functional materials for biosensing and biomedical applications, highlighting how advances in materials chemistry can contribute to the design of next-generation sensing platforms with improved sensitivity, selectivity, and practical applicability.

Master Studen. Soulaf Chenini

University Center of Illizi, Algeria

PRESENTATION: Hand gesture and sign language recognition for Targui Language based on deep learning

Biography

Solaf Chenini, University Center of Illizi, computer science graduate. I am a productive, disciplined, and hardworking Computer Science professional specializing in Artificial Intelligence. Recognized for consistently performing at the top tier of my academic promotion (Class A) while balancing rigorous external responsibilities, I have demonstrated a strong ability to thrive under pressure and multitask effectively. With two years of hands-on experience as a Commercial Agent at a major national enterprise (Sonelgaz) alongside a background in teaching, I bring a well-rounded work ethic, advanced technical expertise, and a proven capacity to excel in both corporate and academic roles.

Abstract

To reduce communication barriers between the deaf population and those who do not use sign language, Targuia Sign Language was created.This research presents the deep learning (DL) framework for Targuia Sign Language recognition. Such a Convolutional Neural Network (CNN) is used to extract the strong and unique spatial features from the sign language image and thus different hand gestures can be classified correctly. The proposed system is trained and tested on a specific dataset which contains several examples of images of Targuia Sign Language. We extensively evaluate the model and show that it achieves high recognition accuracy for multiple gesture classes and different users. Furthermore, the proposed method can be extended to be used for real-time sign language detection in conjunction with a camera-based input device.

Ass.Prof. Ghada Mohamed Taha Ibrahim

Textile Research and Technology Institute, National Research Centre, Egypt

PRESENTATION: Controlled Delivery of Trans- cinnamaldehyde through Chitosan/Alginate Nano capsules for Active Food Packaging Textiles

Biography

Dr. Ghada Mohamed Taha Ibrahim is an Associate Professor at the Textile Research and Technology Institute, National Research Centre (NRC), Cairo, Egypt. She received her B.Sc., M.Sc., and Ph.D. in Organic Chemistry from Ain Shams University. Her Ph.D. research focused on the development of multifunctional finishing technologies for cotton fabrics.

Her research focuses on nanomaterials, smart and functional textiles, green textile finishing, hydrogels, sustainable packaging materials, biomedical applications, and wastewater treatment using advanced adsorbent materials. She has extensive experience in the synthesis and characterization of polymeric and nanostructured materials for environmental and healthcare applications.

Dr. Taha has authored more than 35 peer-reviewed publications in high-impact international journals and has an **h-index of 18**, with over **860 citations**. She has participated in numerous national and international research projects, serves as a reviewer for several leading scientific journals, and is actively involved in scientific conferences, research collaboration, and laboratory quality management according to ISO/IEC 17025 standards.

Abstract

The practical application of natural antimicrobial compounds in active food packaging remains challenging because of their high volatility, susceptibility to degradation, and rapid release, which limit their long-term effectiveness. In this study, biodegradable chitosan/alginate Nano capsules were developed as a bio-based delivery platform for the controlled release of trans-cinnamaldehyde (CINA) and subsequently applied to cotton fabrics to produce multifunctional active food packaging textiles. The Nano capsules were successfully fabricated through polyelectrolyte complex coacervation and immobilized onto cotton substrates using a conventional textile finishing process.

The physicochemical characterization confirmed the successful formation of the Nano capsules, efficient encapsulation of trans-cinnamaldehyde, and its homogeneous distribution on the cotton fiber surface. Encapsulation significantly improved the structural organization of the Nano carrier system while maintaining good colloidal stability and high encapsulation performance. Moreover, the developed Nano capsules provided sustained release of the encapsulated bioactive compound, demonstrating a controlled release behavior that is expected to prolong antimicrobial functionality compared with the free compound.

The functionalized cotton fabrics exhibited pronounced broad-spectrum antimicrobial activity against representative foodborne pathogens together with enhanced antioxidant performance, confirming that Nano encapsulation effectively preserved the biological activity of trans-cinnamaldehyde while improving its stability and availability. The optimized formulation achieved superior antimicrobial efficacy using a relatively low concentration of the active compound, highlighting the efficiency of the developed delivery system.

This study demonstrates that chitosan/alginate Nano capsules provide an effective and environmentally friendly platform for the controlled delivery of volatile natural antimicrobials. The developed bioactive textile system combines sustained release, enhanced antimicrobial protection, and antioxidant functionality, offering a promising strategy for next-generation active food packaging applications aimed at improving food safety and extending product shelf life.

Ass. Lecturer. Sarah Hesham Mohamed Rashed

Borg AlArab Technological University, Egypt

PRESENTATION: Dyeing to Be Clean: How Double Layered Hydroxides are Revolutionizing Industrial Wastewater Treatment

Biography

Sarah Hesham Mohamed Rashed, a Ph.D. researcher at E-JUST, SRTA CITY specializing in green energy and sustainable development. I am Assistant Lecturer at Borg El-Arab Technological University. I am an instructor in AI and IoT sensors of wearable devices. I am also the focal point for the Rally Entrepreneurship Community and was honored as the Best Multidisciplinary Researcher at the 2025 Global Research Conference for "AI for a Better Future." Women Award in Innovation and Technology" in DUBAI 2025.

As an international keynote speaker and mentor in nano science and technology and ICT, I have contributed to global forums like COP29 and COP 30, and specialize in leveraging material science, AI-driven analytics, automation, and connected intelligence to optimize energy systems. By bridging the gap between technical research, technology and entrepreneurial leadership, I aim to empower the next generation of innovators to drive global sustainability goals.

Abstract

Double layered hydroxides sodium dodecyl sulphate (LDH-SDS) and Graphene double layered hydroxides (GO-LDH) adsorbents are synthesized by reconstruction and sonochemical methods, respectively, to adsorb Coomassie brilliant blue G-250 (CBB) from an aqueous solution. The adsorbents are investigated by XRD, SEM, EDS, FTIR, Raman spectra, TGA and Zeta-sizer. According to the characterization results, GO-LDH has better properties than LDH-SDS, resulting in greater CBB adsorption. The batch experiment is done under various parameters such as: contact time, initial dye concentration, adsorbent dose, PH, temperature, and ionic strength effect. It's found that 5 mg of GO-LDH increased the adsorption capacity to 1443 mg/g for 5 minutes at higher initial dye concentration, compared to LDH-SDS of 60 mg at 85 mg for 1 hour at lower initial dye concentration. GO-LDH has a pseudo-second-order initial adsorption rate that is 7 times that of LDH-SDS and has higher boundary layer diffusion, according to the intra-particle diffusion model, than LDH-SDS. Furthermore, GO-LDH obeys the Freundlich isotherm with (1/n) 0.406, whilst LDH-SDS obeys the Langmuir isotherm with (qm) 60 mg/g. The thermodynamic parameters show that the adsorption is restricted, exothermic, and spontaneous. Moreover, GO-LDH has a more effective recycling process than LDH-SDS using the memory effect method.

Prof. Dr. Hanan B. Ahmed (Moderator)

Faculty of Science, Helwan University, Egypt

PRESENTATION: **Green fabricated Nanomaterials for Pharmaceutical and Biomedical purposes**

Biography

Professor Hanan Basioni Ahmed is a full professor at the Faculty of Science, Helwan University, Egypt. She is also an Adjunct Professor at Faculty of Engineering, Chulalongkorn University, Bangkok, Thailand. Professor Hanan Basioni is a member of the National Committee of Pure and Applied Chemistry (2025-2028) that contributes to scientific policy development, interdisciplinary research and the promotion of applied scientific innovation.

Professor Hanan Basioni is a member of several scientific societies, such as the Egyptian Chemical Society, the Egyptian Polymer Society, the Egyptian Society for Textile Industries, and is also a member of the Association for Women in Science in Developing Countries .

In recognition of her scientific excellence, Professor Hanan has received several distinguished awards, including the Egyptian State Encouragement Award (2021), Helwan University Award for Scientific Excellence (2022), and the Helwan University Appreciation Award (2024). Professor Hanan's name was included in Stanford's list of the top 2% of scientists for 2021, 2022, 2023, 2024, and 2025.

She has authored over 75 high-quality publications in prestigious international journals with strong impact factors, has contributed 6 chapters to international books, receiving more than 4,300 citations and achieving H-index of 42. Professor Hanan supervised more than 16 Master's and Doctoral theses at faculty of science-Helwan University .

The field of interest for Professor Hanan Basioni is focused on obeying green chemistry concepts for designing various types of nanomaterials to be applicable in biomedical technologies, environmental sustainability, and various industrial purposes.

Abstract

The growing demand for sustainable and biocompatible materials has driven significant interest in green-fabricated nanomaterials for pharmaceutical and biomedical applications. Green chemistry-based fabrication approaches utilize natural extracts, biopolymers, microorganisms, and environmentally benign reagents to produce nanomaterials with reduced toxicity, lower energy consumption, and minimal environmental impact compared to conventional synthetic methods. These eco-friendly nanostructures exhibit enhanced physicochemical properties, including controlled particle size, improved surface functionality, and superior biological compatibility, making them highly suitable for drug delivery, diagnostics, antimicrobial therapy, tissue engineering, and bio-sensing applications. Green-fabricated nanomaterials have demonstrated remarkable therapeutic efficiency through improved drug solubility, targeted delivery, and controlled release

profiles, leading to enhanced bioavailability and reduced side effects. Moreover, their inherent biological activity derived from natural reducing and stabilizing agents often contributes to synergistic pharmacological effects, particularly in antimicrobial, antioxidant, and anticancer therapies. Despite their promising potential, challenges remain in large-scale production, reproducibility, and regulatory standardization. Addressing these limitations will be essential to translate green nanotechnology from laboratory research to clinical and industrial implementation. Overall, green-fabricated nanomaterials represent a sustainable and innovative platform for advancing pharmaceutical and biomedical technologies, aligning therapeutic development with environmental responsibility and safer healthcare solutions.

Ass. Prof. Moshera S. Abdelaz Youssef

Associate Professor, National Research Centre, Egypt

PRESENTATION: In vitro release and cytotoxicity activity of 5-fluorouracil entrapped polycaprolactone nanoparticles

Biography

Dr. Moshera Samy Abdelaziz Youssef, was born Cairo, Egypt in 1984. She holds a PhD by the Faculty of Science, Ain shams University (2019). She is a Associate Prof. Dr. of Polymers technology at National Research Centre (NRC), Giza, Egypt at 14/04/2025. She has got National Research Center Prize of for the best Master thesis Egypt for 2013. She has been awarded the BSc degree at the Faculty of Science, Cairo University, Egypt (2006). She graduates courses as a partial fulfilment of the Master degree of Science, Organic Chemistry, Department of Chemistry, Faculty of Science, Cairo University, Egypt (2007). She has participated in many national and international conferences. She has participated in six national and international research projects. She is a Member of The Egyptian society of polymers and pigments and young researchers in NRC, Giza, Egypt. She held postdoctoral positions in Deusto Institute of Technology (DeustoTech), Bilbao, Spain (2022) for six months. She focuses her research on Synthesis and characterization of biodegrade Polymers by using blending and casting solvent method and studying their applications in biosensors. This would help in understanding the effect of different methods governing the successful

mechanical properties and improvement of their toughness of the prepared biodegradable Polymers for efficient delivery of anticancer drugs. Her fellowship is funded by pregame Science by Women. Her fellowship is sponsored by Diputación Foral de Vizcaya. Now, I'm an Associate Prof. Dr. of Polymers technology at National Research Centre (NRC), Giza, Egypt at 14/04/2025.

Abstract

In this study, 5-fluorouracil (5-FU) entrapped polycaprolactone nanoparticles (5-FU- PCNPs) have been prepared using double emulsion method. The different factors were examined for assembly to arrive at the best effective formulation of 5-FU-PCNPs formulation for 5-FU–PCNPs, as polymer concentration, stabilizer

concentration. The encapsulation efficiency of PCNPs was in the range of 18.8–45.4%. The prepared nanoparticles showed the spherical shape having an average size of 183–675.5 nm, whereas TEM exhibited the prepared nanoparticles have a spherical shape. FTIR, XRPD, confirmed successful insertion of drug in prepared PCNPs. In vitro release of 5-FU from selected formulations showed sustained release from the nanoparticles where slower release was observed when lower PVA concentration was used. Anticancer activity was examined against cell culture for HCT-116 (human colorectal carcinoma), MCF-7(human breast adenocarcinoma), HepG2 (human hepatocellular carcinoma) and A549 (human lung carcinoma) for six formulations 5-FU–PCNPs nanoparticles. The in vitro cytotoxic activity of the prepared formulations was tested showing that these formulations appeared as promising active anticancer formulations.

Assoc. Prof. Rasha Abdel Baseer Taha Muhammed

Department of Polymers and Pigments Technology at the National Research Centre , Egypt

PRESENTATION: Enhanced electrochemical and optical properties of new conjugated n-p type polymers

Biography

Dr. Rasha A. Baseer is an Associate Professor in the Department of Polymers and Pigments Technology at the National Research Centre (NRC), Egypt. She earned her B.Sc. degree from the Faculty of Science, Benha University, Egypt, and her Ph.D. in Organic Chemistry from the Faculty of Science, Ain Shams University, Egypt.

Dr. Baseer's research expertise encompasses polymer chemistry and its biomedical, electrical, and environmental applications. Her work focuses on the design and development of advanced polymeric materials for biomedical devices, conductive composites, energy storage, and water treatment. She also has extensive experience in the chemical recycling of plastic waste and its valorization for environmental applications, particularly water purification.

In 2019, Dr. Baseer completed specialized training in the pulp and paper industry based on non-wood fibers at the Chinese National Pulp and Paper Research Institute, People's Republic of China.

Dr. Baseer has authored numerous peer-reviewed publications in leading international journals and is an inventor on two awarded Egyptian patents: "Method for Open Cyclic Polymerization of Ethylene Carbonate" (2022) and "Polymeric Derivative of (N-Benzyl-3-acetyl Indole)" (2025). She is also the inventor of two pending Egyptian patent applications: EG/P/2022/1313, New Formula of Gelatin-co-polyacrylamide/Nano Zinc Oxide Hydrogel Based on Cotton Gauze for Wound Healing, and 429/2018, Synthesis of New Poly(indole/thiazole) Derivatives.

Throughout her career, Dr. Baseer has participated in numerous national and international research projects, serving both as a principal investigator and as a research team member. She is an active member of the Egyptian Society for Polymer Science and Technology and the Organization for Women in Science for the

Developing World (OWSD). In addition, she serves as a reviewer for approximately 40 manuscripts across about 15 international scientific journals published by major academic publishers.

Abstract

This research work is intended for the novel synthesis and study of structural, thermal, optical, electrical, and dielectric properties of semiconductor polyindole fused thiazolo-thiadiazole (PAITTD) and its N-substituted indole derivatives with a side chain containing chlorobenzoyl (PACBITTD) and bromobenzene sulphonyl (PABBSITTD). The characterization techniques such as H1NMR, FTIR, SEM, and TGA/DSC ensured the structural and thermal properties of the as-synthesized polymers. The optical band gap as obtained from the absorbance study was found to be in the range of 1.5 eV to 2.58 Ev. Moreover, the ground-state geometries and the absorption wavelengths for the studied compounds were investigated by DFT and TD-DFT, where the theoretical results show coincidence with these of the experimental where the calculated (λmax) values are approximately near to the experimental (λmax) values for the PAITTD, PACBITTID, and PABBSITTD. The impedance ac conductivity and dielectric properties of the PAITTD, PACBITTID, and PABBSITTD were performed in the frequency ranges from 20 Hz–10 MHz at room temperature, where their values were 3.0×10^{-7}, 8.7×10^{-7}, and 2.6×10^{-6}, respectively. Based on the optical and electrical properties, synthetic PAITTD, PACBITTID, and PABBSITTD are promising materials in the design of optoelectronic semiconductor devices.

Ass. Prof. E M. Mahmoud

Associate Professor, National Research Centre, Egypt

PRESENTATION: Recent trends in Advanced Bioceramics for orthopedic Applications

Biography

Dr. Eman Mahmoud obtained her BSc in (Physics and Chemistry), MSC in (Biophysics), and PhD in (Biophysics) from the Faculty of Science, Ain Shams University, in 2006, 2012, and 2016, respectively. She is currently associate Professor of Ceramics Chemistry and Materials Technology in the Refractories, Ceramics and Building Materials Department, Advanced Materials Technology and Mineral Resources Research Institute, National Research Centre. Dr. Mahmoud's research interests include Ceramic Science and Technology, Biomaterials, Nanomaterials, Natural Row Materials, Advanced Bioceramics, Blood analysis and Biochemistry evaluations, Tissue engineering, In-Vitro and In-Vivo evaluations and characterizations of biomaterials, bone graft implant, bone substitutions Materials. Dr. Mahmoud received the Second Cairo Innovation Exhibition Award in 2015. Dr. Mahmoud has published over 17 articles in international journals with a Scopus H-index of 10. She was also a member of several research projects international, national, and a national need project including an industrial project under Title Maximizing the utilization of eggshell wastes resulted from sterilized egg industry for industrialization of high-economic-value products (2023-2024), also Reviewer activity for several international journals.

Abstract

Every year, millions of patients require bone replacement or grafting to replace bone mass lost due to ageing, bone defects, and traumas. There is an increasing need for sustainable prosthetic devices and associated medical systems due to the world's population's longer life expectancy. Utilization of advanced bioceramics in tissue engineering has emerged as a promising approach to treat the injurie or loss bone. Great efforts have been made to improve bone ingrowth. Bioceramics are considered among the essential materials employed in the biomedical field as they were created to be suitable for connection with living tissues. Their biological attitude is based on their chemical constituents and physical features. They are resorted to instead of metallic biomaterials due to their unique properties, such as biocompatible, corrosion resistance and bioactivity. Injectable hydrogel has recently gained greatly of attention in the field of reconstructive bone surgery. Using injectable hydrogels offers several advantages, including providing an easy strategy for healing irregular and complex bone defect areas with an injection needle, as well as reducing recovery time by avoiding away with open surgery procedures. Our study's goal is to shed light on recent trends in advanced bioceramics for orthopedic Applications, properties, Preparation, Characterization, In-Vitro and In-Vivo evaluations. In addition, it discusses and summarizes the considerable advancements in the use of advanced ceramic materials for bone replacement as the perfect choice for orthopedic and maxillofacial applications. Additionally, it highlights of traditional and modern research. It defines various ceramic materials, their bioactivity, clinical applications, advantages, limitations, fabrication techniques, mechanical and physical characteristics, and the criteria for selecting long-lasting implant devices for advanced ceramic materials.

Ass. Prof. Heba-tollah Mahmoud Sweelam

Associate Professor, National Research Centre, Egypt

PRESENTATION: In Vitro Culture as a Sustainable Platform for the Production of Pharmaceutically Important Natural Products from Higher Plants

Biography

Dr. Heba-tollah Sweelam is an Associate Professor in Chemistry of Natural Products, she works at the chemistry of natural compounds department, pharmaceutical and drug industries research institute, National Research Centre, Egypt. She earned her Ph.D. from the botany and microbiology department, college of science, Cairo University. Her research is focused on the isolation and identification of bioactive compounds from plant origin and their potential health promoting properties, as well as practicing different tissue culture techniques to increase the content of bioactive compounds in the regenerated plants.
Her research interest includes metabolomics, tissue culture and antimicrobial activity. She has experience in modern analytical techniques, including Nuclear magnetic resonance spectroscopy (NMR), liquid chromatography tandem mass spectrometry (LC MS/MS), and molecular networking for the identification and characterization of natural products. She actively collaborates with multidisciplinary research groups and has published more than thirteen scientific papers along with book chapters; her research integrates phytochemical analysis, chromatography and spectroscopy. She has participated in collaborative scientific projects related to plant based bioactive compounds. Also, she is a

member of many scientific research projects. She is also a member of many scientific societies such as the society for women in science in the developing countries, the Arab society for medical research and the Egyptian chemical society along with many more. She has contributed scientifically in peer reviewed scientific papers. In addition to her research activities, she assists in undergraduate laboratory teaching and mentor students in phytochemistry techniques, and participates in many workshops focused on advanced laboratory instrumentation.

Abstract

Higher plants, especially wild ones, are considered a source of many important medicinal compounds, fragrances, and dyes. Plants undergo a variety of stresses and harsh conditions to help them flourish and cope with such situations, so they produce a wide range of secondary metabolites that are essential for defense against biotic and abiotic stressors. Notwithstanding their importance and due to their numerous industrial uses, in addition to the limited natural yield of these bioactive chemicals, genetic variability and environmental factors make their extraction and use extremely difficult. Plant tissue culture (cultivation of a small plant part for regeneration of a new plant) has become a competitive substitute in recent years for the regulated synthesis of secondary metabolites through a number of productivity-boosting techniques, such as elicitation and precursor feeding. The main benefit of this technology is to provide a reliable and continuous source of plant pharmaceuticals that could be utilized for the large-scale cultivation of plant cells, from which these metabolites can be extracted. Also, it can be used as a substitute for wild plants due to many reasons, including their slow growth rate, low concentrations of active compounds, and the requirement for biotic and biotic stress to induce biosynthesis of the secondary metabolites. To increase the production of bioactive compounds and boost their synthesis, numerous tactics need to be used including metabolic engineering and up-scaling using effective bioreactor approach. This work presents an updated overview of the use of in-vitro culture for the production of pharmaceutically important plant metabolites.

Ass. Prof. Mona Sayed Ahmed Mohamed

Associate Professor, National Research Centre, Egypt

PRESENTATION: Bioactive Calcium Phosphate - Silicate Composites: Synthesis, Characterization and In-vitro Assessment

Biography

Dr. Mona Sayed Ahmed, Associate Professor of Ceramic Chemistry and Materials Technology in the Refractories, Ceramics, and Building Materials Department, Advanced Materials Technology and Mineral Resources Research Institute, National Research Centre (NRC), Egypt. She has published 26 research articles in international journals, one book, one review article, and is the inventor of one granted patent. Participated as a researcher in many national and international research projects. Former Technical Examiner at the Egyptian Patent Office. Member of the Technology Transfer Office (TICO) at the National Research Centre (2020–2024). Member of the

Office of Technology Commercialization (OTC) at the National Research Centre. Throughout her academic career, she has participated in several international fellowships and scientific visits to Germany, Italy, Poland, and China.

Abstract

Bioceramic materials have attracted considerable attention in regenerative medicine due to their biocompatibility and bioactivity. Among these materials, calcium phosphate ceramics like tricalcium phosphate and hydroxyapatite closely resemble the mineral composition of natural bones. However, their limited mechanical properties and slow degradation rates may restrict their applied in load-bearing application. On the other hand, silicates based materials possess high bioactivity and biodegradability properties. Accordingly, this work focus on development and characterization of bioceramic composites combining calcium phosphate and silicate phases to achieve enhanced in biological and physical properties. Furthermore, the controlled release of calcium, phosphate, and silicon ions contributes to cellular proliferation, differentiation and supporting accelerated bone healing. Preparation methods, including nano-calcium phosphate and silicates powders by precipitation and sol-gel techniques were achieved, respectively. The formed composites powder was pressed into pellets to obtain bioceramic samples and sintering at their optimum temperature. The results demonstrated that Calcium phosphate-silicate bioceramic composite exhibited excellent in vitro bioactivity following simulated body fluid immersion. Cytotoxicity assessment revealed no significant toxic effect. Cell viability studies showed high cell proliferation and viability. The obtained results suggest that the prepared composites have significant potential to be applied as promising biomaterials for bone tissue engineering applications.

Ass. Prof. Samaa M. Faramawy

Associate Professor, National Institute of Standards (NIS), Egypt

PRESENTATION: Metrology and Traceability: Role as a Precondition for Sustainability

Biography

Dr. Samaa M. Faramawy, Associate professor and postdoc researcher at the National Institute of Standards (NIS). My current research interests are, Experienced researcher in radiometric metrology in fields of solar-radiation measurements, UV-Radiation measurements, Phototherapy, Ultraviolet Therapy measurements, and UV-INDEX sensation. I am also Competent in light metrology science (electromagnetic radiation), General principles of measurement and instrumentation, Sensors and sensor systems, Specific applications in measurement and instrumentation, Applications of sensors and instrumentation.

Abstract

This analysis examines the often-overlooked but strictly necessary role of metrology in the successful execution of the Sustainable Development Goals. We argue that because you cannot manage what you cannot

measure, traceable, accurate measurement is a precondition for a sustainable future by 2030. The work presented here focuses on technical applications where standard and certified measurements directly facilitate the realization of specific SDGs. (1) Solar Resource Assessment: The presentation demonstrates how standard and traceable measurements of solar irradiance (W/m^2) are critical for reducing technical uncertainty. This precision facilitates data-driven investment decisions, precise performance modeling, and maximized efficiency for solar generation facilities, thereby directly enabling Goal 7 (Affordable and Clean Energy) and Goal 13 (Climate Action). (2) UVC Disinfection Validation: The metrological calibration of UVC sensors ($\mu W/cm^2$) provides the requisite data to validate adequate radiometric dosing. This validation is essential for confirmed pathogen and microorganism inactivation without wasteful over-dosing, directly supporting Goal 6 (Clean Water and Sanitation) and Goal 3 (Good Health and Well-being). (3) Medical Phototherapy Precision: The presentation will show the essential need for accredited calibration of phototherapy devices and testing different sources. We ensure specific spectral irradiances (($\mu W/cm^2/nm$) for neonatal applications, ($\mu W/cm^2$) for dermatological applications), preventing dangerous dosage errors and serving Goal 3 (Neonatal and Public Health) and Goal 9 (Industry, Innovation, and Infrastructure) through standardized safety protocols.

Ultimately, metrology provides the required bridge from scientific principles to verifiable field impact, serving as the guarantor of human safety in purification and phototherapy, critical water purity validation, the structural reliability of sustainable energy infrastructures, and other essential applications.

Assoc. Prof. Mona Moaness Abdel-Moneam Mohamed

Associate Professor, National Research Centre, Egypt

PRESENTATION: 3D-Printed Reinforced Scaffolds for Bone Regeneration and Controlled Drug Delivery

Biography

Dr. Mona Moaness Abdel-Moneam Mohamed is an Associate Professor of Biomaterials at Refractories, Ceramics and Building Materials Department, Advanced Materials Technology and Mineral Resources Research Institute, National Research Centre (NRC) in Cairo, Egypt, specializing in nanotechnology, advanced drug delivery systems, and regenerative medicine. With a Ph.D. in Biochemistry from Cairo University, her pioneering research centers on the fabrication of bioresorbable electrospun nanofibers, 3D-printed scaffolds, and smart pH-sensitive hydrogels designed to improve bone regeneration and accelerate skin wound healing. Her impactful work has earned her the 2024 NRC President's Award for Best Applicable Research and a co-invented patent for skin regenerative patches. In addition to her extensive record of peer-reviewed publications, international grant collaborations, and journal reviewing duties, Dr. Mona is deeply committed to academic mentorship—supervising graduate theses and student training programs successfully

Abstract

The development of multifunctional scaffolds capable of promoting bone regeneration while preventing implant-associated infections remains a major challenge in bone tissue engineering. In this study, novel 3D-printed sodium alginate/poly (vinyl alcohol) (SA/PVA) scaffolds reinforced with amoxicillin-loaded copper/cerium co-doped 58S bioactive glass nanoparticles (BG-Cu/Ce-Amox) were developed for simultaneous bone regeneration, infection control, and controlled drug delivery. The bioactive glass nanoparticles were synthesized using a sol-gel method and incorporated into the SA/PVA matrix through 3D printing, producing highly porous scaffolds with interconnected architectures suitable for bone tissue growth. Morphological and elemental characterization confirmed the successful incorporation of Cu/Ce-doped bioactive glass and amoxicillin within the polymeric network. Scanning electron microscopy revealed well-defined porous structures and enhanced surface roughness after nanoparticle addition. In vitro bioactivity evaluation in simulated body fluid demonstrated superior apatite formation and calcium-phosphate mineralization compared with pristine SA/PVA scaffolds. The drug-loaded scaffolds retained excellent bioactivity while exhibiting elemental signatures associated with amoxicillin incorporation. Biological assessment using MG-63 osteoblast-like cells showed significantly enhanced cell proliferation and alkaline phosphatase activity after 3 days, indicating improved osteogenic potential. Antibacterial evaluation against Gram-positive and Gram-negative bacteria demonstrated broad-spectrum antimicrobial activity. Drug release studies in PBS (pH 7.4) revealed sustained amoxicillin release for up to 7 days, enabling prolonged local antibiotic delivery and reducing the need for repeated administration. Overall, these multifunctional 3D-printed scaffolds integrate bioactivity, osteogenic stimulation, controlled antibiotic release, and antibacterial performance, making them promising candidates for bone regeneration applications with enhanced resistance to implant-associated infections.

Teaching Assis. Hadeer Mamdouh Ahmed Eldeeb

Faculty of archeology, Cairo University, Egypt

PRESENTATION: **Novel Nanostructured Agarose Gels for the Non-Invasive Cleaning and Protection of Historic Tintype Photographs**

Biography

Hadeer Mamdouh Ahmed Eldeeb, is Egyptian Teaching Assistant at the Organic Conservation Department, Faculty of Archaeology, Cairo University. I hold a Master degree (2022) with a focus on eco-friendly materials for photograph conservation, along with an Excellent grade Bachelorâ€™s degree (2015). My academic portfolio includes published research on fungal effects and private photographic collection conservation. I possess strong technical and practical skills, including proficiency in photography, paper conservation, AutoCAD (2D), Photoshop, and digital transformation. I am a highly adaptable and collaborative professional with excellent communication, teaching, and research abilities.

Abstract

Tintype photographs serve as vital cultural and historical artifacts from the 19th century; however, their long-term preservation is frequently threatened by surface contaminants such as soot and dust. Due to their thin, sheet-like structure, conventional cleaning methods often induce scratching or irreversible damage to the delicate image layer. This study evaluated the cleaning efficacy of alternative approaches, contrasting traditional swab cleaning with agarose gel and nanomaterial-functionalized agarose composite gels incorporating zinc oxide (ZnO), carbon dots (CDots), and graphitic carbon nitride (g-C3N4).The synthesized nanomaterials and composite gels were comprehensively characterized using TEM, DLS, zeta potential, UV-vis, and ATR-FTIR spectroscopy. Cleaning performance was assessed through portable digital microscopy, colorimetric analysis (Î”E), and ATR-FTIR, both before and after treatment, as well as following artificial light aging. The nanomaterial-functionalized gels demonstrated superior cleaning efficiency and protective capabilities compared to traditional methods. Specifically, agarose gels loaded with ZnO (Î”E=3.35), CDots (Î”E=2.7), and g-C3N4 (Î”E=3.09) effectively removed dust without causing perceptible color shifts, while providing an added shield against UV exposure. For soot removal, the agarose + ZnO (Î”E=5.84) and agarose + CDots (Î”E=7.73) formulations achieved the highest performance. Crucially, post-treatment ATR-FTIR analysis confirmed no chemical alterations to the tintype layers, validating these functionalized gels as safe, non-invasive solutions for heritage conservation.

Prof. Dr. Marwa I Wahba

Professor, National Research Centre, Egypt

PRESENTATION: Proteins as grafting materials for the fabrication of covalent immobilizers

Biography

Prof. Dr. Marwa I Wahba, Professor at the Department of Chemistry of Natural and Microbial Products, and at the Advanced Materials and Nanotechnology group, Centre of Scientific Excellence, National Research Centre, Giza, Egypt. Interested in research on biopolymers and their exploitation as immobilization matrices & packaging materials, and also interested in altered methods that can be adopted to promote enzymes stability.

Abstract

Covalent immobilizers are critical for the industrial exploitation of enzymes as they allow enzymes reuse and promote their stability. Nonetheless, covalent immobilizers are usually fabricated via grafting with synthetic polymers, such as polyethylene-imine (PEI), which is expensive and also toxic. Accordingly, safe and cost-effective grafting materials should substitute the PEI. Proteins, such as egg white protein and whey protein isolate were successfully utilized as PEI substitutes. These proteins were utilized to graft altered biopolymers, such as carrageenan and gellan-gum. The grafted protein then allowed the binding of the covalently reactive glutaraldehyde to the immobilizer matrix, and thus, the covalent immobilizers were attained. These covalent immobilizers successfully immobilized altered enzymes, such as β-galactosidase and cellulase while providing them with excellent reusability and escalated stability.

Ass. Prof. Marwa Hany

German International University in Cairo (GUC), Egypt

PRESENTATION: Identification of novel pyrazolines and isoxazolines as antileishmanial agents via a potential antifolate mechanism: synthesis, biological evaluation, and in silico studies

Biography

Dr. Marwa Hany, Assistant Professor of Pharmaceutical and Medicinal chemistry at the German University in Cairo GUC & German International University GIU She is currently the Program Supervisor of both the faculties of Biotechnology and Pharmaceutical Engineering- German International University (GIU). Dr Marwa has earned her MSc. and PhD degrees in Pharmaceutical chemistry from the GUC, she is mainly interested in drug design and development of novel anticancer and anti-infectious small organic molecules. in addition, she is also interested in green chemistry applications in pharma industry and applications of Artificial intelligence in drug discovery and has organized some summer workshops in Berlin to introduce to these areas of interest.

Abstract

Leishmaniasis remains a major neglected tropical disease affecting nearly one million people annually. Due to toxicity and resistance to current therapies, the folate/pterin salvage pathway represents a rational target for antileishmanial inhibitor design. Novel chalcone-derived pyrazolines and isoxazolines were synthesized and evaluated in vitro against Leishmania major. Six compounds (1b-e, 1g, and 1i) showed higher potency than miltefosine. Specifically, the N-formylated pyrazoline derivatives 1g and 1c exhibited exceptional antileishmanial potencies, with IC50 values of 1.41 ÂµM and 1.54 ÂµM against promastigotes as well as 1.71 ÂµM and 1.94 ÂµM against amastigotes, respectively. This represents a significant potency improvement (up to 5.5-fold) over miltefosine (IC50 = 7.83 and 8.07 ÂµM, respectively). The antileishmanial activity was reversed with folic and folinic acid supplementation, supporting an overall antifolate mechanism. Molecular docking and 100 ns molecular dynamics simulations were conducted within the binding pocket of the essential Leishmania major pteridine reductase 1 enzyme (Lm-PTR1; PDB ID: 7PXX), demonstrating stable ligand-protein complexes for the most active candidates. In addition, molecular docking and 100 ns MD simulations of the S-enantiomer of compound 1g against a validated AlphaFold2 model of Lm-DHFR revealed favorable binding scores and stable cofactor-pocket interactions, suggesting a predicted dual-target inhibitory potential against both Lm-PTR1 and Lm-DHFR. Compounds 1g and 1c showed minimal cytotoxicity against Vero cells, with selectivity indices (SI) of 148.7 and 260.9, respectively, compared to miltefosine (SI = 12.6). Based on its superior potency, high selectivity, and stable target binding in silico, the S-enantiomer of compound 1g represents a promising lead for further antileishmanial development.

Ass. Prof. Maha Ahmed Ali Ahmed

Associate Professor, Faculty of Archaeology, Cairo University, Egypt

PRESENTATION: Mount Arafat Through the Lens of Muhammad Sadiq Bey: Documentation and Analysis of a 19th Century Albumen Print

Biography

Dr. Maha Ahmed Ali Ahmed is an Associate Professor of photograph conservation at the Faculty of Archaeology, Cairo University. In 2016, she received her doctorate degree in Conservation from Cairo University. She was a member of a research team in a joint project between Cairo University and University of Catania dedicated to photograph preservation during the period from 2013 2017. She also taught several photograph conservation classes at post graduate level in a joint International Masters Program in Conservation of Antique Photographs and Paper Heritage between Helwan University and University of Catania. She has also held a number of workshops in Egypt, Lebanon, and Morocco with the aim of raising awareness on the significance of photograph heritage in Egypt and the Middle East and their preservation needs. Maha has published several papers discussing different issues related to photograph conservation. Her primary research area is the utilization of advanced analytical techniques and technologies in photograph conservation field. She is an alumna of the Middle East

Photograph Preservation Initiative (MEPPI), a strategic multi-year effort funded by the Andrew W. Mellon Foundation and successfully organized by the Arab Image Foundation (AIF), the Art Conservation Department at the University of Delaware, the Metropolitan Museum of Art and the Getty Conservation Institute to build the capacity of individuals and institutions in the care and preservation of photograph collections in the broad Middle East, from North Africa and the Arabian Peninsula through Western Asia. Maha was also a member of the MEPPI Steering Committee to whom the responsibility to drive the initiative into new grounds and provide a road map for the future of MEPPI was entrusted. She participated in several conservation projects related to photograph conservation (e.g., the photo collection at the library of the University of Alexandria).

Abstract

Muhammad Sadiq Bey, an Egyptian army engineer, surveyor, and treasurer of the Egyptian Hajj caravan, is recognized as the first photographer to document Mecca, Medina, and the Hajj pilgrimage. This study presents the documentation, material characterization, and deterioration assessment of a 19th century albumen print depicting Mount Arafat, photographed by Sadiq Bey. The photograph is mounted on a low-quality secondary support and preserved within a wood frame. Given the inherent instability of albumen prints, a study was conducted to evaluate the photographâ€™s condition and identify the deterioration forms affecting it. The investigation employed visual examination, stereomicroscopy, ultraviolet (UV) and infrared (IR) imaging, scanning electron microscopy coupled with energy-dispersive X-ray spectroscopy (SEM-EDS), pH measurement, Fourier transform infrared spectroscopy (FTIR), and microbiological analysis. The results revealed extensive deterioration, including yellowing and flaking of the albumen binder, fading and discoloration of the silver image, and yellowing and embrittlement of the secondary support. Evidence of insect infestation, including damage associated with Anthrenus pimpinellae larvae, and microbiological activity was also identified. IR imaging revealed evidence of retouching, while SEM examination showed micro-cracking and fungal spores. EDS analysis detected silver sulfide, contributing to image discoloration. The secondary support exhibited an acidic pH of 4.9, while FTIR indicated severe degradation of the albumen binder and oxidative deterioration of the secondary support. Microbiological analysis isolated Aspergillus niger and Emericella nidulans, with A. niger identified as the probable primary contributor to biodeterioration. The findings provide valuable insights into the material composition, deterioration forms, and preservation challenges of this historically significant photographic record of Mount Arafat.

Dr. Sara Sayed Abdel-Haliem Selim

Researcher, Egyptian Petroleum Research Institute,Egypt

PRESENTATION: **Plastic Waste to High-Performance Aerogel Sorbents for Sustainable Oil Spill Cleanup**

Biography

Dr. Sara Sayed Abdel-Haliem Selim is a researcher at the Egyptian Petroleum Research Institute (EPRI), Cairo, Egypt. Her research focuses on the development of advanced materials for environmental remediation, with particular emphasis on adsorption technologies, water and wastewater treatment, and sustainable

waste management. Her work includes the design and application of porous materials, nanocomposites, and carbon-based adsorbents for the efficient removal of pollutants from water and industrial effluents. Her research interests include adsorption science, environmental nanotechnology, oil spill remediation, wastewater purification, nanostructured materials, and the valorization of plastic and industrial wastes into value-added functional materials. She is committed to developing innovative, cost-effective, and environmentally sustainable technologies that support resource recovery and circular economy principles.
Dr. Selim has published extensively in international peer-reviewed journals and has presented her research at numerous national and international scientific conferences. Through multidisciplinary collaborations, she continues to advance sustainable materials and technologies for environmental protection, clean water production, and pollution control.

Abstract

The escalating accumulation of plastic waste and the increasing occurrence of oil spills have emerged as two critical environmental challenges, necessitating innovative and sustainable remediation technologies. This study presents a circular economy strategy for converting discarded plastic waste into ultralight, highly porous aerogel sorbents for efficient oil spill cleanup and wastewater purification. The plastic waste-derived aerogels were fabricated through a facile and environmentally friendly process, producing a three-dimensional interconnected porous network with low density, high porosity, and excellent mechanical stability. Owing to their intrinsic hydrophobicity and superoleophilicity, the aerogels selectively absorb a wide range of petroleum oils and organic solvents while effectively repelling water. Comprehensive structural and physicochemical characterization was performed to correlate the morphology, pore architecture, surface chemistry, and wettability with the sorption performance. Batch sorption experiments demonstrated rapid oil uptake, high absorption capacities, fast sorption kinetics, and excellent selectivity toward various hydrocarbons under different operating conditions. Furthermore, the aerogels exhibited remarkable recyclability, maintaining high sorption efficiency and structural integrity over repeated absorptionâ€“recovery cycles through simple mechanical squeezing or solvent extraction. The interconnected porous framework promoted efficient capillary-driven transport and storage of oils, while the robust structure prevented collapse during repeated use. By transforming persistent plastic waste into high-value functional sorbents, this work simultaneously addresses plastic pollution and oil contamination, providing a sustainable waste-to-resource platform for environmental remediation. The developed plastic waste aerogels offer significant potential for large-scale oil spill response, industrial wastewater treatment, and resource recovery, representing an economically viable and environmentally friendly alternative to conventional oil sorbent materials.

Ass. Prof. Heba Elmotasem

Associate professor, National Research Centre, Egypt

PRESENTATION: **Nanoparticles decorated with targeting ligands for precision medicine.**

Biography

Dr. Heba Elmotasem, Associate Professor, Pharmaceutical Technology Department, Pharmaceutical and Drug Industries Research Institute, National Research Centre, , Egypt . Ph.D, master's degree, and Bachelor's degree from the Faculty of Pharmacy, Cairo University, Egypt. General Certificate of Education. University of London, School Examination Board. Has contributed to 19 scientific papers published in international journals (indexed in Scopus/Web of Science) (H-index 14) and authored two book chapters for Taylor & Francis/CRC Press and Elsevier on functional chemicals and advanced pharmaceuticals. Applied for three patents in the field of diverse drug delivery systems. Additionally, acted as a peer reviewer for leading international publishers including Elsevier, Springer Nature, Bentham, and Dove Press having reviewed over 30 manuscripts. Also co-supervised PhD theses and participated in more than eleven research projects focused on the development of pharmaceutical dosage forms and drug delivery systems. Shared in teaching training courses on cosmetics and drug delivery systems.

Abstract

The absence of effective treatments for chronic diseases can lead to treatment failures. Functionalizing nano-delivery systems with targeting ligands presents a potentially advantageous strategy for improving the diagnosis and treatment of chronic diseases. Nanocarriers, encompassing liposomes and other lipidic and polymeric nanocarriers, can be surface-decorated with targeting ligands. Several ligands, including folic acid, peptides, and antibodies, can be used to precisely recognize receptors overexpressed on the surface of diseased organ cells. This study aims to improve the therapeutic effectiveness of natural therapies with reported protective effects but that suffer from limited bioavailability. Our approach focused on developing surface-decorated nano-drug delivery systems specifically targeting certain chronic diseases. Nano-drug delivery systems were developed. They were evaluated for their physicochemical parameters, including drug entrapment efficiency, particle size, surface charge, and drug release. In addition, biochemical markers and histopathological evaluations were conducted in an animal model of certain chronic ailments. Administration of the functionalized targeting formulation resulted in therapeutic improvements relative to the conventional untargeted nano-carrier counterparts. The therapeutic enhancement can be attributed to the targeting characteristics of functionalizing ligands that are directed to inflammatory receptors overexpressed in certain chronic diseases. This approach facilitates the selective delivery of therapeutic agents to certain organs,

overcoming biological barriers. As a result, it increases the localized concentration of the therapeutic agent while reducing systemic side effects. This highlights the potential of this strategy to improve the prognosis of different chronic diseases. This approach holds significant promise for advancing precision medicine

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Dr. Ahmed Hegazy

National Research Centre, Egypt

PRESENTATION: Beneficiation of Granite Waste in Sustainable Construction Materials

Biography

Dr. Ahmed Hegazy is a dedicated Researcher at the National Research Centre, specializing in the development and characterization of advanced building materials. With a Ph.D. in Analytical and Inorganic Chemistry, his multidisciplinary expertise bridges the gap between fundamental chemical analysis and practical civil engineering applications.

His core research focuses on sustainable construction materials, non-traditional cements, structural ceramics, and advanced composites. A central theme of Dr. Hegazy's work is the transition toward green engineering, specifically through the formulation of innovative geopolymers and eco-friendly cement alternatives.

Over a career spanning more than a decade, Dr. Hegazy has made significant contributions to the valorization of industrial and natural byproducts. His research extensively covers recycling granite sawing sludge, iron slag, and ornamental stone waste into high-performance bricks, sustainable ceramic tiles, and self-leveling mortars. Furthermore, he has pioneered studies on dual-functional polymer-modified magnesium oxychloride cements and supersulphated cement pastes, optimizing their physico-mechanical properties and durability under adverse environmental conditions. His foundational work also deeply explored the modification of gypsum plaster composites using silica and waste additives to enhance water resistance and compressive strength.

Dr. Hegazy's scholarly impact is reflected in his extensive publication record in premier peer-reviewed journals, including the Journal of Building Engineering, Construction and Building Materials, and Scientific Reports. Through his continuous exploration of hybrid geopolymeric composites and pozzolanic materials, Dr. Hegazy remains at the forefront of developing resilient, sustainable construction solutions that address both environmental challenges and the rigorous demands of modern infrastructure.

Abstract

Granite extraction and processing generate large quantities of fragments, powder, and sawing sludge that are commonly landfilled, creating environmental and resource-management challenges. Owing to their quartz, feldspar, mica, and iron-bearing mineral contents, these wastes can be beneficiated as secondary raw materials in ceramic and cementitious products. In fired clay bricks, granite sludge acts as a non-plastic filler and high-temperature flux. At about 15% replacement, it reduces drying shrinkage and, when fired at 1050 °C, markedly improves strength through liquid-phase formation, particle bonding, and densification. At lower firing temperatures, however, incomplete vitrification may limit mechanical performance.

Granite waste is also suitable for ceramic tiles. A body containing 40 wt% granite waste and kaolin, fired at 1200 °C, developed low water absorption, low porosity, high flexural strength, and good chemical durability. Excessive waste content or firing temperature may cause deformation and bloating because of excessive melt formation and gas entrapment. Aplitic granite waste can additionally replace feldspar and quartz, while iron-rich fractions can be used in self-glazed ceramics and temmoku-type glazes.

In cementitious systems, ornamental-stone sludge can replace part of the natural fine aggregate in polymer-modified self-leveling mortars, improving packing, hydration, and compressive strength. Comparable igneous-rock powders may also partially replace cement and enhance durability.

Overall, granite waste can function as filler, flux, aggregate substitute, glaze precursor, and supplementary cementitious material. Effective utilization depends on matching mineralogy, particle size, replacement level, and processing conditions to the intended product, thereby reducing landfill disposal and conserving natural resources, while supporting circular construction-material manufacturing practices.

Ass. Prof. Rasha Mahmoud Abd El-Wahab

Associate Professor, National Research Centre, Egypt

PRESENTATION: Evaluation of a sustainable zeolite/bacterial cellulose nanocomposite membrane as a catalyst for biodiesel production from waste cooking oil.

Biography

Dr. Rasha Mahmoud Abd El- Wahab, Associate Prof. of applied physical chemistry, physical chemistry department, Advanced Materials Technology and Mineral Resources Research Institute, National Research Centre. My research is deeply engaged in materials science and nanotechnology, with a strong focus on environmental sustainability and green applications. Her scientific interests span Recycling environmental waste to produce smart materials and biofuels, the purification of iron-rich ores, the design of advanced nanocomposites for wastewater treatment, and the development of eco-friendly catalysts for biodiesel production. By designing multifunctional materials they contribute to solving pressing challenges in water remediation, renewable energy, and pollution control. Their work reflects a commitment to harnessing innovative nanomaterials not only for energy production and biomedical uses but also for protecting the environment and promoting sustainable technologies.

Abstract

A bacterial cellulose zeolite nanocomposite membrane was successfully prepared, characterized, and evaluated as a heterogeneous catalyst for the transesterification of waste cooking oil into biodiesel. Bacterial cellulose was produced via microbial isolation of Komagataeibacter species, while the zeolite component was synthesized from locally sourced Egyptian raw materials. The prepared catalyst demonstrated efficient catalytic activity, achieving a conversion yield of 77% under optimized reaction conditions: 10 wt% catalyst relative to oil, a methanol-to-oil molar ratio of 40:1, and a reaction time of 9 hours. The physicochemical

properties and morphology of the synthesized catalyst were characterized using X-ray diffraction (XRD), scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDX), and zeta potential analysis. The successful formation of biodiesel was confirmed using thin-layer chromatography (TLC) and high-performance thin-layer chromatography (HPTLC), which were also employed for qualitative and quantitative analysis, respectively. The composition of fatty acid methyl esters (FAMEs) was identified using gas chromatographyâ€“mass spectrometry (GCâ€“MS), revealing the presence of methyl palmitate (40.2%), methyl oleate (53%), and methyl linoleate (5.4%). The produced biodiesel exhibited several fuel properties consistent with international standards, including a saponification value of 193.3 mg KOH/g, an iodine value of 54.6 g Iâ‚‚/100 g, a cetane number of 62.26, and a higher heating value of 42.3 MJ/kg. However, certain properties, such as viscosity and acidity, still require further optimization, despite showing noticeable improvement compared to the original feedstock oil.

Dr. Safaa Hussein Alagamawy

Chemical researcher, Radio Engineering Research Center, Egypt

PRESENTATION: **Evaluating the efficacy of some barium nanocomposites in treating biomicrite limestone**

Biography

Dr. Safaa Hussein Alagamawy (S. H. M. Alagamawy), chemical researcher at the Radio Engineering Research Center. PhD student in the restoration of stone archaeological monuments. Master's degree in the same field, with a thesis entitled "[A Study of Some Multifunctional Nanomaterials in the Restoration and Conservation of Limestone Monuments With Application on a Selected Monument, and a grade of [A+] for [2023].

Abstract

Biomicrite is one of the most common types of limestone used in archaeological architecture, and it is organic carbonate sedimentary rock, as microfossils and metabolic residues of extinct animals are filled with fine-grained calcite and/or dolomite are scattered in the very fine-grained matrix (micrite), which makes it a relatively weak sedimentary stone, and its preservation is a difficult challenge facing conservators, so they resort to using compatible consolidate materials with limestone components. Nanocomposites are a recent trend in the treatment and conservation of monuments, as they aim to perform a number of functions at the same time, with minimal external chemicals interference. This experimental study aims to evaluate the efficiency of a number of multifunctional nanocomposites, namely nano zinc titanate added to nano barium hydroxide or nano barium titanate or nano barium oxide mixed in paraloid B-72 dissolved in high purity acetone, by examining the consolidate stones samples using stero microscope, PM, SEM, XRD and water contact angle, in addition to measuring the mechanical properties and colorimetric measurement, it was found that nano zinc titrate added to nano barium oxide mixed with paraloid B-72 dissolved in high pure acetone gave the best results.

Dr. Ahmed Mahdy Hamed Haggar,

Egyptian Petroleum Research institute, Egypt

PRESENTATION: **Catalytic Upcycling of Plastic Waste into Carbon Nanoblades and Carbon Nanodarts with Enhanced Mechano-Bactericidal Activity**

Biography

Dr. Ahmed Mahdy Hamed Haggar is a researcher at the Egyptian Petroleum Research Institute (EPRI), Egypt, in the Process Development Department. He received his Ph.D. in 2022 for his research on the co-production of hydrogen and carbon nanotubes from polyethylene plastic waste. His research interests focus on the catalytic conversion and upcycling of plastic waste into high-value carbon nanomaterials, including carbon nanotubes, carbon nanofibers, graphene and related nanostructures. His work also covers heterogeneous catalysis, hydrogen production, advanced carbon materials, environmental remediation, energy-storage materials, and antibacterial nanomaterials. Dr. Haggar has extensive experience in the development of catalytic processes for waste-to-value conversion and has investigated the transformation of polyolefin wastes into functional carbon nanomaterials with controlled structures and properties. His recent research emphasizes the design of carbon nanostructures with mechanically active surfaces for antimicrobial applications, contributing to sustainable approaches for simultaneous plastic-waste management and advanced-materials production. He has contributed to scientific publications, research projects, and international conferences in the fields of nanomaterials, catalysis, hydrogen production, waste valorization, and environmental applications.

Abstract

The increasing accumulation of plastic waste and the emergence of antibiotic-resistant bacteria have created an urgent need for sustainable strategies that simultaneously address environmental and healthcare challenges. This study presents a waste-to-value approach for converting plastic waste into high-value carbon nanostructures with potent antibacterial activity. Two distinct carbon nanomaterials, namely carbon nanoblades and carbon nanodarts, were synthesized through catalytic carbonization under controlled thermal and atmospheric conditions. The resulting nanostructures were systematically characterized to establish their morphology, crystallinity, and graphitic nature, and their antibacterial performance was evaluated against representative Gram-positive and Gram-negative bacterial strains. Transmission electron microscopy revealed the successful formation of platelet-like carbon nanofibers with sharp blade-like edges (carbon nanoblades) and elongated tubular carbon nanotubes (carbon nanodarts). X-ray diffraction and Raman spectroscopy confirmed the graphitic structure of both materials, while carbon nanoblades exhibited a slightly higher defect density than carbon nanodarts, suggesting the presence of more active surface sites. Antibacterial activity was assessed against Staphylococcus aureus, Escherichia coli, and Pseudomonas aeruginosa using colony-count and inhibition-zone assays. Both carbon nanostructures demonstrated significant bactericidal effects toward all tested strains. However, carbon nanoblades consistently exhibited superior antibacterial performance compared with carbon nanodarts. The enhanced activity of carbon nanoblades is attributed to their sharp-edged morphology and higher surface roughness, which facilitate direct physical damage to bacterial membranes through a mechano-bactericidal mechanism, leading to membrane disruption and cell death.

These findings demonstrate an effective route for upcycling plastic waste into functional carbon nanomaterials and highlight carbon nanoblades as promising antibacterial agents for environmental, biomedical, and antimicrobial surface applications.

Assi. Researcher Asmaa O. Ahmed

Physical Chemistry Department at the National Research Centre, Egypt

PRESENTATION: Role of Oxygen Vacancies in Enhancing the Hydrogen Evolution Reaction Performance of $SrFeO_{3-x}$ Perovskite Oxides

Biography

Asmaa O. Ahmed is an Assistant Researcher in the Physical Chemistry Department at the National Research Centre and a PhD candidate whose work sits at the intersection of materials science and clean energy technology. Her research centers on electrochemical energy applications, with particular emphasis on water electrolysis for green hydrogen production and the development of high-performance supercapacitors central to the shift toward sustainable energy storage and conversion.

Her approach combines hands-on expertise in material synthesis, using techniques including sol-gel, hydrothermal, co-precipitation, and electrochemical preparation methods, with a strong foundation in structural and electrochemical characterization.

Asmaa's doctoral research investigates perovskite oxide materials, particularly $SrFeO_3$ and $SrNiO_3$, as electrocatalysts for water electrolysis, aiming to improve the efficiency and stability of catalysts critical to scalable green hydrogen production.

Abstract

Perovskite oxides have emerged as promising electrocatalysts for the hydrogen evolution reaction (HER) due to their structural flexibility, tunable electronic properties, and defect engineering capability. In this work, we investigate the influence of spin states and oxygen vacancies on the HER activity of the transition-metal perovskite SrFeO3-X. We successfully synthesized four SrFeO3-X samples by facile sol gel technique through changing either calcination time. XRD revealed that the two samples preheated for 2 h at 500 oC then calcined for 5 and 10 h at 900 oC have a cubic phase mixed with a small percentage of orthorhombic phase. While the other two samples have pure cubic phase. SEM indicated the porous structure of both samples. In contrast to XRD, Electrical properties, Roman spectroscopy, and ESR exposed that all samples contain mixed phases with different percentage. Where, ESR analysis (g of all samples ≈ 2.00) indicates defect-related paramagnetic centers in SrFeO3-X, with optimized oxygen vacancy concentration (S2) correlating with enhanced hydrogen evolution reaction performance. S2 showed the lowest overpotential of 420 mV vs RHE, with best stability over 41 h. Perovskite oxides with mixed-valence Fe3+/Fe4+ states, strong Fe–O–Fe interactions and controlling oxygen vacancy, contribute to coupled electronic and local magnetic behavior, hence, strongly affect electrocatalytic catalytic performance.

Ass. Prof. Adel Abdel Rehiam Ahmed Koriem

Assistant Professor, National Research Centre, Egypt

PRESENTATION: **Interfacial Engineering of CuAlZn Alloy-Reinforced NBR/EPDM Elastomers for Soft Magnetic, Dielectric, and Thermomechanical Applications**

Biography

Dr. Adel A. Koriem is an Egyptian researcher and Assistant Professor in the Department of Polymers and Pigments at the National Research Centre (NRC), Egypt. Born in Giza, Egypt, in 1972, he has built an extensive scientific career spanning more than two decades in polymer science, rubber technology,materials characterization, and nanotechnology.

Dr. Koriem began his professional career at the National Research Centre in 1999 as an Assistant Researcher in the Department of Polymers and Pigments. In 2004, he became a Research Assistant and continued developing his expertise in polymer and rubber materials. Since 2010, he has worked as a Researcher in the Department of Polymers and Pigments within the Division of Chemical Industries, and since June 2025 he has served as an Assistant Professor.

His research activities focus particularly on polymer engineering technology and nanotechnology, including the synthesis and characterization of advanced polymeric and rubber materials. His work involves investigating the chemical structures and properties of synthesized materials, developing antimicrobial and fire-retardant rubber and polymer products, and exploring sustainable materials and nanocomposites. He has also contributed to the development of biochar- and agricultural-waste-derived materials as reinforcing fillers for rubber composites.

Dr. Koriem has an extensive record of scientific publications covering topics such as NBR and EPDM rubber vulcanizates, antimicrobial rubber materials, fire-retardant polymeric materials, sustainable fillers, biochar-based composites, natural antioxidants, cellulose nanocrystals, and advanced rubber nanocomposites. His recent publications include studies published in journals such as *Polymer Composites, Radiochimica Acta, BMC Chemistry, and Materials Science: Materials in Electronics.*

In addition to his research publications, Dr. Koriem is an inventor on a patent entitled "Safety and Antimicrobial Rubber Product and the Method of Preparation," registered under Certificate No. 31131 in 2023.

Dr. Koriem has actively participated in international scientific conferences, including MACRO IUPAC in Istanbul, the European Polymer Congress in Dresden, the Arab International Conference on Polymer Science and Technology in Italy, and an international conference on nanostructured polymers and nanocomposites in Dresden.

He has also contributed to several research projects funded by the National Research Centre, covering polymer-clay nanocomposites, fire-retardant polymeric materials, antimicrobial rubber products for children's safety toys and medical applications, and green antimicrobial rubber products for personal protection.

Beyond academia, Dr. Koriem has served as a scientific consultant for numerous industrial organizations in Egypt, including companies working in plastics, rubber, automotive, chemical, medical-supply, packaging,and manufacturing sectors. He is also a reviewer for academic journals, including Polymer *Bulletin* and The *Egyptian Journal of Chemistry,* and is a member of several professional scientific societies related to chemistry, polymer science, advanced materials, and nanotechnology.

His technical expertise includes Differential Scanning Calorimetry (DSC), Fourier Transform Infrared Spectroscopy (FTIR), tensile testing, Particle Charge Detection (PCD), water and oxygen vapor permeability measurements, and Gel Permeation Chromatography (GPC). He is proficient in Microsoft Office and has experience with several scientific and computing programs.

Overall, Dr. Adel A. Koriem's career reflects a strong combination of academic research, applied materials science, industrial consultancy, scientific publication, and innovation. His work contributes particularly to the development of safer, antimicrobial, fire-retardant, sustainable, and high-performance polymer and rubber materials.

Abstract

This study reports the fabrication of multifunctional NBR/EPDM elastomer composites incorporating a CuAlZn alloy as a functional filler together with 3-(trimethoxysilyl)propyl methacrylate (TMSPM) to improve interfacial compatibility. The designed materials were intended to introduce soft magnetic functionality while preserving favorable curing characteristics and mechanical, thermal, and dielectric performance. Alloy contents ranging from 2.5 to 35 phr were investigated through rheological measurements, stress relaxation analysis, differential scanning calorimetry (DSC), thermogravimetric analysis (TGA), tensile testing, scanning electron microscopy (SEM), vibrating sample magnetometry, and broadband dielectric spectroscopy. The silane coupling agent strengthened the polymer filler interface, resulting in homogeneous particle distribution at intermediate filler concentrations. Composites containing moderate alloy loadings, particularly NE2 and NE3, exhibited superior vulcanization behavior, enhanced tensile properties, lower damping behavior (tan Î´), and reduced solvent uptake, indicating the formation of a more tightly crosslinked and mechanically robust network. Thermal characterization demonstrated improved molecular organization and greater resistance to thermal degradation following alloy incorporation. Magnetic measurements confirmed the successful development of soft magnetic elastomers, with NE3 delivering the most advantageous combination of magnetic response and structural performance. Incorporation of the metallic alloy also enhanced dielectric permittivity and electrical conductivity, where NE3 showed a pronounced temperature-dependent dielectric response, whereas NE8 achieved the greatest conductivity and the highest predicted electromagnetic interference (EMI) shielding capability at elevated frequencies. Nevertheless, excessive alloy loading promoted particle agglomeration, leading to deterioration of mechanical homogeneity. Collectively, these findings demonstrate that the synergistic combination of CuAlZn alloy and TMSPM enables the production of multifunctional elastomeric composites possessing improved mechanical strength, thermal stability, magnetic responsiveness, and

dielectric performance, making them promising candidates for flexible magnetic devices, adaptive vibration-control systems, and lightweight EMI shielding materials.

Dr. Rasha Mostafa Al-Makhlasawy

Digital Signal Processing Laboratory, Computers and Systems Department, Electronics Research Institute (ERI), Cairo, Egypt.

PRESENTATION: Artificial Intelligence for Next-Generation Wireless Communications: Enabling Intelligent, Adaptive, and Sustainable Networks

Biography

Dr. Rasha Mostafa Al-Makhlasawy is a researcher at the Digital Signal Processing Laboratory, Computers and Systems Department, Electronics Research Institute (ERI), Cairo, Egypt. She received her Ph.D. in Electronics and Electrical Communications Engineering from the Faculty of Engineering, Tanta University, Egypt, in 2020.

Her research focuses on artificial intelligence, machine learning, deep learning, digital signal processing, wireless communications, biomedical signal and image processing, Internet of Things (IoT), and intelligent healthcare systems. She has contributed to numerous interdisciplinary research projects that integrate advanced AI techniques with signal processing to address real-world challenges in healthcare, smart communications, and intelligent sensing.

Dr. Al-Makhlasawy has authored and co-authored several peer-reviewed scientific publications in international journals and conferences. Her current research interests include explainable artificial intelligence (XAI), reinforcement learning, computer vision, medical image analysis, digital twins, and next-generation communication systems, particularly 6G-enabled intelligent networks. She is also actively involved in developing AI-driven solutions for disease diagnosis, predictive analytics, and smart decision-support systems.

In addition to her research activities, she is passionate about promoting STEM education and empowering young researchers through scientific training programs, workshops, and collaborative research initiatives. She actively supports interdisciplinary collaboration and encourages the participation of women in science, technology, engineering, and mathematics.

Abstract

Artificial Intelligence (AI) is transforming wireless communication systems by enabling networks to become more intelligent, adaptive, and autonomous. As wireless technologies evolve toward sixth-generation (6G) networks, traditional communication techniques face significant challenges in meeting the increasing demands for ultra-high data rates, ultra-low latency, massive device connectivity, and energy-efficient operation. AI offers innovative solutions by integrating machine learning, deep learning, and reinforcement learning into communication system design and network management.

This presentation highlights the growing role of AI in enhancing the performance of modern wireless communication systems. It discusses how AI-driven algorithms improve channel estimation, resource allocation, spectrum management, beamforming, interference mitigation, modulation adaptation, and network optimization. Special attention is given to reinforcement learning techniques that enable communication systems to learn from dynamic environments and make intelligent real-time decisions without relying solely on predefined mathematical models.

The presentation also explores the integration of AI with emerging technologies, including the Internet of Things (IoT), edge computing, digital twins, reconfigurable intelligent surfaces (RIS), and intelligent transportation systems. These technologies collectively contribute to building self-organizing and self-healing wireless networks capable of supporting future smart cities, healthcare, industrial automation, and autonomous vehicles.

Despite its significant potential, several challenges remain, including data privacy, computational complexity, model interpretability, cybersecurity, and energy consumption. Addressing these challenges requires interdisciplinary collaboration between communication engineers, AI researchers, and industry stakeholders.

Overall, AI is expected to become a fundamental component of future wireless communication infrastructures, driving the evolution toward intelligent, resilient, and sustainable networks while creating new opportunities for scientific research, technological innovation, and real-world applications across diverse sectors.

Ass.Prof. Ahmed F. Ghanem

Packaging Materials Department, National Research Centre,Egypt

PRESENTATION: Development of PVA Composite Films for Meat Preservation

Biography

Dr. Ahmed Ghanem is an assistant professor at the Packaging Materials Department, National Research Centre (NRC), Egypt, and a former Postdoctoral Researcher at the Institute of Hydrogen Energy, Korea Institute of Energy Technology (KENTECH), South Korea. He received his Ph.D. in Polymer Chemistry from Ain Shams University in 2021. Throughout his academic career, he conducted research visits and internships at the University of Arkansas (USA), Université Claude Bernard Lyon (France), the Institute of Physical Chemistry of the Polish Academy of Sciences (Poland), and the University of Science and Technology of China (USTC). Dr. Ghanem has authored more than 33 peer-reviewed publications, is the inventor of two patent applications, H-index 19, and has participated in numerous national and international research projects as a PI and member. He serves as a reviewer for leading scientific journals and has received several awards for outstanding research and scientific presentations. His research interests focus on polymeric materials and nanocomposites for sustainable applications, particularly in food packaging, water treatment, energy storage and conversion, and biomedical technologies.

Abstract

Meat products are the most common sources of protein. However, meat products are highly prone to spoilage during storage or even transportation. Several reports are released annually, presenting promising solutions to overcome meat preservation challenges. In this work, novel composite polyvinyl alcohol (PVA) films have been

developed and investigated for packaging fresh meat. Specifically, active components were extracted from hibiscus flowers and encapsulated using a spray dryer. The well-assembled microcapsules were incorporated into PVA matrices at different concentrations to obtain free-standing composite films. FTIR, SEM, Zetasizer, TGA, mechanical testing, water contact angle measurements, and permeability were performed. Significant changes in the physicochemical and biological properties of PVA films were observed, thanks to the successful formation of extract microcapsules and their inclusion in the polymer matrix. Moreover, the results emphasized the superiority of composite PVA films containing microcapsules over free-form films as packaging materials for real meat samples. The results showed substantial antioxidant activity, along with prolonged antibacterial activity, evidenced by a decline in bacterial growth and a near-pH plateau during refrigerated storage (4 °C) for up to 10 days. These findings support the sustained-release mechanism, which extends their biocidal effect over the storage time. This highlights that adding stable microencapsulated extracts into biopolymers can enhance the applicability and workability of natural extracts for use as active packaging materials, overcoming the limited performance of traditional films incorporating free extracts.

Dr. Basant Aboelsaud Elsebai

National Research Centre, Egypt

PRESENTATION: **Zeolite-Infused Nanofibrous Hydrogel for Water Treatment Applications**

Biography

Dr. Basant Aboelsaud Elsebai is a researcher at the National Research Centre (NRC), Egypt, with expertise in analytical chemistry, electrochemistry, nanomaterials, sensors, and wastewater treatment. She obtained her B.Sc. in Chemistry from Cairo University in 2011, followed by an M.Sc. in Analytical Chemistry in 2020. During her early research career, she contributed to a Marie Curie joint project on electroanalytical sensors for heavy metals and biological species and undertook an international research internship at the University of Coimbra, Portugal, under Prof. Christopher Brett. She received the NRC Best International Publication Award for Young Researchers in 2019 and completed her PhD in Wastewater Treatment Applications at Ain Shams University in 2024. Dr. Elsebai has participated in several national and international research projects and conferences and has published multiple peer-reviewed international articles. Her current research focuses on nanomaterial-based sensors and biosensors, electrochemical technologies, and the development of low-cost, high-efficiency materials for innovative wastewater treatment applications.

Abstract

Heavy metals pose considerable challenges for the global community. Several studies are released annually, presenting new techniques or materials to mitigate this environmental issue. However, many shortcomings are still associated with this kind of research in terms of performance, durability, and cost. Therefore, in this study, a nanofibrous polysaccharide-based hydrogel was produced in a fermentation medium. In an innovative approach, the isolated bio-mat was modified with Na-zeolite particles derived from Egyptian kaolin, resulting in a fully renewable bionanocomposite. This unique assembly is theoretically attractive and exhibits significantly different properties from those obtained by conventional methods. The modified hydrogel was

extensively characterized, and the results showed successful anchoring of crystallized zeolite particles within the hydrogel's open cells, displaying improvements in surface texture and diverse surface charges. Due to its superior properties, the bionanocomposite was tested as an adsorbent for metal ions in synthetic wastewater. Within minutes, the bionanocomposite effectively demineralized the wastewater, achieving high adsorption capacities of 77.33, 76.39, and 24.06 mg g-1 for Pb (II), Cd (II), and Ni (II) ions, respectively. Kinetic and isotherm models indicated that the adsorption followed a pseudo-second-order reaction and obeyed Langmuir isotherms. Dubinin's analysis demonstrated that the removal mechanism is chemisorption. Thermodynamic studies confirmed the exothermic nature of the reaction. Durability tests showed that the modified hydrogel could be reused over five cycles. This research offers an opportunity to develop new adsorbents with both economic and environmental benefits for sustainable water treatment technologies.

Ass.Prof. Ahmed N. E. Osman

Associate Professor, National Research Centre, Egypt

PRESENTATION: Engineered Hybrid Nanocomposites for Safer Healthcare, Cleaner Water, and Sustainable Heritage Preservation

Biography

Dr. Ahmed Nabile Emam Osman, earned his BSc and MSc degrees in medical biophysics in 2006 from Helwan University, and 2010 from Cairo University, respectively. In October 2006, he joined the Center of Excellence for Advanced Sciences - National Research Center (NRC), where he was engaged in Advanced Materials & Nanotechnology Lab as a temporary contract till his graduation of MSc Degree. In November 2017, he was awarded the PhD degree in Laser applications in Photochemistry and Photobiology from the National Institute of Laser Enhanced Sciences – Cairo University. His research work in MSc and PhD focused on both of the biophysical study of semiconductor nanoparticles doped with different magnetic materials to be used in biomedical applications, and development of Fluorescent hybrid nanocomposites for biomedical imaging and early diagnosis of cancer, under supervision of Prof. Amani Mostafa the professor of materials science and Former Dean of Advanced Materials Technology & Mineral Resources Research Institute – National Research Centre, and Prof. Mona Bakr Mohamed, who she was the former postgraduate student of Prof. Mostafa A. El-Sayed, and the Horney professor of nanotechnology and ultrafast spectroscopy and the executive director the Egyptian Nanotechnology Center (EGNC) – Cairo University. In Sep. 2012, he joined to Ultrasound Imaging and Therapeutics Research Laboratory, Biomedical Engineering Department, University of Texas at Austin – USA as a Visiting Researcher till Jan. 2013. In this period, his research work focused on engineering nanomaterials to be used as Photo-acoustic Tomography Imaging (PAT). Dr. Emam published about **46** archival publications (**42** Peer reviewed articles and **4** Book Chapters). Also, he is a reviewer in 30 Scopus indexed journals The main research of interest is based on the material science and nanotechnology, where this research was focused on the fabrications of different categories of nanomaterials such as: Magnetic, Plasmonic, Semiconductor, Metal Oxide, Hybrid Nanocomposites (i.e.; Nanoalloys, Doped Nanomaterials), Carbon –based nanomaterials such as, Carbon quantum dots (C-Dots) and their hybrid nanocomposites. In addition, investigation of the photophysical properties of nanomaterials and their hybrid nanostructures such as, magneto-plasmonic nanoalloys, doped semiconductors, carbon- based nanomaterials. Furthermore, his research focused on the use of nanomaterials in the water treatment, archaeological, antimicrobial, anti-fouling applications. Moreover, Green synthesis of nanomaterials, Energy applications of nanomaterials. Ahmed is also deeply engaged in academic capacity-building through workshops and international programs on nanotechnology, quantum dots, and research ethics across Egypt, Africa, and beyond. His training includes degrees in Medical Biophysics and Laser Applications in Photochemistry and Photobiology, complemented by advanced courses in AI for scientific research, peer review, and statistics. Alongside his research, he mentors graduate students, develops industry-oriented nano-enabled solutions, and contributes book chapters on toxicity and environmental sustainability of hybrid plasmonic nanocomposites.

Abstract

Engineered hybrid nanocomposites are emerging as multifunctional platforms that connect nanomedicine, environmental remediation, and cultural heritage conservation to enable a safer and healthier life-style. By integrating plasmonic, magnetic, semiconductor, metal-oxide, carbon-based, and biopolymer building blocks, these nanoarchitectures are tailored for cancer theranostics, antimicrobial control, bone regeneration, and advanced imaging and sensing. Representative systems include gold-doped wollastonite for osteogenic repair, multifunctional silver, copper oxide, and zinc oxide hybrids to suppress multidrug-resistant pathogens, and carbon dot–based nanocomposites for high-contrast diagnostic imaging and latent fingerprint visualization. In parallel, water-desalination and photocatalytic nanomaterials, nanostructured consolidants for limestone and stucco monuments, and nano-enabled inks and coatings for industrial and biomedical interfaces collectively target generous quality-of-life improvements. Emphasis on green synthesis, toxicological evaluation, and application-relevant performance underpins their translation into theranostic healthcare, clean-water technologies, and sustainable conservation strategies. Altogether, engineered hybrid nanocomposites constitute an integrated toolbox for advancing health security, environmental safety, and heritage durability, thereby supporting more generous and healthier lifestyles at individual, societal, and ecosystem levels.

Environment & Agriculture

SCIENTIFIC PROGRAM & SPEAKER PROFILES

Section: Environment & Agriculture

--- SESSION 1- Environment & Agriculture ---

Prof. Dr. Nour Shafik El-Gendy

Egyptian Petroleum Research Institute (EPRI), Egypt

PRESENTATION: Marine Macroalgae Recruitment for Achieving Food, Energy and Water Nexus

Biography

Prof. Dr. Nour Shafik El-Gendy is a Professor in the field of Environmental Sciences, Sustainable Development, Clean Energy and Nano-Biotechnology. She is the chairman of the Board of Directors of Central Analytical Labs – Egyptian Petroleum Research Institute (EPRI). She is the Former Head Manager of Petroleum Biotechnology Lab., the Former Acting and Vice Head of Process Design & Development Department, (EPRI). El-Gendy is Advisor to the Chairman of the Board of Trustees of October University for Modern Sciences and Arts, MSA University. She is a visiting professor in the School of Chemical Engineering of Sathyabama Institute of Science and Technology, India and Trivandrum Engineering Science & Technology (TrEST) Research Park, India. She is the Former Advisor for the Egyptian Minster of Environment. She is member of Scientific Research, Technology, and Cyber-security Committee, National Council for Women (NCW) of Egypt. El-Gendy is a member of the Higher Scientific Committee of the Arab Union for Sustainable Development and Environment. She is also a member of the Arab Council for Creativity and Innovation. El-Gendy is expert in the assessment of environmental pollution and its remediation, wastewater treatment, biofuels, petroleum bioupgrading, green chemistry, nanobiotechnology, valorization of wastes, biocorrosion and its mitigation. She has published 12 chapters, 10 books and 142 research papers, and supervised 29 MSc and PhD theses. Dr. El-Gendy is also an editor and reviewer in 92 and 252 international journals, respectively. She participated also as PI, Co-PI or research member in national and international research projects concerning with; biovalorization of wastes, solid waste management, bioethanol from lignocellulosic wastes; biofuels and value-added products from algae, application of nanobiotechnology in upgrading of petroleum and bioremediation of polluted environment. Dr. El-Gendy participated in 70 international workshops, 68 international training courses, 186 international conferences and 27 international seminars. Dr. El-Gendy established 4 ongoing research schools; Petroleum and Environmental Biotechnology, Petroleum and Environmental Nanobiotechnology, Biofuels, and Valorization of Solid Waste Biomass. Dr. El-Gendy is a member in many national and international committess, associations and organizations concerned with petroleum industry, sustainable development, environmental health and sciences. She is in international collaboration with different foreign international universities, research institutes, and centers. She is listed in the Stanford–Elsevier global list of top 2% scientists 2025. She was honored in many scientific forums. She was awarded from Egypt the Women Appreciation Award in the field of "Water, Energy and Environmental Sciences" - 2024. She was also awarded the Regional Award of the Arab Union for Sustainable Development and Environment in the field of "Environment and Sustainable Development" - 2023. She was awarded from India the Research Excellence Award, Science father Women Researcher Award 2022 for the contribution and honorable achievement in innovative research, and the International Scientist Award 2021 on Engineering, Science and Medicine. El-Gendy has also been awarded the Best Women Scientist Award 2021 from College of Agriculture and Food Sciences, Florida Agricultural & Mechanical University, USA. She has been honored by the Official Portal of Department of Environment – Ministry of Environment and Water – Malaysia and University of Putra Malaysia – 2021. Her biography is recorded in Who's Who in Science and Engineering.

Abstract

Intensive fishing practices, climate change, nutrient enrichment, waste discharge, sedimentation, and other forms of anthropogenic pollution contribute to the excessive proliferation of marine macroalgae, eutrophication, coral bleaching and mortality, and reduced light penetration for photosynthesis. These impacts disrupt marine ecosystems, diminish biodiversity, damage coastal environments, and adversely affect food webs, tourism, economic stability, and progress toward the Sustainable Development Goals (SDGs).

To advance the three pillars of sustainability—environmental protection, social well-being, and economic development—while addressing the food–energy–water nexus, it is essential to valorize surplus aquatic biomass through integrated, zero-waste, circular economy approaches. Our findings demonstrate that converting this underutilized biomass into value-added products is both feasible and environmentally beneficial. Specifically, it can be applied in (1) wastewater treatment; (2) the production of biodegradable plastics, biocides, probiotics, biofertilizers, and green nanocatalysts for diverse applications; and (3) the generation of various types of biofuels.

Dr. Chadene Diyab

International expert in climate change and sustainable development, France .

PRESENTATION: Environmental Diplomacy and Women: Issues of Stability and Climate Peace in the Arab World and the EMENA Region

Biography

Dr. Shaden Diab, research professor and trainer at several academic and training institutions in France and the Euro-Mediterranean region, PhD in Environmental Sciences. She is an international expert in climate change and sustainable development. A chemical engineer, she holds a PhD in Environmental Sciences from Pierre and Marie Curie University (Paris 6). She has over fifteen years of experience in higher education, scientific research, training, and international consulting, working as a research professor and trainer at several academic and training institutions in France and the Euro-Mediterranean region. Her work focuses on climate change, climate justice, corporate social responsibility, educational innovation, artificial intelligence for climate, and community resilience. She is also a Climate Peace Ambassador for the Institute for Economics & Peace (IEP), where she focuses on the relationship between climate change, peace, and human security. She has received several international awards and honors related to environmental innovation and climate action, including contributions and recognition related to COP21. She was also selected for the Overseas Elite program in China in recognition of her contributions to innovation and international cooperation. Dr. Diab is considered a leading figure in the field of climate change and sustainable development. Shaden Diab is a member of the Scientific Committee of the European Federation of European Universities (FEDE) and contributes to developing visions related to education, quality training, and future skills. She also works as an international trainer in environmental journalism and climate communication, having delivered training sessions and lectures to journalists and media organizations, including national radio stations, the French Ministry of Foreign Affairs, the Labomed Laboratory, and the

Arab States Broadcasting Union (ASBU), on the role of media in understanding climate challenges and communicating environmental issues to the public.

She is the founder of the Green Education and #GreenRefugees for Education project, which links environmental education, climate action, and the inclusion of vulnerable groups. This project has been developed in France, Greece, and Turkey. Through her writings, publications, and articles, she contributes to the global debate on climate change, climate migration, environmental security, artificial intelligence, and the transfer of green technologies between the North and South. Dr. Shaden Diab combines scientific expertise, academic work, and communication skills. Media, and climate diplomacy, with the aim of raising awareness and building innovative solutions to address major environmental challenges.

In addition to her scientific and academic career, Dr. Shaden Diab is a writer and novelist who is interested in combining science, literature, and human thought.

She has published several novels and intellectual works, including novels that address the relationship between humanity and the world and its transformations, such as the novel "A Chance with," where human questions meet scientific and philosophical reflections. In her latest book, "The Box... Yahya Who Saw the World: The Wonder of Creation and the Experience of Creation," she seeks to build a bridge between science and literature by exploring the relationship between creativity and knowledge, and between science fiction and human experience, presenting a vision that combines the physics of writing, the chemistry of creativity, and the embodiment of the text. This path reflects the distinctiveness of her experience, which combines scientific research, literary writing, and intellectual communication, with the aim of rethinking the relationship between science, culture, and humanity.

Abstract

The world is confronting an acceleration of environmental crises — biodiversity loss, rising global temperatures, and the multiplication of climate "hotspots" Pfenning-Butterworth, A., Buckley, 2024) Populations are growing risks of vulnerability and forced environmental migration. According to the World Bank's Groundswell reports, up to 216 million people could be displaced within their own countries by 2050, with Sub-Saharan Africa and North Africa among the most exposed regions. In this context, environmental diplomacy is emerging as a central issue for regional stability and security(Clemet 2021). This paper examines the role of women in environmental and climate diplomacy, with a particular focus on the Arab world and the EMENA/Africa region, where women are among the populations most vulnerable to climate change impacts. These vulnerabilities are reflected in increasing water insecurity, climate-induced displacement, food insecurity, and socio-economic pressures, which often exacerbate existing gender inequalities (IPCC, 2022; UN Women, 2022). Climate change affects women and men differently due to unequal access to resources, decision-making processes, land ownership, and economic opportunities. In many climate-vulnerable regions, women are often responsible for household water and food security, making them particularly exposed to environmental degradation and resource scarcity (UN Women, 2022; FAO, 2023). 1 At least one-third of refugee households in the Arab States are female-headed households, reflecting the growing responsibility placed on women in contexts of forced displacement. Female-headed households are often disproportionately affected by climate-related shocks because they have fewer economic resources, weaker social protection mechanisms and limited access to adaptation capacities (UN Women, 2020; UNHCR, 2023). The contribution of women to diplomacy has historical roots. Long before women gained formal access to diplomatic institutions, they played important informal diplomatic roles as salonnières, court intermediaries, political advisers, and spouses of ambassadors, using social networks and cultural spaces to facilitate communication, information exchange, negotiation and mediation between political actors (Calvié, 2015; Hamilton & Langhorne, 2011). These informal diplomatic practices demonstrate that women have historically contributed to international relations through forms of influence that operated outside traditional state structures and official diplomatic channels. The intersection between climate change, displacement and gender inequality highlights the importance of integrating women’s perspectives into environmental diplomacy, climate negotiations and regional security strategies.

Dr. Pulane Hannah Koosaletse-Mswela

Senior Lecturer at the University of Botswana, Botswana

PRESENTATION: Environmental Quality-Based Assessment of a Small Dam for Sustainable Recreational Use in Semi-arid Botswana

Biography

Dr. Pulane is an environmental scientist and Senior Lecturer at the University of Botswana, with expertise in environmental quality, water and wastewater management, waste management, and resource recovery. She holds a PhD in Environmental Science and an MSc in Applied Microbiology, and has built a strong academic and professional career at the intersection of science, sustainability, and policy-relevant research in semi-arid environments.

Her research focuses on environmental quality assessment, ecological risk analysis, and sustainability-oriented evaluation of water resources, waste systems, and anthropogenic activities, with particular emphasis on small dams, surface water bodies, and community-linked ecosystems in arid and semi-arid regions of Southern Africa. Her work supports evidence-based decision-making for environmental protection, public health, and sustainable recreational and productive water use.

Dr. Koosaletse-Mswela has supervised and graduated numerous postgraduate students at MSc and PhD levels, and has published in peer-reviewed journals while actively participating in national and international scientific conferences. She is experienced in interdisciplinary research, project coordination, and applied environmental assessment that bridges science and practice.

Beyond academia, she has contributed to environmental governance and science-policy interfaces, including serving on the technical committee that advised Botswana's National Waste Management Policy. She is the Secretary General of the Botswana Academy of Science and the inaugural Vice Chair of the OWSD-Botswana Chapter, where she advocates for mentorship, inclusive excellence, and leadership development for women in science.

Her work aligns strongly with global efforts to empower science, innovation, and entrepreneurship in addressing environmental challenges and building resilient, sustainable futures for society.

Abstract

Tourism is one of the fastest-growing sectors globally and the Botswana Tourism Organisation has come up with the Dams Tourism initiative to diversify and grow the tourism sector in Botswana. The absence of national recreational water quality standards complicates the assessment of these environments for safe tourism use, hence the need for baseline data to guide sustainable planning and policy development. This study, therefore, assessed the environmental quality of Mogobane Dam, to determine its suitability for recreational eco-tourism. Water, sediment, and fish samples were analyzed for physico-chemical parameters, heavy metals, and fecal coliforms, while qualitative data were gathered through semi-structured interviews and questionnaires administered to community leaders, farmers, and tourists. Secondary data sources, such as rainfall and temperature records, were obtained from the Department of Meteorological Services. The results indicated that all analyzed water samples contained detectable levels of lead, iron, and aluminium .

The mean Pb concentration (4.17 mg/L) exceeded WHO, BOBS, and SANS permissible limit.. Fecal coliform testing recorded values beyond the WHO and USEPA guidelines for safe recreational waters, deeming Mogobane Dam unsuitable for direct contact activities such as swimming or consumption. Fish samples (Clarias gariepinus) contained elevated heavy metal levels, particularly in gills and fins. Interviews revealed limited awareness of water quality issues and the absence of formal management strategies among local users. No structured environmental monitoring or waste management practices were found in place. The Geo-accumulation Index classified Mogobane sediments as moderately to highly polluted, while Enrichment Factor values suggest significant anthropogenic contribution to heavy metal accumulation. Bioaccumulation Factor analysis indicated strong uptake of Pb and Fe in fish tissues, highlighting potential risks of metal transfer through the aquatic food chain. The study emphasises the necessity of regular water quality monitoring, watershed management and stakeholder awareness raising to facilitate the safe and sustainable utilisation of dams in Botswana and similar environments.

Prof. Dr. Seham Ali Shaban

Egyptian Petroleum Research Institute, Egypt

PRESENTATION: Biochar – Sustainable Material for A Green Future

Biography

Prof. Dr. Seham Ali Shaban is a professor at the Egyptian Petroleum Research Institute. She has held significant leadership roles, including serving as the former head of the Refining Department, and the former head of the Catalyst Laboratory. Professor Shaban has supervised approximately 25 scientific theses. She has received numerous research grants and has published around 60 research papers in various fields, with a focus on green energy and waste treatment. In addition, she is a member of numerous international arbitration panels for scientific research and numerous local and international associations.

Abstract

Biochar has created a lot of interest because of its unique properties of sustainable agricultural production and environmental protection. Biochar is known to reduce nitrogen loss from soil in terms of nitrous oxide emission and ammonia volatilization, improve the nutrient retention capacity and structural and chemical properties of soil, and increase plant growth and productivity. Biowaste materials such as biosolids, municipal waste, and paper mill sludge are effective raw materials for biochar production. These materials are rich in nutrient content, and biochar produced using these waste products is highly effective for agricultural production and environmental remediation.

Dr. Safaa Saeid Mohamed Ali,

Physics Research institute, National Research Centre, Egypt

PRESENTATION: Spatial Distribution and Health Risk Assessment of Heavy Metals in the Road Dust Deposited on Tree Leaves in the main Overcrowded Squares of Greater Cairo.

Biography

Associated Professor of Spectroscopy and its application. I had experience in Spectroscopic plasma diagnostics, Qualitative/quantitative determination of the concentration of elements in different kinds of samples, Trace and ultra trace analysis of liquid samples using the inductively coupled plasma (ICP-AES), Analysis of gaseous samples (hydride generation) using wall stabilized plasma arc and Elemental Analysis using atomic absorption and graphite furnace .

I am a stuff member in number of research projects such as Egyptian-Spanish Joint Program of Co-operation on Industrial Research and Development “New Strategies for Reducing the Costs of the Steel Industry; Ladle Slag Recycling and Raw Material Refining”•. As well as the Center of Training and Developing Skills, National Research Center, Cairo, Egypt, (Lecturer for Inductively coupled plasma and Atomic Absorption courses), also the Unit of Analysis and Central Scientific Service, National Research Center, Cairo, Egypt.

Abstract

Rapid urbanization and heavy vehicular traffic in Greater Cairo have led to increasing concerns regarding over heavy metal accumulation in urban dust and its potential impacts on public health. This study assessed the contamination levels and human risks of heavy metasl in roadside dust deposited on tree leaves collected from highly crowded public squares across Greater Cairo, Egypt. Elemental concentrations were determined using inductively coupled plasma mass spectrometry (ICP-MS) and high-resolution continuum source graphite furnace atomic absorption spectrometry (HR-CS-GF-AAS). Contamination status was evaluated using enrichment factors (EF) and the Pollution Load Index (PLI). Non-carcinogenic and carcinogenic health risks for adults and children were estimated according to the US EPA risk assessment model. Results showed that the average concentrations of all studied heavy metals exceeded local background soil values, indicating strong anthropogenic influence. PLI values greater than unity confirmed significant multi-element pollution across the sampled squares. Ingestion was the dominant exposure pathway, followed by dermal contact. Children exhibited higher non-carcinogenic risks than adult, particularly for cadmium. Although contamination levels were elevated, the carcinogenic risks values for Cr, Cd, and Pb remained within the acceptable limits (10-6 to 10-4). These findings highlight the serious environmental burden of traffic-related heavy metals pollution in Greater Cairoâ€™s urban squares and underscore the urgent need for continuous monitoring and effective mitigation measures to protect public health in densely populated urban environments.

Takya Mohammad Ahmed Elnady

El-Sadat STEM School, Egypt

PRESENTATION The Role of Green Chemistry in Achieving Sustainable Development Goals: A Modern Educational Vision

Biography

Takya Mohammad Ahmed Elnady is a Chemistry Teacher and Capstone Projects leader at El-Sadat STEM School, Egypt, and a PhD Researcher in the Department of Curriculum and Instruction at Sadat City University, where she combines classroom teaching with active research in science education. Her research interest in green chemistry education grew directly out of her classroom experience. While supervising students' capstone projects and practical laboratory sessions, she observed the significant volume of chemical waste generated by conventional experiments and the long time required for its decomposition. This observation shaped her research focus on integrating green, environmentally responsible alternatives into secondary science curricula. Building on this, she designed a structured program of green chemistry activities organized around five thematic axes, each paired with a distinct active-learning strategy, aimed at developing two key learner outcomes: observable environmental practices and strategic thinking skills across five dimensions planning, generating creative alternatives, decision-making, systems analysis, and future foresight. At this forum, she will present her work title "The Role of Green Chemistry in Achieving Sustainable Development Goals: A Modern Educational Vision."

Beyond her doctoral work, Takya has also developed "EcoChem Lab," a poster presenting her green chemistry program for a school-level innovation competition at Cairo University, reflecting her continued effort to translate research into practical, classroom-ready tools.
Takya's work sits at the intersection of chemistry education, environmental sustainability, and the Sustainable Development Goals, with particular focus on preparing STEM students to connect scientific knowledge with responsible environmental action.

Abstract

Amid escalating environmental challenges and the global drive toward achieving the Sustainable Development Goals, there is a growing need to reconsider science teaching curricula, moving beyond the transmission of theoretical knowledge toward building environmental practices and strategic thinking capable of addressing sustainability issues. The idea for this research emerged from field observations of the volume of chemical waste generated by students' experiments and practical projects, and the long time such waste takes to decompose — highlighting the need to move toward green alternative materials that achieve the same reactive purpose while reducing environmental impact. Within this framework, the researcher adopts an educational vision that employs the principles of green chemistry as an effective pedagogical entry point, rather than as isolated chemical content, aiming to link classroom learning with real-world environmental behavior among school students in general, and students of STEM (Science, Technology, Engineering, and Mathematics) schools of excellence in particular.

To address this problem, the researcher proposed a program based on green chemistry principles, from which a practical handbook to a number of alternative green chemistry experiments can be developed. This presentation discusses this program, which is designed around five thematic axes, with each activity linked to an active learning strategy, reflecting an intentional methodological diversity. The program aims to develop two variables among learners: environmental practices, understood as observable behaviors within the classroom and laboratory setting, and strategic

thinking skills across their five dimensions (strategic planning, generating creative alternatives, strategic decision-making, systems thinking, and future foresight.

This modern educational vision positions green chemistry as a multidimensional pedagogical tool that contributes to achieving several Sustainable Development Goals — foremost among them Quality Education (Goal 4), Responsible Consumption and Production (Goal 12), and Climate Action (Goal 13) by preparing a generation of learners capable of connecting scientific knowledge with responsible environmental action.

Prof. Dr. Hamdy F. El-Mowafi

Research Institute, Agricultural Research Center (ARC), Egypt

PRESENTATION: New Egyptian innovations to develop the production of hybrids , super and basmati smart rice resilien

Biography

Prof. Dr. Hamdy F. El-Mowafi is an Emeritus Research Professor at the Field Crops Research Institute, Agricultural Research Center (ARC), Egypt, and the National Project Leader for the Development of Hybrid, Super, and Basmati Rice under Water Scarcity and Climate Change. He is one of Egypt's leading rice breeders, with over four decades of experience in rice genetics, breeding, and sustainable crop improvement.

Prof. El-Mowafi pioneered Egypt's hybrid rice breeding program and has developed numerous high-yielding, climate-resilient rice varieties, including hybrid, super, and Basmati rice cultivars. His research focuses on breeding for drought and salinity tolerance, water-use efficiency, and food security. He has authored more than 60 scientific publications and has received several prestigious awards, including the Gold Medal at the 48th Geneva International Exhibition of Inventions (2022) and the Arab Food Security Award (2023). Throughout his distinguished career, he has played a leading role in national and international rice research initiatives, collaborating with organizations such as the International Rice Research Institute (IRRI) and FAO, and has trained researchers and agricultural professionals across Africa and Asia.

Abstract

Climate change, water scarcity, and increasing food demand pose significant challenges to sustainable rice production worldwide. Egypt has made remarkable progress in developing climate-resilient super, hybrid, and Basmati rice varieties through integrated conventional and modern breeding approaches. These varieties exhibit high yield potential, early maturity, tolerance to salinity, drought, heat, and limited irrigation, while maintaining superior grain quality. The developed cultivars, including Sakha Super 300, Sakha Super 301, Sakha Super 302, Sakha Super 303, Giza Basmati 201, and Egyptian Basmati Hybrid Rice 11, achieve yield advantages of up to 30% over traditional varieties and can save approximately 30–40% of irrigation water through improved water management practices. The use of EGMS and CMS breeding systems has further enhanced hybrid seed production and genetic purity. This chapter highlights Egypt's achievements in developing and disseminating climate-adapted rice cultivars and discusses their contribution to food security, sustainable agriculture, efficient water use, and climate change adaptation. These innovations provide a practical model for increasing rice productivity while strengthening the resilience of agricultural systems under changing environmental conditions.

Dr. Jenny Carolina

Researcher in postdoctorade, National University of Colombia, Colombia

PRESENTATION: Smart Vermicomposting: Automating Biofertilizer Production Through Community Knowledge and Low-Cost IoT

Biography

Jenny Carolina lives in Cali, state of Valle del Cauca, in the country Colombia, located in South America. She is an Agricultural Engineer, holds a Master's degree in Biological Sciences and is PhD in Agroecology; currently is undertaking postdoctoral research within a program with the Colombian government that supports research projects by women in artificial intelligence, science and quantum technologies and is head of her house as single mom. A scientist with experience in public-private partnerships in agriculture and food systems programs. She has worked on research projects with multiethnic rural communities in Colombia and Spain. She has also contributed to the design, evaluation, and implementation of productive projects. In addition, she teaches at universities both nationally and internationally. She has pursued her postgraduate studies with scholarships and grants due to her high performance in research work. As part of the Agroecology Research Group at the National University of Colombia - Palmira campus, she has represented them in spaces such as the Academic Dialogue Platform on food and nutritional security as part of the Territorial Council for Food Sovereignty and Security for Santiago de Cali; she was a member of the Food and Nutritional Security Advisory Council, collaborating in the construction of the Food and Nutritional Security Action Plan of Valle del Cauca; she participated in micro-region workshops for the participatory construction of the "Departmental Plan of Agroecology of Valle del Cauca" and in the Agroclimatic Technical Table of the agricultural sector of Valle del Cauca.

Abstract

Synthetic chemical fertilizers have caused various environmental impacts, which is why there is increasing interest in the use of biofertilizers. However, the production of these fertilizers requires the incorporation of strategies that contribute to stabilizing their production. Therefore, this research project was proposed, consisting of two vermicomposting pilot projects: one at the biofertilizer plant of the Universidad Autónoma de Occidente (UAO) and the other on the ancestral territory of an Afro-Colombian community. The project focuses on automation, which allows for the monitoring and regulation of variables determining the quality of biofertilizer production within the process. The project's approach to biofertilizer production involves biotic component, utilizing red wiggler worms (Eisenia foetida) in vermicomposting processes to obtain solid and liquid humus. The automated system uses low-cost IoT devices to monitor crop variables (bed) such as temperature, humidity, and pH, and to generate regulatory actions. It also incorporates AI algorithms that process the variable values to generate estimates related to production levels.The social appropriation of IoT in the control of biofertilizer production variables is an ongoing process that requires the collaboration of multiple stakeholders and is a community-centered approach. Through the co-construction of knowledge mediated by participatory methodologies, significant benefits are generated for the Afro-Colombian community belonging to the "Sinecio Mina" Historical and Cultural Research Movement, associated with the project, as well as for the GEAS Research Group and the UAO's food security research group, by fostering an inclusive and creative use of these technologies, improving quality of life, and promoting sustainable development.

Ms. Marwa Sulaiman Mohsin Al-Hinai

Sultan Qaboos University, Oman

PRESENTATION: Nanotechnology-Enabled Smart Agriculture for Enhanced Drought Resilience and Sustainable Agroecosystems

Biography

Marwa Sulaiman Mohsin Al-Hinai is a PhD researcher in Plant Sciences at Sultan Qaboos University, Oman, specializing in plant biotechnology, physiology, and abiotic stress tolerance. Her research focuses on the application of nanotechnology to enhance crop performance under drought and salinity stress, with particular emphasis on nano-trehalose formulations. She holds a Master's degree in Plant Production (Plant Sciences) from Sultan Qaboos University, where her research centered on the characterization and improvement of wheat genotypes of Omani origin for salt tolerance. She also earned a Bachelorâ€™s degree in Applied Biology from the Higher College of Technology, Muscat. Her academic training has provided her with hands-on experience in plant stress physiology, antioxidant enzyme analysis, molecular techniques, tissue culture, and nanoparticle synthesis and characterization.

Abstarct

Climate-induced drought has emerged as one of the most severe threats to global food security, disrupting soil–water dynamics and limiting crop productivity in arid and semi-arid regions. In this context, the convergence of nanotechnology and smart agriculture offers an unprecedented opportunity to develop climate-resilient and resource-efficient food systems. This review critically synthesizes recent advances in nano-enabled drought mitigation—focusing on nanosensors, nano-fertilizers, and nano-hydrogels—alongside digital technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), and autonomous systems for precision water and nutrient management. The analysis reveals that integrating nanotechnology with smart farming frameworks enhances drought tolerance through molecular-level monitoring, controlled nutrient delivery, and data-driven irrigation optimization. Despite remarkable progress, significant challenges persist regarding nanoparticle safety, data interoperability, and scalability in resource-limited contexts. A conceptual Nano–Smart integration framework and a research roadmap are proposed to bridge current gaps and guide future innovation toward eco-safe, self-regulating, and ethically governed agroecosystems. This synthesis underscores that Nano–Smart agriculture is not merely a technological evolution but a transformative paradigm—linking scientific intelligence with ecological sustainability and social equity to secure food production under a changing climate.

Ass.Prof. Ahmed R. Henawy

Faculty of Agriculture, Cairo University, Giza, Egypt

PRESENTATION: Gut Microbiomes of Black Soldier Fly Larvae: Approaches for Bioconversion, Degradation of Antibiotics, Detoxification of Mycotoxins, and Production of Biofertilizers for Sustainable Waste Management

Biography

Dr. Ahmed R. Henawy , Assistant Professor ,Cairo University & Huazhong Agriculture University. He holds a BSc in Biotechnology from Cairo University 2012, followed by an MSc in Microbiology from Huazhong Agricultural University in China 2019, and a PhD in Microbiology from Cairo University 2023. He has also completed a MicroMasters in Bioinformatics through the University System of Maryland 2020, along with specialist training in metagenomics and whole-genome sequencing of bacteria. Currently, he serves as Assistant Professor in the Microbiology Department at Cairo University, a position he has held since 2023. Concurrently, he holds the role of Postdoctoral Fellow and Group Leader at Huazhong Agricultural University in Wuhan, China, where his research career first took an international dimension. His career reflects a strong commitment to bridgirg research and academia, to leading collaborative research groups at one of China's premier agricultural universities. He also serves as coordinator for academic collaboration between Cairo University and leading Chinese institutions, including HUST and HZAU.

Abstract

This study presents an innovative biotechnology framework that integrates Black Soldier Fly (Hermetia illucens) larval bioconversion with targeted microbial biodegradation to address critical challenges in organic waste management, food safety, and the development of a sustainable bioeconomy. The approach leverages the synergistic interaction between insect metabolism and specialised microbial consortia to transform agro-industrial residues into high-value products, including protein-rich biomass and biofertilisers.

A key focus of this research is the detoxification of mycotoxins contaminated substrates, wherein microbial strains and larval gut microbiota collaboratively facilitate degradation through enzymatic pathways such as oxidation, reduction, and hydroxylation. By combining culture-dependent microbial isolation with meta genomic and functional analyses, this research identifies key microbial players and metabolic pathways responsible for enhanced bioconversion efficiency and toxin mitigation.

The results demonstrate that integrating engineered or selected microbial communities with Black Soldier Fly systems significantly improves substrate conversion rates, detoxification efficiency, and system resilience under variable environmental conditions. These findings underscore the potential for scalable, cost-effective biotechnological solutions that can be translated into industrial applications.

Importantly, this research supports the development of circular economy models by converting waste streams into commercially valuable outputs, thereby opening opportunities for startup innovation and sustainable agribusiness. The study contributes to bridging the gap between laboratory research and market-oriented solutions, positioning insect-microbe systems as a promising platform for future bio-based entrepreneurship and green technology development.

Ass.Prof. Asmaa Ahmed Mostafa Halema

Faculty of Agriculture, Cairo University, Giza, Egypt

PRESENTATION: From Contaminated environment to Sustainable Solutions: Microbial Genomics for Heavy Metal Bioremediation and Plant Growth Promotion

Biography

Dr. Asmaa Ahmed Mostafa Halema is an Assistant Professor in the Genetics Department, Faculty of Agriculture, Cairo University, Egypt, a position she has held since June 2023. She earned her Ph.D. in Genetics from Cairo University (2019“2023) and her M.Sc. in Microbiology from Huazhong Agricultural University, China (2016“2019), under a Chinese Government Scholarship (CSC). She also holds a B.Sc. in Biotechnology from Cairo University, where she graduated first in her class.

Dr. Halema's research centers on microbial genetics, genomics, and bioinformatics, with a particular focus on heavy metal bioremediation, plant-microbe interactions, and omics-driven approaches to understanding bacterial resistance mechanisms. Her work spans whole-genome sequencing and analysis of environmentally significant bacterial strains, including Enterobacter, Raoultella, and plant growth-promoting rhizobacteria, contributing to sustainable agricultural and environmental biotechnology.

She has authored or co-authored numerous peer-reviewed publications in journals such as World Journal of Microbiology and Biotechnology, AMB Express, Frontiers in Bioengineering and Biotechnology, and Biodegradation, alongside a book chapter on soil biodiversity and climate change. She has served as Co-PI on national research projects, including an ASRT-NBNE grant on COVID-19 and biotechnology research, and has collaborated on several international projects with institutions in Saudi Arabia.

Recently, she conducts research on Black Soldier Fly Larvae (BSFL)-associated microorganisms and their applications in sustainable agriculture and environmental biotechnology with Prof. Longyu Zheng.

Dr. Halema has authored and co-authored numerous publications in high-impact international journals and has collaborated with researchers from China, Saudi Arabia, and German. She has participated in several nationally and internationally funded research projects and actively contributes as a reviewer for scientific journals. Her long-term research vision is to integrate multi-omics technologies, bioinformatics, and AI-driven approaches to develop innovative solutions for precision agriculture, environmental sustainability, food security, and the circular bioeconomy.

Abstract

Heavy metal pollution from industrialization, mining, and improper waste disposal threatens ecosystems, agriculture, and human health, making sustainable microbial bioremediation an attractive alternative to conventional remediation methods. This work integrates microbial isolation, physiological characterization, electron microscopy, whole-genome sequencing (WGS), comparative genomics, structural bioinformatics, and gene expression analysis to elucidate heavy metal resistance and plant growth promotion mechanisms in environmentally important bacteria isolated from industrial wastewater and rhizospheric soil. Among the characterized strains, Enterobacter roggenkampii FACU2 removed 62% of cadmium, with significant induction of czcA, cadA, czcC, and czcD resistance genes under cadmium stress. E. kobei FACU6 achieved 83.4% lead removal (571.9 mg/g biomass) and tolerated up to 3,000 mg/L Pb, with resistance genes upregulated under optimal growth conditions. E. cloacae FACU1 (Ars9) removed 85.5% of arsenic (1,415 mg/g biomass), while a second E. cloacae strain tolerated up to 15,000 ppm arsenic; structural modeling (SWISS-MODEL, AlphaFold2) and docking confirmed ArsC-mediated arsenate reduction and ArsB/Acr3-mediated arsenite efflux. Raoultella planticola FACU3 combined high lead tolerance with plant growth-promoting (PGP) genes for IAA, phosphate solubilization, and phenazine production. Separately, Escherichia coli FACU2024 was identified as a dual-function PGPR, harboring 142 growth-promotion genes and 15 heavy-metal resistance genes. SEM/TEM imaging confirmed metal adsorption, intracellular accumulation, and stress-related morphological changes across strains. Genome sequencing (4.8–5.7 Mb) revealed diverse metal-resistance genes, efflux systems, genomic islands, and mobile genetic elements underlying environmental

adaptability. Collectively, these findings establish a genomic framework identifying dual-function bacterial strains with strong potential for environmental bioremediation, sustainable agriculture, and next-generation biofertilizer development.

Ass.Prof. Sameh Mohamed Mohamed El-Sawy

Agricultural and Biological Research Institute, National Research Centre, Cairo, Egypt.

PRESENTATION: Precision Agriculture and Smart irrigation: Mitigating Water Scarcity and Enhancing the crop productivity under climate Change

Biography

Dr. Sameh Mohamed Mohamed El-Sawy is an Associate Professor at the Agricultural and Biological Research Institute, National Research Centre (NRC) in Cairo, Egypt. He holds a Master's degree in Strawberry Cultivation and Productivity and a PhD in Rationalizing Water Consumption for Tomato Plants under Arid/Sandy Soil Conditions (2016). His international academic record features significant participation in China, including three professional training programs and a prestigious tenure under the Talented Young Scientist Program (TYSP) at Ningxia University (2017-2018), where he focused on smart saving-irrigation technology in arid regions and conducted field trials in Tongxin County. A seasoned researcher, Dr. El-Sawy has contributed to 15 local and 6 international projects, serving as the Principal Investigator for many projects; 1- the NRC-funded project on deficit irrigation techniques, 2- using modern technology in greenhouses for producing seedlings and vegetable crops in soilless cultures and 3- sustainable development for ASRT New Valley farm using new technology funded by Academy of Scientific Research and Technology (ASRT). Furthermore, Dr. Sameh worked as Corresponding Investigator for aquaponics systems in saline arid regions (ASRT-funded), and climate-smart practices project funded by ANSO (China). He also Executive Manager for a multi-year Mediterranean greenhouse optimization initiative international project (PRIMA-funded). Moreover, he is authored 38 papers in peer-reviewed international journals, contributions to 15 national and international conferences, and active supervision of postgraduate students in Egypt and China, he also leads agricultural operations as the Head of Vegetable Production at the NRC Farm in El-Noubaria and Director of the Model Research Farm in Al-Moghra under the Academy of Scientific Research and Technology.

Abstract

Global agriculture faces an unprecedented dual challenge: meeting the food demands of a rapidly growing global population—projected to reach nearly 10 billion by 2050—while simultaneously mitigating the adverse impacts of climate change, resource depletion, and water scarcity. Traditional farming practices are no longer sufficient to sustain productivity under these volatile environmental dynamics. Egypt faces severe challenges to its food security, primarily driven by its arid climate and low annual rainfall. These natural constraints are further exacerbated by climate change, which negatively impacts crop productivity and plant growth. Consequently, maximizing plant water-use efficiency through advanced irrigation technologies and water-saving methods has become imperative. Agriculture currently accounts for approximately 70% of global freshwater withdrawals. However, traditional irrigation techniques, such as flood or standard overhead sprinkler systems, often result in significant water loss through evaporation, runoff, and deep percolation. To overcome these structural inefficiencies, the agricultural sector is undergoing a digital transformation often referred to as Agriculture 4.0. By leveraging advanced digital tools, sensor networks, and predictive algorithms, smart agriculture shifts farming from a reactive, intuition-based activity to a proactive, data-driven science. Among these technological innovations, precision irrigation stands out as a crucial intervention for climate adaptation and resource conservation. Smart agriculture and precision irrigation are no longer technological luxuries; they are essential

frameworks for ensuring global food security under mounting environmental pressures. By combining real-time IoT monitoring, satellite-based remote sensing, and predictive AI, modern agriculture can produce significantly more yield while consuming drastically fewer resources.

Ass.Prof. Sahar Abd EL-Aziz Othman

National Water Research Center (NWRC), Egypt

PRESENTATION: Mitigating Severe Salinity Stress in Vicia faba L. via Exogenous Proline

Biography

Dr. Sahar Abd EL-Aziz Othman is a Researcher at the National Water Research Center (NWRC), Central Laboratory for Environmental Quality Monitoring (CLEQM), Egypt. She earned her Ph.D. in the field of wastewater treatment and has over three decades of professional experience in water and environmental research.

She began her career at the Drainage Research Institute (DRI), NWRC, where she worked from 1996 to 2001. Since 2001, she has been working at the Central Laboratory for Environmental Quality Monitoring (CLEQM), NWRC, contributing to research and environmental monitoring programs.

Her expertise includes chemical, physical, and biological analyses of water, soil, and plant samples, with extensive experience in environmental quality assessment and laboratory analysis. Her current research focuses on water and wastewater treatment using innovative chemical and biological materials, as well as water quality assessment through the application of mathematical water quality models.

Her research interests include sustainable water treatment technologies, environmental monitoring, water quality modeling, wastewater reuse, and the development of environmentally friendly treatment approaches to support water resource management

Abstarct

Climate change and its impacts on soil salinization threaten faba bean (Vicia faba L.) production in the Nile Delta. Although the osmotic benefits of exogenous osmoprotectants are well recognized, long-term, tissue-specific partitioning mechanisms of repeated foliar proline under unmitigated severe saline irrigation (12 dS m-1) are poorly quantified. In the study, a holistic hydrochemical and agronomic approach was used to evaluate the efficacy of weekly foliar proline (500 mg L-1) as an intervention for climate-resilience. The hydrochemical modeling confirmed severe irrigation constraints at 12 dS m-1 indicating strong reverse base-exchange phenomena.. Weekly proline applications reprogrammed the stress response, with pod counts returning to 33.67 and 91.75% reproductive potential preserved. Under moderate salinity (6 dS mâ•»Â¹), proline application recovered 70.91% of the yield potential lost due to osmotic stress. Metabolic recovery was governed by a coordinated Nâ€“Kâ€“protein trait axis, as indicated by a multivariate principal component analysis (97.47 % of the variance explained). The study offers a robust data-driven framework to transform hazardous saline water into an agricultural input and a scalable climate-adaptation approach aligned with SDGs 2, 6, and 12.

Dr. Heba Abdelaziz Moussa AbdAlla

Botany Department, Agriculture and Biological Institute at National Research Centre, Egypt

PRESENTATION: Reproductive-Stage Heat Stress Tolerance in Rice: Implications for Climate-Resilient Agriculture under Global Warming

Biography

Dr. Heba Abdelaziz Moussa AbdAlla is a Researcher at the Botany Department, National Research Centre (NRC), Egypt, specializing in plant biotechnology, tissue culture, molecular biology, and sustainable agricultural innovation. Her research integrates advanced biotechnology, genomics, and molecular breeding to develop climate-resilient crops capable of withstanding heat, drought, and salinity stresses, contributing to global food security and sustainable agricultural systems.

She has established an active international research profile through strategic collaborations with leading institutions, particularly the Chinese Academy of Sciences (CAS), and has played a key role in developing joint Egypt–China research initiatives and international scientific partnerships. Her work spans plant tissue culture, crop genetic improvement, controlled-environment agriculture, carbon footprint assessment of plant production systems, and the sustainable utilization of plant resources.

Beyond research, Dr. Abdalla is actively engaged in science diplomacy, research management, and the organization of international scientific conferences that foster interdisciplinary collaboration and knowledge exchange. She has successfully contributed to numerous competitive research proposals, international collaborative projects, and scientific publications. Driven by a passion for translating scientific discoveries into practical solutions, she is dedicated to advancing sustainable agriculture, strengthening global research partnerships, and promoting innovation that addresses the pressing challenges of climate change and food security.

Abstract

Climate change represents one of the most critical challenges to global agricultural sustainability, particularly through rising temperatures, increasing drought frequency, and soil salinization, which severely threaten crop productivity and food security. My recent research on rice investigates the physiological, cellular, and molecular mechanisms underlying reproductive-stage heat stress tolerance, a critical determinant of yield stability under warming climates.

Using an integrated approach that combines tissue culture-based screening, controlled environmental stress assays, and molecular analyses, I evaluated genotype-specific responses to elevated temperature, focusing on pollen viability, spikelet fertility, and yield-related traits. The results revealed substantial genetic variation among Egyptian and international rice germplasm, enabling the identification of heat-resilient genotypes with superior reproductive performance under thermal stress. Heat exposure significantly affected pollen development, anther dehiscence, and fertilization efficiency, which were closely associated with differential expression patterns of stress-responsive and chloroplast-associated genes.

These findings provide valuable molecular targets and physiological markers for breeding programs aimed at enhancing heat tolerance in rice. The study further highlights the importance of integrating advanced phenotyping platforms, molecular breeding strategies, and data-driven approaches to accelerate the development of climate-smart crop varieties. Overall, this work contributes to sustainable agricultural strategies by improving crop resilience to climate extremes, enhancing yield stability, and supporting food security in vulnerable regions, particularly in arid and semi-arid environments. Moreover, the research underscores the critical role of international collaboration in advancing climate adaptation strategies through scientific innovation, technology transfer, and knowledge exchange.

Dr. Mai Nossair Amer

Pharmaceutical and Drug Industries Research Institute, National Research Centre, Egypt

PRESENTATION: From Lab Innovation to Agricultural Entrepreneurship: Lactic Acid Bacteria for Sustainable Farming and Healthy Food Systems

Biography

Dr. Mai Nossair Amer is a Researcher of Chemistry of Natural and Microbial Products at the National Research Centre (NRC), Egypt, with over 15 years of scientific expertise in microbiology, biotechnology, and functional probiotics. Holding a BSc in Biotechnology, an MSc in Prophylactic and Therapeutic Probiotics, and a PhD focused on anticancer bioactive compounds from probiotics, she has built a distinguished research career spanning microbial genetics, natural bioactive molecules, food microbiology, and sustainable bio-based solutions. Dr. Amer has contributed to numerous national and international projects on probiotics, microbial therapeutics, environmental biodegradation, and biological plant protection, and has authored multiple indexed scientific publications in high-impact journals. Alongside her research, she teaches food microbiology and infection diseases at Helwan University and is an active member of OWSD and several scientific networks. As an emerging entrepreneur CEO and founder of AgriFood Serv, she bridges science with innovation by transforming microbial research into sustainable agricultural and food-industry applications.

Abstract

Transforming scientific research into real-world agricultural and food solutions is essential for building sustainable and healthy communities. This presentation explores how lactic acid bacteria (LAB) can move beyond laboratory research to become a driving force in agricultural entrepreneurship and functional food innovation.

Lactic acid bacteria are well known for their applications in food fermentation and probiotic products. However, recent advances demonstrate their powerful role in agriculture as biofertilizers, biostimulants, and biological control agents. Through the production of organic acids, antimicrobial metabolites, enzymes, and plant-growth–promoting compounds, LAB improve soil fertility, enhance nutrient availability, suppress plant pathogens, and increase crop resilience to environmental stress.

By integrating microbiological research, strain development, and scalable production systems, LAB-based technologies can support sustainable farming practices while simultaneously contributing to healthier food chains with reduced chemical residues. This approach connects agricultural productivity with food safety, environmental protection, and market competitiveness.

The presentation highlights how science-driven entrepreneurship in agricultural biotechnology can empower women scientists to create innovative enterprises that strengthen food security, promote healthy nutrition, and advance sustainable development.

Dr. Maissara Mohamed Kamal Elmaghraby

Agriculture Research Center (ARC),Egypt

PRESENTATION: Lactic Acid Bacteria-Based Bio-Stimulant Production from Low-Quality Date Waste: A Sustainable Circular Bioeconomy Approach

Biography

Dr. Maissara Mohamed Kamal Elmaghraby is a dedicated Researcher at the Central Laboratory of Organic Agriculture, Agricultural Research Center (ARC), Giza, Egypt. With over 15 years of professional experience in organic and agricultural microbiology, he has contributed significantly to advancing sustainable agricultural practices in Egypt.

He earned his B.Sc. in Agricultural Cooperative Sciences (Agribusiness and Marketing Department) from the Higher Institute for Agriculture Cooperation (2006). He completed his Pre-M.Sc. studies in Agricultural Microbiology at Cairo University (2009, Excellent), followed by a Master's degree (2015) focusing on "PGPR as Enhancer of Mint under Organic Agriculture System." He later obtained his Ph.D. in Agricultural Microbiology (2021) from Benha University, with research on organic farming applications to increase vegetable crop productivity.

Since 2008, he has worked extensively in soil and plant microbiology, compost and vermicompost production, greenhouse and field applications, biofertilizers, and biological control of nematodes and soil-borne fungi.

In addition to research, Dr. Elmaghraby is actively involved in academic training and community outreach. He supervises practical training for agriculture students, provides technical consultations for graduation projects, trains farmers and organic companies, and participates in national initiatives promoting organic agriculture, biofertilizers, and composting technologies across Egyptian governorates."

Abstract

The accumulation of low-quality date palm (Phoenix dactylifera L.) fruits represents a significant environmental and economic challenge in major date-producing regions, including Egypt and the Middle East. Large quantities of downgraded or postharvest-damaged dates are often discarded, contributing to organic waste buildup, greenhouse gas emissions, and loss of valuable bioresources. This study proposes an innovative biotechnological approach utilizing lactic acid bacteria (LAB) as a bio-stimulant product to recycle low-quality date waste into value-added agricultural inputs. Selected LAB strains, were employed to ferment date waste under controlled conditions. The fermentation process enhanced the release of bioactive compounds such as organic acids, amino acids, phenolic compounds, and soluble nutrients, producing a biologically enriched formulation suitable for plant application.

The resulting LAB-based bio-stimulant demonstrated potential to improve nutrient availability, stimulate root development, enhance microbial soil activity, and increase plant tolerance to abiotic stress. Recycling date waste through LAB fermentation not only reduces environmental pollution and landfill burden but also transforms agro-industrial by-products into sustainable agricultural solutions. Economically, this approach supports circular bioeconomy principles by creating new revenue streams from otherwise discarded materials, lowering production costs for bio-stimulant manufacturing, and increasing farm productivity.

Overall, the integration of LAB biotechnology with date waste valorization offers an eco-friendly, cost-effective strategy that aligns with sustainable agriculture goals, promotes waste reuse, mitigates environmental impact, and enhances agricultural resilience in arid and semi-arid regions.

Prof. Dr. Salwa K. Hassan

Environment and Climate Change Research Institute, National Research Centre, Egypt

PRESENTATION: Assessing Indoor Pollution: Organic and Carcinogenic compounds; Heavy Metal Exposure, and Health Risks in Indoor Environments

Biography

Prof. Dr. Salwa K. Hassan is a professor of Air Pollution Chemistry and the head of Air Pollution Research Department, Environment and Climate Change Research Institute, National Research Centre, Dokki, Giza, Egypt. She got B.Sc. degree in 1991 from Department of chemistry, Faculty of Science, Cairo University. She joined the National Research Centre, Air pollution Research Department, Environment Research Division, as a Researcher Assistant in 1994. In 2000, she promoted to Assistant Researcher after obtaining her M.Sc. from Department of chemistry, Faculty of Science, Cairo University in the field of Air Pollution Chemistry. She was promoted to Researcher position in 2007 from Department of chemistry, Faculty of Science, Cairo University. In 2013, she has been promoted to Associate Professor in the field of Air Pollution Chemistry. In 2019 she was promoted to a full Professor in the same field. On 2024 she becomes the head of air pollution research department. Her main field of interest is air pollution monitoring and measurements; atmospheric photochemical reactions; study the chemistry of the atmosphere and its chemical composition, formation of fine aerosol and its physical characterization and impact of pollution and environmental exposures on health. She has about 34 years of experience in the atmospheric chemistry field. She has organized workshops and participate in national and international conferences. She has published 51 scientific articles cited in SCI journals. She has co-edited a series of books on air pollution, human health and the environment. She has participated in national and international projects as PI and as member. She also reviewer to many international journals and is member of several international scientific committees. Finally, she received the Academy of Scientific Research Award in the field of Environmental Research and Environmental Education.

Abstract

This presentation highlights the pivotal role of women researchers in environmental health by examining indoor pollution and its health implications for families. Drawing from empirical studies on domestic and indoor spaces, this research evaluates heavy metals and organic pollutants to which women and children are daily exposed.
The first study in primary schools revealed that urban environments exhibit elevated levels of polycyclic aromatic hydrocarbons (PAHs), driven primarily by traffic emissions. While ingestion remains the primary exposure pathway for children, cumulative cancer risks remained within acceptable baselines. The second study in residential houses assessed indoor dust loading rates and heavy metal concentrations (lead, cadmium, and nickel). Indoor dust loading averaged 1.58 g/m2/week, with unpaved side streets significantly increasing dust accumulation, whereas main-street proximity amplified toxic metal loads. Building age and paint degradation further escalated lead concentrations. Notably, environmental tobacco smoke severely degraded indoor air quality, increasing dust loading by 1.7-fold and heavy metal levels up to 2.7-fold, causing over 77% of smoking households to exceed US EPA lead safety guidelines.
These findings confirm that domestic spaces pose hidden environmental health risks. Through this field research, practical public health strategies (such as regular wet cleaning, banning smoking indoors, and using lead-free materials) can be developed, enabling scientists to make crucial progress in protecting domestic environments and community health.

Ass. Prof. Alyaa Ibrahem Salim

School of biotechnology, Nile University, Egypt

PRESENTATION: Synergistic CO_2 Capture: A Novel Chitosan-Metal Oxide Nanoparticle-MOF Hydrogel Composite from Waste-Derived Materials

Biography

Dr. Alyaa Ibrahem Salim is a full-time assistant professor at the Faculty of Engineering and Applied Science, Nile University (NU). She received the B.Sc. (2006) in chemistry, faculty of science, Mansoura University, Egypt. She got a diploma in biochemistry, Alexandria University (2007-2008) and diploma in Analytical Chemistry, Cairo University (2010). She received the M.Sc. (2015) and Ph.D. (2018) degrees in analytical chemistry, chemistry department, faculty of science, Cairo University, Egypt. She has publications distributed between high-impact journals, conferences, and book chapters. Her research tracks cover health care, climate changes and solutions, different applications of nanotechnologies, food spoilage, and environmental science. Her research interests include pharmaceutical analysis, electrochemical analysis, spectroscopic techniques, HPLC techniques, quantum analysis, and sensors fabrication. She is a member in the Egyptian Society of Analytical Chemistry, Cairo university, Egypt.

Abstract

This study presents a novel composite hydrogel bead system for efficient CO_2 capture, synthesized entirely from green, waste-derived materials. Chitosan serves as the biodegradable matrix, integrated with metal oxide nanoparticles (MO NPs) from organic waste and metal-organic frameworks (MOF) as active nano-adsorbents. Beads were fabricated via dropwise coagulation. Characterization confirmed successful integration: XRD verified crystallinity, FTIR showed chemical bonding, and SEM revealed a transformation from smooth chitosan to highly porous, granular composites, indicating increased surface area. EDX quantitatively confirmed nanomaterial incorporation.

CO_2 capture performance, evaluated using a custom laboratory system with colorimetric monitoring, demonstrated exceptional results. The composite beads achieved 100% capture efficiency for 50 mL CO_2, outperforming chitosan-only (15%), MO NP-only (55%), and MOF-only (75%) beads. Post-capture analysis provided direct evidence of chemisorption: FTIR showed carbonate species formation, SEM indicated surface densification, and EDX confirmed a carbon content increase from 14.66 wt% to 27.81 wt%.

The integration of waste-derived MO NPs and MOF into chitosan creates a synergistic, high-capacity, reusable sorbent. This system combines MO NP chemisorption, MOF physisorption, and chitosan's structural versatility. These green-synthesized hydrogel beads offer a scalable, sustainable approach for industrial point-source CO_2 capture, supporting circular economy principles and decarbonization pathways.

Ass. Prof. Shymaa Sabry Zaher

National Institute of Oceanography and Fisheries, Egypt

PRESENTATION: Assessing the Therapeutic efficiency of Spirulina platensis on ethanol-Induced Oxidative Markers, DNA Damage and some Organs histopathology in Rat

Biography

Asssociate prof. Shymaa Sabry Mohamed Zaher, Assistant Professor of Phycology, Fresh water and lakes division, Inland water branch, National Institute of Oceanography and Fisheries (NIOF), Egypt. her research interests concern the Botany specially phycology. Algal biotechnology, Usage of applied algae in the medical field, cosmetics and different scientific fields. Special expert in Algal Identification in marine environment. Special concern on toxic algal blooms. Membership in The Egyptian phycological society, Egyptian net for climate change CAN EGYPT 2030. Participating in over than 15 research projects at various aquatic environments.

Abstract

The therapeutic antioxidant properties of Spirulina platensis conquered scientific interest. The present study assesses the possible antioxidant activity and DNA fragmentation of Spirulina platensis powder against ethanol-induced gastric ulcers in rats. To explore the possible efficiency, antioxidant defense enzymes (e.g., superoxide dismutase, and glutathione peroxidase), were examined as well as the level of lipid peroxidation (MDA) and the DNA damage in the gastric mucosa by comet Assay. Intragastric administration of Spirulina (500 mg/kg body weight) before (protected) or after (treated) injection with ethanol significantly lowered the ulcer index value compared to the ulcer control group ($p < 0.05$). Moreover, the alcohol-induced decrease in stomach antioxidant enzyme activity found in the ulcer control group was prevented by Spirulina powder post-treatment or pretreatment. Furthermore, it reversed the ethanol-induced increase in lipid peroxidation and DNA fragmentation. Also, the focus was on assessing the impact of these compounds on histopathological lesions induced by ethanol in the liver, kidney, and lung of rats. On conclusion, Spirulina recorded, anti-ulcerative, anti-inflammatory, antioxidant, and diminish DNA damage. Its healing action may be used as alternative treatment gastric ulcer instead using chemical drugs and has similar effect as organic tablets.

Dr. Douaa Hussein Abdel Aziz

National Research Centre, Egypt

PRESENTATION: Bioremediation and Our Lives: Restoring the Environment We Depend On

Biography

Dr. Douaa Hussein Abdel Aziz started working at the National Research Center by 2008. She had her MSc in Microbiology in 2014 from Ain Shams University and her PhD in Microbiology from Ain Shams University in 2019. She focused on working in microbial ecology, bioremediation for pollution reduction, and microbial treatment of environmental hazards to reduce their harmful effect on the environment and humans as well. She published five papers; three of them are indexed in Scopus. She was awarded the Best Master Thesis—Agricultural Professions Syndicate, 2015. She was a speaker at the International Conference of the Faculty of Science, Ain Shams

University, “Scientific Innovations and Sustainable Development (SISD)" – 2018. She has been a reviewer in the International Journal of Microbiology and Biotechnology (IJMB) since 2021. She is a member of the American Society of Microbiology and the Society of Applied Microbiology, 2025.

Abstract

Reducing contamination in our environment is a crucial step to improve our health and well-being. Coal is considered one of the world's primary energy sources and a priceless natural resource. It has played a key role in economic growth by serving as a major source of energy. Although coal mining operations have contributed to economic growth in many countries, it is one of the most dangerous activities in the world, due to high environmental and safety risks. These risks affect soils, plant communities, wildlife, and water sources and release significant amounts of CO_2 and CH_4 into the atmosphere. This contributes to increased greenhouse gases, global warming, and environmental damage that affect human life. In addition, coal mining leads to adverse health effects, including a lot of diseases like ischaemic heart disease, cerebrovascular disease, chronic obstructive pulmonary disease, lower respiratory infections, and lung cancer. The white-rot fungi have garnered significant interest due to their remarkable ability to degrade lignocellulosic materials and decompose some environmental contaminants, such as charcoal. This study aims to evaluate the biodegradation of the fungal strain Phanerochaete chrysosporium for the pollutant charcoal and its extreme tolerance to cope with such a pollutant and use it as a sole carbon source to reduce its bad impact on the environment.

Dr. Elham Ahmed Abdelrahim Mohamed

Conservation Department, Faculty of Archaeology, Fayoum University, Egypt,

PRESENTATION: The Role of Artificial Intelligence in the Preservation of Mural Paintings Under Climate Change: Opportunities, Challenges and Emerging Frameworks

Biography

Elham Ahmed Abdelrahim Mohamed is an archaeological conservator and academic researcher specializing in the conservation and restoration of antiquities, with a focus on mural paintings. She is a recipient of the prestigious "Scientists for Next Generation" (SNG) grant from the Academy of Scientific Research and Technology (ASRT) in Egypt. Elham graduated with a Bachelor's degree (B.A.) in Archaeology with a grade of "Excellent with Honors" from Fayoum University. She subsequently obtained her Master's degree (M.Sc.) in Conservation from the same institution with a grade of "Excellent with a formal recommendation for publication." Her research interests lie at the intersection of cultural heritage preservation, nanotechnology, and advanced computer applications, particularly the integration of Artificial Intelligence and Machine Learning in preventive conservation and climate-adaptive heritage management.

Abstract

This study provides a structured analytical framework to elucidate how AI-driven methodologies can fundamentally shift preventive conservation strategies from a paradigm of reactive repair to a proactive, sustainable stewardship approach under climatic stress. By conducting a systematic examination of the integration of Machine Learning (ML), Deep Learning (DL) and the Internet of Things (IoT), the research paper critically evaluates the efficacy of these technologies within real-world heritage scenarios. The analytical results highlight the transformative capability of Convolutional Neural Networks (CNNs) for identifying microcracks and surface defects within controlled environments with remarkable precision, achieving accuracy rates of 94%. Concurrently, it explores the potential of digital twins in simulating non-invasive restoration interventions, thereby minimizing physical risks to the artwork. findings demonstrate that adopting AI-driven predictive maintenance strategies offers significant economic benefits, potentially reducing long-term conservation costs by approximately 35% through the implementation of early warning systems. Consequently, this paper establishes a strategic roadmap for integrating AI into heritage management, emphasizing the need to adopt Â«Green AIÂ» principles and establish ethical governance frameworks. It concludes that fostering interdisciplinary collaboration is essential for building resilient, climate-adaptive systems capable of safeguarding cultural heritage for future generations.

Dr. Arig Wahid Elkhouly

Faculty of Industrial and Energy Technology, Borg Al Arab Technological University (BATU), Egypt

PRESENTATION: Hazelnut Shell Powder as a Sustainable Novel Functional Ingredient for Quality Improvement of Crackers

Biography

Dr. Arig Wahid Elkhouly is a researcher and lecturer in Food Processing Technology at Borg El Arab Technological University (BATU), Egypt. She holds a Ph.D. in Environmental Studies, a Master's in Dairy Science and Technology, and a Bachelor's in Food Science and Technology from Alexandria University.

Her research focuses on sustainable food systems, valorization of agro-industrial wastes, and upcycling processing by-products into high-value functional ingredients within circular economy frameworks. Dr. Elkhouly's work is published in leading international journals, including Nature Scientific Reports and Springer's Discover Food, reflecting her dedication to zero-waste food technologies and sustainable nutrition.

Dr. Elkhouly is a Certified ISO 9001 Lead Auditor with eight years of industrial experience as a production engineer in the packaging sector (2012â€“2020). She has also served as an Exam Invigilator for the British Council in Alexandria since 2012. Recognized as an ambassador for climate action, water conservation, risk management, and governance, she actively participates in international scientific forums and environmental initiatives, bridging academic research, industrial application, and sustainable development goals.

Abstract

To address the challenges of agro-industrial wastes, agri-food by-products represent valuable, underutilized sources of bioactive compounds for sustainable processing. This study investigated the valorization of hazelnut shells a primary agro-industrial waste by incorporating hazelnut shell powder (HSP) into baked crackers through partial wheat flour substitution at levels of 0%, 2.5%, 5%, 7.5%, and 10% (w/w). Physicochemical evaluations revealed that HSP fortification significantly ($P < 0.05$) enhanced crude protein, ash, crude fiber, cellulose, hemicellulose, lignin, and essential minerals (Ca, Zn, Mn, Cu). Furthermore, HSP integration markedly increased total phenolic and flavonoid contents, leading to elevated DPPH and ABTS radical scavenging antioxidant activities. Texture profile analysis demonstrated an increase in hardness, gumminess, cohesiveness, and chewiness alongside higher substitution levels. Sensory evaluation confirmed that moderate incorporation levels preserved overall consumer acceptability while delivering a nutrient-dense, fiber-enriched functional snack. These findings demonstrate the strong potential of upcycling agro-industrial processing waste into functional food ingredients, offering an effective strategy for waste reduction and the development of value-added healthier food products. This research directly supports sustainable food systems and aligns with Sustainable Development Goal 12 (Responsible Consumption and Production).Keywords: Agro-industrial waste, Upcycling, Hazelnut shells, Functional food, Crackers, Circular economy, SDG 12 also can select SDG 9: Industry, Innovation, and Infrastructure for sustainable food innovation).

General Session: From Knowledge to Impact

---Session 1 : General Session: From Knowledge to Impact---

Prof. Dr. Hala Yousry (Moderator)

Desert Research Center, Egypt

PRESENTATION: Bridging the Gap between Scientific Research and Practical Application Causes, Challenges, and Strategic Interventions

Biography

Prof. Hala Yousry is a Professor of Rural Sociology and former Head of the Social Studies Department at the Desert Research Center in Egypt. She is a member of the Rural Women Committee at the National Council for Women and serves as Chair of the National Steering Committee for the Small Grants Programme (SGP) under GEF/ UNDP.

Prof. Yousry has played a significant regional and international role, serving for eight years as the Country Focal Point, North Africa Representative, and Board Member of the African Forum for Agricultural Advisory Services, part of the Global Forum for Rural Advisory Services. In 2012, she was selected by UN Women to represent Egyptian and Arab women at the Rio+20 Conference, where she was further chosen by the Women Major Group (WMG) under United Nations Environment Programme to deliver a plenary speech.

She is also an founding member of the Board of Trustees of the Egyptian Sustainable Development Forum. Bridging academia, civil society, and government, Prof. Yousry has participated in hundreds of conferences, workshops, and seminars at national, regional, and international levels.

Her research portfolio includes numerous scientific publications and working papers in both Arabic and English, focusing on the social dimensions of desert and rural communities, Bedouin and rural women, climate change, desertification, water and food security, and sustainable development in Egypt and the Arab region. She has received multiple national and international awards and certificates of recognition.

In addition to her academic work, Prof. Yousry is a published writer, a frequent guest in television and radio interviews, and an experienced coach working in Egypt and internationally. She has been active and in good relations with national, regional and international organizations such as UN Women, UNESCO, UNEP, UNDP, ESCWA, IUCN, ICARDA, CEDARE, AARDO and FAO.

Abstract

The continuous growth of scientific research has resulted in an unprecedented accumulation of peer-reviewed publications, research reports, and postgraduate theses across diverse disciplines. Despite this substantial body of knowledge, a significant proportion of research outcomes remains inadequately translated into practical applications, limiting their contribution to economic growth, technological advancement, and societal well-being. This disconnect is particularly evident in strategic sectors such as agriculture, industry, healthcare, environmental management, and other productive and service domains, where evidence-based innovations are urgently needed.

The persistence of this research–practice gap represents not only an inefficient utilization of intellectual capital but also a considerable loss of financial resources and human effort. Scientific research demands substantial investments in funding, infrastructure, expertise, and time. However, in the absence of effective mechanisms for knowledge transfer, technology commercialization, institutional collaboration, and evidence-informed policymaking, many valuable research outputs remain confined to academic archives rather than generating measurable societal and economic impact.

This situation raises a fundamental question: Why does a substantial proportion of scientific research fail to progress beyond publication and into practical implementation? Addressing this question requires a comprehensive understanding of the multifaceted barriers that impede research translation, including institutional constraints, limited academia–industry partnerships, inadequate innovation ecosystems, regulatory challenges, insufficient funding for technology transfer, and the misalignment between research priorities and national or market needs.

This presentation provides a systematic analysis of the underlying causes and challenges that contribute to the gap between scientific knowledge generation and its practical utilization. Furthermore, it proposes an integrated framework of strategic interventions operating at the individual, institutional, industrial, and societal levels. The proposed framework emphasizes strengthening research governance, promoting interdisciplinary and cross-sector collaboration, enhancing knowledge-transfer mechanisms, supporting innovation and entrepreneurship, and aligning scientific research with sustainable development priorities and market demands.

By fostering stronger linkages between researchers, policymakers, industry, and society, these interventions aim to maximize the societal value of scientific research, improve national innovation capacity, enhance economic competitiveness, and accelerate progress towards sustainable development.

Prof. Dr. Heba Soliman (Moderaor)

Port Said University, Training manager, Egypt

PRESENTATION: The Future of Science: Training, Employability Skills, and Entrepreneurship

Biography

Prof.Dr. Heba Soliman Training Director at Port Said University. Professor of Antennas at the Faculty of Engineering. Certified Professional Trainer from the Career Development Center of the American Chamber of Commerce. Certified Professional Manager from the Institute of Certified Professional Managers at James Madison University, USA. Over twenty-five years of experience in training and teaching. Expertise in global university rankings and sustainable development in higher education.

Abstract

The future of science is being shaped not only by rapid technological advances but also by evolving workforce demands that extend beyond traditional disciplinary expertise. We will talk about the critical role of training, employability skills, and entrepreneurship in preparing scientists for diverse and dynamic career pathways. Emphasis is placed on the integration of transferable skills—such as critical thinking, communication, collaboration, digital literacy, and adaptability—into scientific education and research training. We will highlight entrepreneurship as a powerful mechanism for translating scientific knowledge into societal and economic impact, fostering innovation, and addressing global challenges. By examining emerging trends in academia, industry, and the innovation ecosystem, we will advocates for a future-oriented training model that equips scientists to thrive as researchers, innovators, and leaders. Ultimately, the talk underscores the need for holistic capacity-building approaches that align scientific excellence with employability and entrepreneurial mindsets in an increasingly interdisciplinary and innovation-driven world.

Mrs. Shaimaa ElBanna

Director of Change Leaders Scholarship, Egypt

PRESENTATION: The Hidden Costs of Women's Careers in STEM: Challenges and Pathways to Inclusion

Biography

Shaimaa ElBanna is a senior international development and education leader with over 20 years of experience in program management, strategic planning, and high-level stakeholder engagement. She has led and managed multi-million-dollar initiatives advancing education reform, workforce development, and institutional collaboration across government, international organizations, academia, and the private sector.

She currently serves as Director of the Change Leaders Scholarship at the American University in Cairo (AUC), where she oversees a flagship national program supporting talented youth from underrepresented communities. In this role, Shaimaa focuses on enhancing scholars' employability, leadership capacity, and social impact through integrated academic, professional, and community engagement pathways aligned with Egypt's development priorities.

Previously, Shaimaa was Chief of Party for the USAID Egyptian Pioneers Program at AUC, providing strategic leadership for an $86 million initiative aimed at strengthening education systems, skills development, and economic growth. She led high-level stakeholder engagement, regulatory compliance, and program governance, ensuring alignment with both national priorities and USAID objectives.

Before joining AUC, Shaimaa served as Director of Education at the British Council Egypt, managing a broad education and science portfolio encompassing transnational education, student mobility, and research partnerships. She played a key role in leading the £55 million Newton-Mosharafa Fund and acted as Country Director during 2020 and 2021, overseeing financial and operational management during a critical period.

Across her career, Shaimaa has worked extensively in policy influence, gender equity, business development, and impact assessment. Passionate about capacity building and knowledge exchange, she has contributed to research and publications and continues to shape education and development initiatives with lasting impact in Egypt and beyond.

Abstract

This research delves into the experiences of women in STEM careers, specifically focusing on the higher costs they face compared to men. Through employing a mixed methods approach, combining quantitative and qualitative methodologies, with a specific focus on incorporating in-depth focus group discussions, the study examines the personal costs resulting from role stereotyping and sex discrimination. This study draws on social career cognitive theory. It investigates the challenges women scientists encounter in balancing their scientific careers with family life and explores the strategies they employ to navigate these conflicts. The study aims to develop a comprehensive understanding of the intersectionality between women's scientific careers and their

personal and family lives. The findings shed light on women's experiences in STEM, factors that impact their retention in these fields, and propose intervention strategies for creating a more inclusive STEM workforce. The research addresses two key questions: how the unequal distribution of responsibilities and freedoms outside of the workplace contributes to the gender gap in the STEM labor force, and what mechanisms have proven successful in retaining women in STEM fields.

The main findings of the study reveals that the unequal distribution of responsibilities and societal expectations outside the workplace contribute significantly to the gender gap in the STEM labour force. Women's caregiving burdens limit their time and energy for pursuing STEM careers, reinforcing stereotypes and discouraging their full commitment. Retaining women in STEM requires robust support systems, including parental support, mentorship, and inclusive environments. Addressing gender biases and barriers to advancement is crucial for promoting women's retention in STEM.

Overall, the research reveals the impact of contextual factors, the significance of support systems, the existence of barriers, and the effectiveness of strategies in promoting women's success in STEM careers. It underscores the need to address societal expectations, biases, and gender disparities to create a more inclusive and supportive environment for women in STEM. Further research can delve into the effectiveness of these strategies and identify additional tactics to enhance success and well-being in STEM careers.

Dr. Bothaina Mahmoud Fouad

New Cairo Technological University (NCTU), Egypt

PRESENTATION: Envisioning the Future: The Importance of Entrepreneurship for Women in Science

Biography

Dr. Bothaina Mahmoud Fouad is an accomplished economist, entrepreneurship and innovation expert, and university lecturer with over 12 years of experience spanning higher education, scientific research, project management, and international collaboration. She holds a Ph.D. in Economics from Ain Shams University, specializing in entrepreneurship, innovation, international relations, technology transfer, and R&D competitiveness.

She currently serves as Lecturer of Entrepreneurship, Soft Skills, and Project Management at New Cairo Technological University (NCTU), Egypt, and as an International Lecturer at Midocean University, UAE. Throughout her academic career, Dr. Bothaina has mentored and supervised numerous student startups and entrepreneurial initiatives, preparing teams for prestigious international competitions such as the Hult Prize, Enactus, and the China–Africa Tech Challenge. She previously held the position of Acting Director of the Research Projects Department at the Electronics Research Institute (ERI), Egypt, where she led strategic research and innovation projects, developed feasibility studies, and contributed to national policy initiatives on technology transfer, innovation, and sustainable economic development.

Dr. Bothaina's professional expertise spans entrepreneurship education, innovation ecosystem development, startup incubation and acceleration, curriculum design, project management, feasibility studies, and technology commercialization. She has built international partnerships with universities, research institutions, and innovation organizations across the United Kingdom, China, South Korea, Japan, the United Arab Emirates, and beyond. She is a Certified Incubator Manager and an active member of leading international professional networks, including the Marie Curie Alumni Association and Daedeok Innopolis (Republic of Korea). In recognition of her expertise in entrepreneurship and vocational education, Dr. Bothaina was honored to represent New Cairo Technological University as an official speaker at the Seminar on Vocational Training, Innovation, and Entrepreneurship for Belt and Road Countries, held in Ningbo, China (11–25 June 2026). There, she presented Egypt's experience in technological education, innovation, entrepreneurship, and vocational training, while engaging with representatives from Belt and Road countries to foster international collaboration and knowledge exchange.

Driven by a strong commitment to empowering young people, Dr. Bothaina is passionate about bridging academia and industry, championing innovation-driven entrepreneurship, and advancing sustainable economic development through education, research, and international cooperation.

Abstract

For many women scientists, the hardest part isn't the discovery—it's what comes next. Turning years of research into a real product, company, or service that changes lives often means navigating a world that wasn't built with them in mind. This abstract looks at why supporting women's entrepreneurship in science matters so deeply, not just for fairness, but for the future of innovation itself. Too often, women scientists hit invisible walls: investors who overlook them, networks that exclude them, and biases that make commercializing their work an uphill climb. When we remove these barriers, something powerful happens—research turns into real-world solutions, whether that's a healthcare breakthrough, a clean energy innovation, or a new agricultural technology that helps farmers thrive. What's remarkable about women-led ventures is their ripple effect. They tend to give back to their communities, hire more inclusively, and open doors for the next generation of women coming up behind them. And when more diverse voices lead innovation, the solutions we create become richer, more thoughtful, and more responsive to the people they're meant to serve. That's why this work pushes for real investment—in incubators, mentorship, and funding designed with women in mind—alongside policies that finally level the playing field. Because when women have the tools, connections, and confidence to build their own ventures, everyone benefits. At its heart, this is about imagining a future where science truly knows no borders—and where every woman scientist has the chance to shape what comes next.

Ms.Gulzar Karybekova

Centre of Actual Problems and Development, Director, Russia

PRESENTATION: Imagining What is Possible for the Future; Developing Evidence-Based Governance and Innovation Through Women's Leadership

Biography

Ms. Gulzar Karybekova is the Director of the Centre of Actual Problems and Development, a public fund based in Kyrgyzstan. She earned a degree in International Relations from the Diplomatic Academy of the Ministry of Foreign Affairs of the Russian Federation in 2011 and a Bachelor's degree in Law from Kyrgyz State National University in 2001, completing an additional year for a 5-year Specialist diploma. In 2013, she received a diploma from the Inter-Parliamentary Assembly of the Commonwealth of Independent States (Saint Petersburg, Russia). She has several scientific publications and has participated as a speaker and moderator in international conferences, workshops, seminars, and forums across Asia-Pacific and Europe.

Abstract

The Kyrgyz Republic's public sector is challenged by the effects of an economic transition, increasing regional differences, and sustainable development. In order to create sustainable, transparent, and citizen-oriented institutions, it is necessary to integrate evidence-based governance with innovative management approaches and solutions. Young female scientists are leading the change and using political science, management, and technology to innovate. This abstract describes the leadership roles that young female researchers have in developing innovations for the Kyrgyz Republic's public sector. Through their work, they are expanding digital access to municipal services, using data-driven solutions for implementing policy, and creating participatory engagement opportunities for citizens to be involved in decision-making. The use of science and research-based decision-making is helping young female researchers to improve efficiency and create gender equality in the public sector.

Collaboration at the international level strengthens the impact of the work of young female researchers. Organizations such as Women in Science Without Borders (WISWB) support these initiatives by providing mentorship, global networks, and platforms to create solutions together with peers in other countries.

Empowering women scientists to assume leadership roles in evidence-based governance encourages entrepreneurial thinking within public administration and supports the creation of innovative solutions to social, economic, and environmental issues. By supporting the next generation of women leaders in the Kyrgyz Republic, we can create a public sector that develops, participates in, and co-creates sustainable and equitable policies through innovation, evidence-based decision-making, and collaboration with the global community.

Biotechnologist Natalia Montellano Duran

Universidad Católica Boliviana San Pablo, Organization for Women in Science in Deceloping World, Director, Bolivia

PRESENTATION: Gender-neutral Scientific Leadership in Bolivia: Evidence-Based Insights on Women's Participation Across the STEM Education and Research Pipeline

Biography

Natalia Montellano Duran, PhD, is a Bolivian biotechnologist and academic leader dedicated to strengthening STEM education, scientific capacity, and gender equity in science. She is currently the Director of the Biotechnology

Engineering Program at the Universidad Católica Boliviana “San Pablo” (UCB), Santa Cruz, where she leads academic development, curriculum innovation, and research-based training aligned with national and regional challenges.

Her career in STEM integrates biological sciences, applied research, and higher education leadership. Through teaching, laboratory work, and supervision of undergraduate research projects, she has actively contributed to building scientific skills, critical thinking, and practical problem-solving in contexts where scientific infrastructure and funding remain limited. Her research output includes peer-reviewed scientific publications across multiple areas of STEM, reflecting an interdisciplinary trajectory that bridges biotechnology, molecular and biological sciences, bioactive compounds, soft matter, and STEM education.

A defining pillar of her trajectory is her commitment to the empowerment and visibility of women in STEM. She served as Chair of the Organization for Women in Science for the Developing World (OWSD) – Bolivia National Chapter (2021–2025), leading the consolidation and expansion of a national network of women scientists and strengthening collaboration with universities, policymakers, and civil society.

Her scientific and academic contributions have been recognized through multiple awards, competitive fellowships, and international distinctions that reflect both her research excellence and leadership in science and education. She was selected as a member of the Global Young Academy (GYA) in 2025, and she is also affiliated with the TWAS Young Affiliates Network (TWAS-TYAN)."

Abstract

Reliable, gender-disaggregated data are essential for strengthening scientific systems in countries where structural inequalities limit the full development of human capital. This study presents the first coordinated national mapping of women’s participation in science, technology, engineering, and mathematics (STEM) in Bolivia, conducted by the IDRC, OWSD-UNESCO and the Bolivian National Chapter between 2024 and 2025. Using a mixed-methods approach, the project combined quantitative surveys (n = 1,962) across three key levels; secondary students, university STEM students (including dropouts), and STEM professionals; with qualitative interviews, focus groups, and institutional and policy mapping.

Results show that while women achieve parity in overall tertiary education enrollment and exceed men in graduation rates, they remain significantly underrepresented in STEM fields, comprising only 34.2% of STEM students and graduates. At the same time, women represent a majority of researchers nationally (52.6%), revealing a critical disconnect between educational pathways and scientific workforce participation. Persistent data gaps, limited mentorship, cultural stereotypes, uneven access to resources, and institutional barriers; particularly outside major urban centers; continue to constrain women’s entry, retention, and leadership in STEM.

By systematically quantifying where women are present, absent, or lost along the STEM pipeline, this study demonstrates that “counting women in STEM” is not merely descriptive, but a strategic tool for evidence-based policy, institutional reform, and targeted intervention. In a country like Bolivia (low income and multicultural), where scientific capacity remains underdeveloped, understanding gender participation patterns is fundamental to improving STEM enrollment, strengthening research ecosystems, and advancing inclusive and sustainable national scientific development."

Anthropologist Ralitsa Savova

Member of the Editorial Committee of Cultural route "Longobard Ways across Europe, Italy",

PRESENTATION: Is Africa really the poorest continent on Earth?

Biography

Ralitsa Savova is an anthropologist whose main research interests are in the field of migration, labour mobility, cultural heritage, cultural heritage in migration, cultural and scientific diplomacy, international relations, and African studies. She is a PhD Candidate in International Relations, Interdisciplinary Doctoral School, University of Pécs, Hungary; has a Master's degree in the UNESCO Chair in Cultural Heritage Management from the Corvinus University of Budapest, Hungary; a Master's degree in Marketing from the Economic Institute of the Bulgarian Academy of Sciences, Sofia, Bulgaria. She is a Post-graduate in Marketing and Management, the Open University, the Open Business School, MK, England; holds a diploma in Tourism and Travel Agency Management from Cambridge International College, Jersey, Britain; and a Pedagogical Competences diploma from Teacher's Training Department in Varna, Bulgaria. Ralitsa Savova is an Official Representative for Central and Eastern Europe of the European cultural route "Longobard Ways across Europe", Cividale del Friuli, Italy; Member of the International Scientific Committee "Amelio Tagliaferri", Brescia, Italy; Member of the Editorial Committee of the Online journal for History and Historiography "Storia e Futuro - Rivista di Storia e Storiografia", Università di Bologna, Bologna, Italy; Member of the European Network on Regional Labour Market Monitoring (EN RLMM); Member of the European Consortium for Political Research (ECPR), United Kingdom (UK); former Guest-researcher at the Institute for Political Science, Centre for Social Sciences, Hungarian Academy of Sciences. Ralitsa Savova is an editor, co-editor, co-author, or author of 19 books, and 16 scientific journal articles, published in 9 countries: Italy (in Rome, and Pisa); Germany (in Baden-Baden); The Netherlands (in Amsterdam); Estonia (in Tartu); Greece (in Ioannina); Hungary (in Budapest, and Debrecen); Romania (in Iasi); Russia (in Moscow, and St. Petersburg); Bulgaria (in Sofia).

Abstract

Africa is often called "the poorest continent on Earth." For some of those who claim this, it is an assumption, for others it is a prejudice, and for still others it is an association with the problems and challenges faced by the African people. These challenges include unemployment, hunger, poverty, disease, military conflicts, forced migration, migration due to climate change and natural disasters, etc. Despite the challenges, however, Africa is not a poor continent. It is rich in area and population – it is the second largest and most populous continent in the world after Asia. The African population is growing by 80 people every minute and between 2015 and 2050, over half of the world's population growth will take place in Africa. Further, the African continent is young in terms of the age of its inhabitants – their average age is 18.8 years and is therefore a continent rich in human resources especially young. Also, Africa is a very rich continent in natural resources and is home to a large part of the planet's mineral resources. Finally, the continent is the fastest growing economy in the world after Asia. Despite Africa's vast resources, the aforementioned challenges exist, including a large portion of the African population living in poverty, and it is the task and cause not only of Africa itself, but also of

organizations from other continents, as well as global organizations, to help the people of Africa, and for organizations that are already doing so to continue this work, so that Africa can develop its natural and human resources, so that the population can live better and Africa can be and be perceived not as a poor, but as a rich continent, which it actually is. (My lecture is based on a chapter from a book titled "Is Africa Really the Poorest Continent on Earth?", which was published in 2025 by one of the European Institutes for African Studies).

Online

---Session 1 Online---

Panel 1:Genaral Panel Discussion

The session scenario will be as the following:

-What kind of future do we want to create—and how can science, technology, innovation, and entrepreneurship help us achieve it?

This inspiring panel brings together accomplished women leaders from different fields and backgrounds to share their experiences, ideas, and visions for the future. Through an open and engaging conversation, the speakers will explore how scientific knowledge can be transformed into innovative solutions, how technology can create new opportunities, and how entrepreneurship can turn bold ideas into meaningful impact. The discussion will also highlight the important role of women as scientists, innovators, entrepreneurs, decision-makers, and leaders. By sharing lessons from their professional journeys, the panelists will encourage greater collaboration, creativity, and inclusion while inspiring the next generation to think beyond traditional boundaries and contribute to a more sustainable and prosperous world.

Eng. Hoda Mansour (Moderator)

President Egypt- AngloGold Ashanti, Board Member-Sukari Gold Mines

Engineer Hoda Mansour is a highly experienced, creative, and self-motivated executive with a proven track record of delivering innovation and sustained business value across multiple industries and geographies. In March 2025, she was appointed Managing Director, General Manager and Vice Chair of Sukari Gold Mines, a Tier-1 gold mine, representing AngloGold Ashanti, the world's fourth-largest gold producer headquartered in Denver, USA. Her appointment marks her as the first women to hold such role in the Middle East and North Africa. Prior to this, Eng. Mansour served on the Board of Directors of Centamin PLC, the former operator of Sukari Gold Mines, which was listed on the London and Toronto Stock Exchanges and acquired by AngloGold Ashanti in November 2024. During her tenure, she was a member of Centamin's Audit & Risk Committee and Sustainability Committee. In May 2026, she got promoted to President AngloGold Ashanti Egypt.

Ms. Mansour joined AngloGold Ashanti from IFS, a global provider of industrial AI and enterprise software solutions, where she served as Chief Operating Officer for Asia Pacific, Japan, the Middle East, and Africa. Earlier in her career, she held senior regional and global leadership roles at SAP, Oracle, and Microsoft, living and working across three continents; an experience that has shaped her deep understanding of diverse cultures, business models, and complex markets. Ms. Mansour spent nearly 11 years at SAP, where she led Business Process Transformation for Southern Europe, the Middle East, and Africa (EMEA South), covering 75 countries. Eng. Mansour was also a member of SAP's Global Executive Leadership Team and is a frequent speaker at global and regional forums. During this period, she was:

- The first woman appointed Managing Director at SAP across the Middle East and Africa
- The first Chief Operating Officer for SAP UAE and Oman
- The first woman to lead a multinational software company in Egypt

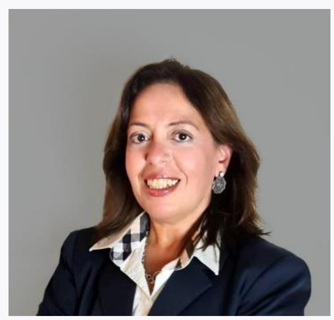
Recognizing her impact on the business, Forbes Middle East named her one of 100 Most Powerful Businesswomen since 2018 till date and one of the Top 5 Women in Technology in 2022. In October 2025, she became the only woman appointed as a Non-Executive Board Member of the Banking Reform and Development Fund, chaired by H.E. the Central Bank Governor. In November 2024, she was appointed by Presidential Decree as a board member of Egypt's National Council for Women. Hoda is also serving as a Board Member of Commercial International Bank (CIB) since March 2023, where she is Chair of the Sustainability Committee and a member of the Board Audit Committee and Strategy & Technology Committees. She was also a member of AmCham Egypt's Board of Governors from June 2021 to May 2025 and previously served as Vice President and Board Member of the German Chamber of Commerce.

Eng. Mansour holds a B.Sc. in Engineering with Distinction and Honors from Alexandria University, and an MBA with Distinction with a focus on Information Technology & Management from Maastricht School of Management.

Eng. Hanan Abdel Meguid

Entrepreneur and leading figure in technology and innovation

Hanan Abdel Meguid is a pioneering entrepreneur and one of the Middle leading figures in technology and innovation, with over 25 years of experience driving digital transformation and startup growth. She has held executive leadership positions at LINK Development and Orascom Telecom Ventures and later served as Vice President of Innovation at The American University in Cairo. She also served as Head of Audio Innovation & Partner at Iqraaly, the first Arabic audiobook platform, and continues to mentor entrepreneurs through leading incubators including Flat6Labs, Tahrir2, and Google's Ebda2 initiative, helping shape the next generation of innovators in Egypt and the region. In 2014, she founded Kamelizer, an angel investment studio dedicated to accelerating Egypt's startup ecosystem. She later launched Consoleya in 2020, a dynamic business and coworking hub, followed by Kamelizer Spaces in 2023, an innovative hybrid workspace that seamlessly combines hospitality with flexible offices and coworking spaces.



Prof. Dr. Samhaa el Beltagy

Chief AI Officer at Nawy

Samhaa R. El-Beltagy is the Chief AI Officer at Nawy, where she leads the company's AI strategy and innovation agenda, driving advanced AI solutions that transform real estate technology and customer experience. She is also a Professor of Computer Science and the Founding Dean of the School of Information Technology at Newgiza University (NGU). Her career spans academia, industry, and entrepreneurship, with a consistent focus on bridging research and practice: she co-founded AIM Technologies, a Middle East–based startup specializing in NLP and advanced analytics, and previously served as Head of Research and Development at Optomatica, a company focused on AI solutions for real-world complex problems. Dr. El-Beltagy has played a pivotal role in national AI initiatives and is a member of the technical committee of the Egyptian National Council for AI, as well as the advisory committee of the ACL Special Interest Group on Arabic NLP (SIGARAB). She was named one of the Top 30 AI Leaders in the Middle East by MIT Technology Review (2022) and has been featured in Stanford University's global list of the top 2% of scientists (2019, 2021). She has authored over 90 international publications and served in senior roles at leading conferences, including Tutorial Chair at EMNLP 2022, Program Chair at ArabicNLP 2026(co-located with EMNLP 2026) and Program Chair at ArabicNLP 2023 (co-located with EMNLP 2023). She is a long-standing jury member for the L'Oréal-

UNESCO For Women in Science program, and co-founder and organizer of the International ACLing Conference Series (initiated in 2015, published by Elsevier). Her research interests include Artificial Intelligence, Natural Language Processing, Large Language Models, and Arabic NLP.

Eng. Dawlat Hashem

Strategic Energy Consultant at Mideast Communication Systems (MCS)

Dawlat Hashem is a Strategic Energy Consultant at Mideast Communication Systems (MCS), former Vice Chairman of EGAS, and former Board Member of Eagle Gas Shipping Company. With over 30 years of experience in digital transformation and information technology, particularly in the oil and gas sector, she has led major ICT, cybersecurity, and digital innovation initiatives that have contributed to the modernization of Egypt's energy industry. She is Country Advisory Council member for G100 Youth Engagement. A recognized leader and international speaker, Dawlat is passionate about driving innovation, advancing sustainability, and empowering women in leadership. She actively supports initiatives that foster inclusion, inspire future generations, and create meaningful social impact.

Key Areas of Discussion

How science and technology can shape a better future

Turning scientific knowledge and creative ideas into practical solutions

The opportunities and challenges created by artificial intelligence and emerging technologies

Women's leadership in innovation, technology, and entrepreneurship

Building stronger and more inclusive innovation ecosystems

Supporting women-led startups through investment, mentorship, and collaboration

Creating partnerships that connect local knowledge with global opportunities

Inspiring young women to become future scientists, innovators, and entrepreneurs

Panel 2: Science diplomacy-Policy, management, Industry perspective

Prof. Dr. Paul Arthur Berkman

Founder of the Science Diplomacy Center Inc. (SDCI)- USA

PRESENTATION: Informed Decision-making for Lasting Peace

Biography

Professor Paul Arthur Berkman is a Fellow of the International Science Council and Fulbright Specialist 2024-2027 with the United States Department of State. He founded the Science Diplomacy Center, Inc. (SDCI) nonprofit in the United States in 2022. He is a Faculty Associate with the Program on Negotiation at Harvard Law School; and a Distinguished Visiting Professor with the International Institute of Science Diplomacy and Sustainability (IISDS) at UCSI University in Malaysia. Paul wintered in Antarctica, where he set a Guinness World Record on a SCUBA research expedition in 1981, becoming a Visiting Professor at the University of California Los Angeles the following year at the age of 23, when he began his global journey as a science diplomat to balance national interests and common interests for the benefit of all on Earth across generations.

Abstract

The challenges of our world operate across generations, which means we must learn to address a 'continuum of urgencies' from security to sustainability time scales. This brief intervention will ponder how to enhance informed decisionmaking with science diplomacy for the benefit of all on Earth across generations, building common interests with global inclusion to balance national interests as a choice

Dr Angela Liberatore

Former Head of the Scientific Department at the European Research Council

PRESENTATION: Science as global public good to safeguard other global public goods: the role of science diplomacy.

Biography

Angela Liberatore is pro bono Visiting Fellow on Science Diplomacy at the European University Institute.

Previously she was Head of the Scientific Department at the European Research Council (ERC).

Earlier on she worked in various roles in the European Commission's Directorate General for Research and Innovation, in the research programmes on environment and climate, on social sciences and humanities and then international cooperation.

Among her professional 'highlights', she was one of the authors of the European Framework on Science Diplomacy, co-chaired the Social Protection group for the UN Research Roadmap for COVID-19 Recovery, participated in the making of EC White Paper on European Governance (with focus on democratising expertise) and in the negotiation of the Kyoto Protocol on Climate Change.

Angela holds a PhD in Political and Social Sciences (European University Institute) and a degree in Philosophy (University of Bologna). She is also an alumna of the Kennedy School of Government -Harvard University that she visited with a Fulbright Fellowship. Among her publications, the book 'The Management of Uncertainty. Learning from Chernobyl.

Abstract

In a world where geopolitical tensions strongly impinge on international scientific collaboration and where some scientific information has significant commercial value, the notion of science and practice of it as a global public good is under pressure and so are the ways to pursue it.

Science diplomacy acknowledges that science impacts diplomacy and diplomacy impacts science, and that both cooperation and competition shape the diverse practices of science diplomacy: from science fostering international relations and serving global public goods to foreign and security policy interests conflicting with the notion of science as an open collaborative enterprise.

Three challenges and opportunities for safeguarding science as a global public good by science diplomacy seem especially worth considering. They relate to strategy, data and people.

Strategy: some key issues and 'balancing' need to be addressed. How can research security be tackled and academic freedom be preserved? Are scientific sanctions a good response to conflicts? Is technological sovereignty compatible with or a threat to the notion of science (closely interlinked with technology) as a global public good?

Data: scientists have been developing and relying on large datasets to address the causes and impacts of climate change, pandemics etc. These are at risk due to both government deletion policies and private companies' appropriation processes. How can science as a global public good tackle this challenge?

People: science is done by people (even in an increasingly 'AI world'). Researchers –and teachers, students, universities and scientific institutions – have been increasingly affected by violent conflicts. Academic freedom is being restricted in many countries. What are the implications for science as a global public good?

Dr Amal Kasry

Chief of Basic Sciences, Engineering, Technology & Innovation (BETI) Sciences (SCS) Sector, UNESCO

PRESENTATION: Connecting Science Across Borders: Access, Capacity and Shared Action

Biography

Dr. Amal Kasry is the Chief of the Basic Sciences, Engineering, Technology and Innovation (BETI) Section in UNESCO's Sciences Sector, where she leads the development and implementation of UNESCO's global programmes in basic sciences, engineering, technology and innovation. Her work focuses on strengthening scientific capacities, advancing science policy, fostering international scientific cooperation, and promoting equitable access to science and emerging technologies in support of sustainable development.

She currently leads several flagship UNESCO initiatives, including the implementation of the International Decade of Sciences for Sustainable Development (2024–2033), proclaimed by the United Nations General Assembly, and the UNESCO Global Quantum Initiative (GQI), which aims to promote inclusive global cooperation and capacity development in quantum science and technology. She also leads UNESCO's work on expanding equitable access to advanced scientific infrastructure, including remote access to research equipment, helping strengthen scientific capabilities in developing countries. Through these initiatives, she works closely with governments, international organizations, scientific institutions, industry and funding partners to build strategic partnerships and mobilize collective action for science.

Before joining UNESCO, Dr. Kasry built an international career spanning academia, industry and research. She was awarded a fellowship at the Max Planck Institute for Polymer Research (Germany) and received her PhD in Materials Science and Biophysics from Johannes Gutenberg University Mainz in 2006, with research focused on biosensing technologies. She subsequently held research and academic positions at the University of Connecticut Health Center (USA), Cardiff University (United Kingdom) and the Austrian Institute of Technology (Austria), and worked in industrial research at IBM (USA) and Nitto Denko (Singapore).

In 2015, she joined the British University in Egypt as Assistant Professor in the Faculty of Engineering before being appointed Director of the Nanotechnology Research Centre (NTRC). There, she established the university's research services and innovation support system while leading research on biosensing technologies and carbon nanomaterials.

Dr. Kasry is the author of approximately 40 peer-reviewed scientific publications, seven book chapters, and four patents. She serves as a reviewer for leading scientific journals and research funding agencies and has extensive experience in building international scientific partnerships, science diplomacy, research capacity development, and translating scientific excellence into global policy and development impact.

Abstract

Today's scientific challenges are increasingly global, while access to knowledge, infrastructure and emerging technologies remains deeply unequal. Addressing this gap requires more than individual projects. It requires sustained cooperation that connects scientists, institutions and countries around shared objectives.

This talk will highlight how UNESCO is fostering such connections through several major initiatives. The International Decade of Sciences for Sustainable Development (2024–2033) provides a global framework for bringing together disciplines, countries and sectors to address common challenges. The Global Quantum Initiative seeks to build inclusive international cooperation around emerging technology before existing divides become more deeply entrenched. UNESCO's work on remote access to scientific equipment shows how the sharing of facilities, expertise and training can create practical opportunities for researchers who would otherwise remain excluded from advanced scientific infrastructure.

Together, these initiatives demonstrate how shared scientific action can strengthen capacities, widen participation and build lasting relationships across borders. By enabling countries not only to access knowledge but also to produce and apply their own knowledge, such cooperation can help create trust, mutual understanding and more resilient partnerships in an increasingly fragmented world.

Dr Natisha Dukhi

a Senior Research Specialist at the Human Sciences Research Council-GYA cochair

PRESENTATION: Science Diplomacy from the Global South: Bridging Evidence, Youth Voice, and Policy

Biography

Dr Natisha Dukhi is a Senior Research Specialist at the Human Sciences Research Council and PhD-qualified public health scientist with over a decade of experience across Africa and globally. Her expertise spans nutrition, child and adolescent health, non-communicable diseases, and science diplomacy. A Principal Investigator and prolific published researcher, she informs public health policy in underserved communities. She serves on the South African Young Academy of Science, Co-Chair: Global Young Academy, and the International Science Council's Global Roster of Experts. Dr Dukhi is a recipient of the prestigious Gro Brundtland Award for outstanding contributions to Public Health and Sustainable Development.

Abstract

As Co-Chair (LMIC) of the Global Young Academy, I bring together the perspectives of a public health researcher and an early-career advocate for LMIC representation in global science governance. This talk explores what science diplomacy looks like when practiced from the Global South, under resource constraints and unequal access to global platforms. I reflect on how young and early-career researchers can shape science-policy dialogue, strengthen South-South collaboration, and ensure that science diplomacy serves not only geopolitical interests but also local health and development priorities.

Mrs. Rebone Seleoane

Head of P&C Africa MCO

PRESENTATION: Supporting women in Sanofi beyond boundaries

Biography

Mrs Rebone Seleoane is the Head of P&C Africa MCO within Sanofi's P&C International department. She leads pharmaceutical and consumer health operations across the Africa Market Cluster Organization, driving strategic initiatives and operational excellence in the region.

Abstract

In a world where talent knows no borders, Sanofi is committed to creating an inclusive environment where women can thrive at every stage of their careers. Join Rebone Seleoane, Head of People & Culture, as she shares Sanofi's transformative approach to supporting women across our global organization. This session will explore innovative programs, leadership development initiatives, and cultural shifts that break down barriers and unlock potential.

Mrs. Alice Berson

Chief Finance Officer

PRESENTATION: Empowering women through sustainable procurement

Biography

Alice Berson, a Dean's List graduate with a Master in Management from IESEG School of Management in Paris, brings nine years of progressive finance and internal audit experience in the pharmaceutical industry to her current role as Chief Financial Officer for Africa MCO and Finance Business Partner for Pharma at Sanofi, where she leverages her expertise in financial strategy, business partnering, and operational excellence to ensure fiscal discipline and drive sustainable growth across the African region in support of Sanofi's mission to chase the miracles of science.

Alice Berson is the CFO for Africa MCO and Finance Business Partner for Pharma within the CFO International Markets department. She provides strategic financial leadership and business partnership support for Sanofi's pharmaceutical operations across the Africa Market Cluster Organization.

Abstract

Sustainable procurement is more than a business strategy—it's a catalyst for social impact and gender equity. Alice Berson, Chief Financial Officer at Sanofi, will illuminate how strategic procurement decisions can create meaningful opportunities for women entrepreneurs, suppliers, and communities worldwide. This session explores the intersection of financial stewardship, sustainability, and social responsibility, demonstrating how Sanofi leverages its supply chain to drive inclusive economic growth

Dr. Ibtissame Azzaoui

President and Founder of Parliamentarians for Climate (P4C)

PRESENTATION: Parliamentary Science Diplomacy: Bridging Evidence, Policy and International Cooperation

Biography

Dr. Ibtissame Azzaoui is the President and Founder of Parliamentarians for Climate (P4C), an independent global network bringing together more than 160 current and former parliamentarians from over 43 countries to advance international cooperation on climate and sustainable development.

She is a former Member of the Moroccan Parliament (2016–2021), where she served on the Committee on Foreign Affairs, National Defence, Islamic Affairs and Moroccan Expatriates, actively contributing to parliamentary diplomacy and international engagement. She currently serves as an elected City Councillor in Rabat, having won one of the most competitive constituencies in Morocco.

Dr. Azzaoui holds a PhD in Economics and Management from Mohammed V University, where her doctoral thesis, "Moroccan Parliamentary Diplomacy in Africa: A New Approach to Strategic Governance," was awarded the National Prize for the Best Parliamentary Thesis in Morocco. She is also an IT Engineer from the Hassania School of Public Works (EHTP) and holds a Specialized Master's Degree in Information Systems Management from École Centrale Paris, France.

She has over 15 years of international professional experience and serves as an international consultant. Her work lies at the intersection of governance, science, technology and public policy, with a particular focus on evidence-informed policymaking, parliamentary diplomacy, digital transformation and international cooperation. She speaks Arabic, French, and English fluently.

Abstract

My work has focused on strengthening the interface between science, public policy, and international cooperation through parliamentary diplomacy and evidence-informed governance. As a former Member of the Moroccan Parliament, President of Parliamentarians for Climate (P4C)—a global network of more than 160 current and former parliamentarians from over 43 countries—and an international policy expert, I have worked to promote the integration of scientific evidence into legislative processes and public decision-making, particularly in the areas of climate governance, digital transformation, artificial intelligence, and sustainable development.

My academic background has also shaped this commitment. My PhD thesis, "Moroccan Parliamentary Diplomacy in Africa: A New Approach to Strategic Governance," was awarded the National Prize for the Best Parliamentary Thesis in Morocco, recognizing research that bridges academic knowledge and policymaking. Throughout my career, I have authored numerous scientific publications, policy papers, policy briefs, and

opinion articles that translate research into practical recommendations for governments, parliaments, and international organizations.

Through international parliamentary networks, multistakeholder dialogue, and policy advocacy, I have sought to strengthen collaboration between scientists, policymakers, and diplomatic actors. My work demonstrates how parliamentarians can serve as effective science diplomats by fostering international cooperation, promoting evidence-based legislation, and building governance frameworks capable of addressing complex global challenges. I believe that science diplomacy is most effective when it is institutionalized through democratic processes, supported by strong legislative institutions, and grounded in collaboration across disciplines, sectors and borders.

Panel 3: Science diplomacy-networking, cultural aspects and Cooperation across borders

Prof .Dr. Mohd Basyaruddin Abdul Rahman

Chair of the ISTIC-UNESCO Governing Board

PRESENTATION: Diplomacy New Currency of Science Diplomacy Empowering the Global South through STI Partnerships

Biography

Mohd Basyaruddin is a Senior Professor (VK5) of Chemistry at Universiti Putra Malaysia and was appointed Distinguished Visiting Scholar at the University of California, Berkeley. He holds a PhD in Catalysis Chemistry from the University of Southampton and completed a postdoctoral fellowship in Structural Biology at the University of Edinburgh. His research covers biocatalysis, chemical biophysics, computational chemistry, and nanodelivery, with recent work focusing on reticular materials for sustainability, particularly their applications in biomedical and agricultural innovation. He has published more than 350 papers, 400 proceedings, and 30 intellectual properties. He is actively involved in engaging young talents in STEM and science policy development. He is a Fellow of the Academy of Sciences Malaysia and the World Academy of Sciences. Currently, he is the Chair of the ISTIC-UNESCO Governing Board. These appointments have enabled him to play a more significant role in STI capacity building, particularly in science advocacy, diplomacy, and policy in the Global South, and to bridge the North-South gap.

Abstract

Science diplomacy is becoming an essential mechanism for addressing global challenges such as climate change, health security, food sustainability, and digital transformation. In today's evolving geopolitical landscape, the new currency of science diplomacy is no longer scientific excellence alone, but trust, equitable partnerships, knowledge co-creation, and shared leadership. These principles enable the Global South to become an equal partner in shaping global science, technology, and innovation (STI) agendas. Strategic STI partnerships strengthen scientific capacity, accelerate innovation, support evidence-informed policymaking, and advance the Sustainable Development Goals (SDGs). South–South and triangular cooperation provide effective pathways for building resilient research ecosystems and delivering solutions with local relevance and global impact. The International Science Technology and Innovation Centre for South–South Cooperation under the auspices of UNESCO (ISTIC–UNESCO) serves as a trusted platform for science diplomacy by connecting governments, academia, industry, and international organisations. Through capacity building, science policy dialogue, leadership development, and collaborative networks, ISTIC–UNESCO empowers the Global South to foster sustainable partnerships and contribute to a more inclusive, resilient, and equitable global STI ecosystem.

Prof.Dr.Yoko Shimpuku,

Professor of Global Health Nursing at Hiroshima University, Japan

PRESENTATION: Science Diplomacy for Distributive Justice: Lessons from the World Forum for Women in Science – Japan

Biography

Yoko Shimpuku, PhD, CM, RN, is a midwife and Professor of Global Health Nursing at Hiroshima University, Japan. She serves as a member of the WHO Midwifery Expert Group and the ICM Regional Professional Committee for the Western Pacific. Her research focuses on maternal and newborn health, midwifery, digital health (mHealth), and global health, with collaborative projects across Asia and Africa. She has chaired the World Forum for Women in Science – Japan and is committed to strengthening midwifery education, advancing women-centered care, and improving maternal and newborn health through international research and partnerships.

Abstract

Science diplomacy plays an increasingly important role in addressing global challenges such as climate change, pandemics, conflict, and health inequities. However, international scientific collaboration must also address distributive justice by ensuring that knowledge, resources, opportunities, and leadership are shared equitably across countries and communities.

This keynote reflects on lessons learned from the World Forum for Women in Science – Japan, where researchers, policymakers, innovators, and students from diverse backgrounds came together to build new partnerships. The forum highlighted that effective science diplomacy is founded on trust, mutual respect, shared leadership, and co-creation, rather than one-way knowledge transfer.

Drawing on experiences in global health, midwifery, and digital health, this presentation discusses how distributive justice can guide more equitable international collaboration. It argues that science diplomacy should not only advance scientific excellence but also promote inclusion, peace, and sustainable development by empowering diverse voices and fostering long-term partnerships.

Mrs Alexandra Kinias

Women of Egypt

PRESENTATION: Egyptian Women: The Power of Purposeful Networking

Biography

Alexandra Kinias holds a BSc. in Mechanical Engineering. After moving to the U.S., she became a screenwriter, novelist, travel advisor, and freelance market research consultant on Egyptian women's issues. A women's rights advocate and public speaker, her work appeared in various outlets in MENA and the U.S.

Her movie Cairo Exit won the Best Non-European Film award at the Independent European Film Festival.

Kinias founded WoEgypt Initiative to empower Egyptian women, highlight key issues to drive positive change.

Celebrating her 60th Birthday, she summited Mountain Kilimanjaro, to challenge age stereotypes.

Abstract

How does a simple idea turn into a movement? In this 10-minute talk, I will share the story behind Women of Egypt's inception—one of the region's fastest-growing networks—and show how purposeful networking creates real social impact.

I will discuss how we built a community of over one million followers by prioritizing collaboration over competition, and connecting women across industries, generations, and backgrounds.

The talk will highlight the key principles behind our growth: authentic storytelling, strategic partnerships, and digital engagement.

Through real stories from the network, I will show how a single meaningful connection can spark entrepreneurship, mentorship, and economic growth, creating a ripple effect that extends far beyond the individuals involved.

Ms Paloma Marín-Arraiza

Director of Engagement at ORCID

PRESENTATION: Data-driven decisions based on ORCID trust markers

Biography

Paloma Marín-Arraiza holds a degree in Physics and a master's in Scientific Information and Communication, both from the University of Granada (Spain). She obtained her PhD in Information Science at São Paulo State University (Brazil), with a thesis on data modelling for enhanced publications. Before joining ORCID, she worked in research libraries in Germany and Austria, dealing with non-textual research outputs and persistent identifiers, respectively. Since March 2020, she has been part of the ORCID team, where she is currently the Director of

Engagement. Apart from that, she is a guest lecturer on courses about Research Data Management and a member of the editorial board and reviewer committee of four diamond open-access journals in the field of information science.

Abstract

In an era of rising research integrity challenges, scholarly stakeholders need reliable mechanisms to verify researcher identity and contributions. This presentation explores how organisations can make high-fidelity, data-driven workflow decisions using ORCID "trust markers".

At the core of this approach is ORCID's distributed trust model, which balances researcher autonomy with rigorous data provenance. While researchers retain complete control over their profiles, the trust model relies on authenticated member organisations—such as universities, publishers, and funders—to assert metadata directly into a record.

These organizationally validated assertions (e.g., peer reviews, funding awards, and verified institutional email domains) act as visible trust markers. By analysing these digital signals, editorial and funding systems can visualise an individual's "scholarly bona fides" at a glance. Attendees will learn how to leverage these automated provenance indicators to reduce manual vetting, mitigate misrepresentation, and confidently accelerate screening and submission workflows.

Dr Nerina Finetto

Nerina Finetto Founder and CEO Traces&Dreams AB

PRESENTATION: "Not an Accessory: Why Communication Must Be Foundational to Science"

Biography

Dott. Nerina Finetto is a visionary strategist, narrative expert, and interdisciplinary connector dedicated to shaping sustainable and equitable futures through the power of knowledge, culture, and collective transformation.

She is the Founder and CEO of Traces&Dreams, where she developed the Futures Narrative Framework to drive cultural change and strategic foresight, and the Managing Director of Science 4 Humanity Global Society, bridging science, policy, business, and civil society to advance global inclusion.

Rooted in a background spanning gender history, journalism, and work as an award-winning documentary filmmaker, Nerina transforms complex ideas into shared narratives that inspire collective action. Deeply committed to legacy and future generations, she creates spaces where science, imagination, and human-centered values intersect to build a wiser tomorrow.

Abstract

We are drowning in knowledge and starving for wisdom. Information and disinformation now occupy the same feeds, the same platforms, the same fraction of a second's attention — and science has to find its voice inside that noise, not above it. This session is a reflection on what it means to treat communication not as something added to science once the work is done, but as part of science itself: part of how a question is framed, how evidence is tested, how understanding is shared and, sometimes, lost along the way. What does it mean to communicate responsibly in a world flooded with data but short on discernment? What gets harder, and what becomes possible, once we stop treating communication as separate from the practice of science itself?

Prof. Dr. Roberta D'Alessandro

Netherlands- Head of the Linguistics Division at Utrecht University

PRESENTATION: Your Academic Auntie: mentoring and supporting early career researchers globally

Biography

Roberta D'Alessandro is Chair Professor of Linguistics — Syntax and Language Variation and Head of the Linguistics Division at Utrecht University. She studies theoretical syntax, with a focus on how language contact shapes grammar, particularly in minority and heritage languages.

Roberta is also active in science governance and research policy. She co-authored the Dutch Code of Conduct for Research Integrity, advises the European Commission through JRC initiatives, and serve on the ISC Committee for Freedom and Responsibility in Science. She is Editor-in-Chief of Isogloss, a diamond open-access journal for Romance linguistics with broad participation of scholars from the global south, and is series co-editor of Language Science Press's Open Generative Syntax.

D'Alessandro founded Your Academic Auntie, a free mentoring initiative for early-career researchers worldwide, complemented by a Microfund for researchers in low- and middle-income countries.

Abstract

The Your Academic Auntie initiative is a voluntary mentoring effort designed to support early career researchers (ECRs) as they navigate the initial stages of their academic careers. Over the past year, ECRs have been able to book informal "brainstorming chats" through my website, creating a confidential space for open discussion of professional challenges, uncertainties, and strategic decisions. Notably, the concerns raised are remarkably consistent across countries and institutional contexts, highlighting shared structural and cultural features of contemporary academia.

In this talk, I present an overview of the most common issues discussed during these mentoring sessions. These include balancing research with caregiving responsibilities, particularly for academics with young children; evaluating one's academic standing and coping with feelings of inadequacy; managing periods of low productivity; sustaining a research agenda in intellectually isolating environments; navigating non-linear career paths and presenting them effectively in CVs; listing publications under revision; responding constructively to rejection; and articulating research impact in theoretical fields. Two especially recurring themes are learning how to learn to say no and how to ask for help.

This initiative has also revealed structural gaps in academic support. At many institutions, formal mentoring ends with the PhD, leaving postdoctoral researchers and junior faculty without guidance during a critical transition. By sharing these insights, this initiative aims both to empower ECRs and to raise awareness of the need for sustained mentoring.

Dr. Noushin Arfatahery

Founder & Director, WISEA Germany

PRESENTATION: Empowering Women Scientists as Drivers of Inclusive Innovation and Sustainable Development

Biography

Noushin Arfatahery is a postdoctoral researcher and independent scientist based in Berlin, Germany. She holds a Doctorate in Biology from Freie Universität Berlin, with a research background in environmental microbiology, sustainable aquaculture, and food safety. Her doctoral and postdoctoral work has focused on microbial interactions, antibacterial compounds from marine organisms, and nature-based solutions for resilient food systems.

She is currently an independent postdoctoral researcher (PI-equivalent) affiliated with the Leibniz Institute of Freshwater Ecology and Inland Fisheries (IGB) in Germany and a research collaborator with the Institute of Agrifood Research and Technology (IRTA) in Spain. Her interdisciplinary research integrates microbial ecology, circular aquaculture systems, and climate resilience, with a strong emphasis on translating scientific knowledge into practical and socially relevant outcomes.

In parallel with her academic work, Dr. Arfatahery is the Founder and Director of WISEA (Women in STEM & Environmental Advocacy), a Berlin-based initiative dedicated to promoting gender equity, inclusive innovation, and environmental sustainability in science. Through WISEA, she leads mentorship programs, workshops, and international collaborations supporting women researchers, early-career scientists, and refugee scholars.

She is a TWAS Young Affiliate (UNESCO), an active member of several international scientific societies, and has participated in global conferences and policy-oriented forums across Europe, Asia, and the Americas. Her work lies at the intersection of science, innovation, and social impact, with a particular focus on empowering women scientists as key contributors to sustainable development, science diplomacy, and inclusive research ecosystems."

Abstract

Gender equality is a critical enabler of scientific innovation and sustainable development, yet women scientists continue to face systemic barriers that limit their leadership, visibility, and participation in decision-making processes. Addressing these inequities is essential not only for social justice, but also for strengthening the quality, relevance, and impact of scientific research worldwide.

This presentation draws on interdisciplinary research and leadership experience in environmental microbiology, sustainable aquaculture, and gender-inclusive scientific initiatives across Europe and the Global South. It highlights how empowering women scientists contributes to innovation-driven solutions in areas such as food security, environmental resilience, and public health, while fostering more inclusive and responsive research ecosystems.

Through applied research projects and international collaborations integrating microbial ecology, circular food systems, and community engagement, the presentation demonstrates how women-led scientific initiatives can translate knowledge into socially relevant outcomes. Particular emphasis is placed on mentorship, capacity building, and cross-sector collaboration as mechanisms for enabling women researchers to move from scientific training to leadership and innovation roles.

The contribution argues that gender equality in science must be recognized as a strategic investment in sustainable development rather than a peripheral objective. Embedding gender-inclusive policies within research institutions, funding frameworks, and innovation ecosystems enhances scientific excellence and societal impact simultaneously. By positioning women scientists as key actors in innovation and entrepreneurship, science can more effectively respond to global challenges and contribute to resilient, equitable, and sustainable futures.

Technical sessions

1-Biology-Medical sciences

Prof Folasade M. Olajuyigbe

Federal University of Technology, Akure, Nigeria

PRESENTATION: Enzyme Biotechnology: Catalyzing Sustainable Solutions for a Better Future

Biography

Professor Folasade Mayowa Olajuyigbe is an internationally recognized biochemist, structural biologist, and research leader whose work in enzyme biotechnology and environmental bioremediation is advancing sustainable solutions to global health and environmental challenges. She is Professor of Applied Biochemistry at the Federal University of Technology, Akure (FUTA), Nigeria, where she currently serves as Director of the University Central Research Laboratory. She is also the National Chairperson of the Organization for Women in Science for the Developing World (OWSD), Nigeria Chapter, and Project Coordinator of the United States Government Exchange Alumni Association of Nigeria (USGEAAN), Ondo State Chapter. Professor Olajuyigbe leads the Enzyme Biotechnology and Environmental Bioremediation Research Unit (E-Biotech Lab) as Principal Investigator. She has successfully supervised more than 50 postgraduate researchers, including 10 Ph.D. and over 40 M.Tech. graduates. Her multidisciplinary research integrates structural biology, enzyme biotechnology, microbial biocatalysis, bioenergy, and environmental bioremediation, with applications in structure-based drug discovery, industrial biotechnology, agricultural waste valorization, renewable energy production, and sustainable pollution control. Her research has contributed significantly to the development of enzyme-based technologies for biomass conversion and environmental remediation, resulting in over 50 peer-reviewed publications in reputable international journals and more than 1,100 citations. A Fulbright Senior Research Scholar at Rutgers University, USA (2017–2018), Professor Olajuyigbe has secured competitive research grants from the International Centre for Genetic Engineering and Biotechnology (ICGEB), Italy; Nigeria's Tertiary Education Trust Fund (TETFund) National Research Fund; and the International Foundation for Science (IFS), Sweden. She has also won international conference and travel grants from Nature, IFS, and the American Society for Biochemistry and Molecular Biology (ASBMB). Her distinguished career has earned her numerous honours and awards, including two Schlumberger Foundation Faculty for the Future Fellowships, Associate Award of the International Centre for Theoretical Physics (ICTP), Italy, the 2024 Seeding Labs Instrumental Access Award for the FUTA Central Research Laboratory, and multiple FUTA Excellence and Productivity Awards under the Vice-Chancellor's Special Recognition category. Widely respected as a mentor and advocate for women in STEM, she has received national and international recognition for her leadership in advancing gender equity and scientific capacity development. Professor Olajuyigbe is an active member of several professional organizations, including the Nigerian Society of Biochemistry and Molecular Biology (NSBMB), the American Society for Biochemistry and Molecular Biology (ASBMB), the West African Research Association (WARA), and the American Association for the Advancement of Science (AAAS). She is a regular invited speaker at national and international scientific conferences, where she has delivered keynote, plenary, and lead papers. Outside academia, Professor Olajuyigbe enjoys creative cooking and window shopping. She is happily married to Revd. Canon (Dr.) Ebenezer Olajuyigbe, and they are blessed with children.

Abstract

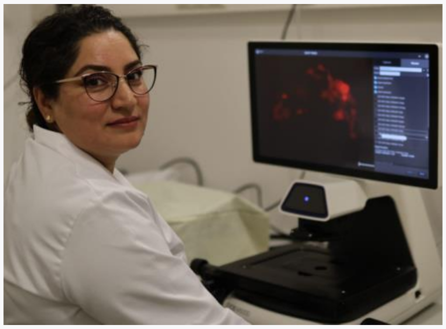
Enzyme biotechnology is transforming the global pursuit of sustainable development by replacing energy-intensive and environmentally hazardous processes with efficient, renewable, and eco-friendly biocatalytic systems. This paper highlights the pivotal role of enzyme biotechnology in driving the circular bioeconomy through innovative applications in biomass valorization, environmental remediation, and renewable energy production. Drawing on our research, we demonstrate how the discovery, characterization, and engineering of robust microbial enzymes have significantly enhanced the bioconversion of lignocellulosic agricultural residues into fermentable sugars for biofuel production. Particular emphasis is placed on lignocellulolytic enzymes, including cellulases, β-glucosidases, laccases, and other lignin-degrading oxidoreductases, as well as multifunctional enzyme systems that enable efficient delignification and saccharification of recalcitrant biomass. Our findings further demonstrate the remarkable potential of free and immobilized enzymes in the degradation of persistent pollutants, including synthetic dyes, bisphenol A, and petroleum hydrocarbons, thereby advancing sustainable waste management and ecosystem restoration. Furthermore, innovations in enzyme immobilization and recombinant DNA technology have substantially enhanced enzyme stability, catalytic performance, and industrial scalability, addressing key challenges associated with process efficiency and economic viability. The exploration of indigenous microbial biodiversity has also expanded the portfolio of industrially relevant biocatalysts adapted to diverse environmental conditions. Collectively, these advances establish enzyme biotechnology as a powerful catalyst for integrating waste valorization, pollution mitigation, and clean energy generation into sustainable bioprocesses. Continued investment in enzyme discovery, protein engineering, and bioprocess innovation will accelerate the development of resilient, cost-effective, and environmentally responsible technologies capable of supporting global sustainability goals and a thriving bio-based future.

Dr. Amna Mehmood

Senior Scientist, Martin Luther University, Germany

PRESENTATION: Clinico-Genomic Characterization of Breast Cancer: From Whole-Exome Sequencing to Precision Oncology

Biography



Dr. Amna Mehmood is an Afghan molecular biologist and Senior Scientist at the Institute of Microbiology, Martin Luther University Halle-Wittenberg, Germany. As a child, she fled Afghanistan with her family to Pakistan, where she completed her Bachelor's degree in Biotechnology. She later moved to Germany to pursue an M.Sc. in Molecular Genetics and subsequently earned her Ph.D. in Biochemistry at Jacobs University Bremen through a prestigious DAAD scholarship.

Before returning to academia, Dr. Mehmood served as a Senior Molecular Biology Specialist at the Ministry of Public Health, Afghanistan, where she worked on Global Fund and WHO projects supporting the establishment and strengthening of national reference laboratories for infectious disease diagnostics.

Since 2017, she has been conducting interdisciplinary research at Martin Luther University Halle-Wittenberg, initially as a Project Leader and currently as a Senior Scientist. Her research integrates cancer genomics, molecular imaging, protein engineering, and microbial biochemistry, with applications ranging from whole-exome sequencing of breast cancer and photoacoustic reporter proteins to the molecular mechanisms of [NiFe]-hydrogenase maturation.

Beyond her scientific work, Dr. Mehmood is a committed advocate for women in STEM, science diplomacy, and equitable access to education. Through international collaborations and community initiatives, she works to strengthen scientific capacity and expand opportunities for women and displaced scholars worldwide.

Abstract

Breast cancer is the most frequently diagnosed cancer among women worldwide; however, many populations remain underrepresented in genomic research, limiting the implementation of precision medicine. Expanding genomic studies to diverse populations is essential for identifying population-specific genetic variants and improving personalized cancer care.

In this presentation, I will highlight our work on the clinico-genomic characterization of breast cancer in women from Khyber Pakhtunkhwa (KP), Pakistan, one of the first comprehensive whole-exome sequencing studies in this understudied population. We identified both somatic and germline variants in key DNA damage response genes, including ATM, CHEK2, PALB2, and XRCC2, with several recurrent mutations demonstrating potential as population-specific biomarkers. To investigate their biological significance, genomic analyses were complemented by structural bioinformatics, protein interaction network analysis, molecular docking, and molecular dynamics simulations, providing insights into the functional consequences of these variants and their potential implications for targeted therapy.

This work demonstrates how integrating genomics with computational structural biology can improve our understanding of cancer biology and contribute to more equitable precision oncology for underrepresented populations. The study was conducted as part of a DAAD-supported international collaborative research project between Germany and Pakistan, highlighting the importance of international scientific partnerships in strengthening research capacity and advancing global health research.

In addition to this work, I will briefly present how my research has expanded into molecular imaging through the development of photoacoustic reporter proteins, protein engineering, and the molecular mechanisms of [NiFe]-hydrogenase maturation in Escherichia coli. Together, these projects illustrate how interdisciplinary and collaborative research can bridge fundamental molecular biology with translational biomedical applications while fostering scientific cooperation across borders.

Ms Hayat Daeia

Dalhousie University, Canada

PRESENTATION: Brain-Computer Interfaces (BCI): Machine Learning Classification of Neural SSVEP Signals

Biography

Hayat Daeia is an upcoming third-year Bachelor of Science student majoring in Medical Sciences at Dalhousie University in Halifax, Nova Scotia, Canada. Her academic interests center on women's health and wellness, with a long-term goal of specializing in this area within clinical medicine. In her first year, she was selected to join the Dalhousie Integrated Science Program, a competitive, research-focused cohort. Through this program, she completed two scientific research papers, including one conducted during her winter-term placement at the Neurocognitive Imaging Lab (NCIL) at Dalhousie University. She continues to volunteer with the lab, contributing to ongoing research through data collection and by assisting with literature screening for a graduate student's review paper. Hayat has presented her findings at several + conferences. She gave presentations at the Brain Recovery Center's Research Day in Halifax, Nova Scotia, and the Canadian Undergraduate Conference on Healthcare in Kingston, Ontario. Additionally, Hayat actively participates in student advocacy and leadership. In addition to holding executive positions in a number of university societies, she represents students from the Middle Eastern and North African (MENA) diaspora as the Medical Sciences Society's Interpersonal Diversity and Equity Representative. Hayat takes part in local environmental initiatives. She was awarded a grant to create and present composting workshops at local libraries that encourage composting and lessen food waste after taking part in multiple programs focused on girls' entrepreneurship and climate change.

Abstract

Brain-computer interfaces (BCI) utilize machine learning classifiers to categorize specific brain signals into recognizable commands. These signals are typically recorded using electroencephalography (EEG) that measure various neural responses such as steady-state visually evoked potentials (SSVEPs). SSVEP signals are brain responses primarily generated in the visual cortex, where neural activity synchronizes with the frequency of continuously flickering visual stimuli. These signals are processed and classified using machine learning algorithms, with classification accuracy measured. A major challenge in SSVEP-BCIs is visual fatigue, prompting research to enhance signal strength and accuracy while maintaining user comfort. Previous studies on optimal stimulus parameters have yielded conflicting results, often involving trade-offs between stronger responses and user comfort. In this study, we investigate whether a novel aperiodic shape, the hat tile, can improve SSVEP signal to noise ratios (SNR) and classification accuracy compared to a square stimulus. We hypothesized no significant differences. We tested this by collecting data from seven participants (ages 18-56), comparing amplitudes of SSVEP responses, and evaluating accuracy. Our findings refute our hypothesis; the square yielded higher SNR and accuracy. Given limitations such as small sample size, electromagnetic interference, non-normally distributed data, and not testing a checkerboard-like stimuli, further research is warranted to explore the potential of aperiodic stimuli in SSVEP-BCI applications.

Dr Zohra Aloui-Zarrouk

Institute of Pasteur , Tunisia

PRESENTATION: Bridging Science and Entrepreneurship: The Emergence of Zora Pharm as a Scientific Spin-off for innovative Plant-based Dermaceuticals

Biography

Dr Zohra Aloui-Zarrouk, phD in biochemistry, is a Research Scientist at Institut Pasteur de Tunis (IPT) and ex-lecturer in biochemistry at Tunis ElManar University. At the Molecular Epidemiology and Experimental Pathology Laboratory (MEEP) at IPT, she contributed and lead several projects mainly the structural and pharmacological characterization of novel proteins from snake venom related to VEGF and the valorization of natural plant products for the development of novel intervention tools and strategies against infectious diseases mainly drugs for prevention or therapy.

Zohra is an African Academy of Sciences (AAS) Affiliate since 2017, and she lead the Working Group in “Basic and Applied Sciences” With the aim to strengthen more collaborations between African researchers.

With a particular interest in Scientific Entrepreneurship and the valorization of research outcomes, Zohra is a recipient of a PAQ-PAES project funded by the World Bank to create an Spin-off related to the R&D environment. This Spin-off is incubated at the Institut Pasteur of Tunis, Tunisia.

Abstract

Skin lesions represent a wide spectrum of pathological conditions and constitute a major clinical and socio-economic burden, particularly in low- and middle-income countries. These lesions are frequently associated with inflammation, infection, delayed healing, and permanent scarring, highlighting the need for therapeutic strategies that simultaneously control inflammation and promote tissue regeneration. Limited access to healthcare services and the high cost and side effects of conventional pharmaceuticals further exacerbate this burden. Medicinal and aromatic plants are a valuable yet underutilized source of bioactive compounds with demonstrated antioxidant, anti-inflammatory, anti-infectious, and regenerative properties. Africa’s rich botanical diversity offers significant opportunities for the development of innovative and affordable therapeutic solutions; however, these resources remain insufficiently translated into standardized pharmaceutical and dermo-cosmetic products. Our Spin-off project focuses to develop and characterize novel skin-healing formulations based on natural active ingredients, including essential oils and plant extracts previously identified and studied in our laboratory. To enhance bioavailability, skin deposition, and therapeutic efficacy, these actives will be incorporated into advanced-based delivery systems, such as bio-polymeric carriers and lipid nanoparticles. Quality control and biological activity are evaluated using appropriate in vitro cellular models according to pharmacopeia and ISO standards. Our ultimate goal is to produce safe, effective, and affordable skin-care formulations derived from natural sources, with reduced reliance on harmful chemicals. By leveraging readily available raw materials, this project seeks to add value to medicinal and aromatic plants while addressing a critical healthcare need at both national and international levels.

Dr Bishayr Hassan Aljaffar

King Fisal General Hospital, Saudia Arabia

PRESENTATION: Predictors of Exclusive Breast feeding in Employed Mothers

Biography

Bishayr Hassan Aljaffar is a dedicated Nursing Specialist from Saudi Arabia, recognized for her commitment to patient'centered care and professional excellence. Born on September 4, 1991, she has built her career on compassion, resilience, and a strong sense of responsibility toward improving healthcare outcomes.

Abstract

This study examined the gendered predictors of exclusive breastfeeding among employed mothers in Saudi Arabia using Bronfenbrenner Ecological Systems Theory as a guiding framework. A crossâ sectional design was employed, with a random sample of 201 employed mothers recruited from two hospitals and two primary healthcare centers in Alâ Ahsa. Data were collected through validated Arabic questionnaires addressing individual, microsystem, exosystem, and macrosystem factors. Hierarchical logistic regression was used to analyze predictors across ecological levels. Findings revealed that 36.8% of employed mothers did not practice exclusive breastfeeding at six months postpartum. The final regression model explained 42.9% of the variance in breastfeeding practices. Significant predictors included perception of milk supply (OR = 1.029), maternal mental health status (OR = 0.931), workplace support (OR = 1.024), and social norms (OR = 2.009). These results highlight the multiâ€'layered influences shaping maternal health behaviors, where individual perceptions, psychological well 'being, organizational policies, and societal expectations intersect to determine breastfeeding outcomes. The study underscores the importance of genderâ€'sensitive interventions that address maternal confidence, mental health, workplace flexibility, and community attitudes. Healthcare providers should routinely assess mothers perceptions of milk sufficiency and mental health, while advocating for supportive workplace policies. Public health campaigns targeting social norms are also essential to empower women in sustaining exclusive breastfeeding. In conclusion, exclusive breastfeeding among employed mothers is not merely a personal choice but a gendered health behavior shaped by ecological systems, requiring integrated strategies across clinical, organizational, and societal levels.

Dr Hameeda Matooq AlJanabi

Dammam Health Network Eastern Health Cluster, Saudia Arabia

PRESENTATION: Knowledge and Attitudes Toward PICC Line Care Before and After Specialised Training in the Dammam Health Network, Saudi Arabia

Biography

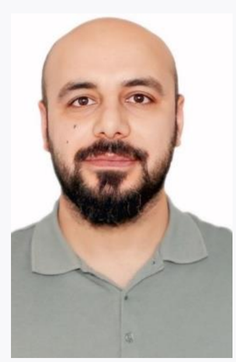
Hameeda Matooq Aljanabi is a highly accomplished Senior Nurse Educator and Nursing Professional Development Specialist, currently serving within the Dammam Health Network " Eastern Health Cluster. With a strong academic foundation, she holds a Master of Science in Nursing and is pursuing her PhD, reflecting her dedication to advancing nursing education, leadership, and evidence ‘based practice. Her career spans extensive experience in clinical nursing, academic instruction, and professional development. As a Registered Nurse and Advanced Clinical Nurse Educator (ACNE), Hameeda has played a pivotal role in shaping nursing standards, mentoring staff, and implementing innovative educational strategies that enhance patient care and workforce readiness. She is recognized for her ability to integrate theory with practice, ensuring that nurses under her guidance are equipped with both technical competence and compassionate care values. Hameeda expertise includes curriculum design, competencyâ€‘based training, and the development of professional growth pathways for nurses across diverse healthcare settings. She has contributed to initiatives that strengthen nursing leadership, improve patient safety, and align clinical practice with international accreditation standards. Her work emphasizes continuous improvement, collaboration, and the empowerment of nurses to meet evolving healthcare challenges. As a PhD candidate, Hameeda is committed to advancing research in nursing education and professional development, with a vision of transforming healthcare delivery through knowledge, innovation, and leadership. Her career reflects excellence, resilience, and a passion for elevating the nursing profession in Saudi Arabia and beyond.

Abstract

Peripherally Inserted Central Catheters (PICC) are widely used for long‘term intravenous therapy, yet their safe and effective management depends on nurses knowledge, attitudes, and adherence to evidenceâ€‘based practices. Inadequate training can result in serious complications, including infection, thrombosis, and catheter occlusion, highlighting the need for structured educational interventions. This study assessed the effectiveness of a specialized training program in improving nursesâ€™ knowledge and attitudes toward PICC care within the Dammam Health Network (DHN), Kingdom of Saudi Arabia. A quasi‘experimental pre and post ‘test design was employed, involving 180 registered nurses who completed a two ‘week program consisting of six sessions integrating theoretical instruction and practical demonstrations. Knowledge and attitudes were measured before and after training using validated assessment tools. Results revealed statistically significant improvements ($p < 0.05$) across multiple domains. Notable gains were observed in understanding aseptic techniques (from 63.3% to 87.2%, $p = 0.001$), flushing protocols, infection prevention strategies, and documentation standards. In addition, participants reported increased confidence in their ability to manage PICCs and expressed readiness to engage in continuous professional development. The findings confirm that targeted training substantially enhances nurses competencies in PICC care, reduces risks of catheterâ€‘related complications, and fosters a culture of safety and accountability. This study underscores the importance of regular, tailored educational programs as part of professional development initiatives, ensuring that nurses remain equipped to deliver high ‘quality patient care. Integrating such training into institutional practice can strengthen clinical outcomes and advance nursing standards across healthcare systems.

Dr Anwar Omar Dawod

University of Dohuk - College of Pharmacy, Iraq



PRESENTATION: Monitoring hospitalized elderly patients and the use of polypharmacy in Dohuk: a biochemical evaluation

Biography

I am Anwar Omar Dawod, a Pharmacist and researcher with a strong academic and clinical interest in biochemical monitoring and polypharmacy effects. I graduated from the

University of Duhok, College of Pharmacy, in 2015. Since graduation, l have gained extensive experience working in hospital and community pharmacy settings, including clinical practice at Heevi Pediatric Hospital, Dohuk and other pharmaceutical centers. I am also involved in community pharmacy practice through my own pharmacy, Norsi Pharmacy. In addition, I have attended many scientific conferences including Pluxes (Croatia, 2024), Kurdistan Hematology Summit (2023) and others. I have participated with WHO (2016) in two periods in the implementation of ECAP in Duhok. I have participated in many pharmaceutical scientific and selling skills trainings including: BD phaSeal system (2023), training program of portfolio medical review and selling skills in 2018 and others. I am currently conducting a Master's degree research project entitled "Biochemical Monitoring of Hospitalized Elderly Patients Subjected to the Use of Polypharmacy in Duhok." My research focus on evaluating biochemical parameters such as liver function tests, renal markers, inflammatory markers, oxidative stress indicators, and drug–drug interactions among elderly patients exposed to multiple medications. My work integrates clinical pharmacy, laboratory biomarkers, and statistical analysis using SPSS to assess the impact of polypharmacy on patient health outcomes.

Abstract

Aging and polypharmacy in hospitalized elderly patients are associated with physiological stress, decline in renal and hepatic function, oxidative damage, chronic inflammation, and increased susceptibility to drug–drug interactions. Objective: This study evaluated the biochemical effects of aging and polypharmacy on renal, hepatic, inflammatory, oxidative stress, and nutritional biomarkers, while examining gender-specific differences and interaction frequency. Methods: A cross-sectional study enrolled 250 elderly patients admitted to Azadi Hospital, Duhok. Participants were classified into two age groups: 50–64 years and ≥65 years. Polypharmacy was defined as concurrent use of 5, 6–9, or ≥10 medications. Measured biochemical parameters included renal markers (urea, creatinine), hepatic enzymes (AST, ALT, ALP, bilirubin), oxidative stress marker malondialdehyde, inflammatory marker C-reactive protein, and nutritional indicators (albumin, hemoglobin). Drug–drug interactions were evaluated using the Medscape Drug Interaction Checker. Results: 30.8% of patients were receiving ≥10 medications. Increasing medication burden significantly elevated malondialdehyde levels (+84% to +139.8%), reduced albumin concentrations (−18.2% to −23.3%), and increased pancreatic amylase activity (+5.9%). Patients aged ≥65 years demonstrated renal impairment, particularly females, with elevated urea (+62.6%) and creatinine (+168.2%). C-reactive protein remained elevated (87.21±81.17 mg/L), indicating systemic inflammation. Drug–drug interaction increased with medication number, with major interactions predominating among patients receiving ≥10 medications. Females showed broader age-related biochemical deterioration, whereas males demonstrated greater susceptibility to oxidative stress and hypoalbuminemia. Conclusion: Aging with polypharmacy worsens organ dysfunction and inflammation, emphasizing the need for individualized pharmacotherapy and routine biochemical monitoring to improve patient safety and clinical outcomes overall.

Prof. Jibin Sun

Professor, Chinese Academy of Sciences, China

PRESENTATION: Industrial Synthetic Biology, Biofoundry Platforms, and Systems Biotechnology

Biography

Dr. Sun is an expert in industrial synthetic biology and systems biotechnology from the Chinese Academy of Sciences, with over three decades of academic research, industrial transformation and international cooperation experience. He earned his PhD in Biotechnology and Bioinformatics from the German Research Centre for Biotechnology and Technische Universität Braunschweig in Germany between 2000 and 2004. Over there, he continued his postdoctoral research until 2008, before returning to China to join Tianjin Institute of Industrial Biotechnology (TIB), CAS as a full professor.

From 2012 to 2023, Dr. Sun served as Deputy Director General of TIB. During this pivotal tenure, he made contributions to China's synthetic biology infrastructure: he spearheaded the establishment of the National Center of Technology Innovation for Synthetic Biology and the core high-end biofoundry platform at TIB, driving the institute into China's top 1% ranking for technology transfer. Currently, he holds multiple roles including Principal Investigator and Director of the Center of Systems Biology at TIB, alongside key leadership positions in China's national microbiology, biotechnology and bioindustry associations. Recently, he was elected as Chairperson of the Coordinating Council of COMSTAS.

Academically, Dr. Sun has published more than 150 peer-reviewed international papers and filed nearly 100 invention patents worldwide, many of them successfully licensed to industry. His achievements have won numerous prestigious provincial and national science and technology awards across China.

Abstract

Biomanufacturing offers a promising pathway toward sustainable development by using biological systems and biotechnology to produce valuable products with reduced environmental impact. This presentation highlights how advances in plant biotechnology, genomics, metabolic engineering, and bio-based production can support sustainable agriculture, improve resource efficiency, and generate high-value products from renewable biological resources. Particular attention is given to innovative approaches that connect scientific research with practical applications, contributing to climate resilience, circular bioeconomy, and sustainable economic development. The integration of biotechnology with emerging technologies can accelerate the transition toward greener and more sustainable production systems.

Dr Iman Abdulmanaf Taha

University of Dohuk, College of Science,,Iraq

PRESENTATION: Biochemical and Developmental Outcomes in Iraqi Children with Phenylketonuria Following Late Diagnosis and Dietary Intervention.

Biography

My name is Iman Abdulmanaf Taha, and I was born on 14 April 1999. I live in Kurdistan, Iraq. I'm currently a researcher studying a Master's at the University of Duhok, College of Science, Chemistry Department. The theme of my MSc study is "Evaluation of biochemical markers and anthropometric parameters in children diagnosed with phenylketonuria in Northern Iraq and Nineveh." Driven by a deep curiosity and passion for science, after earning my Bachelor's degree (2021-2022), I continued my academic journey with determination and a strong commitment to research and learning. In addition to my studies, I have been volunteering at the Hivi Pediatric Hospital for more than two years. This experience has strengthened my practical skills and enhanced my sense of responsibility, compassion, and dedication to serving others. Through both my academic and volunteer work, I have grown professionally and made a positive impact in the community. I have joined the International Conference on the Role of Universities in Achieving Sustainable Development Goals (ICRU-SDGs), Theme: "Advancing The SDGs Through Education, Research, and Innovation" held on the 23rd and 24th of September, 2025.

Abstract

Phenylketonuria (PKU) is an autosomal recessive disorder requiring early detection via newborn screening (NBS) to prevent neurological impairment. In Northern Iraq, conflict has disrupted healthcare, leading to delayed diagnoses and inadequate management of PKU patients. Objectives: To evaluate the clinical and demographic profile and outcomes of children with PKU. Methods: a cohort of 19 children diagnosed with PKU in Kurdistan Region of Iraq (KRI) and Nineveh Governorate was evaluated through phenylalanine level quantification and application of questionnaires. Results: children's age ranged from 1 month to 15 years with male predominance (58%). Balanced ethnic distribution between Iraqi Kurdish non-Yazidi and Iraqi Kurdish-Yazidi (47% each) was identified. High consanguinity rate was found (84%) with 62.7% of marriages among first-cousins. 54% of the parents did not perform pre-marital medical exams. Moderate PKU appeared as the main phenotype (53%) followed by classic PKU (41%). Delayed diagnosis was found, with 53% having received the PKU diagnosis at the age of 1-5 years., followed by the age range of 6-10 years (16%). High incidence of PKU misdiagnosis appeared (79%), where the most prevalent type of misdiagnosis was cerebral palsy (47%). The incidence of seizures (81%) was found associated with cerebral palsy (60%) and microcephaly (20%). Conclusions: the findings emphasize the need for implementation of comprehensive newborn screening programs, integration of metabolic testing into pediatric neurology evaluations, integration of genetic counseling into premarital services, besides the need for specialized metabolic professionals in the region. Strengthening these preventive and diagnostic strategies is essential to reduce PKU mortality.

Prof.Dr. Abdallah A. Shaltout

Spectroscopy Department, Physics Research Institute, National Research Centre, Egypt

PRESENTATION: **Steel Slags Recycling**

Biography

Prof. Dr. Abdallah A. Shaltout is a distinguished researcher at the National Research Centre in Giza, Egypt (Spectroscopy Department, Physics Research Institute), where he has been affiliated since 1994 . He earned his doctorate in Solid State Physics from the Technische Universität Wien in Austria. His research expertise centers on X-ray Fluorescence (XRF), analytical atomic spectroscopy, and air pollution. He has published over 115 documents with an h-index of 32, receiving more than 3150 citations. His research spans materials characterization, environmental monitoring, medical geology, and energy applications, with frequent international collaboration involving scholars from Egypt, Saudi Arabia, Sweden, Germany, and Jordan. Prof. Shaltout is actively involved in international research collaboration. His work is supported by various international funding bodies, including the COST Action CA18130 (European Network for Chemical Elemental Analysis by TXRF), King Saud University, the Synchrotron Light Research Institute, and IUPAP . He collaborates with scholars based in Egypt, Saudi Arabia, Sweden, Jordan, Brazil, Germany, Austria, and France. His research frequently appears in prestigious journals such as Food Chemistry, Environmental Pollution, Electrochimica Acta, X-Ray Spectrometry, and Journal of Analytical Atomic Spectrometry.

Abstract

To prevent the deterioration of natural resources and support sustainable development, the valorization and recycling of various steel slags have become increasingly important. In the present work, six different types of steel slags were collected, characterized, analyzed, and partially recycled namely: electric arc furnace (EAF) slag, ladle refining furnace slag treated with Si and Al (LRF-Si and LRF-Al), steel fume dust, mill scale flakes, and refractory materials waste. High resolution wavelength dispersive X-ray fluorescence (WDXRF) spectrometry with multi-dispersive crystals was used for precise quantitative elemental analysis. X-ray diffractometry was employed to identify and quantify the crystalline phases within the collected samples. The analysis revealed that the mill scale flakes and EAF slags contain high concentrations of iron oxides, at approximately~95% and ~40%, respectively, making them attractive for various applications. Due to the high concentration of calcium oxides and calcium compounds in LRF slag (~ 50%) makes it suitable for various applications in the cement and concrete industry. Successful implements of EAF and LRF was investigated in the cement and concert application. In the case of mill scale flakes, a reduction from iron oxides to zero-valent iron using hydrogen gas was established. X-ray photoelectron spectroscopy (XPS) was implemented for monitoring the oxidation states of iron before and after reduction. Furthermore, the steel fume dust was found to contain remarkably high concentration of zinc oxides (ZnO) reaching to ~40 %. The high ZnO content makes the steel fume dust a promising candidate for many applications such as zinc extraction, fertilizers, and cosmetics productions.

Ass. Prof. Samar Said Mohamed

Egyptian Petroleum Research Institute, Egypt

PRESENTATION: WO3 Anchored on FAU(Y)/AlSBA-15 Binary Supports: Impact of Hierarchical Structure on Catalytic Dehydrogenation

Biography

Samar Said is a dedicated researcher and academic with a strong background in catalysis and sustainable chemical processes. Her scientific interests focus on catalytic reactions relevant to energy, environment, and cleaner fuels, with particular expertise in oxidative desulfurization, ethanol dehydrogenation, and CO_2 hydrogenation to value-added chemicals. Over recent years, her work has emphasized the design and application of heterogeneous catalysts, including metal oxides supported on mesoporous materials, aiming to improve reaction efficiency, selectivity, and environmental performance.

Samar has actively engaged in both experimental research and scientific writing, contributing to literature reviews, research proposals, and technical reports intended for international journals and edited books. Her research approach combines careful experimental design with a clear understanding of reaction mechanisms and structure–performance relationships in catalytic systems. She is especially interested in translating laboratory-scale studies into practical solutions relevant to fuel upgrading and green chemistry.

In addition to her research activities, Samar is committed to professional development and international scientific collaboration. She is an approved member of the Organization for Women in Science for the Developing World (OWSD), reflecting her dedication to advancing science and supporting women researchers globally. She regularly seeks opportunities for training, networking, and participation in competitive scientific programs.

Beyond her professional life, Samar is known for her perseverance, sense of responsibility, and ability to balance academic ambitions with personal commitments. Her career path reflects resilience, continuous learning, and a strong motivation to contribute meaningful scientific knowledge that addresses real societal and environmental challenges.

Abstract

Much work has been done on synthesizing materials that enhance the conversion and selectivity of catalysts in industries by combining the stability and acidity benefits of zeolites with those of mesoporous materials. This study describes the synthesis of novel WO3- supported micro-mesoporous composites made of AlSBA-15 (Si/Al = 5) and HY (FAU) zeolites type, using a simple deposition technique. The experimental results indicated that this composite is effective catalysts for the non-oxidative dehydrogenation of ethanol to acetaldehyde. Several spectroscopic techniques were used to emphasize the physicochemical characteristics of the produced micro-mesoporous composite. Additionally, its catalytic performance was evaluated when compared to the non-oxidative dehydrogenation of ethanol, utilizing a flow system to produce more valuable products. Several spectroscopic methods demonstrated that this simple synthesis approach successfully included the used zeolite type seeds into the matrix bulk of the AlSBA-15(5). In contrast to its constituent components, the

synthesized WO_3-supported micro-mesoporous catalyst exhibit finer dispersion, basicity, a newly formed peak in the XPS O1S spectra due to W–O–W bridging oxygens, and comparatively higher amounts of isolated tetrahedrally monomeric WO_4^{2-} (W—O) species to oligomeric ones.

With an ethanol conversion of about 85 % at 450 ◦C, the WO_3-supported micro-mesoporous composite catalyst exhibits highest catalytic activity in terms of conversion. The catalyst's basicity, the presence of W–O–W bridges, and the comparatively higher amounts of isolated tetrahedral monomeric to oligomeric WO_4^{2-} species all had a significant impact on the ethanol conversion and product selectivity's during the non-oxidative dehydrogenation of ethanol.

Dr. Elzahraa Ahmed Ramadan Elgohary

Photochemistry Department, Chemical Industries Research Institute, National Research Centre (NRC), Egypt

PRESENTATION: Photocatalytic studies on using nanocomposites based on modified titania for CO2 fixation environmental applications

Biography

Dr. Elzahraa Ahmed Ramadan Elgohary is a Researcher at the Photochemistry Department, Chemical Industries Research Institute, National Research Centre (NRC), Egypt.

She earned her Ph.D. in Physical Chemistry from the Faculty of Science, Cairo University (2022). Her doctoral research focused on the synthesis and optimization of high-efficiency nanomaterial-based photocatalysts dedicated to environmental remediation.

Dr. Elzahraa has demonstrated an active research footprint through international and national collaborative projects including UK-Egypt Collaborations: Partnered with the University of Birmingham on projects co-funded by the British Council and the Science, Technology & Innovation Funding Authority (STDF), focusing on industrial wastewater treatment, water desalination, and photocatalytic nanomaterials, ANSO – Chinese Academy of Sciences (CAS): Contributed to sustainable environmental solutions under the Alliance of International Science Organizations (ANSO) framework., Served as a project member and co-principal investigator on internal projects at the NRC aimed at developing cost-effective photocatalytic materials derived from natural resources for water purification.

She awarded the best Oral Presentation Award at the 6th Autumn School on Physics of Advanced Materials (PAMS-6) during the 15th International Conference on Physics of Advanced Materials (ICPAM-15) in Sharm El-Sheikh, Egypt (November 2023) for her presentation on Plasmonic Photocatalytic Nanocomposites for Advanced Environmental Remediation.

She presented research at international conferences across Egypt, Greece, Canada, and Italy. She has authored multiple peer-reviewed papers in high-impact journals and contributed a book chapter published by Elsevier.

Abstract

Irradiation of a series of aromatic carcinogenic epoxides and CO2 greenhouse gas in the presence of TiO2, TiO2/MWCNT, Pt//TiO2/MWCNT and Pd/TiO2/MWCNT photocatalysts was investigated with visible light- induced cycloaddition reaction for the production of high value products, cyclic carbonates and formic acid, with high activity and selectivity. It was revealed that the catalytic activity order of these catalysts was as follows: Pd/TiO2/MWCNT > Pt//TiO2/MWCNT > TiO2/MWCNT > TiO2, respectively. The main features of this protocol are the transformation of CO2 to cyclic carbonates at ambient temperature and under atmospheric pressure. The band gap energies for Pd/TiO2/MWCNT and Pt/TiO2/MWCNT nanocomposites were estimated from the Kubelka-Munk plot to be ca~ 2.39 and 2.33 eV, respectively that was proved to be active catalysts in comparison with bar TiO2 or TiO2/MWCNT that possess band gab value ca~ 3.02 and 2.46 eV, respectively. A plausible mechanism explaining the behavior of the CO2 with Pd/TiO2/MWCNT nanocomposite was proposed.

3-Environment -Agriculture

Prof. Amna Jrra

Independent Researcher, Climate Modeling Expert, Jourdan

PRESENTATION: Projecting Climate Hazards in Azraq, Jordan: Multi-Scenario Changes in Extreme Events and Socioeconomic Vulnerability

Biography

Dr. Amna Jrrar is a climate scientist specializing in atmospheric modelling, the development and validation of coupled global climate models—with a particular focus on sea-ice variability—and the analysis of regional climate model outputs, including climate extreme indices. She earned her PhD in 2004 from the Centre for Atmospheric Science at the University of Cambridge, where she investigated the dynamical variability of stratospheric ozone under the supervision of Prof. John A. Pyle.

She is a member of the Task Force for the Eastern Mediterranean and Middle East (EMME) Climate Change Initiative: The Scientific Basis, which supports the development of a regional action plan to address climate impacts and advance mitigation strategies. Findings from this work contributed to the review paper Climate Change and Weather Extremes in the Eastern Mediterranean and Middle East (https://doi.org/10.1029/2021RG000762).

Dr. Jrrar has contributed to several national and regional projects, providing climate change projections of extreme temperature and precipitation indices using dynamically and statistically downscaled climate simulations.

She currently holds a National Geographic Explorer Level 2 grant for a project focused on the Azraq region, where she is co-producing climate-actionable information and climate storylines in collaboration with the local community.

Abstract

Azraq, in eastern Jordan, is a uniquely fragile socio-ecological system shaped by its arid climate, groundwater-dependent oasis, and rapidly shifting land-use patterns. Decades of groundwater over-abstraction has led to the collapse of the Azraq Oasis in the late 1980s/early 1990s, triggering profound and lasting consequences for local livelihoods of the multi-ethnic communities living there, and leading to pronounced changes in both socioeconomic activities and the demographic composition of the region.

In recent decades, the area has experienced intensifying climate-related stresses, including droughts, flash floods, and increasingly frequent extreme heat events.

This study investigates the intensifying threat of climate-induced hazards in Azraq, focusing on recent extreme events and multi-decadal future projections. Utilizing high-resolution (10km) regional climate models (RCMs) simulations over the Mashreq domain, we analyze two Shared Socioeconomic Pathways: SSP2-4.5 (moderate emissions) and SSP5-8.5 (high emissions).

Projections for the near future to mid 21st century show a significant increasing trend in the average temperature, and in the frequency of ""very hot days"" (Tmax > 40°C) particularly under the SSP5-8.5 scenario. While total annual rainfall does not show a clear significant trend, the amount of highest precipitation rain and the rain intensity is projected to increase, highlighting the risk of flash floods .

The study further explores how these physical hazards intersect with current socioeconomic trends in Azraq and their vulnerability under these extremes. In particular we focus on how past experiences of the local population may shape their resilience strategies and what initiatives can support their adaptation efforts.

Pof. Dr. Jianwei Zhang

National Key Laboratory of Crop Genetic Improvement Huazhong Agricultural University Wuhan, China

PRESENTATION: Toward Green Nutritious Super Rice --
A Rice Pan-genome System and Its Implementation in Genomic Breeding

Biography

Prof. Dr. Jianwei Zhang is a dual-appointed professor at the College of Life Science and Technology and the College of Informatics at Huazhong Agricultural University, and a member of the National Key Laboratory of Crop Genetic Improvement. He has long been engaged in genomics and bioinformatics research, focusing on complete genome assembly, bioinformatics analysis, and integrated data platform development. His work includes the development of sequence assembly and editing tools GPM and DEGAP, the determination of gap-free reference genome sequences for crops such as rice and soybean, the use of high-quality pan-genomes to analyze the genetic diversity of Asian rice, and the establishment of the world's first Rice Gene Index platform for comparative genomics research.

Abstract

Asian rice is central to global food security and remains a primary target for breeding Green Nutritious Super Rice (GNSR). The rapid accumulation of high-quality rice genomes over the past five years has propelled the field into a true pan-genomic era, enabling unprecedented exploration of natural variations, structural variants, and organelle-derived sequences such as nuclear organelle DNA (NORG). However, most existing databases curate only a limited subset of genomes or focus on narrow thematic topics, which restricts their utility for genomic breeding and multi-omics integration.

To address this need, we developed the Rice Gene Index (RGI)—a comprehensive and user-friendly rice pan-genome system, which dynamically integrates genes, transcripts, and genome features across sixteen platinum-standard reference genomes that represent the full population structure of Asian rice. The system supports extensive search, comparative genomics, and multi-dimensional visualization, making it well suited for studying complex traits associated with sustainability, nutrition, stress resilience, and yield. RGI thereby establishes a scalable informatics foundation for multi-omics analysis, identification of functional natural variations, and implementation of genomic breeding practices aimed at designing future GNSR.

RGI 1.0 (and the expanded RGI 2.0 Beta incorporating >50 high-quality assemblies) is publicly accessible at https://riceome.hzau.edu.cn, providing the rice research and breeding community with a powerful resource for genomics, genetics, evolution, and synthetic biology applications.

Dr. Sarra Hechmi

Water Research and Technology Center (CERTE), Tunisia

PRESENTATION: Scaling Safe Water Reuse in Tunisia: Balancing Agricultural Imperatives against Emerging Contaminant Risks for SDG 6

Biography

Dr. Sarra Hechmi is a research scientist at the Water Research and Technologies Center (CERTE) in Tunisia. Operating at the academic and practical intersection of soil and water science, and environmental research, she specializes in circular economy resource management and sustainable agricultural development in water-scarce regions. Her primary research focuses on the tracking, characterization, and environmental risk assessment of persistent micropollutants including microplastics and other emerging pollutants across critical terrestrial and aquatic ecosystems. Beyond academic research, Dr. Hechmi is committed to translating laboratory findings into environmental policy and public advocacy. She actively collaborates with non-governmental organizations to promote chemical safety, agricultural sustainability, and plastic pollution awareness. Through her integrated work on treated wastewater reuse, soil health, and contaminant pathways, she strives to develop resilient, nature-based engineering solutions that protect Mediterranean agro-ecosystems and safeguard long-term public health.

Abstract

In water-scarce Tunisia, reusing treated wastewater (TWW) is a strategic necessity aligned with SDG 6 (Clean Water and Sanitation). Tunisia generates approximately 320 Mm3/year of raw wastewater, treating around 292 Mm3/year across 125 conventional wastewater treatment plants (WWTPs). However, agricultural reuse remains limited to roughly 7-20%, irrigating about 8,000 hectares primarily dedicated to arboriculture (olives, orchards) and fodder crops under the regulatory framework of Tunisian Standard NT 106.03. While standard regulation mandates the monitoring of conventional parameters like biochemical oxygen demand, chemical oxygen demand, heavy metals and fecal coliforms, emerging contaminants are largely discarded from routine tracking. Conventional activated sludge WWTPs exhibit low elimination efficiencies for recalcitrant microplastics, persistent pesticides, pharmaceuticals, and endocrine-disrupting compounds. When unmonitored TWW is discharged into environmental receptors (rivers and seas) or applied directly via surface irrigation, it triggers complex fate dynamics. These include horizontal accumulation in topsoils, vertical migration to underlying aquifers, and trophic transfer into seafood and crop tissues, creating chronic eco-toxicological risks for human health.To bridge the gap between water security and chemical safety, Tunisia must upgrade its infrastructure and governance. While advanced membrane processes provide exceptional filtration, their high operational costs remain prohibitive at scale. Conversely, integrated nature-based solutions (NBS) such as constructed wetlands offer cost-effective, decentralized options for polishing effluent

quality. Prioritizing the modernization of national regulations alongside targeted WWTP technology upgrades is critical to establishing resilient, non-conventional water cycles for sustainable agriculture.

Ass. Professor . Dina Mohamed Salama Ahmed

Vegetable Research Department, National Research Centre (NRC), Egypt

PRESENTATION: Nanotechnology in Vegetable Production: Enhancing Productivity and Quality for Sustainable Agriculture

Biography

Dr. Dina Mohamed Salama Ahmed is an Assistant Research Professor at the Vegetable Research Department, National Research Centre (NRC), Egypt. She obtained her B.Sc. in Vegetable Science from the Faculty of Agriculture, Cairo University, in 2006, followed by an M.Sc. in Arid Land Agricultural Studies and Research from Ain Shams University in 2010. She received her Ph.D. in 2015 from the same university.

Dr. Salama has developed extensive research experience in vegetable production, sustainable agriculture, nanotechnology, nanofertilizers, plant nutrition, and the application of nano-enabled technologies to improve crop productivity and quality. Her research has particularly focused on the application of nanomaterials in vegetable crops, including their effects on growth, yield, nutritional quality, physiological responses, and genomic stability.

She has participated in several international and national research projects and has served as Principal Investigator in projects addressing nanotechnology and its impacts on vegetable productivity, quality, and genetic traits. She also contributes as a reviewer for several international scientific journals.

She has received distinguished recognition for her scientific contributions, including the State Encouragement Award in recognition of her outstanding achievements in agricultural and scientific research. She has also received the Dr. Amal Saber Award for Individuals, presented by the Academy of Scientific Research and Technology (ASRT), Egypt.

She has published 32 scientific papers. Her work contributes to the development of innovative, efficient, and environmentally sustainable approaches for modern vegetable production.

Abstract

Conventional fertilizers and pesticides play a major role in maintaining productivity in vegetable production, but their excessive and ineffective use can reduce nutrient-use efficiency, degrade soil quality, increase environmental pollution, and raise production costs. To increase agricultural productivity and environmental sustainability, these difficulties have spurred the development of novel, sustainable technologies, such as nanotechnology. Through the effective use of nano-enabled agricultural inputs, nanotechnology has become a promising technique for improving vegetable output, quality, and resilience. Depending on their physicochemical characteristics and the intended agronomic response, nanomaterials can be applied by foliar spraying, root application, or seed priming. The presence of essential elements, including potassium, phosphorus, zinc, iron, and copper, etc, in the form of nanoparticles can improve nutrient availability and

uptake, boost photosynthetic activity, encourage biomass accumulation, and boost yield. Additionally, nanoparticles can increase a plant's resistance to abiotic stressors like heat, drought, salt, and heavy metal toxicity by fortifying antioxidant defense mechanisms. Opportunities for the targeted and prolonged release of active chemicals are provided by nano-enabled pesticides and controlled-delivery systems, which may enhance the control of pests and diseases while lowering environmental losses. By modifying the build-up of minerals, vitamins, antioxidants, and other health-promoting bioactive chemicals, nanotechnology may also improve the nutritional and functional quality of vegetables. However, the size, concentration, chemical makeup, surface properties, application technique, crop species, and environmental factors all affect how well and safely nanoparticles work. Therefore, it is important to thoroughly assess any potential issues with nanoparticle accumulation, environmental persistence, and interactions with soil and plant-associated microbes. All things considered, combining nanotechnology with sustainable crop-management techniques is a viable approach to creating vegetable production systems that are robust, productive, resource-efficient, and ecologically conscious. Long-term field validation, dose optimization, environmental safety, and the creation of efficient and long-lasting nano-enabled technologies should be the main focus of future research.

Dr. Heba Kamal El-Kholly

Water Pollution Research Department at the Environmental and Climate Change Research Institute, National Research Centre, Egypt

PRESENTATION: Synergistic Fe_2O_3 nanoparticles and chemical pretreatment for enhanced biohydrogen/biogas production from pulp sludge

Biography

Dr. Heba Kamal El-Kholly is a Researcher in the Water Pollution Research Department at the Environmenta l and Climate Change Research Institute, National Research Centre, Egypt. She received her Ph.D. in Organic Chemistry from the Faculty of Science, Helwan University, in 2022. Her doctoral thesis focused on the preparation of eco-friendly nanomaterials and their applications in environmental remediation.

Dr. El-Kholly has gained international research experience through competitive research grants and scientific missions. In 2019, she was awarded an Egyptian Missions Pre-Doctoral Study Grant to conduct research at the Faculty of Engineering, University of Naples Federico II, Italy. In 2017, she received a grant from the Ministry of Industry and Trade to undertake research at the College of Environmental Science and Engineering, Soochow University, China.

She has also obtained several internationally recognized certificates in environmental management, sustainability, occupational safety, and carbon footprint assessment, including NEBOSH-EMC, IOSH, ISEC, TAIEX, and OSHA certifications. Her research expertise includes the synthesis and characterization of nanomaterials and materials derived from agricultural wastes, with applications in water treatment, pollutant removal, and heavy-metal remediation. She has more than ten years of experience in the physical and chemical analysis of drinking water and industrial wastewater and in developing sustainable, environmentally friendly, and cost-effective treatment approaches.

Abstract

Bioenergy plays a crucial role in mitigating climate change by offering a renewable alternative to fossil fuels. Through the integration of chemically synthesized Fe_2O_3 nanoparticles and targeted pretreatment strategies, this study aimed to enhance biohydrogen and biogas recovery from pulp and paper waste sludge (PPS). Mineral acids e.g., hydrochloric acid (HCl), nitric acid (HNO_3), sulfuric acid (H_2SO_4), and phosphoric acid (H_3PO_4) and alkaline sodium hydroxide (NaOH) treatment were applied to improve PPS biodegradability. Structural and morphological changes in PPS post-treatment were analyzed using various advanced analytical techniques and they showed surface alteration, potentially to improve hydrolysis. The bioenergy potential was evaluated through a two-stage process: biohydrogen production (BHP) followed by biomethane production (BMP). Pretreatments significantly influenced energy yields, with BHP and BMP values varying across treatment types. Mass balance analysis of hydrogen yield and volatile fatty acid (VFA) production showed differences between 2.0 % and 19 %, confirming data reliability. Fe_2O_3 nanoparticle addition further enhanced BMP performance, reaching up to 283 mL $gCOD^{-1}$ (g-chemical oxygen demand of substrate) at 200 mg/gVS (g-volatile solids of substrate) dosage. Kinetic modeling of the data indicated a strong fit to first-order kinetics ($R^2 > 0.95$) for both control and pretreated PPS. These findings demonstrate that combining chemical pretreatment with nanomaterials effectively improves substrate solubilization, microbial accessibility, and

methane conversion efficiency. The proposed strategy offers a scalable and efficient approach to valorize PPS into bioenergy, contributing to sustainable waste management and renewable energy development.

WORLD FORUM
FOR WOMEN IN SCIENCE

Envisioning the Future: Empowering Science Innovation and Entrepreneurship

EGYPT 2026

INVITED CHAIRS
(ALPHABETICALLY ARRANGED)

PROF. AMANI MOSTAFA
Former Dean of Advanced Materials Technology Research Institute, , National Research Centre, Egypt

PROF. AMIN ABDELLATIF ELMELIGI
Institute of Advanced Materials Technology Research and Mineral Resources, , National Research Centre , Egypt

PROF AMR M ABDOU
Professor of Microbiology and Immunology, National Research Centre ,Egypt, ASM Ambassador

PROF. FAGR ABD EL-GAWAD
Dean of Environment and Climate Change Research Institute, National Research Centre, Egypt

PROF. INAS K. BATTISHA
The Solid state physics department, Physics Research Institute, National Research Centre , Egypt

PROF. NAGWA SAYED ATA
Professor of Microbiology and immunology, Veterinary Research Institute, National Research Centre, Egypt

PROF. SALMA NAGA
Institute of Advanced Materials Technology Research and Mineral Resources, National Research Centre ,Egypt

PROF. WAFAA MOHAMED ABD EL-RAHIM
Dean of the Agricultural and Biological Research Institute, National Research Centre ,Egypt

NATIONAL RESEARCH CENTRE

SEPTEMBER 8–10 , 2026

FORUM PROGRAM

WORLD FORUM
FOR WOMEN IN SCIENCE, Egypt 2026

Envisioning the Future: Empowering Science Innovation and Entrepreneurship

- **Date:** 8–10 September 2026
- **9–8 Sept Venue:** National Research Centre, Egypt
- **10 Sept Venue:** Virtual Online Day

MASTER PROGRAM OVERVIEW

8–10 SEPTEMBER 2026

Date	Time	Program Block	Details
Tuesday, 8 September 2026	09:30-10:00	REGISTRATION	Registration
	10:00–10:30	Opening Remarks.	
	10:30–12:30	GOVERNING SESSION	Envisioning the Future: Empowering Science, Technology, Innovation, Entrepreneurship and the Governance Interface for Sustainable Development
	12:30–12:45	Breakout sessions	
	12:45–14:15	PARALLEL SCIENTIFIC SESSIONS	Biology Session 1 | Materials–Engineering Session 1 | Environment & Agriculture Session 1
	14:15–15:00	Lunch Break	
	15:00–17:00	PARALLEL SCIENTIFIC SESSIONS	Biology Session 2 | Materials–Engineering Session 2 | Environment & Agriculture Session 2
Wednesday, 9 September 2026	09:30–11:00	General Session: From Knowledge to Impact	
	11:00–13:00	PARALLEL SCIENTIFIC SESSIONS	Biology Session 3 | Materials–Engineering Session 3 | Environment & Agriculture Session 3
	13:00–14:30	PARALLEL SCIENTIFIC SESSIONS	Biology Session 4 | Materials–Engineering Session 4
	14:30–15:00	Lunch Break	
	15:00–17:00	PARALLEL SCIENTIFIC SESSIONS	Biology Session 5 | Materials–Engineering Session 5
Thursday, 10 September 2026	10:00 -5:00	ONLINE DAY	Online scientific program

TUESDAY, 8 SEPTEMBER 2026 | OPENING Remarks& Opening Session

DAY 1 - MORNING

Registration & Opening Ceremony -09:30–10:30

Hani El –Nazer Hall

09:30–10.00: Registration-Hani El-Nazer Hall

10:00-10:30: Opening Remarks

Hani El –Nazer Hall

Prof . Amal Amin

Professor for nanotechnology/polymer technology at National Research Center, Egypt, founding chair of World forum for women in science.

Prof. Ahmed Youssef

Dean of Chemical Industrial Research Institute, National Research Centre, Egypt

Judge Mrs Amal Ammar

President of the National Council for Women (NCW)

HRH Princess Dr. Nisreen El-Hashemite

Executive Director of the Royal Academy of Science International Trust (RASIT),

Prof. Quarraisha Abdool Karim

President of The World Academy of Sciences (TWAS)

Mr. Adrien Delamare-Deboutteville

International Pharma MCO Head for Africa, Sanofi's.

Prof. Mamdouh Moawad Ali Hassan

President of the National Research Centre (NRC), Acting President of the Academy of Scientific Research and Technology (ASRT)

TUESDAY, 8 SEPTEMBER 2026 | OPENING Remarks& Opening Session

DAY 1 - MORNING

Hani El –Nazer Hall

Opening Plenary Session |10:30-12:30

🏛Envisioning the Future: Empowering Science, Technology, Innovation, Entrepreneurship and the Governance Interface for Sustainable Development

Session Chairs: Prof. Ahmed Youssef & Prof. Hossam El-Nazer

Moderators: Mr. Ashraf Amin & Mr. Mohamed El-Hawary

Panel 1 — Turning Science and Innovation into Sustainable Solutions

- **Prof. Adla Ragab**
  Member of Parliament, Tourism and Aviation Committee
- **Prof. Maged Ghoneima**
  Founder and General Partner of M Empire Angels Chairman of Startup Egypt
- **Mrs. Ghada Michel**
  Head of Communications and CSR at Sanofi
- **Mrs. Nagwa A. ElSayed**
  Head of the Scientific Research and Innovation Sector at Misr-elkheir Foundation

Panel 2 — Building Stronger Governance for Science, Innovation and Sustainable Development

- **Mrs. Rania Sadeky**
  Member of the Egyptian Senate
  Member of the Defence and National Security Committee
- **Prof. Nevine Makram Labib**
  Chair of the Computer and Information Systems Department, Sadat Academy for Management Sciences (SAMS)
- **Prof. Heba M. Sharobeem**
  Member of the Egyptian Senate
  Vice chair of the Committee of Human Rights and Social Solidarity
- **Ms. Mona keshta**
  Egyptian House of Representatives, Member of the Foreign Affairs Committee

Concluding Discussion — From Ideas to Action

12:30–12:45 | Break Out Sessions

12:45–14:15 | PARALLEL SCIENTIFIC SESSIONS

14:15–15:00 | Lunch Break

15:00–17:00 | PARALLEL SCIENTIFIC SESSIONS

Tuesday, 8 September 2026 | Biology and Medical sciences — Session 1

12:45–14:15

Biology and Medical sciences — Session 1

Hani El –Nazer Hall

Chairs: Prof. Mohamed Salama & Prof. Nadia Hamdy

Speaker / Role	Affiliation / Role	Presentation Title
Invited Talks (15 minutes each)		
Prof. Mohamed Salama (Moderator)	The American University in Cairo (AUC), senior fellow for the Global Brain Health Institute (GBHI), Egypt	Risk and Resilience in Dementia: Towards a more proactive approach
Prof. Nadia Hamdy (Moderator)	Biochemistry Dept., Faculty of Pharmacy, Ain Shams University, Cairo, Egypt	Epi genetics Integration into Medicine for Advancing Precision via Chem/Bio-informatics Piloting/Validation
Regular talks (10 minutes each)		
Prof. Dalia Osama Saleh	Pharmacology Department, Medical Research and Clinical Studies institute, National Research Centre, Egypt	Natural Products as Multi-Target Therapeutics for Polycystic Ovary Syndrome
Dr. Mawell Sameh, MD, CMD	Head of Vaccines Egypt and Sudan, Sanofi	Enhancing women’s healthcare by vaccinations
Dr. Sherehan Khalil, MD, CMD	Medical Operations Lead Egypt, Sanofi	Diversity and Inclusion in clinical trials
Dr. Eman Abd El-Razek Metawea Abbas	Researcher, National Research Centre, Egypt	Host Genetic Biomarkers Associated with Disease Progression in HCV-Related HCC

Tuesday, 8 September 2026 | Materials and Engineering— Session 1

12:45–14:15

Materials and Engineering— Session 1

Main Hall

Chairs: Prof. Ahmed A. Abdel Moneim & Prof. Nour F. Attia

Speaker / Role	Affiliation / Role	Presentation Title
Invited Talks-15 minutes		
Prof. Ahmed A. Abdel Moneim (Moderator)	Founding Dean of the Faculty of Basic and Applied Sciences at the Egypt-Japan University of Science and Technology (E-JUST), Egypt	Turning CO_2 into Opportunity: The New Carbon Economy
Prof. Nour F. Attia (Moderator)	Gas analysis and fire safety lab, chemistry division, National Institute of standards (NIS(	Sustainable and Scalable Nanomaterials for Energy and Environmental Applications
Prof. Marwa S. M. I. Shalaby	Professor of chemical engineering, National Research Centre, Egypt	Next-Generation Composite Membranes for Water and Wastewater Treatment
Regular talks (10 minutes)		
Prof. Ahlam M. Fathi	Physical chemistry dept., NRC, Egypt	Progress in using metal oxide-based electrode materials as safe and sustainable electrode for electrochemical supercapacitors
Ass. Prof. Nihal El-Mehalawy	Institute of Advanced materials technology and mineral resources researches, National Research Centre, Egypt	Dielectric Evaluation of the fabricated Alumina Composites doped with Different Oxides for Electrical Insulation Applications
Ass. Prof. Heba Elsayed Ali Hassan	Associate Professor, National Research Centre, Egypt	Advanced Ternary Nanocomposite Heterojunctions for Solar-Driven Hydrogen Evolution and Environmental Remediation

Tuesday, 8 September 2026 | Environment and Agriculture— Session 1

12:45–14:15

Environment and Agriculture— Session 1

Hall A

Chairs: Prof. Nour Shafik El-Gendy & Prof. Seham Ali Shaban

Speaker / Role	Affiliation / Role	Presentation Title
Invited talks (15 minutes)		
Prof. Nour Shafik El-Gendy (Moderator)	Egyptian Petroleum Research Institute (EPRI), Egypt	Marine Macroalgae Recruitment for Achieving Food, Energy and Water Nexus
Dr .Chadene Diyab	International expert in climate change and sustainable development, France.	Environmental Diplomacy and Women: Issues of Stability and Climate Peace in the Arab World and the EMENA Region
Regular talks (10 minutes)		
Dr. Pulane Koosaletse-Mswela	Senior Lecturer at the University of Botswana, Botswana	Environmental Quality-Based Assessment of a Small Dam for Sustainable Recreational Use in Semi-arid Botswana”
Prof. Seham Ali Shaban (Moderator)	Egyptian Petroleum Research Institute, Egypt	Biochar – Sustainable Material for A Green Future
Ass. Prof. Safaa S. M. Ali	Physics Research institute, National Research Centre, Egypt	Spatial Distribution and Health Risk Assessment of Heavy Metals in the Road Dust Deposited on Tree Leaves in the main Overcrowded Squares of Greater Cairo
Ms. Takya Mohammad Ahmed Elnady	El-Sadat STEM School, Egypt	The Role of Green Chemistry in Achieving Sustainable Development Goals: A Modern Educational Vision

14:15–15:00 | lunch BREAK

Tuesday, 8 September 2026 | Biology and Medical Sciences— Session 2

15:00–17:00

Biology and Medical Sciences— Session 2

Chairs: Prof. Reham Dawood & Prof. Menha Mahmoud Swellam

Hani El –Nazer Hall

Speaker / Role	Affiliation / Role	Presentation Title
Regular talks (10 minutes each)		
Prof. Reham Dawood (Moderator)	Professor, National Research Centre, Egypt	Evaluation of lncRNA PVT1 Variants and Serum E-Cadherin Levels in Breast Cancer
Prof. Menha Mahmoud Swellam (Moderator)	Head of High Throughput Lab, NRC, Egypt	Clinical Impact of MiRNAs in Women Malignancies: Current Insights and Validation
Ass. Prof. Ghada Salum	Associate Professor, National Research Centre, Egypt	Genetic Dysregulation in Breast Cancer: Impact on Immune Surveillance and Infection Control
Ass. Prof. Ahmed Hassan Hassan Afifi	Associate Professor, National Research Centre, Egypt	Marine Environment as a Potential Source for Small Molecule Targeted Anti-Cancer Agents
Ass. prof Mona Mohammad Agwa	Associate Professor, National Research Centre, Egypt	Active Tumor Targeting Exploiting Overexpressed Cellular and Subcellular Receptors
Dr. Basma Elsayed Fotouh	Researcher, National Research Centre, Egypt	Impact of KRAS and NOXA Gene Expression on Ovarian Cancer Development and Progression
Dr. Tânia dos Reis	PhD Candidate, IHMT & University of Cape Verde, Cape Verde	Nasal Microbiota Profile and Its Correlation with Pediatric Asthma and Atopic Conditions
Dr. Walaa Abdallah Gad	Researcher, National Research Centre, Egypt	Epigenetics and Gene Therapy Technologies for Precision Veterinary Medicine
Dr. Amal Abdel-Bassir Draz	Researcher, National Research Centre, Egypt	Integration of DNA Barcoding, Metabolite Profiling, and Antimicrobial Evaluation of *Zygophyllum scabrum*
Ass. Prof. Mai Abd El Meguid Ali Dawoud Oada	Associate Professor, National Research Centre, Egypt	Evaluation of PNPLA3 genetic variant in Egyptian HBV-infected patients concomitant with metabolic-associated steatotic liver disease (MASLD)
Ass. Prof. Ola Mohamed Mousa	Associate Professor, Faculty of Nursing, Minia University, Egypt	Exploring Risk Perception and Behavioral Patterns of Antibiotic Misuse Among Students

Tuesday, 8 September 2026 | Materials and Engineering — Session 2

15:00–17:00

Materials and Engineering — Session 2

Main Hall

Chairs: Prof. Shimaa El-Hadad & Prof. May M. Eid

Speaker / Role	Affiliation / Role	Presentation Title
Regular talks-10 minutes		
Prof Giovanna Avellis	Director of International Consortium of Research Staff Associations (ICoRSA), Italy	Entrepreneurship and Science innovation by tackling the Women Digital Divide Gap
Prof. Shimaa El-Hadad (Moderator)	Central Metallurgical Research and Development Institute (CMRDI)), Egypt	Titanium Alloys: Recycling Technologies and Challenges
Prof. May M. Eid (Moderator)	Physics Research institute, National Research Centre, Egypt	Biomolecular Tracking of SPION Nanocomposites: Hyperspectral FTIR Microscopy in Preclinical Safety Evaluation
Ass. Prof. Tamer Sami Mansour	Associate Professor, National Research Centre, Egypt	Diatomite-based electromagnetic wave absorption and shielding materials: Current research and future possibilities
Ass.Prof. Rim Gasmi	University Center of Illizi, Algeria	Artificial Intelligence as a Catalyst for Science-Based Innovation and Entrepreneurship in Desert Regions
Assis. Researcher Mona S. H. Gaber	British University in Egypt (BUE), Egypt	Advanced Functional Materials for Biosensing and Biomedical Applications
Master Studen. Soulaf Chenini	University Center of Illizi, Algeria	Hand gesture and sign language recognition for Targui Language based on deep learning
Ass.Prof. Ghada Mohamed Taha Ibrahim	Textile Research and Technology Institute, National Research Centre, Egypt	Controlled Delivery of Trans- cinnamaldehyde through Chitosan/Alginate Nano capsules for Active Food Packaging Textiles
Ass. Lecturer. Sarah Hesham Mohamed Rashed	Borg AlArab Technological University, Egypt	Dyeing to Be Clean: How Double Layered Hydroxides are Revolutionizing Industrial Wastewater Treatment

Tuesday, 8 September 2026 | Environment and Agriculture— Session 2

15:00–17:00

Environment and Agriculture— Session 2

Hall A

Chairs: Prof. Hamdy F. El- Mowafi & Prof. Wafaa Mohamed Abd El-Rahim

Speaker / Role	Affiliation / Role	Presentation Title
Invited Talks (15 minutes)		
Prof. Hamdy F. El- Mowafi (Moderator)	Crops Research Institute, Agricultural Research Center (ARC), Egypt	New Egyptian innovations to develop the production of hybrids , super and basmati smart rice resilien
Regular talks (10 minutes)		
Dr. Jenny Carolina Orjuela Garcia-Agri	Researcher in postdoctorade, National University of Colombia, Colombia	Smart Vermicomposting: Automating Biofertilizer Production Through Community Knowledge and Low-Cost IoT
Ms. Marwa Sulaiman Al-Hinai	Sultan Qaboos University, Oman	Nanotechnology-Enabled Smart Agriculture for Enhanced Drought Resilience and Sustainable Agroecosystems
Ass.Prof. Ahmed R. Henawy	Faculty of Agriculture, Cairo University, Giza, Egypt	Gut Microbiomes of Black Soldier Fly Larvae: Approaches for Bioconversion, Degradation of Antibiotics, Detoxification of Mycotoxins, and Production of Biofertilizers for Sustainable Waste Management
Ass.Prof. Asmaa Ahmed Mostafa Halema	Faculty of Agriculture, Cairo University, Egypt	From Contaminated environment to Sustainable Solutions: Microbial Genomics for Heavy Metal Bioremediation and Plant Growth Promotion
Ass. Pof. Sameh Mohamed Mohamed El-Sawy	Agricultural and Biological Research Institute, National Research Centre, Cairo, Egypt.	Precision Agriculture and Smart irrigation: Mitigating Water Scarcity and Enhancing the crop productivity under climate Change
Ass. Pof. Sahar Abd EL-Aziz Othman	National Water Research Center (NWRC), Egypt	Mitigating Severe Salinity Stress in Vicia faba L. via Exogenous Proline
Dr. Heba Abdelaziz Moussa AbdAlla	Botany Department, Agriculture and Biological Institute at National Research Centre, Egypt	Reproductive-Stage Heat Stress Tolerance in Rice: Implications for Climate-Resilient Agriculture under Global Warming
Dr. Mai N. Amer	Pharmaceutical and Drug Industries Research Institute, National Research Centre, Egypt	From Lab Innovation to Agricultural Entrepreneurship: Lactic Acid Bacteria for Sustainable Farming and Healthy Food Systems
Dr. Maissara M.K. Elmaghraby	Agriculture Research Center (ARC),Egypt	Lactic Acid Bacteria-Based Bio-Stimulant Production from Low-Quality Date Waste: A Sustainable Circular Bioeconomy Approach

Wednesday, 9 September 2026 | General Session: From Knowledge to Impact

09:30–11:00

General Session: From Knowledge to Impact

Hani El –Nazer Hall

Chairs: Prof. Hala Yousry & Prof. Heba Soliman

Speaker / Role	Affiliation / Role	Presentation Title
Invited Talk-15 minutes		
Prof. Hala Yousry (Moderator)	Desert Research Center, Egypt	Bridging the Gap between Scientific Research and Practical Application Causes, Challenges, and Strategic Interventions
Regular Talks- 10 minutes		
Prof. Heba Soliman (Moderaor)	Port Said University, Training manager, Egypt	The Future of Science: Training, Employability Skills, and Entrepreneurship
Mrs. Shaimaa ElBanna	Director of Change Leaders Scholarship, Egypt	The Hidden Costs of Women's Careers in STEM: Challenges and Pathways to Inclusion
Dr. Bothaina Mahmoud Fouad	New Cairo Technological University (NCTU), Egypt	Envisioning the Future: The Importance of Entrepreneurship for Women in Science
Ms.Gulzar Karybekova	Centre of Actual Problems and Development, Director ,Kyrgyzstan	Imagining What is Possible for the Future; Developing Evidence-Based Governance and Innovation Through Women's Leadership
Biotechnologist Natalia Montellano Duran	Universidad Católica Boliviana San Pablo, Organization for Women in Science in Deceloping World, Director, Bolivia	Gender-neutral Scientific Leadership in Bolivia: Evidence-Based Insights on Women's Participation Across the STEM Education and Research Pipeline
Anthropologist Ralitsa Savova	Member of the Editorial Committee of Cultural route "Longobard Ways across Europe, Italy،"	Is Africa really the poorest continent on Earth؟

Wednesday, 9 September 2026 | Biology and Medical Sciences — Session 3

11:00–13:00

Biology and Medical Sciences — Session 3

Chairs: Prof. Nesrine Hegazi & Prof. Amani Mostafa

Hani El –Nazer Hall

Speaker / Role	Affiliation / Role	Presentation Title
Regular talks (10 minutes each(		
Prof. Nesrine Hegazi (Moderator)	Professor, National Research Centre, Egypt	From Sea to Table: The Power of Metabolomics in Marine Food Science and Facing Challenges
Ass. Prof. Mona Abd El-Mouty Moustafa Raslan	Associate Professor, National Research Centre, Egypt	Beyond Random Screening: MS/MS Molecular Networking and the Evolution of Natural Product Drug Discovery
Ass. Prof. Rehab Fikry Taher Fareed	Associate Professor, National Research Centre, Egypt	The Molecular Mechanisms Underlying the Antineoplastic Profile of Vitexin 2″-O-α-L Rhamnopyranoside From Cordyline australis (G.Frost.): Isolation and Constituent Profiling
Ass. Prof. Zeinab Abd Elkhalek Ali Elshahid	Associate Professor, National Research Centre, Egypt	Investigation of Endophytic Fungi Secondary Metabolites with Potential Biological Activity
Ass. Prof. Gehan Fawzy Abdel Raoof Kandeel	Associate Professor, National Research Centre, Egypt	From Waste to Wellness: Bioactive Profiling and Multi-Target Therapeutic Potential of Vicia faba Peel By-Products
Ass. Prof. Marwa N. Ahmed	Associate Professor ،Nile University, Egypt	Evolutionary Trajectories of Antibiotic Resistance in *Pseudomonas aeruginosa*:Mechanisms, Dynamics, and Strategies for Constraint
Ass. Prof. Rasha Mohamed Hassan	Associate Professor, National Research Centre, Egypt	Synthetic Anti-Biofilm Agents as a Strategy in Combating Bacterial Resistance
Ass. Prof. Doaa Khalaf	Associate Professor, National Research Centre, Egypt	Antimicrobial Resistance in the Animal Sector: The 'One Health' Approach as a Sustainable Framework
Ass. Prof. Islam Hany Ali Haridy	Associate Professor, National Research Centre, Egypt	New Microtubule Modulators: Design, Synthesis and Anticancer Evaluation
Founder & CEO. Yosra Morsy	OASIS Pysychology Creations, Founder & CEO Mental Healthcare in 2035: From Clinics to intelligent Care, Egypt	Mental Healthcare in 2035: From Clinics to Intelligent Care

Materials and Engineering: — Session 3

Main Hall

Chairs: Prof. Salma Naga & Prof. Hanan B. Ahmed

Speaker / Role	Affiliation / Role	Presentation Title
Regular talks-10 minutes each		
Prof. Hanan B. Ahmed (Moderator)	Faculty of Science, Helwan University, Egypt	Green fabricated Nanomaterials for Pharmaceutical and Biomedical purposes
Ass. Prof. Moshera S. Abdelaz Youssef	Associate Professor, National Research Centre, Egypt	In vitro release and cytotoxicity activity of 5-fluorouracil entrapped polycaprolactone nanoparticles
Assoc. Prof. Rasha Abdel Baseer Taha Muhammed	Department of Polymers and Pigments Technology at the National Research Centre , Egypt	Enhanced electrochemical and optical properties of new conjugated n-p type polymers
Ass. Prof. E M. Mahmoud	Associate Professor, National Research Centre, Egypt	Recent trends in Advanced Bioceramics for orthopedic Applications
Ass. Prof. Heba-tollah Mahmoud Sweelam	Associate Professor, National Research Centre, Egypt	In Vitro Culture as a Sustainable Platform for the Production of Pharmaceutically Important Natural Products from Higher Plants
Ass. Prof. Mona Sayed Ahmed Mohamed	Assistant Professor, National Research Centre, Egypt	Bioactive Calcium Phosphate - Silicate Composites: Synthesis, Characterization and In-vitro Assessment
Ass. Prof. Samaa M. Faramawy	Associate Professor, National Institute of Standards (NIS), Egypt	Metrology and Traceability: Role as a Precondition for Sustainability
Assoc. Prof. Mona Moaness Abdel-Moneam Mohamed	Associate Professor, National Research Centre, Egypt	3D-Printed Reinforced Scaffolds for Bone Regeneration and Controlled Drug Delivery
Teaching Assis. Hadeer Mamdouh Ahmed Eldeeb	Faculty of archeology, Cairo University, Egypt	Novel Nanostructured Agarose Gels for the Non-Invasive Cleaning and Protection of Historic Tintype Photographs

Environment & Agriculture— Session 3

Hall A

Chairs: Prof. Fagr Abd El-Gawad & Prof. Salwa Kamal Mohamed Hassan

Speaker / Role	Affiliation / Role	Presentation Title
Regular Talks (10 minutes)		
Prof. Salwa Kamal Mohamed Hassan (Moderator)	Environment and Climate Change Research Institute, National Research Centre, Egypt	Assessing Indoor Pollution: Organic and Carcinogenic compounds; Heavy Metal Exposure, and Health Risks in Indoor Environments
Ass.Prof. Alyaa Ibrahem	school of biotechnology, Nile University, Egypt	Synergistic CO_2 Capture: A Novel Chitosan-Metal Oxide Nanoparticle-MOF Hydrogel Composite from Waste-Derived Materials
Ass. Prof. Shymaa Sabry Zaher	National Institute of Oceanography and Fisheries, Egypt	Assessing the Therapeutic efficiency of Spirulina platensis on ethanol-Induced Oxidative Markers, DNA Damage and some Organs histopathology in Rat
Dr. Douaa Hussein Abdel Aziz	National Research Centre, Egypt	Bioremediation and Our Lives: Restoring the Environment We Depend On
Dr. Elham Ahmed Abdelrahim Mohamed	Conservation Department, Faculty of Archaeology, Fayoum University, Egypt،	The Role of Artificial Intelligence in the Preservation of Mural Paintings Under Climate Change: Opportunities, Challenges and Emerging Frameworks
Dr. Arig wahid Elkhouly	Faculty of Industrial and Energy Technology, Borg Al Arab Technological University (BATU), Egypt	Hazelnut Shell Powder as a Sustainable Novel Functional Ingredient for Quality Improvement of Crackers

Wednesday, 9 September 2026 | Biology and Medical sciences— Session 4

13:00–14:30

Biology and Medical sciences— Session 4

Hani El –Nazer Hall

Chairs: Prof. Amr Hussein & Prof. Eman E. El Shanawany

Speaker / Role	Affiliation / Role	Presentation Title
Regular Talks-10 minutes		
Prof. Eman E. El Shanawany (Moderator)	Professor, National Research Centre, Egypt	Platyhelminth Glycans: Host–Parasite Interactions, Immunomodulation, and Vaccinology
Dr. Aya abdallah Seleem	Lecturer, Cairo University, Egypt	The Rise of Pan-Drug Resistance: Implications for Public Health and Zoonotic Diseases
Dr. Rana Ahmed Mohamed Eissa	Research Assistant, Badr University, Egypt	Antibacterial Film-Coated Stainless Steel Trays for Controlling Klebsiella pneumoniae and Pseudomonas aeruginosa in Intensive Care Units
Ass. Prof. Reham Amgad Mohamed-Ezzat	Associate Professor, National Research Centre, Egypt	Synthetic Strategies Towards Novel Pyrimidine Sulfonamides as Potent Antiviral Agents
Ass. Prof. Iman Ahmed Youssef Ghannam	Associate Professor, National Research Centre, Egypt	Development of Benzenesulfonamide-based PPARγ Agonists for the Treatment of Type II Diabetes Mellitus
Lecturer Afnan Hamdy Elgowily	Lecturer and Postdoctoral Researcher, Chemistry Department, Biochemistry Section, Faculty of science, Tanta University, Egypt	Pathophysiological Significance of Mallory Denk Bodies: Addressing a Century-Old Mystery
Ass. Prof. Amira Samy Talaat Abou Taleb	Assistant Research Professor, Electronics Research Inst., Egypt	Beyond the Black Box: How Explainable AI (XAI) Can Support Trustworthy Healthcare Decisions
Dr. Dina Magdy Ghanem	Researcher, National Research Centre, Egypt	Integrated Phytochemical Profiling and Biological Evaluation of Egyptian Tribulus terrestris L.: A Promising Source for Therapeutic Product Development

Wednesday, 9 September 2026 | Materials and Engineering: Session 4

13:00–14:30

Materials and Engineering: Session 4

Main Hall

Chairs: Prof. Marwa I. Wahba & Prof. Inas K. Battisha

Speaker / Role	Affiliation / Role	Presentation Title
Regular talks-10 minutes each		
Prof. Marwa I Wahba (Moderator)	Professor, National Research Centre, Egypt	Proteins as grafting materials for the fabrication of covalent immobilizers
Ass. Prof. Marwa Hany	German International University-GIU- German University in Cairo-GUC , Egypt	Identification of novel pyrazolines and isoxazolines as antileishmanial agents via a potential antifolate mechanism: synthesis, biological evaluation, and in silico studies
Ass. Prof. Maha Ahmed Ali Ahmed	Associate Professor, Faculty of Archaeology, Cairo University, Egypt	Mount Arafat Through the Lens of Muhammad Sadiq Bey: Documentation and Analysis of a 19th Century Albumen Print
Dr. Sara Sayed Abdel-Haliem Selim	Researcher, Egyptian Petroleum Research Institute,Egypt	Plastic Waste to High-Performance Aerogel Sorbents for Sustainable Oil Spill Cleanup
Ass. Prof. Heba Elmotasem	Associate professor, National Research Centre, Egypt	Nanoparticles decorated with targeting ligands for precision medicine.
Dr. Ahmed Hegazy	National Research Centre, Egypt	Beneficiation of Granite Waste in Sustainable Construction Materials
Ass. Prof. Rasha Mahmoud Abd El-Wahab	Associate Professor, National Research Centre, Egypt	Evaluation of a sustainable zeolite/bacterial cellulose nanocomposite membrane as a catalyst for biodiesel production from waste cooking oil.

14:30–15:00 | lunch BREAK

Wednesday, 9 September 2026 | Biology and Medical sciences— Session 5 A

15:00–17:00

Biology and Medical sciences— Session 5 A

Hani El –Nazer Hall

Chairs: Prof. Nagwa Sayed Ata & Prof. Rasha Mohamed Abdelraouf

Speaker / Role	Affiliation / Role	Presentation Title
Regular talks-10 minutes		
Prof. Rasha Mohamed Abdelraouf (Moderator)	Professor, Cairo University, Egypt	From Static to Smart: 4D Printing Applications in Dentistry
Dr. Tamer Mohamed Hamdy	Associate Researcher Professor in the Department of Restorative and Dental Materials Department, Oral and Dental Research Institute, National Research Centre, Egypt	Intelligent Dentistry: How Smart and 4D Materials Are Transforming Oral Care
Ass. Prof. Eman Samy Shalaby	Associate Professor, National Research Centre, Egypt	Bilosomes as a Promising Approach for Topical Application of Allantoin Intended Against Skin Aging
Dr. Salma Elmaaz	Founder, Solvira Pharma / Menoufia University, Egypt	Synergistic Topical Cream Formulation for Burn Scar Reduction Using Shell Nanoparticles
Ass. Prof. Yousra Aly Ibrahim Sabry	Associate Professor, National Research Centre, Egypt	Innovative Trends in Dentistry: A Shift Toward Conservative Concept
Ass. Prof. Salma Ahmed El Marasy	Associate Professor, National Research Centre, Egypt	Anti-Osteoporotic Effect of Sitagliptin in an Osteoporosis Model of Ovariectomized Rats: Role of RUNX2 and RANKL/OPG Ratio
Dr. Toka A. Ahmed	Faculty of biotechnology, Badr University, Cairo, Egypt	Human Adipose-Derived Pericytes in the Pathophysiology of Type 2 *Diabetes Mellitus*
Dr. Ragaa Fikry Fathy	Veterinary Research Institute, National Research Centre, Egypt	Ecotoxicological Effects of Emerging Pharmaceuticals on African Catfish: Biomarkers for Neurotoxicity, Immunotoxicity, and Histopathology
Ass. Prof. Radwa Hassan El-Akad	National Research Center, Egypt	Marine Metabolomics for Bioactive Compound Discovery in Sea Cucumbers"

Wednesday, 9 September 2026 | Biology & Medical — Session 5 B (young researchers)

15:00–17:00

Biology and Medical sciences— Session 5 B (young researchers)

Hani El –Nazer Hall

Chairs: Prof. Nagwa Sayed Ata & Prof. Rasha Mohamed Abdelraouf

Speaker	Affiliation / Role	Presentation Title
Short talks-5 minutes		
Dr. Naglaa Ramadan Atta Mohamed	Researcher, Cairo University, Egypt	Exploring the Therapeutic Potential of Postbiotics: Antimicrobial and Anticancer Properties from Sheep and Goat Sources.
Ms. Gehad kamel AbdalAl	Undergraduate Student, Minia University, Egypt	Exploring Future Healers' Views: Knowledge and Attitudes Toward Stem Cells Among Minia University Health Sector Students
Mr. Amr Abdelsattar gamal	Undergraduate Student, Minia University, Egypt	Evaluating the Accuracy of Blood Pressure Measurement Knowledge and Skills Among Nursing Students
Ms. Aya Dahy	Undergraduate Student, Minia University, Egypt	Bridging the Knowledge Gap: PCOS Awareness Among Female University Students at Minia University
Ms. Basmalaa Ahmed Thabet Hassan	Egyptian Biomedical Engineering researcher specializing in artificial intelligence for healthcare, computational medicine, and digital twin technologies, Egypt	An In-Silico Computational Simulation Framework for Glioblastoma Progression Modeling: Integrating 3D Transformer Segmentation, Computational Radiogenomic Feature Fusion, Simulation-Based Treatment-Response Modeling and Phantom-Based In-Silico Geometric Optimization
Mr. Mohammad Ahmed Ismael Ali Teleba	4th Year Student, Fuclty of nursing / Minia University, Egypt	Awareness of Mental Health Disorders among Students at Minia University, Egypt: A Cross-Sectional Study
Ms. Nada Hamada Elsayed	Bioechnology student, Ain shams university, Egypt	Integrating 16srRNA with multi-omics for high resolution and reproductible microbiome profiling
Dr. Sayeda Abdelrazek Abdelhamid	Researcher, National Research Centre, Egypt	Ecofriendly silver nanoparticles biosynthesis from Penicillium commune NRC 2016-3 as antimicrobial agents for textile materials
Dr. Ikram Benmakhlouf	PhD student in Computer Science from Algeria, Algeria	Toward Accessible AI Healthcare: A Roadmap for Offline-First, On-Device Deep Learning in Medical Image Screening

Wednesday, 9 September 2026 | Materials and Engineering— Session 5

15:00–17:00

Materials and Engineering— Session 5

Main Hall

Chairs: Prof. Amin El- Meligi & Prof. Mahmoud Essam

Speaker	Affiliation / Role	Presentation Title
Regular talks-10 minutes		
Dr. Safaa Hussein Alagamawy	Chemical researcher, Radio Engineering Research Center, Egypt	Evaluating the efficacy of some barium nanocomposites in treating biomicrite limestone
Dr. Ahmed Mahdy Hamed Haggar،	Egyptian Petroleum Research institute, Egypt	Catalytic Upcycling of Plastic Waste into Carbon Nanoblades and Carbon Nanodarts with Enhanced Mechano-Bactericidal Activity
Asst .Researcher Asmaa O. Ahmed	Physical Chemistry Department at the National Research Centre, Egypt	Role of Oxygen Vacancies in Enhancing the Hydrogen Evolution Reaction Performance of $SrFeO_{3-x}$ Perovskite Oxides
Ass. Prof. Adel Abdel Rehiam Ahmed Koriem	Assistant Professor, National Research Centre, Egypt	Interfacial Engineering of CuAlZn Alloy-Reinforced NBR/EPDM Elastomers for Soft Magnetic, Dielectric, and Thermomechanical Applications
Dr. Rasha Mostafa Al-Makhlasawy	Digital Signal Processing Laboratory, Computers and Systems Department, Electronics Research Institute (ERI), Cairo, Egypt.	Artificial Intelligence for Next-Generation Wireless Communications: Enabling Intelligent, Adaptive, and Sustainable Networks
Ass.Prof. Ahmed F. Ghanem	Packaging Materials Department, National Research Centre,Egypt	Development of PVA Composite Films for Meat Preservation
Dr. Basant Aboelsaud Elsebai	National Research Centre, Egypt	Zeolite-Infused Nanofibrous Hydrogel for Water Treatment Applications
Ass.Prof. Ahmed N. E. Osman	Associate Professor, National Research Centre, Egypt	Engineered Hybrid Nanocomposites for Safer Healthcare, Cleaner Water, and Sustainable Heritage Preservation

THURSDAY, 10 SEPTEMBER 2026 | ONLINE DAY

DAY 3 - VIRTUAL

Panel 1 Online | 10:00–11:00

Virtual Zoom Platform

Women National Council Session

The session is organised by Women National Council

Panel Title	**Envisioning The Future: Inspiration in Leadership and Technology**
Moderator	Eng. Hoda Mansour, Egypt President Egypt- AngloGold Ashanti, Board Member-Sukari Gold Mines
Panelists	**Eng. Hanan Abdel Meguid** Entrepreneur and leading figure in technology and innovation **Prof. Samhaa el Beltagy** Chief AI Officer at Nawy **Eng. Dawlat Hashem** Strategic Energy Consultant at Mideast Communication Systems

THURSDAY, 10 SEPTEMBER 2026 | ONLINE DAY — SCIENCE DIPLOMACY

11:00–12:30

Panel 2: Science diplomacy-Policy, management, Industry perspective

11:00–12:30

Chairs: Dr. Natisha Dukhi & Dr. Ibtissame AZZAOUI

Speaker Name	Affiliation / Role	Presentation Title
Prof. Paul Arthur Berkman	Founder of the Science Diplomacy Center Inc. (SDCI)- USA	Informed Decision-making for Lasting Peace
Dr Angela Liberatore	Former Head of the Scientific Department at the European Research Council	Science as global public good to safeguard other global public goods: the role of science diplomacy.
Dr Amal Kasry	Chief of Basic Sciences, Engineering, Technology & Innovation (BETI) Sciences (SCS) Sector, UNESCO	Connecting Science Across Borders: Access, Capacity and Shared Action
Dr Natisha Dukhi (Moderator)	a Senior Research Specialist at the Human Sciences Research Council-GYA cochair	Science Diplomacy from the Global South: Bridging Evidence, Youth Voice, and Policy
Mrs. Rebone Seleoane	Head of P&C Africa MCO	Supporting women in Sanofi beyond boundaries
Mrs. Alice Berson	Chief Finance Officer	Empowering women through sustainable procurement
Dr. Ibtissame AZZAOUI (Moderator)	President and Founder of Parliamentarians for Climate (P4C)	Parliamentary Science Diplomacy: Bridging Evidence, Policy and International Cooperation

Panel 3: Science diplomacy-networking, cultural aspects and Cooperation across borders

Chairs: Prof. Dr. Mohd Basyaruddin Abdul Rahman & Dr. Nerina Finetto

12:30-14:00

Speaker Name	Affiliation / Role	Presentation Title
Prof Mohd Basyaruddin Abdul Rahman (Moderator)	Chair of the ISTIC-UNESCO Governing Board	Diplomacy New Currency of Science Diplomacy Empowering the Global South through STI Partnerships
Prof Yoko Shimpuku,	Professor of Global Health Nursing at Hiroshima University, Japan	Science Diplomacy for Distributive Justice: Lessons from the World Forum for Women in Science – Japan
Mrs Alexandra kinias	Women of Egypt	Egyptian Women: The Power of Purposeful Networking
Ms Paloma Marín-Arraiza	Director of Engagement at ORCID	Data-driven decisions based on ORCID trust markers
Dr Nerina Finetto (Moderator)	Nerina Finetto Founder and CEO Traces&Dreams AB	"Not an Accessory: Why Communication Must Be Foundational to Science"
Prof. Roberta D'Alessandro	Netherlands- Head of the Linguistics Division at Utrecht University	Your Academic Auntie: mentoring and supporting early career researchers globally
Dr. Noushin Arfatahery	Founder & Director, WISEA Germany	Empowering Women Scientists as Drivers of Inclusive Innovation and Sustainable Development

THURSDAY, 10 SEPTEMBER 2026 | TECHNICAL SESSIONS — BIOLOGY

14:00-17:00

1-Biology-Medical sciences

Prof. Folasade M. Olajuyigbe & Prof. Jibin Sun

14:00-17:00

Speaker Name	Affiliation / Role	Presentation Title
Prof Folasade M. Olajuyigbe (Moderator)	Federal University of Technology, Akure, Nigeria	Enzyme Biotechnology: Catalyzing Sustainable Solutions for a Better Future
Dr. Amna Mehmood	Senior Scientist, Martin Luther University, Germany	Clinico-Genomic Characterization of Breast Cancer: From Whole-Exome Sequencing to Precision Oncology
Ms Hayat Daeia	Dalhousie University, Canada	Brain-Computer Interfaces (BCI): Machine Learning Classification of Neural SSVEP Signals
Dr Zohra Aloui-Zarrouk	Institute of Pasteur , Tunisia	Bridging Science and Entrepreneurship: The Emergence of Zora Pharm as a Scientific Spin-off for innovative Plant-based Dermaceuticals
Dr Bishayr Hassan Aljaffar	King Fisal General Hospital, Saudia Arabia	Predictors of Exclusive Breast feeding in Employed Mothers
Dr Hameeda Matooq AlJanabi	Dammam Health Network Eastern Health Cluster, Saudia Arabia	Knowledge and Attitudes Toward PICC Line Care Before and After Specialised Training in the Dammam Health Network, Saudi Arabia
Dr Anwar Omar Dawod	University of Dohuk - College of Pharmacy, Iraq	Monitoring hospitalized elderly patients and the use of polypharmacy in Dohuk: a biochemical evaluation
Prof. Jibin Sun (Moderator)	Professor, Chinese Academy of Sciences, China	Industrial Synthetic Biology, Biofoundry Platforms, and Systems Biotechnology
Dr Iman Abdulmanaf Taha	University of Dohuk, College of Science,,Iraq	Biochemical and Developmental Outcomes in Iraqi Children with Phenylketonuria Following Late Diagnosis and Dietary Intervention.

2- Materials & Engineering

Speaker Name	Affiliation / Role	Presentation Title
Prof.Dr. Abdallah A. Shaltout (Moderator)	Spectroscopy Department, Physics Research Institute, National Research Centre, Egypt	Steel Slags Recycling
Ass. Prof. Samar Said Mohamed	Egyptian Petroleum Research Institute, Egypt	WO3 Anchored on FAU(Y)/AlSBA-15 Binary Supports: Impact of Hierarchical Structure on Catalytic Dehydrogenation
Dr. Elzahraa Ahmed Ramadan Elgohary	Photochemistry Department, Chemical Industries Research Institute, National Research Centre (NRC), Egypt	Photocatalytic studies on using nanocomposites based on modified titania for CO2 fixation environmental applications

3 -Environment & Agriculture

Speaker Name	Affiliation / Role	Presentation Title
Prof. Amna Jrra (Moderator)	Independent Researcher, Climate Modeling Expert, Jourdan	Projecting Climate Hazards in Azraq, Jordan: Multi-Scenario Changes in Extreme Events and Socioeconomic Vulnerability
Pof. Dr. Jianwei Zhang	National Key Laboratory of Crop Genetic Improvement Huazhong Agricultural University Wuhan, China	Toward Green Nutritious Super Rice -- A Rice Pan-genome System and Its Implementation in Genomic Breeding
Dr. Sarra Hechmi	Water Research and Technology Center (CERTE), Tunisia	Scaling Safe Water Reuse in Tunisia: Balancing Agricultural Imperatives against Emerging Contaminant Risks for SDG 6
Ass. Professor . Dina Mohamed Salama Ahmed	Vegetable Research Department, National Research Centre (NRC), Egypt	Nanotechnology in Vegetable Production: Enhancing Productivity and Quality for Sustainable Agriculture
Dr. Heba Kamal El-Kholly	Water Pollution Research Department at the Environmental and Climate Change Research Institute, National Research Centre, Egypt	Synergistic Fe2O3 nanoparticles and chemical pretreatment for enhanced biohydrogen/biogas production from pulp sludge

THURSDAY, 10 SEPTEMBER 2026 | ONLINE DAY CLOSING

VIRTUAL FORUM

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- Virtual poster presentations and interactive Q&A sessions.
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In-Person Venue Details (8–9 Sept)

Location: National Research Centre, Egypt

Key Halls: Hani El-Nazer Hall | Main Hall | Hall A

SCAN

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